The decision about a person: who looks, who signs, who pays

Lecture 7

Today, someone's scan will be read before a person ever looks at it.

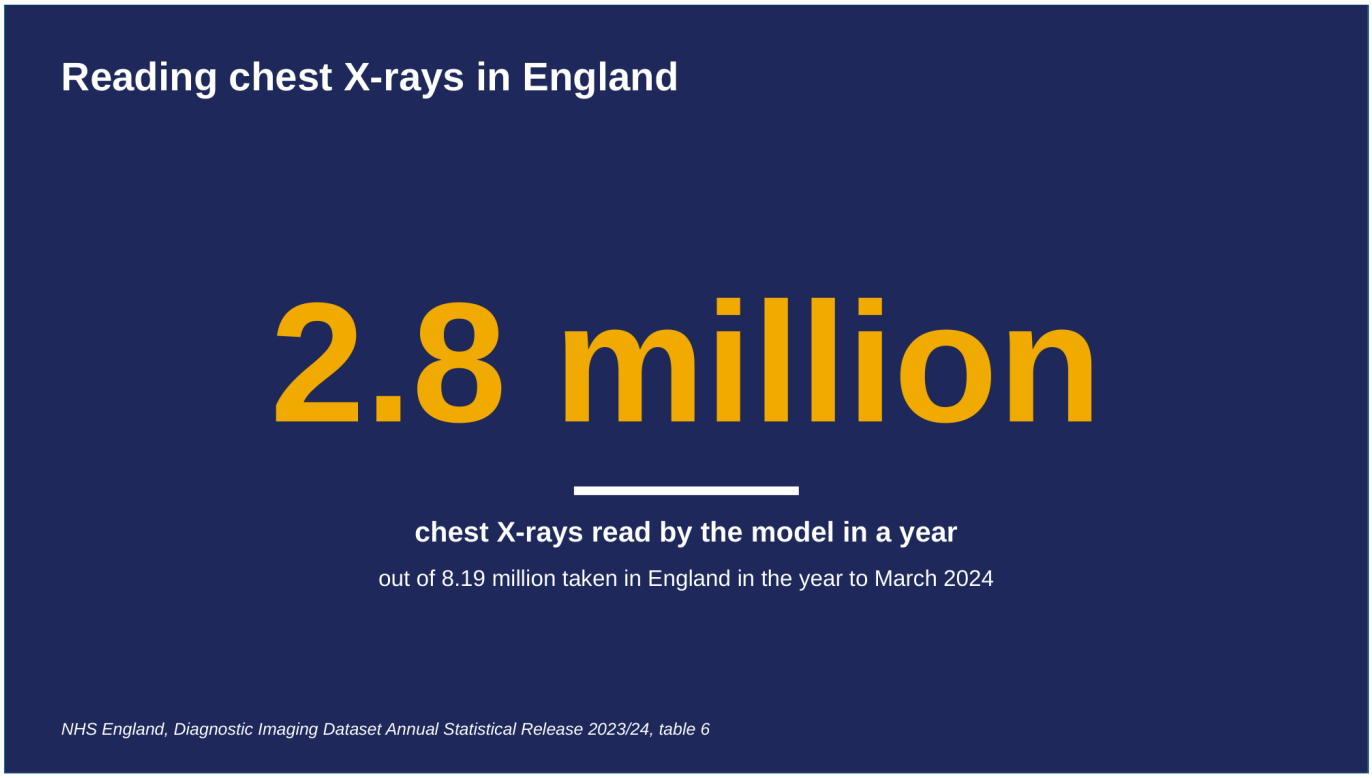
The lecture examines medicine: eight stages of the care-delivery cycle, a trained model at each one, and the person who signs off after it. The decision made here is a decision about a person, and that is what sets the cost of error.

How the subject differs from the last lecture: there, what accumulated was counted in units of product; here, it is counted in people. A batch of defective parts gets recalled; a clinical decision, once made, cannot be recalled, even when the mistake is caught right after. That changes the very framing of the question of the cost of error, and the course moves into medicine second, so that the cost of refusal becomes visible by contrast with the industry already covered. The subject itself stays the same: the same pair of questions — what a miss costs, what a false alarm costs — the same method of analysis through the cost of both errors. The boundary of acceptable risk is drawn in a different place, and abstract reasoning does not show that.

What "the procedure holds down the cost of error" means: double reading of a scan, a colleague's second opinion, a case conference for a disputed case, informed consent before an intervention — not one of these devices was invented for artificial intelligence. All of them existed long before it and solved one problem: so that one person's mistake would not become the final decision without a check. The model enters a procedure that is already in place; it does not build one of its own. That is why the lecture's question is about the procedure itself and about the person inside it: what happens to them when a participant appears beside them who does not get tired, does not doubt, and does not ask for a pause.

What is worth taking away, even if every number from tonight is forgotten by the end: the question has to be asked concretely — what exactly does the person do, in each specific case, after the model has already looked. The lecture spends the whole evening

answering exactly that question, case by case. It does not hand out grades to the tools — it shows what each tool did to the work around it.



In England, in the year ending March 2024, 8,194,405 chest X-rays were taken — across all referral routes, not just suspected-cancer cases. A trained model read about 2.8 million scans out of that flow. Roughly every third chest X-ray in the country is now opened first by something other than a person. That is the evening's first number, and it settles the question of whether this is already happening at all.

The numerator and the denominator come from different hands, and that is worth knowing before stating the share. 8.19 million is official health-service statistics, an annual report on diagnostic imaging, a table with a published counting method. 2.8 million is the figure named by the tool's developer. The government does not give its own numerator at all: it states something else — the tool is in use in roughly half of England's trusts, and four million people got either a diagnosis or the lifting of a suspicion faster.

And the denominator itself has a second official version. A government announcement about the same rollout speaks of "over seven million chest X-rays a year." The gap between the two government figures is more than a million scans, and the resulting share moves between thirty-four and forty out of a hundred. The figure on the slide is the pair whose denominator is published as a table; there are, in fact, two official denominators, and the room has a right to know that. These two bases are examined in more detail in the conversation about instrumental diagnostics — along with why the share of trusts and the share of scans do not have to match: a trust can be connected and still send through only a small part of its flow.

From here, the whole evening is about one thing: what the person does within this picture. The scan has been read, and not by them. The model's accuracy will get its own number with its own base in every breakdown; the evening's question is what

happens to the procedure that has, until now, held down the cost of error — double reading of a scan by two independent readers, a colleague's second opinion, arbitration on disagreement, a signature under the report, a case review after the outcome. Not one of these devices was invented for machine reading, and none was invented against it; all of them exist because one person makes mistakes, and the procedure already knows that.

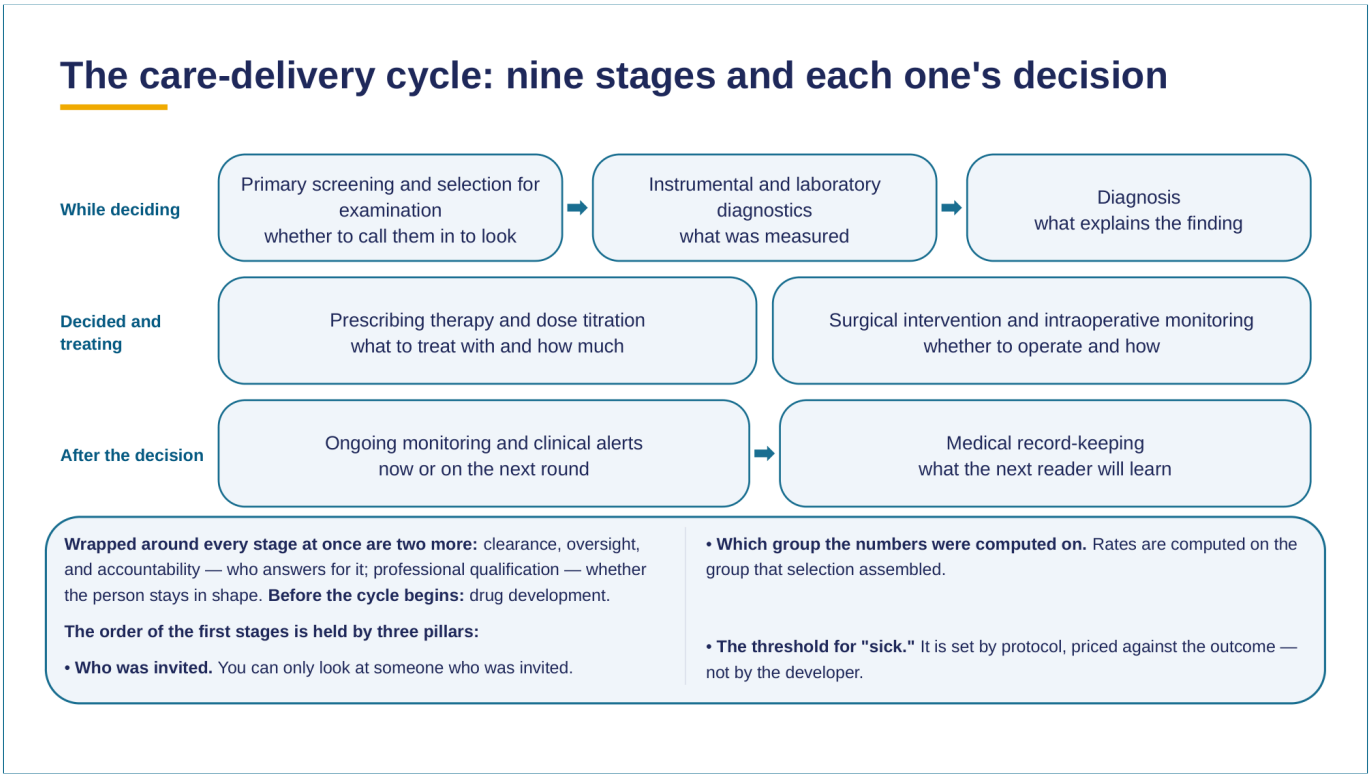
A large number, on its own, says nothing about either benefit or harm. Two-plus million scans read is not "the model saved two million people" and not "two million people were left without a doctor." It is a volume, and only a volume. What exactly changed for the doctor, the patient, and the department — the volume does not say; it says that the time has come to answer that question.

Russia belongs in the same picture, and its numbers are worth stating alongside. Moscow's radiology diagnostic services have, in cumulative total, processed more than eighteen million studies, about 1.4 million a month, and roughly eighty percent of them come from the regions. Nobody here has a share of the nationwide flow: no one publishes the number of scans done per year across the country, and the "about a quarter" estimate circulates in official statements with no denominator at all. An estimate from a denominator that does not exist is not an estimate, which is why England stands first on the slide: it has both halves of the fraction.

If the room asks about the United States — figures of this kind are not published there at all. Two other quantities are published: how many devices with a trained model have been cleared over the entire history, and what share of doctors say they use a model. Neither has anything to do with the share of the flow actually read, and they cannot be added to the English share.

What to take from this slide: the industry is already living this way. The evening's question is what the person does now, and who measured that.

Sources. NHS England, Diagnostic Imaging Dataset Annual Statistical Release 2023/24, table 6 · UK government announcement, 10 June 2026 · Vasilyev et al., *Digital Diagnostics* 2025;6(1):5–22



The care-delivery cycle is laid out here in full: nine stages, each with its own decision about a person, and three pillars that hold the order of the first of them. The stage names used are the ones the industry itself uses: an engineer walking into a medical project will hear exactly these words, and the conversation will start with them.

The first band is while deciding. **Primary screening and selection for examination** answers one question: whether to call this person in to look at all. Screening is examining people with no complaints, by a formal criterion, most often age and the interval since the last exam. **Instrumental and laboratory diagnostics** answers a different question: what has been measured about the person, and with what accuracy. **Diagnosis** brings together complaints, examination, and test results, and answers which illness explains what was observed and which test to order next.

The second band is decided and treating. **Prescribing therapy and dose titration** answers what to treat with, how much, and whom it's fine not to treat; the dose is refined against measurements, and this adjustment is repeated many times until the indicator settles within the target range. **Surgical intervention and intraoperative monitoring** decides whether to operate, by what approach, and what to do on the table when the picture turns out to be different from what was expected.

The third band is after the decision. **Ongoing monitoring and clinical alerts** answers a question that repeats: intervene now, or on the next round. What's read here is the direction of the measurement trend: pulse on its own says little; a rising pulse with falling pressure says a lot. **Medical record-keeping** answers a question about the next reader: what should the person who wasn't at the visit, and who will be the one making the next decision from the record, learn.

Two stages don't fit into this row, and lining them up in it would be a misrepresentation.

Clearance, oversight, and accountability is a procedure that stands around every stage at once: a signature under a report as a legal act, informed consent, a case review after the outcome, clearing a device for use and overseeing it after clearance. The decision this stage makes sounds like this: who is answerable for what was done, and what changes in the order of work when a review finds a discrepancy. It cannot be placed as a separate step in the cycle. **Professional qualification: upkeep and wear** is a measurement of the person themselves: whether they're staying in shape. The profession doesn't ask the doctor about this — it counts on their behalf, through practice volume, accreditation, and recertification. No step of the work itself shows this measurement at all.

And one more stage stands before the whole cycle begins. **Drug development** answers the question of where the thing being prescribed at the prescribing stage even comes from in the first place: a target in the body, a molecule that acts on it, preclinical work on cell cultures and animals, then phases of trials on people.

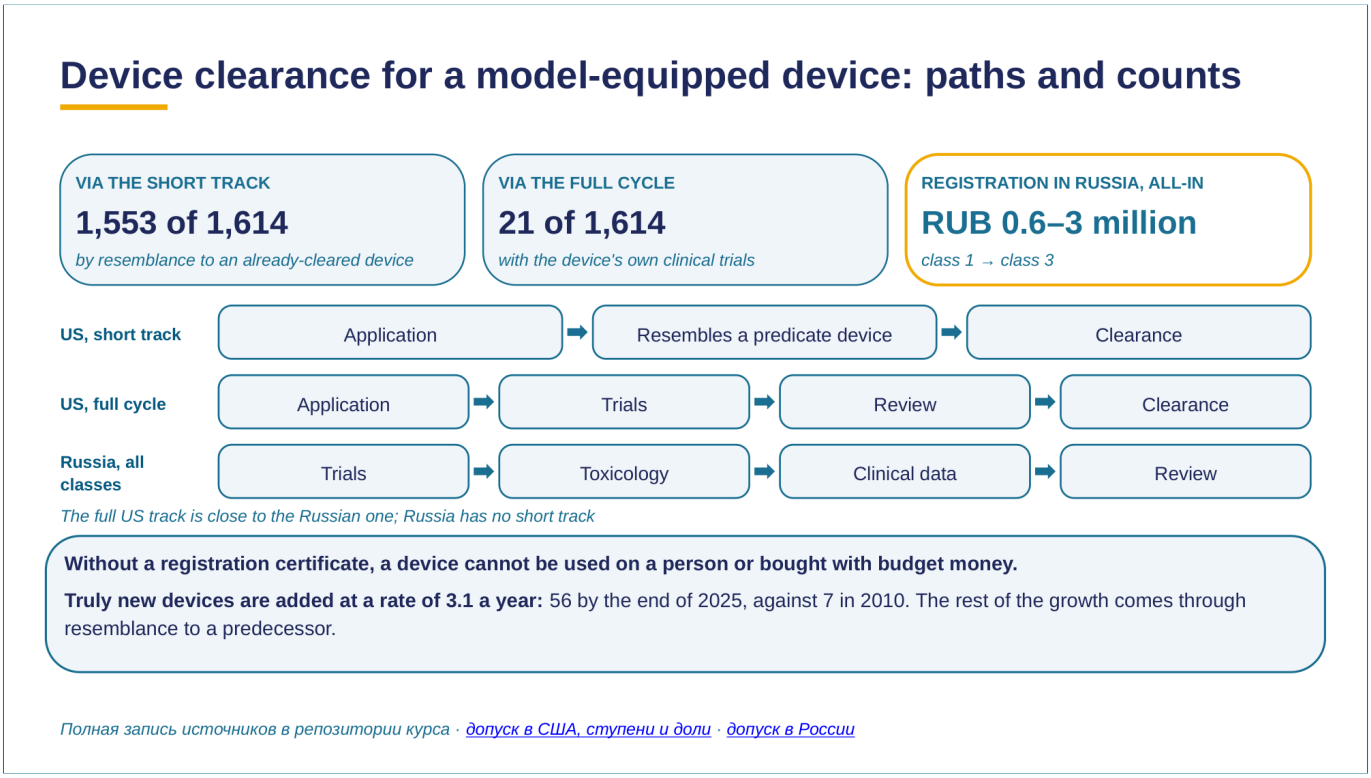
Now about the order, because it isn't decorative and it isn't a convenience of exposition. The order of the first stages is held by three pillars, and each can be checked.

Who was invited. You can only look at someone who came into view. A person who wasn't invited is seen by no system, however accurate it is. So the decision about whom to invite is made before all the others, and it determines who even becomes the subject of the decisions that follow.

Which group the numbers were computed on. The group on which a measurement's properties are computed is assembled by the selection stage. The rates used to describe any "sick or not" system are computed on different groups of people: one on those who have the disease, another on those who don't. The share of true alerts among all the alerts raised depends on how prevalent the disease is in the group examined, and it is not a property of the device at all. The same system, in a mass screening and in a specialized department, will place a completely different picture on the doctor's desk.

The threshold for "sick." The point past which the system declares "sick" is chosen by protocol, priced against the outcome — against a price set before any accuracy at all: which is worse, missing a finding or calling in someone unnecessarily. That choice is made before the developer, and the developer does not overrule it.

Further on, the order holds the same way. You can only decide on what has been measured. Monitoring only makes sense after a decision has been made. You can only read a record after it has been made. There is no reordering under which this still works.



Device clearance is a question an engineer on a medical project hears before the question of the model's accuracy. Until a device is cleared, it cannot be applied to a person at any level of accuracy; after clearance, nobody revisits the accuracy until the next change to the device.

Let's start with the American side, because more machine-learning devices have gone through it than through any other. There, devices are split into three classes by risk to the person; the class is assigned to the type of device, and it is what determines the mandatory clearance track — the path is chosen long before the developer writes the first line of code. On the short track, the manufacturer does not prove safety and effectiveness from scratch: it proves that its device is substantially equivalent to an already legally marketed one. The comparison covers intended use and technical characteristics, sometimes bench data, rarely minimal clinical data; this track almost never requires prospective trials, nor a long-term patient outcome. The full cycle requires the device's own clinical data and a full manufacturing dossier, and for significant applications, also a public hearing before an outside panel.

The Russian track is built differently at both ends. There are four classes, and they are assigned by duration of use and invasiveness, not by intended purpose: two devices with the same intended use can get different classes. Toxicology studies are mandatory at every class, including the first. Clinical trials come in two forms, and this part is worth reading straight from the order's own wording: trials involving people are required when the device is of a new type, when new, complex, or unique methods are used, and when an analysis of clinical data has not confirmed effectiveness and safety. Further on, verbatim: "In all other cases, clinical trials of medical devices are conducted in the form of an analysis and evaluation of clinical data." The gate here is the novelty of the device or method; there is no mention of risk class in this clause at all, nor of whether a

registered analogue exists. And there is a last stage the short American track does not have at all: a state review of quality, effectiveness, and safety, commissioned by the oversight body, not by the manufacturer's own application. The basis for registration is phrased with the conjunction "and": trial results *and* review.

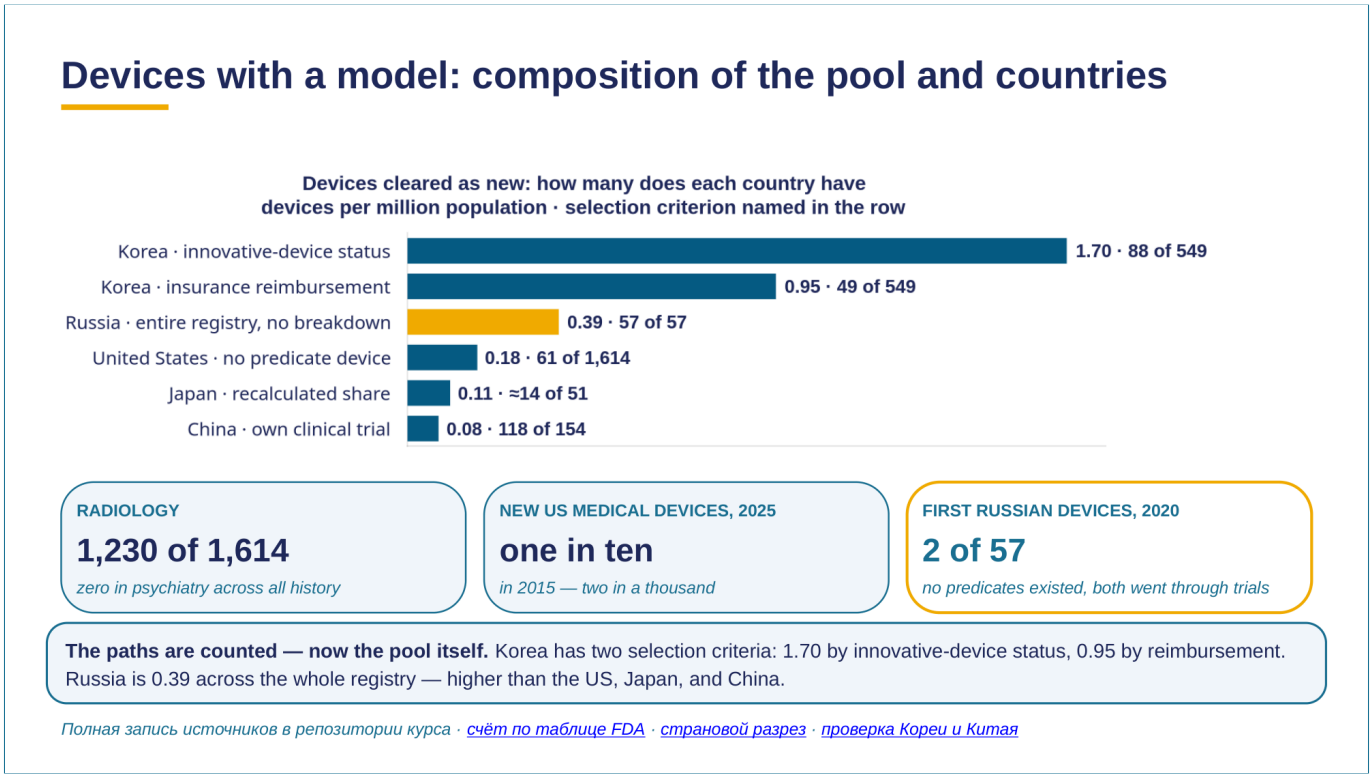
The price in money. For review on the short track in the US, the applicant pays $26,067; for the full cycle, $579,272. The Russian review fee is tied to the class: from 72,000 to 184,000 rubles plus 11,000 for issuing the certificate. The figure on the slide, 0.6 to 3 million, is not the fee — it is the full "turnkey" cost of registration: an estimate from registration intermediaries, in which the fee is only one part, the rest being the lab, toxicologists, and dossier preparation.

Now the count of devices, and the figures are taken by direct reading of the regulator's own table, not from an industry digest: digests on this subject disagree with each other by hundreds of devices. In the list as of 17 October 2026 there were 1,614 devices with a trained model, and every share from here on is counted from that number. 1,553 went through the short track — 96.2%. 40 got a new class with no predecessor. 21 devices went through the full cycle with their own clinical trials — 1.3%, while across all medical devices over the system's history, the share going through the full cycle is about 1.87%: for devices with a model, it is even lower than average. The practical meaning of this fraction is this: a typical model reaches the clinic by proving resemblance to an already-cleared device, while nobody measures patient outcomes along the way.

And about the figure on the slide — 3.1 truly new devices a year. If you remove the short track and count only devices cleared with no reference to a predecessor, the cumulative count gives 7 by 2010 and 56 by the end of 2025. Plus 49 over sixteen years, 3.1 a year on average, and the line does not grow year over year: 7 · 2 · 7 · 10 · 6. The short track adds to the pool fast; genuine novelty adds about three devices a year, and it is this second line that is worth keeping in mind when thousands of cleared devices are quoted.

If asked right now why the countries can't be lined up against each other: no single international definition of a "truly new device" exists, and each country counts by its own criterion.

Sources. FDA, AI/ML-Enabled Medical Devices · Ministry of Health Order No. 2n, published in Rossiyskaya Gazeta · Order No. 885n, Ministry of Justice document · Russian Tax Code, art. 333.32.2 · FDA, MDUFA Performance Goals · Korea, count of 549 · Korea, reimbursement 49 of 549 · China, clinical evaluation pathways · Webiomed



In the FDA's list as of 17 October 2026, there are 1,614 devices with AI or machine learning, and this slide breaks down what the pool is made of: by specialty, by share of the flow of new medical devices, and by country. The figures are taken by direct reading of the regulator's own table: industry digests on this subject disagree with each other by hundreds of devices. Every share from here on is counted from 1,614.

The breakdown by specialty explains what "AI in medicine" actually means. Radiology — 1,230 devices, 76.2%. Cardiology — 154, that's 9.5%. Neurology — 73, 4.5%. After that come anesthesiology, gastroenterology, hematology, pathology, ophthalmology, surgery, and clinical chemistry — each with 9 to 30 devices. Psychiatry has not a single device in the list across its entire history. In other words, AI in medicine as of 2026 is predominantly work with images, and the rest of the specialties together don't add up to even a quarter.

Now the share of the flow. The whole pool of AI-enabled devices is growing fast: 6 devices in 2014, 335 in 2025 — meaning one in ten new medical devices in the US in 2025 has AI, against two in a thousand in 2015. Alongside that sits a counterfactual, without which this growth reads wrong: the entire flow of new US medical-device clearance decisions grew only 1.07-fold over ten years — in other words, stayed in place, at about 3,063 decisions in 2015 and about 3,265 in 2025. AI-enabled devices did not replace something else; they were added on top.

Now the country row, and it has to be read row by row, not by the length of the bars. No single international definition of a "truly new device" exists, so each country is counted by its own criterion, and the criterion sits right in the row's name — it is not a caveat, it is part of the data.

Korea: 88 devices out of 549 cumulatively as of Q1 2026, criterion — voluntary

innovative-device status, which the regulator assigns itself; against a population of 51.7 million, that comes to 1.70. The second Korean row is the same registry, a different criterion: whether the device has gone through insurance reimbursement. There are 49 of those out of 549, and the rate drops to 0.95. Two honest figures for one country, and the gap between them is bigger than the distance between the US and China in this same row. If the room remembers one thing from this slide, let it be this: it is not medicine that sets the order of the countries here, it is the chosen criterion.

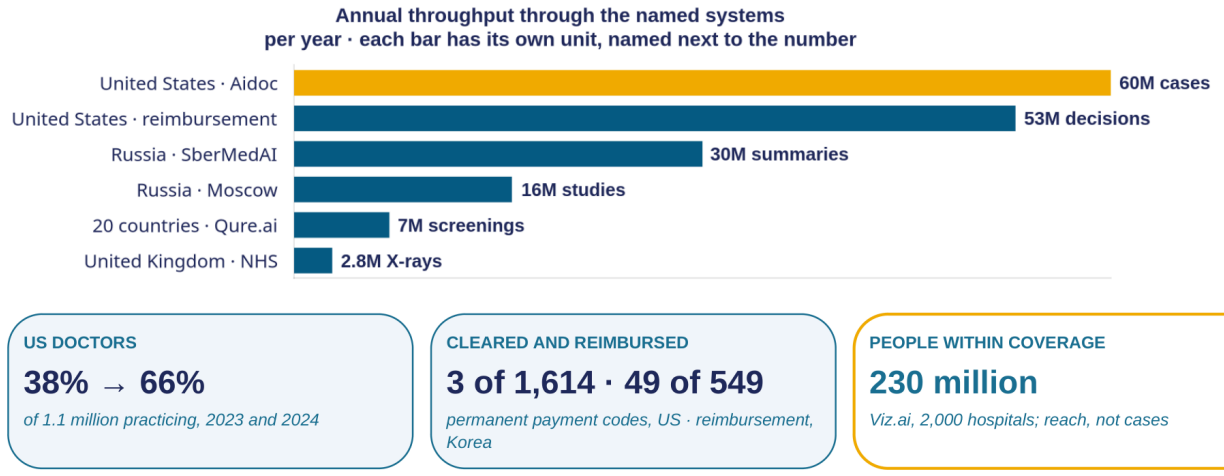
China: 118 of 154 devices went through the company's own clinical trial as the clearance path, and separately for the high-risk class, 116 of 123; against a population of 1,409 million, that is 0.08. Japan: about 14 of 51, and this is an extrapolation of the overall share across all software medical devices onto the part involving AI, not a direct count — which is why the row reads "share recalculated." US: 61 of 1,614 against a population of 342 million — 0.18.

The Russian row stands on a different criterion, and that is stated right on the slide. Roszdravnadzor does not publish a breakdown of how many devices went through trials involving people versus a document-based evaluation, so the numerator holds all 57 registered devices, and against a population of 146 million that gives 0.39. The novelty criterion isn't applied to them — there's nothing to apply it to. What is reliably known: the first two Russian devices, from 2020, Webiomed and Botkin.AI, had no predicate device in the registry, and both went through full clinical trials with patient participation. By 2026, the registry had accumulated more than ten medical-image-analysis products, and for new applications in the same product category, a document-based evidence base already exists. So equating the 57 Russian devices with 57 "truly new" devices in the American sense would be wrong — and that is exactly why the row names its own criterion out loud instead of hiding it in a footnote.

And the reason this row is on the slide at all. Russia sits above the US, Japan, and China on it — that holds under any of the criteria counted here, and it is not a trivial point: in absolute terms it has 57 devices against America's 1,614. But a second thing also holds: its row is the only one to which the novelty criterion is not applied at all. Both statements live on the same slide, and neither cancels the other.

Sources. FDA, AI/ML-Enabled Medical Devices · Korea, cumulative count of 549, industry digest · Korea, reimbursement 49 of 549 · China, clinical evaluation pathways, JMIR Medical Informatics 2026 · Japan, final report on AI-based software medical devices · Webiomed, first Russian AI-enabled device

Using the model: doctors and the flow of studies



US DOCTORS

38% → 66%

of 1.1 million practicing, 2023 and 2024

CLEARED AND REIMBURSED

3 of 1,614 · 49 of 549

permanent payment codes, US · reimbursement, Korea

PEOPLE WITHIN COVERAGE

230 million

Viz.ai, 2,000 hospitals; reach, not cases

Doctors worldwide: 3 million of 13–15 million. Russia: 15–25% of 770,000. Who has a model standing in their loop — and who looks into its failures?

Полная запись источников в репозитории курса · агрегаты внедрения · использование и эффект

How many doctors in the world actually work with a trained model: about three million out of thirteen to fifteen million practicing. In the US the share rose from thirty-eight to sixty-six percent of about 1.1 million doctors; in Russia, fifteen to twenty-five percent of 770,000 apply it regularly. Clearing a device is not yet using it, and the counters involved are different: cleared, installed, in use, cases processed per year. Every figure from here is labeled with which counter it is, because they keep getting added together, producing a quantity nobody has actually measured.

How many doctors use it. For the US there's a series taken with one method: 38 percent in a 2023 survey, 66 percent in a November 2024 survey. The denominator is practicing US doctors, and it itself does not converge to one number: a licensing census gives 1,082,000, the association of medical colleges gives 1,032,000, an independent 2025 estimate gives about 1.11 million. We take 1.1 million as a working average; the eight-percent spread does not change the order of magnitude. In absolute terms that is a rise from roughly 410,000 to 720,000 doctors in a year. Alongside sits a second, stricter count — a survey by the professional network Doximity, winter 2025–2026: 63 percent on the same base. The gap between the two counts is not an error: one counts any professional contact with AI, including a one-off use; the other counts regular use within the clinical process. Both figures are honest.

For Europe there is no direct survey of all doctors; there are two indirect measurements. The European Society of Radiology, 2024 survey: 47.9 percent of respondents use AI in practice, and the denominator matters here — 572 people answered out of 28,000 invited, meaning this is self-selection, not a random sample. The British radiology department census of 2025: AI in at least one application is present in 75 percent of departments, but the counter here is the department, not the doctor. There's a curious reversal in the same report: a drop in workload from AI-drafted

reports is noted by 24 percent of departments, an increase by 37 percent, and the remaining 56 percent see no change at all.

For Russia there is no single annual survey with one method, and that is a property of the data itself. There are one-off surveys: more than half of doctors have worked with neural networks at least once, 15–25 percent apply them regularly. The denominator is 771,700 doctors in 2025, and the series over the years is not monotonic; the Ministry of Health separately names 614,000 doctors in state organizations, a narrower basket. Over the range of the share, that's 115,000–190,000 people, and this is an estimate from a share, not a direct count of the absolute number.

How many cases pass through the model. These deployments are different categories of work, and that's worth stating before the numbers. Aidoc — 60 million cases of diagnostic triage a year; Qure.ai — 7 million tuberculosis screenings in one fiscal year; Annalise in the British system — 2.8 million chest X-rays a year across forty-plus trusts; the Moscow platform — 16 million studies by the 2026 report (an earlier report the same year gives 9–10 million, and the method isn't disclosed in either). These four are one category, diagnostic imaging, and only they can be honestly added: 60 plus 7 plus 2.8 plus 16 — 85.8, that is, 79–86 million studies a year accounting for the range in the Moscow platform's figures. Beyond that are two categories of a different kind of work, and they are not part of this sum: the digital doctor's assistant from SberMedAI — 30 million status summaries, which is documentation, not diagnosis; payment decisions in Medicare Advantage — 53 million, which is an administrative decision. Adding all six into one figure would be exactly what this lecture's screening chapter warns against: the same phrase, "passes through the model," here means different procedures with no common unit. Viz.ai is not included in the diagnostic sum for a different reason: it sits in the same American hospitals as Aidoc, and adding it would double-count part of the same scans. I'll name Viz.ai's own figure separately: two thousand hospitals and 230 million people within its coverage area — that is the reach of the service area, not cases per year.

Share of the total flow. The global volume of diagnostic imaging is 3.6–4.2 billion procedures a year: X-ray over 2.5 billion, ultrasound over 500 million, CT about 375 million, MRI 100–150 million. Against that, the diagnostic subtotal — 79–86 million — is 1.9–2.4 percent of the global flow. And this is the floor of the count, not the ceiling: it does not include China, where more than 1,200 top-tier hospitals use AI diagnostics but the volume flowing through them is not disclosed; it does not include most of automated appointment scheduling — a category with about sixty companies and not one industry-wide total; it does not include Korea or Japan.

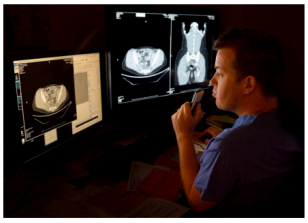
And the category that touches more people at once than any diagnostic platform: administrative. 53 million prior-authorization payment decisions in 2024, against 34.1 million people insured under Medicare Advantage — that's 1.55 decisions per person a year. Not all of them went through an algorithmic check equally, but the order of magnitude is clear: AI in medicine does not reduce to diagnosis.

The last two figures are about the gap between clearance and payment, and

everywhere it is measured in orders of magnitude, not percentages. US: 1,614 cleared algorithms, some payment code exists for 26, a permanent code with guaranteed payment exists for three. Korea: 549 cleared, 49 got reimbursement — meaning 500 devices are cleared and stay outside insurance. Germany: 56–59 digital applications are paid permanently or provisionally, but there's no published denominator of submitted applications in open sources. France: three telemonitoring devices got reimbursement, and all mobile apps were rejected — for a whole class, that's zero. I state the German and French rows without a share deliberately: their denominator isn't published, and a share of an unknown denominator measures nothing.

Sources. AMA, AI use among US doctors · Doximity, State of AI in Medicine 2026 · European Society of Radiology, 2024 survey · British departmental census, 2025 · Russia, overview of physician surveys · Aidoc, 60 million cases a year · Viz.ai, two thousand hospitals and 230 million people · Global imaging flow · Medicare Advantage, insured population · Reimbursement for AI devices in the US · Korea, reimbursement 49 of 549

Six distortions at the front door: the names



Physician

Patient

- **Anchoring.** The conclusion shifts toward the model's suggestion, even when it is plainly wrong.
- **Narrowing.** Where the model stays silent, the eye does not linger.
- **Skill decay.** Once used to the model, people find less without it.
- **Alert stream.** Alerts come one after another, and so does dismissing them, one after another.
- **Disagreement.** A mismatch with the model is now handled as a separate step, and that step is new.
- **Correction.** The one that works in our favor: the doctor catches the model's mistake and fixes its report.

Полная запись источников

The first thing worth saying, looking at the two photos: the model doesn't take a seat beside the visit, it takes a seat inside it — between the doctor and the decision — and the patient either knows that or doesn't, and their behavior changes with it either way. That is why there are two sides, and they were measured by different means: for the doctor, behavior on the task; for the patient, agreement with the decision. These two measurements cannot be added together.

This slide hands out the vocabulary that marks up the rest of the evening. There are six names, and deliberately no numbers under them yet: every number will be presented where the mechanism is first encountered, with its own sample and its own base, and tallied in the section summaries — which of the six a given section ran into. Naming them in advance is so that, by the time the mechanism is encountered, it is recognized rather than explained from scratch in the middle of a breakdown.

The frame behind these mechanisms is not ours. Three of them come from a study of distortions in human–model interaction, verified across three different fields on more than two thousand people in total, and the entire evening's medical measurements fall onto two of the three as a special case. Worth saying outright: medicine is not special here, and the objection "that's how doctors are, because it's medicine" is not knocked down by one medical measurement — it's knocked down by the fact that the same mechanisms are measured outside medicine too.

First of the three is bias amplification, and it is the only one with a control arm. People labeled facial expressions, their answers were fed to a convolutional network, it amplified the original bias, and alongside it a fresh group of people worked with the model's output: the share of biased answers rose from 49.9 to 56.3 out of a hundred. In the control arm, other people stood in for the model — there the share did not change:

50.6 against 51.5, no difference. What amplifies is not agreement with someone else's opinion in general — it is specifically the model. And the amplification grows as the work goes on: 50.7 out of a hundred in the first block, 61.4 in the last. Second is anchoring: 775 managers rated employees' work after seeing a suggestion, and their ratings pulled toward it. That study doesn't print a magnitude, and substituting a similar number from a different study is not allowed. What it does name is the move that breaks anchoring: requiring the person to consider the opposite. That is not the same as a warning: a warning is a message that gets noted and then ignored while work continues as before, while considering the opposite is an action the person is required to carry out. Third is inheritance: 666 participants took a critical-thinking test with no access to the model, and the link between how often they used it and their score turned out negative and significant.

Why this isn't visible from the inside is a question the room asks itself. In that same first study, participants were asked how much each algorithm had influenced them, and they rated the biasing one as less influential than the accurate one — exactly backwards. And with no model at all, a person corrects their own bias on their own: over the course of the trial the share of biased answers fell from 56.7 to 51.9. The capacity for self-correction is there; next to a model it stops working. And the honest other half of the same measurement: an accurate algorithm raised participants' own accuracy. The slide isn't saying the model is harmful — it's saying where to look.

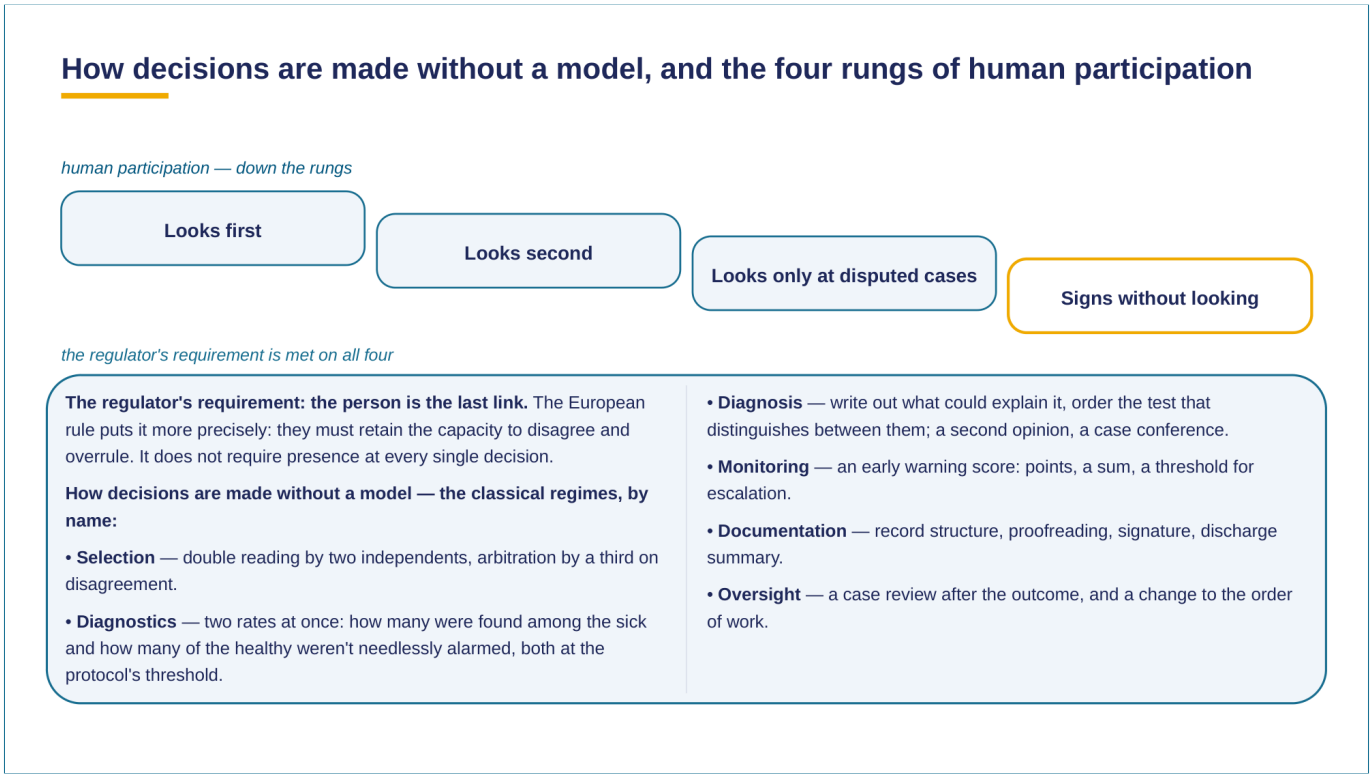
Now the other side — the patient. The evening does not give these measurements a breakdown of their own, which is why they stand here. For the doctor, what was measured was behavior on the task: reads, concludes, finds or misses. For the patient, what is measured is agreement — how they respond to the bare fact that a model was involved in their case. And first: agreement is not one number. Disclosure — 1,762 adults, representative of the US population: when a patient is told the doctor uses a model, trust in the doctor falls from 0.63 to 0.30 on the scale used, and a doctor's flat refusal to use the model earns them 13 points on top of what they'd have if it were never mentioned at all. Loneliness — 518 women who came in for a breast scan: only 23 people agreed to accept the model as the sole reader, 368 of the same 518 accepted it as a second reader — a sixteenfold difference. The patient is against not the model, but its being alone. Asymmetry — 359 participants, both questions asked: a worry raised by the model, let a doctor look — 319 said yes; one raised by the doctor, let the model look — 184. Double-checking is eagerly requested upward and barely requested downward. Price — 248 adults with type 1 diabetes: removing a $50 co-payment raised the choice of a model-based exam from 43 to 81 out of a hundred, and those who chose it still wanted a specialist's double-check even more than before. Price buys agreement; it does not buy trust. Anonymity — 45 people with a cancer diagnosis rated mixed, unsigned answers: they rated the machine's answer's empathy at 4.11 out of five, the doctor's at 2.01. These five cannot be added together: the samples and the questions are different.

And the overall boundary of this whole block, better named going in than excused coming out: not one of these measurements reaches the patient's actual outcome. What was measured was the doctor's work and the patient's agreement; what

happened to the person's health, none of the sources we cite measured. Separately: the effect on the doctor was almost always measured in a lab, where the share of deliberately corrupted suggestions is higher than in real life, while the effect on the patient was measured by a survey about an imagined situation. Both methods get the direction right; they overstate the magnitude.

Six names in front of you. Which of the six, in your view, will turn out the most costly for the doctor — and which for the patient?

Sources. Full record — breakdown of the study on distortions in human–model interaction (the frame of three mechanisms and all 18 sources) · compilation of medical measurements · compilation on patient trust in the model — which also names the two pairs deliberately not combined, and lists what simply was not found at all.



All the work below existed before any automation, and exists now in the same form. The model enters an order that is already in place; it does not build one of its own, and that is why the conversation about it begins with how that order itself is built.

Briefly, about where data on a person even comes from, because there are six ways, and they give different things. A visit to a doctor gives complaints, history, and examination — what the person said and what the doctor saw with their own hands. An instrumental study gives values taken by a device: a plain scan, a tomogram, an endoscopy. A lab test gives blood, urine, tissue, and culture readings — and some of them come with their own built-in delay that nothing removes: a sputum culture takes three weeks. A remote study is set up so that the person is scanned in one place and the image is read by a specialist somewhere else; this is what radiology diagnostics rests on wherever there is no radiologist of their own. Reviewing documents already on file adds no new measurement at all — it extracts from what's already recorded and becomes a source of decisions in its own right. And a sixth way, which over the past two decades has become commonplace: measurements the patient takes themselves — a glucose monitor, a home blood-pressure cuff, a symptom diary. These same six ways are how data is obtained wherever a model works alongside too: the model doesn't change how the data is obtained, it changes who decides on it, and how.

Now the classical regimes, and they have to be called by name: the model will have to be compared against exactly these, and a model compared against an empty space always wins, and that win means nothing.

In selection, double reading is at work. The scan is read by two people, and the word "independently" is load-bearing here: the moment the second reader sees the first one's conclusion, two opinions stop being two — the second reader starts answering "do

I agree," which is a different task with a different error rate. The payoff of double reading comes from the mismatch of errors: the two people's misses land on different scans. Arbitration resolves a disagreement — a third reader, brought in only on disagreement, who wrote neither of the two conclusions. In diagnostics, a pair of rates is at work: of everyone who has the disease, what share did the system call sick, and of everyone who doesn't, what share it didn't needlessly alarm. These two rates are computed on different groups of people and are always stated as a pair, because they move together: shift the threshold, and one rises while the other falls. That's why they're given for one threshold, and the threshold is chosen by protocol, priced against the outcome. In establishing a diagnosis, differential diagnosis is at work: writing out the diseases that could explain what's observed, and ordering the test that will tell them apart; the order of the tests is a separate decision, because some of them are themselves dangerous or expensive. Alongside sit a second opinion, a case conference, and clinical guidelines. In monitoring, an early warning score is at work: every measured value gets points for deviating from normal, the points are added up, the sum is compared against a threshold, and the threshold dictates the action. Two of its properties: the points pass between shifts, while the judgment "I think he's taken a turn" does not; and the weights in the table are set by people and published. In documentation, record structure is at work — complaints, history, examination, diagnosis, plan — proofreading, and a signature. And over all of it stands oversight: a case review after the outcome, and a change to the order of work as a result.

What all these regimes have in common is one thing, and it's worth carrying into any other industry. Each one exists because one person makes mistakes, and the procedure already knows that. The procedure doesn't make people more attentive — it's built so that one person's mistake is caught before it becomes irreversible, or reviewed after it already has.

The ladder on the slide is the criterion by which all the evening's breakdowns are sorted: who looks first. Looks first — the person reads the data themselves, and the model is brought in afterward, or not at all; this is the classical regime the model gets compared against. Looks second — the first pass is made by the model, and the person reads its result and either confirms or rejects it. Looks only at disputed cases — the person is brought in only where the model and a second source disagree; they never see the rest of the flow. Signs without looking — the requirement is formally met, the signature is there, but there's no actual participation behind it. The ladder doesn't rank the rungs by quality: each one has both a payoff and its own blind spot. But the gap between the first rung and the fourth is bigger than the gap between any two models: an argument about whose model is more accurate happens inside one cell, while the move from "looks first" to "signs without looking" changes the procedure itself.

Why it is the person who signs, and not the model — three reasons, and accuracy isn't among them: it's measured that they are often not more accurate, and a case for them resting on their accuracy would collapse at the very first breakdown. The right to sign is granted to them: in none of the countries examined does a model hold the right to sign, and in a case review it is the person's actions that get evaluated. They err differently: the payoff of a pair comes from the mismatch of errors, not from one side's superiority.

A decision already made cannot be undone: a device gets recalled, and recalled for real, while decisions made on it before the recall stay as they were.

And three checks that work beyond medicine too; the evening will come back to them at the end. First: if a conclusion is checked against the model's own self-explanation, that's not a check — the explanation is built by the same system and is aligned with the conclusion by construction. Second: you need two rates at once — the share of suggestions rejected, and the share of correct rejections; without the second, the first confirms nothing, since you could reject everything. Third: can the person physically keep up — seconds available for the decision against the volume of the flow; sometimes the answer to this is "no," and later it will be given a number. A loop that fails these three checks is exactly the fourth rung: the signature is there, and no participation stands behind it.

Example of a human–AI system at work: 50 doctors read case write-ups

the gap between the pair and the solo human is 2 points, not significant

Three seconds and zero rubles, against 48 seconds of human reading. *The units are different — the comparison is only on order of magnitude.*

How it was measured. Fifty practicing doctors each read five real case write-ups and built out their reasoning: hypotheses, justification, which tests to order. They were scored blind, on a structured-reasoning scale. The two human conditions were randomized; "model alone" was not part of the randomization.

Reading the data got cheaper — it's paid for by a step the human does.

What pays for it: attention — looking after the model · skill — stops doing it themselves · signature — put down without reading · **arbitration — now costs more than the old reading did**. And the check of the suggestion itself has been measured: **1.99 seconds** by median.

The model looked first. What does the person do now — and who measured that?

Goh et al., JAMA Network Open 2024

There's one number I'm naming here that I'll name three more times tonight, because it measures the fourth payment — the one that looks the most harmless. Arbitration now costs more than the old reading did; we'll get to that at the screening stage. But how much the check of the suggestion itself costs has been measured — one and ninety-nine hundredths of a second by median. I'll give the base right away, because without it the number sounds stronger than it is: the measurement is from a lab, and taken outside medicine entirely — twenty-one programmers, one thousand ninety-six code-assistant suggestions shown, behavioral telemetry. This is the time the person keeps the suggestion on screen BEFORE deciding to accept it. The same study has a second median alongside it, six and forty-eight hundredths of a second, and the two must not be confused: that one counts the check AFTER acceptance. The question "does the person have time to read before they sign off" is answered by the first. Carrying this over to a doctor is done by analogy, and I'm calling it an analogy.

Reading medical data has come to cost three seconds and zero rubles — against 48 seconds of human reading on average, 25 by median. The units behind these two figures are different: a chest X-ray and a mammogram are read differently, so the comparison only holds at the order-of-magnitude level, and that's exactly how it's labeled on the slide. The tool opens in a browser, the computation runs on the same laptop, the scan never leaves for a server, and there's no barrier to entry at all. What follows is what sits on the other end of this cheapening, and there are four addresses where the payment lands.

The trio on the row of plates is one measurement, and it has to be named in full. Fifty practicing doctors read written descriptions of real cases — complaints, history, test results already in hand — five cases per doctor, and built out their reasoning: hypotheses, justification, which tests to order. They were scored blind, on a

structured-reasoning scale, and two human conditions were randomized: the doctor as usual — seventy-four out of a hundred; the doctor on top of the model — seventy-six. The two-point difference is not statistically significant: p equals zero point six zero, meaning a gap this size happens freely by chance, and it cannot be called a gain. The model running alone — ninety-two — was not a randomized arm and carries less weight as evidence. From here comes the short formula by which the whole lecture reads: ninety-two was bought, seventy-six is what actually works, and the difference between them doesn't show up in the reporting. The trial names the reason for the null result itself: the suggestion arrived ready-made, and no one taught the doctors how to question it and when not to trust it. The tool was handed over; the procedure for working with it was not.

Attention. The person looks after the model, and their visual search narrows — this has been measured by eye movement, and in both directions: with a correct suggestion, the area of the scan scanned by the eye shrinks, and reading quality goes up; with a suggestion that missed a finding, the area scanned shrinks the same way, and reading quality collapses.

Skill. The person stops doing on their own what they'd done for years. This was measured by disabling the assistant: the detection rate in procedures done without it, among those accustomed to it, turned out lower than it had been before the assistant ever appeared. The body of evidence here is weaker than for the other three, and it's worth naming that weakness plainly: in real work, the consequence has been measured once, and it's disputed; the same phenomenon has been measured by random assignment in a training setting, but on real work there isn't a single replication.

Signature. The person confirms a decision they didn't read. The requirement that a person remain the last link is written down by the regulator; a check of that requirement itself was found at 56 of 717 radiology devices — eight percent.

Arbitration. Resolving one disagreement now costs more than the old reading did: a scan takes 48 seconds on average, 25 by median — resolving a disagreement on arbitration takes two minutes twelve seconds, that is, five readings. And as first human readings were cut in half, the volume of arbitration grew by more than half.

Half of this axis is already familiar to the course from the last lecture: a step gets cheaper, and the bill is paid somewhere else. What's new here is one thing, and it's the whole reason the axis is announced — the address of the payment. It's paid in the step a human does, not in money, and not in compute time.

Now about how these four words keep living on, because hearing them once isn't enough. The course already knows the vocabulary of forms the process can be affected by, from the last lecture, and its nine names carry almost every breakdown that follows: "the load moved to a different step," "the operator changed," "the cost of error shifted." The four payment addresses don't replace or repeat that vocabulary — they answer a different question about the same case. A form says what happened to the work: where it now sits, how much of it there is, whose hand does it, what an error ends in. An address says at which human step this happened: do they look second now, have they

stopped doing it themselves, did they sign something unread, are they handling a disagreement as a step that's new.

From here comes the order every breakdown follows from now on. Each one ends with a form — or with a named refusal to assign one, and that refusal is as much a verdict as a name is. In the summary, forms are mapped back onto the four addresses, and that mapping is done by count: against every address stands a measurement with its stated order of magnitude. Both halves need to be held in mind: a form measures the work, an address measures the person. They must not be mixed up: work can move in a way no human step ever feels, and a person can pay where the work never moved at all.

Sources. Goh E. et al., JAMA Network Open 2024;7(10):e2440969, 28 October 2024 · photo: NIAID, Wikimedia Commons, CC BY 2.0

The relocation of work has nine forms, and the full list will be presented at the first breakdown.

1. Primary screening and selection for examination

Who to look at in the first place. Screening is examining people with no complaints, by a formal criterion. Before anyone gets treated, someone decides who's even worth looking at.

In order:

Example 1, without a model: a Swedish program calls people in by age

Example 2: a first-line chatbot replaces a call to the practice

Example 3: the retinal fundus is photographed at the visit

Example 4: Moscow, Russia — a model stands in as the second reader for mammograms

The front door ended on an axis and a vocabulary: reading the data got cheaper, and the bill is paid in attention, skill, signature, and arbitration; nine forms of relocated work were handed out before their first use, so they could be called by one word from here on. And it ended on a question that will now be asked at every stage: the model looked first — what does the person do now, and who measured that. Here that question is asked for the first time, and it's asked at the very first stage of the path, because there's nothing before it: you can only look at someone who was invited. The woman with the letter by age, the one the lecture opened with, stands right here — she's been invited, and what follows is a look at who decides how to call people in, and whom to look at more closely.

The makeup of the section: four breakdowns of one piece of work from four different angles, and one of them is a failure — the first-line chatbot, the only case in the lecture where the test "take the model away" answered "the service keeps running." Each of the three takes up more than one slide: for the first, the cost of arbitration is counted separately; for the second, how it's built; for the third, both.

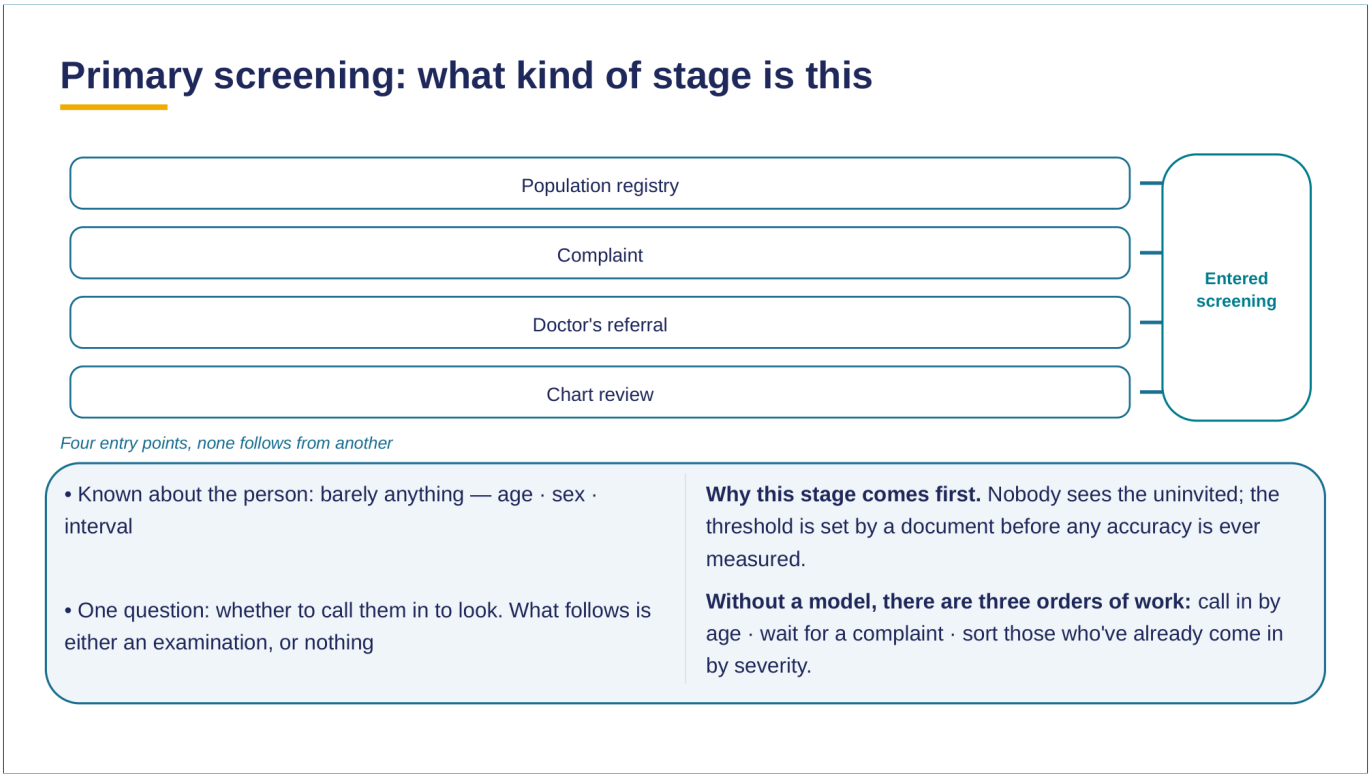
Further on the lecture examines decisions about a person who has already come into view, while here it's being decided whether they come into view at all — and that decision is made before any diagnosis, often with no doctor, and always with no participation from the patient at the moment it's made.

Four breakdowns show one piece of work from four sides. The Swedish program stood inside an already-running invitation channel and decided not "refer or not" but "who gets the second human reading"; it's examined through what a randomized trial measured, and through what it didn't measure at all — the construction of arbitration itself and the asymmetry of trust in a finding, as a separate slide right after. The

first-line chatbot stood at the entry point of the flow of inquiries and decided whom to route where; it's examined through what was publicly claimed, and through what a cheapened entry point did to the channel's economics. The retinal fundus at the visit is a case where the payoff is measured not in reading accuracy but in turnout: 22 out of a hundred against a hundred out of a hundred. The numbers and bases are on the slides themselves.

What's common across the three breakdowns isn't the topic or the industry — it's what all three show: selection decides not just "who to treat" but, before that, "who's even worth noticing," and a mistake at this step isn't visible where it's usually looked for, because it happens before any diagnosis and before any accuracy measurement.

The photo alongside — a technician positioning a person at the mammography machine: the exact minute an invitation by age turns into a procedure. The woman with the letter stands right here, right now: the decision about her was made not by her and not by a doctor, but by a program — by her year of birth. The work of selection looks like this: not dramatic, but routine — and it's in that routineness that it most often goes unnoticed.



Primary screening and selection for examination — that's what the industry calls this stage, and an engineer walking into a medical project will hear exactly these words. What's known about the person going in is scant: age, sex, and the interval since the last examination. The stage asks one question — whether to call this person in to look at all, and "no" is followed by nothing. The stage has to be described before the tool: otherwise it's unclear what the tool is even intervening in.

Screening is examining people who have no complaints, by a group-membership criterion: age, sex, occupation, a prior diagnosis. The benefit of screening rests on one assumption: a finding made earlier than the complaint changes the outcome. If the disease is treated the same way whether found early or late, screening remains pure cost plus the harm of false alarms.

Data about the person at this stage is scant, and that's its main property. Age. Sex. Interval since the last exam. Sometimes a questionnaire or a single complaint the person brought in themselves. None of what's usually called medical data exists yet: no scan, no test, no examination. And the question decided on this data is also just the one: whether to call this person in to look at all.

There are four ways to obtain this data, and they give different things. An invitation by population registry is the case where there's no data on the person yet, and they're called in by a formal criterion: year of birth, sex, interval. A person's own request with a complaint runs the opposite way: they came in because they felt something, and a symptom drives what follows; their probability of having the disease going in is entirely different, and in the next section that probability will determine what any positive result even means. A referral from another doctor brings someone else's hypothesis along with the person. Reviewing documents already on the chart adds no new measurement

at all — it extracts from what's already recorded and becomes a source of its own decision.

The decision this stage makes is short: the person either comes into view, or doesn't. What follows the decision is either an examination, or nothing — until the next interval, or until the day the person feels it themselves.

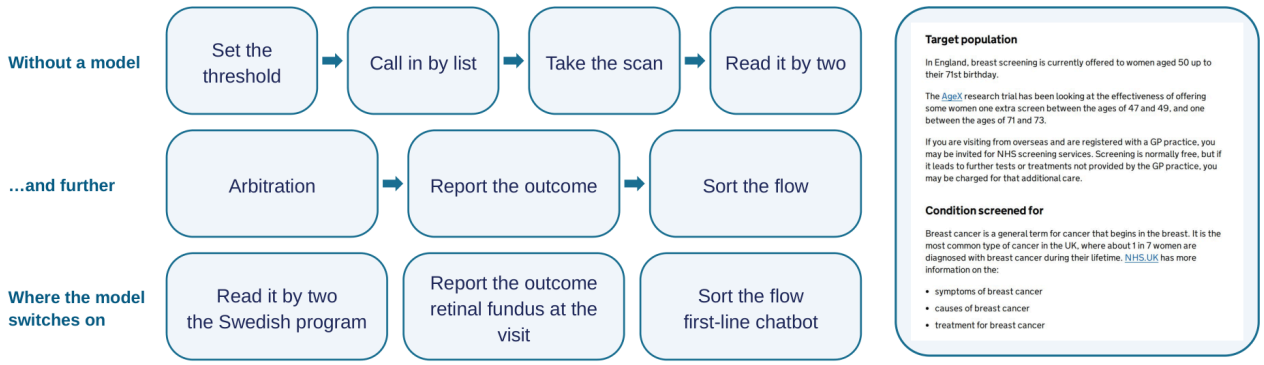
Now about why this stage comes first, because the order here isn't a convenience of exposition. There are three pillars, and each can be checked. First — who was invited: you can only look at someone who came into view, and a person who wasn't invited is seen by no system, however accurate it is. Second — which group the numbers were computed on: the group on which any measurement's properties are computed is assembled by this very stage, and the share of true alerts among all raised depends on how prevalent the disease is in the assembled group. Third — where the threshold for "sick" sits: the point past which the system declares "sick" is set by protocol, priced against the outcome, and that price is set before any accuracy at all — by whoever decided what's worse, missing a finding or calling in someone unnecessarily. All three pillars sit on this stage, and that's why it can't be moved anywhere else.

And the last thing before the breakdowns — how this decision is made with no trained model at all. This work has to be called by name, because the model will be compared against exactly this: a model compared against an empty space always wins, and that win means nothing. There are three orders of work here. First — call everyone in by age and interval: a program runs, the person takes no part in the decision. Second — wait for a complaint: do nothing until the person comes in on their own. Third — sort the flow of people who've already come in by severity: who gets seen today, who in two weeks; the classical tool for this sorting is five severity categories assigned by a nurse under protocol, and each category gets a mandated time to be seen by a doctor. Of these three, the classical regimes of selection, costed in one price, are two — the first two; their figures, base, and caveats stand on a separate slide in this section. The third order is counted differently, in a different unit. Inside screening reading itself stand two more regimes devised before any automation at all: double reading, where two people read the scan without seeing each other's conclusion, and arbitration — a third reader, brought in only on disagreement.

A mistake at this step isn't visible where people usually look for it: it happens before any diagnosis and before any accuracy measurement.

Occupation is a fourth group criterion alongside age, sex, and prior diagnosis: it's used to call people in for occupational exams, and that's the same no-complaints screening. What follows the stage's decision is either an examination, or nothing — until the next interval, or until the day the person feels it themselves. The threshold set by a document is the age at which people get called in, the interval between invitations, and the criterion for recalling someone for further testing.

Seven actions of the screening stage — and three where the model switches on



Seven actions of the stage — and three steps where the model switches on

The letter the lecture opened with came out of exactly this: the program takes the list of residents, selects by age and sex, and sends out invitations. The person didn't come in on their own — they were called.

The threshold, the invitation, the scan, and arbitration stayed with people. The new step is getting a usable scan.

Whose working loop has an automatic filter in it — and who reviews what got filtered?

NHS England breast screening program, GOV.UK, accessed 13 October 2026

The stage has been named, and now, step by step: what actions it's made of, and who does each one. The woman with the letter the lecture opened with has arrived — and from here on a system decides about her in which she takes part at no step at all. On the slide, seven actions of that system are named in one line each; here, each gets its own.

Set the threshold. The age at which people get called in, and the interval between invitations, are written into a health-authority document. They're set not by the developer and not by the doctor at the monitor, but by whoever decided what's worse — missing something or calling in someone unnecessarily.

Call in by list. The letter reaches a person who hasn't complained about anything: they were picked by a group criterion — age, sex, occupation, prior diagnosis.

Take the study. The scan is taken by a technician, and that's its own qualification, its own workstation, its own queue.

Read it by two. Double reading was invented before any automation at all, for one reason: so that one person's mistake doesn't become the decision. The payoff comes from the mismatch of errors — two people's misses land on different scans — and it rests on the second reader not seeing the first one's conclusion.

Resolve a disagreement. A third reader is brought in only where the two disagreed, and resolving the disagreement falls to someone who wrote neither of the two conclusions.

Report the outcome. Either a "normal" letter, or a call inviting further testing — and weeks of waiting in between.

Sort those who came in with no appointment. Here it's decided who gets seen today

and who in two weeks. Someone walking into an emergency department is assigned to one of five severity categories: a nurse asks a few questions, takes a few readings, and checks them against a table. Each category gets a mandated time to be seen by a doctor. Underestimating it — someone with a developing heart attack sits in the waiting room; overestimating it — a resource gets tied up that someone else needed. Both errors are paid for by the same shift.

Now the same field with a model, and it has to be said precisely at which step it switches on in each of this section's breakdowns — otherwise the examples and the stage end up living on two separate layers. The third row of the diagram names this by name.

The step "read it by two" — the Swedish screening program. The first reading stays human; no study bypasses the radiologist. The model scores the scan for risk, and that score decides whether the study needs a second independent reading. It doesn't work as a second reader: for most of the flow, the second-reading step is simply removed.

The step "report the outcome" — retinal fundus photography with no ophthalmologist. There, the model switches on right after the photograph is taken and takes over both the reading and the outcome: "refer or not" gets decided on the spot, in under a minute, and there's no arbitration step in this order at all.

The step "sort the flow" — the first-line chatbot. And here there's an important caveat the diagram can't state: the chatbot sits at a different entry point of the stage. Nobody is called in here by a list — people come in on their own, with a complaint, and the slot the model now occupies was previously occupied by a phone triage line with an operator working from protocol. The stage has no single pipeline, and that's not nitpicking: the cost of error at the no-complaint entry and the complaint-driven entry is different, while the construction of the step is the same.

What the model doesn't take over. It doesn't set the threshold: behind that sits the cost of both errors and a signature under it. It doesn't touch the invitation, the scan, or arbitration. And it adds exactly one step: getting a usable scan — an operation the specialist never had before, since they used to compensate for shooting conditions themselves.

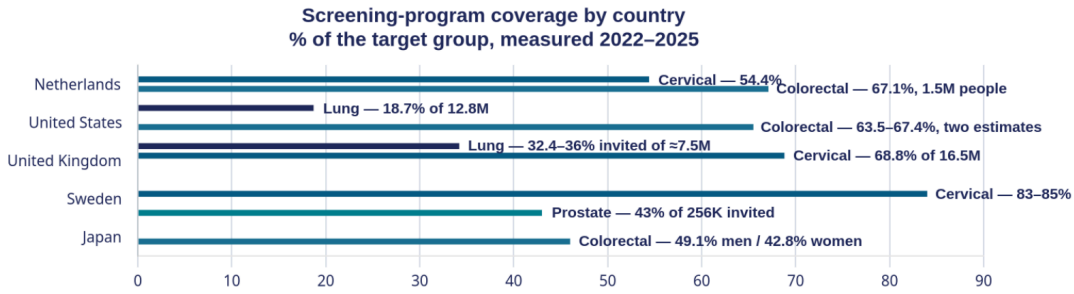
Nothing follows "no," and nowhere is it written down: an event that didn't happen leaves no trace in the chart.

And a property of this scale that will come up more than once: its weights are set by people and published. A trained model, standing at the same spot, produces the same output — a category or a priority — and doesn't present its own weights. Comparing it becomes possible only by result.

That woman's path through this system looks like this: a technician takes the scan; it's read by two radiologists who have never seen her and don't talk to each other about her; if they disagree, the scan goes to a third; a few weeks later a "normal" letter arrives, or a call inviting further testing. At none of these steps is she asked anything.

The woman with the letter the lecture opened with has arrived — and from here on a system works around her in which she takes part at no step at all: a technician takes the scan, it's read by two radiologists who have never seen her and don't talk to each other about her, if they disagree the scan goes to a third, and a few weeks later a "normal" letter arrives, or a call inviting further testing.

Screening programs by country, and where the model takes part



MODEL IN THE ARCHITECTURE

Poland · Croatia

lung, since 2020; sources give no coverage figure

MODEL WITH SEPARATE PAYMENT

Japan, since 2024

colorectal: 43% and 28% of hospitals

EFFECT ON PATIENT OUTCOME

measured nowhere

participation, coverage, and payment are measured

Row = country, column = program. No column, no program: prostate in the US, lung in the Netherlands — a decision both times, not a data gap.

National screening-program reports, 2022–2025 — ten sources are listed in the notes

Coverage of classical screening programs by country: a row is a country, a column is its program. Where there's no column, there's no program — prostate in the US, lung in the Netherlands, and both times that's a decision, not a gap in the data. There are four programs here besides mammography: lung, cervical, colorectal, and prostate, and each has its own answer to who gets called in and whether a trained model takes part in selection. The chart is laid out by country, not by program, deliberately: the question an engineer brings to a project like this is "how is this built here," not "where in the world is the percentage higher."

The rates are computed from different bases — invited, attending, or the whole target group by registry — and the denominator is labeled at the end of each column, right next to the number. They can't be combined into one ranking, and the chart doesn't offer that: within a country, the columns stand in the same program order every time, so the eye compares country to country along the same band.

Lung screening is an annual low-dose CT for people in a smoking-risk group. In the US, 12.8 million people meet the criteria, and 18.7 percent of them get screened — up from 4.5 percent in 2022, almost fourfold in two years, even as absolute coverage stays low. In the Netherlands there is no national program at all — even though it was the Dutch-Belgian NELSON trial, more than ten years ago, that proved a mortality reduction from exactly this kind of screening. Proven method effectiveness and a built program are two different things, and here the gap between them exceeds a decade. Meanwhile, in Poland and Croatia the model is built straight into the national program's architecture from the start, not layered on top of it by a vendor: Croatia, since 2020, is the first EU country with a fully integrated lung-screening program. Sources for neither country give a target-group denominator, so they don't appear on the coverage chart: the fact of the model's deployment is there; a coverage figure to compare it against is not.

Cervical screening shows a mirror image. Coverage here has been settled for decades and is higher everywhere: Sweden 83–85 percent, the UK 68.8 out of 16.5 million, Australia 74.2 and falling, the Netherlands 54.4. The model, meanwhile, is nowhere built into the program itself — only into separate commercial lab systems (Hologic in the US, Techcyte in Austria) and into a non-mandatory recommendation from a professional association in the UK.

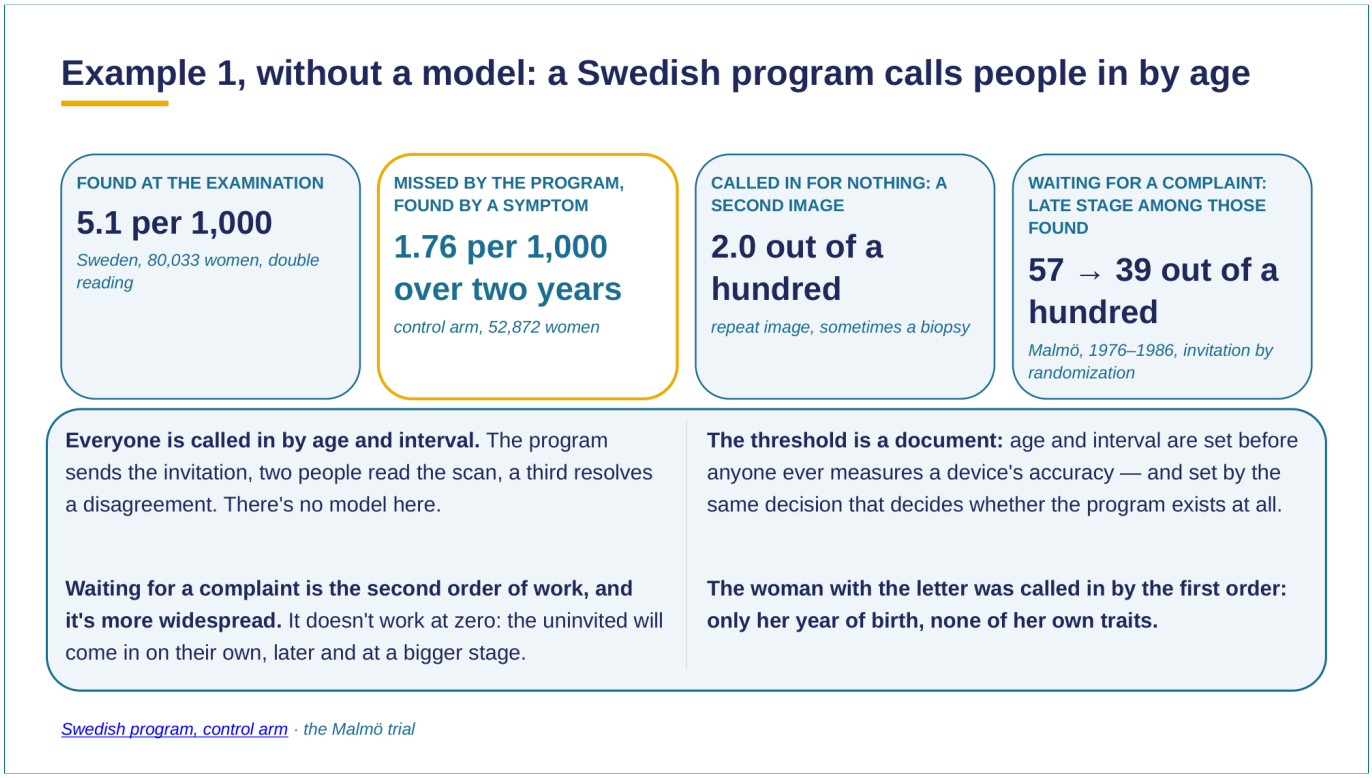
Colorectal cancer is the strongest case of model involvement in this block. The Netherlands holds one of the highest coverage rates in the world, 67.1 percent, one and a half million people on a home fecal test — noticeably higher than the same country's cervical coverage in the same year: turnout depends on the type of test, home versus a visit, not just on a general screening culture. The US holds 63.5–67.4 percent across two 2023 measurements. Japan shows a persistent gap between men and women: 49.1 against 42.8 percent; on the chart this is one column, and the label at its end names both source figures, not an average. And since February 2024, Japan has been paying a separate insurance surcharge specifically for the AI component of colonoscopy — EndoBRAIN-EYE, the world's first case of paying not for the device as a whole, but specifically for its model-driven part. The condition narrows the field: the surcharge only applies to an endoscopist with at least five years of experience. By early 2025, the system had been deployed in 43 percent of university hospitals and 28 percent of regional hospitals; sources give no absolute count of such hospitals, and I'm naming that as a gap, not padding it with an estimate.

Two countries didn't make the chart: Korea holds 51.2–58.6 percent on lung screening across two 2023 measurements, Australia holds 74.2 and falling on cervical. Both carry just one column each, and both programs are already shown by other countries on the chart; the slide's band holds five groups so the labels stay readable, not seven. Separately — Korea's colorectal coverage is missing for a different, substantive reason: the available figure of 74.4 percent counts coverage across six types of screening combined, not colorectal on its own. Putting it in the colorectal column would mean substituting someone else's denominator — exactly the kind of conflation this whole section is examining.

Prostate is a case where the question about the model isn't even first in line. The current recommendation from the US task force, since 2018: the decision on a PSA test for men 55–69 is individual, after a conversation with a doctor; outside that window, including 70 and older, population-wide screening isn't recommended at all. This isn't a measurement gap but a considered regulatory position: prostate cancer grows so slowly in many men that it would never have surfaced in their lifetime, and the cost of overdiagnosis — surgery or radiation therapy, with incontinence and erectile dysfunction — outweighs the benefit of the finding for such cases. There's exactly one confirmed exception, and it's on the chart: since 2020, Sweden has run a registry-based invitation for men 50–74 across sixteen of twenty-one regions, 256,000 invited, 43 percent participation. The program, though, is built around the PSA test, not the model; a narrow pilot of MRI reading by a model inside it hasn't been measured as a separate figure.

And the last, most important point on this slide. The effect of the AI component itself on patient outcome — finding disease earlier, mortality — isn't measured by sources in any of these countries. What's measured is participation, coverage of the target group, or the fact of payment — that is, what happened to the organization, not what this changes for the person. From here comes a direct question for a vendor: if they say "deployed," ask into what exactly — into the program's architecture or as a separate layer on top of it — whether anyone pays for it separately, and whether there's any measurement of effect at all, or whether it stopped at regulatory clearance.

Sources. American Cancer Society — US lung-screening coverage, 2025 · NHS England — cervical screening, 2023–24 · Netherlands National Institute for Public Health — cervical-screening monitor, 2024 · Netherlands National Institute for Public Health — colorectal-screening monitor, 2024 · Australian Institute of Health and Welfare — cancer screening programs report, 2025 · US Preventive Services Task Force — prostate cancer screening · Swedish Organised Prostate Cancer Testing, 2025 guidelines · Cybernet — insurance surcharge for EndoBRAIN-EYE, February 2024 · WHO European Observatory — national lung-screening program in Croatia, 2025 · Lancet Western Pacific — lung screening in Poland00066-0/fulltext)



The Swedish program calls everyone in for a mammogram by age and interval: the program sends the invitation, two people read the scan, a third resolves a disagreement. There's no model here at all, and the program is counted in people — out of a thousand examined, 5.1 cases were found, 1.76 were missed until the next invitation, and 2.0 out of a hundred were called back for a second scan. The threshold inside the program is set by the same decision that sets its existence: whom to call in, from what age, and how often. In other words, both the program and its threshold come from the same place — a document, not a device.

And one historical foundation under this. The question of when screening is justified at all was formalized by the World Health Organization in 1968: James Wilson and Gunnar Jungner laid out ten criteria that still hold today. The disease must have a recognizable early stage; a recognized treatment must exist, with the capacity to deliver it; the test must be acceptable to the population; the screening itself must be an ongoing process, not a one-off campaign. Not one of these ten criteria mentions an algorithm's accuracy — all ten were written decades before the first trained model. This is the lesson that holds the whole evening together: the question "is screening justified" doesn't reduce to the question "how accurate is the model inside it," and the first was posed and settled long before the second. As for where the idea of examining a person with no complaints even comes from: in the 1920s in the US, there was about one doctor for every eight hundred people, and there was physically nothing to examine everyone with; that hidden disease exists in a huge share of people who look healthy was shown by the draft boards of two world wars — forty-seven out of more than two million conscripted into the First World War were found to have a diagnosis never made before; and the first organized program for an entire city was the mass chest-X-ray campaign in Glasgow in the spring of 1957, seven hundred fifteen thousand people in five weeks,

seventy-six out of a hundred adults in the city.

The threshold for intervention is a document. The age at which people are called in. The interval between invitations. The criterion by which a person is recalled for further testing. All of this is written into a health-authority document, and written long before anyone measures any device's accuracy. The model's developer doesn't set the threshold; the doctor at the monitor doesn't set the threshold; it's set by whoever decided what's worse — missing something or calling in someone unnecessarily. That's where the breakdown of any case starts: until the cost of both errors is named, no accuracy figure reads as anything.

The cost of a miss, in the first regime, is counted as the share of cancer found between two invitations: the program didn't catch it, a symptom did. The unit is honest, because it's counted on the same people the program examined — one and seventy-six hundredths of a case per thousand examined, over two years of follow-up. This figure has a second half — the same figure in the model arm — and it's presented where the reading of the scan is examined, together with the caveat about significance: without it, the improvement reads stronger than it's actually measured.

The cost of a false alarm is two people out of a hundred examined called back. Behind that sits a repeat scan, sometimes a biopsy, and a few weeks the person spends living with an open question. The last part doesn't make it into the reports, and it's a cost too — paid not by the program and not by the vendor, but by whoever was called in.

These figures share one base, and it has to be named along with them: the control group of the Swedish randomized screening program, fifty-two thousand eight hundred seventy-two women, double reading with no model at all, two years of follow-up. How much this order of work finds: five point one cases per thousand examined — this figure is counted on a different, smaller sample from the same program, eighty thousand thirty-three randomized women, and it's labeled with that sample on the slide separately. Further on, on the slide with the model, both of these figures will appear in the same units, and the comparison can be made directly. Assignment to groups is random, outcomes are written down in advance — nothing in this lecture will be stronger than this, and that's exactly why the control group works as a reference point for everything else.

The second classical regime is doing nothing and waiting for the person to come in on their own. It's harder to count than the first, because it doesn't work at zero: whoever wasn't called in will still show up eventually — later, and at a different stage. So the comparison has to be not "found or not found" but "when found." The one time this regime was measured by random assignment was when screening was first being introduced. The Malmö trial enrolled participants from the mid-1970s to the mid-1980s and randomly assigned the invitation: the two groups differed by exactly one thing — whether the person was called in or not. The share of stage II and higher among the cases found was ninety-one out of a hundred fifty-nine in the uninvited group — fifty-seven out of a hundred — against sixty-two out of a hundred sixty in the invited group, that is, thirty-nine out of a hundred.

A caveat, without which this second figure shouldn't be used. The trial ran nearly half a century ago: a different device, different treatment, a different era. This is a measurement of the regime, not of the equipment's properties, and it can't be transferred to today's technology. It can't be combined with the Swedish program's figures either: two different programs, two different eras, and together they give exactly one thing — two sides of a price set before any model at all. A trial like this can't be run today at all, because a program doesn't get suspended for the sake of a measurement. Replication: by direction of the effect — many times over, by different programs in different countries; by magnitude — no, and the magnitude depends heavily on the era.

The main thing worth taking from this slide: both prices and both regimes are set and counted before any model and with no model at all. When someone later says "the system reduces the number of misses by such-and-such percent," the first question won't be about the percent — it'll be about which of the two costs this system was supposed to reduce, who decided that, and exactly which "before" it was compared against.

The woman who got the letter fell entirely under the first regime. There isn't a single trait of her own in the decision to call her in: no complaint, no finding, no family history — only her year of birth and the interval, written into a document. From here, the lecture will track what the model does with her case; the first decision about her was made without her and before any model existed.

Two caveats, cut from the slide but required to be said out loud. First: the two regime figures come from different programs and different eras — together they give not a sum, but two sides of a price set before any model. Second: the share of late stages among cases found doesn't depend on how many cases were found — only on when they were found; that's exactly why it's suited to comparing two orders of work, whereas "how many were found" would simultaneously compare coverage, program power, and group size.

What to ask a vendor right on this slide: they say they're comparing "against how it was before" — ask exactly which "before" they counted.

The plates' units are stated in full: a false recall is two people out of a hundred examined, called back in the control arm of the Swedish program, where the scan was read by two people with no model at all and follow-up ran two years. Late stages are the share of stage II and higher among cases found in the Malmö trial, which assigned the invitation at random.

Sources. MASAI safety analysis, Lancet Oncology 2023 · MASAI final analysis, The Lancet 2026

Example 1, with a model: who gets the second reading

OUT OF A HUNDRED WOMEN WITH CANCER — FOUND

74 → 81

105,934 women, randomized into two arms

CALLED IN FOR NOTHING: A SECOND IMAGE

2.0 → 2.2 out of a hundred

80,033 women from the same program

LESS SECOND HUMAN READING

by 44.3 percent

the same 80,033 women

The same program and the same invitation. One thing changed.

What they did. Women were randomized. One arm: two readers, a third on disagreement. The other: a radiologist reads first and issues their own conclusion, while the model scores the scan for risk from 1 to 10: at 1–9 there's no second reading, at 10 double reading stays.

The conclusion is issued by the human. The model selects who gets read twice.

How it's built: input — a mammogram, output — a risk score. Publications don't give the system's name; standing in for it are Transpara, Lunit INSIGHT MMG, and iCAD ProFound. *Image recognition.*

Swedish program, final analysis

The Swedish program set up a randomized trial: one arm kept reading scans the old way — two people, with arbitration on disagreement — in the other, the model scored each scan for risk on a ten-point scale and used it to sort the flow: at a score of one to nine, the second independent reading doesn't happen at all; at ten, double reading stays. The first reading remained human: the radiologist reads the case themselves and issues their own conclusion, and no study bypasses them — but most of those examined no longer get a second independent conclusion. Outcomes were written down in advance. The model did not work as a second reader, and this has to be stated precisely: the second-reading step was removed for most of the flow — which is why the volume dropped by 44.3 percent, not by half.

The program has two publications, and their samples differ: an interim safety analysis on eighty thousand randomized women, and a final analysis on a hundred six thousand. The figures from them can't be combined into one list: the volume of human reading belongs to the interim analysis, while the pair of 74 and 81 out of a hundred women with cancer belongs to the final one. The share of healthy women spared unnecessary alarm is the same in both arms, 98.5 out of a hundred. And a third pair that doesn't appear on the slide: 6.1 against 5.1 found per thousand examined in the interim analysis, p equals zero point zero five two — meaning it didn't reach significance, which is why it isn't shown on screen, and is spoken only with that caveat. A figure placed under someone else's sample becomes a different claim.

This breakdown's answer to the section's question: most of those examined no longer get a second independent human reader. Double reading as a mechanism of mismatched errors has been replaced by one human reading on top of the model plus arbitration — and the cost sits right here, in the construction of the procedure, not in the device's accuracy. And the stage here is still exactly what it was: the model doesn't

make a diagnosis and doesn't read in the person's place — it decides who gets read twice, meaning it selects, the same as the whole stage does.

The German program stands here for contrast, and it deserves less trust than the Swedish one: this is an observational deployment, each radiologist decided for themselves whether to use the model, and there was no random assignment to groups. So the direction is confirmed, but the magnitude can't be credited to the model. The construction of the work has been published — double reading with model support against ordinary double reading, meaning the second reader in Germany was never replaced. From here follows a conclusion that a single deployment can't give on its own: the phrase "the model was deployed in screening" means two different procedures, and the question "did detection go up" says nothing on its own until the order of work it went up inside is named.

And here's a case where coverage is reported but the construction of the work isn't: the domestic radiology platform. More than sixteen million studies across seventy-five regions, with the counting method undisclosed; how many radiology studies are done in the country per year in total isn't published, so a coverage share can't even be computed. There's no independent protocol in the open comparing it against a doctor with no model. This is a claim about one single source, and it doesn't extend to the screening programs.

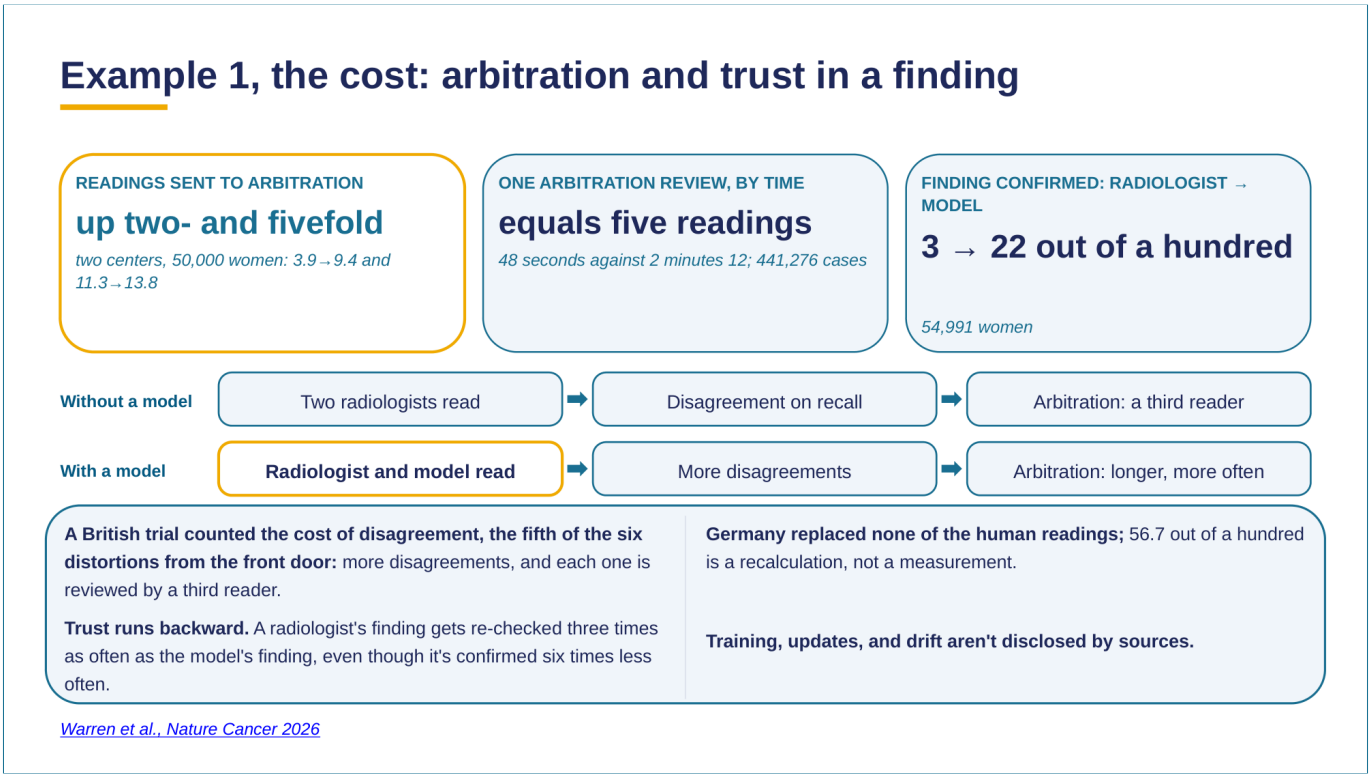
The system's construction is named on the slide to exactly the extent the publications name it: the input is a digital mammogram, the output is a risk score and the study's place in the reading queue. What the system was trained on isn't published by any of the three programs, and it has no name of its own in these studies either — they say "a commercially available system." Standing in for it are Transpara, Lunit INSIGHT MMG, and iCAD ProFound; which one exactly stood in the Swedish program doesn't follow from what's published, and a name can't be substituted by guesswork. A breakdown that can't be checked against the device's name can only be checked against the construction of the procedure — which is why the procedure is what's examined here.

Significance levels for both pairs of figures. The pair of 74 and 81 out of a hundred: sensitivity of 80.5 percent with an interval of 76.4 to 84.2 against 73.8 with an interval of 68.9 to 78.3, significance level 0.031 — the difference holds up. The pair of 5.1 and 6.1 per thousand: a rate ratio of 1.2 with an interval of 1.0 to 1.5, level 0.052 — not significant, and the interval's lower bound touches one. The difference between these two pairs is exactly why they're placed side by side.

The slide's two figures sit on different samples, and they can't go into one list: the volume of human reading belongs to the interim safety analysis, the pair of 74 and 81 out of a hundred women with cancer belongs to the final one. A figure placed under someone else's sample becomes a different claim. What the system was trained on isn't published by any of the three programs, and which of the named systems actually stood in the Swedish one doesn't follow from what's published; a name can't be substituted by guesswork.

Sources. MASAI safety analysis, Lancet Oncology 2023 · MASAI final analysis, The

Lancet 2026 · Eisemann et al., PRAIM, Nature Medicine 2025



A British trial, in two centers of England's National Health Service: 50,000 women, the model placed as second reader, and the cost of a new step was measured — arbitration, resolving a disagreement with a third reader. Disagreements with the model went up: 5,283 arbitrations against 3,449 without it, on the same 50,000 women. This is what disagreement, the fifth of the six distortions presented at the front door, looks like.

The Swedish trial measured the volume and accuracy of reading; what happens to arbitration itself, it didn't measure. That was done by the British study, and the two trials must not be confused: the model doesn't remove arbitration — it changes its composition.

Without the model, 3.9 percent of readings went to arbitration in one center and 11.3 percent in the other; with the model as second reader — 9.4 percent and 13.8 percent: a 142 percent rise in the first center and a 22 percent rise in the second. The spread between centers isn't measurement noise, it's a difference in protocol: the first center only sends disagreements on recall to arbitration, the second sends every case of recall at all, and that choice of protocol sets the final load no less than the model itself does. A separate Norwegian measurement, on 441,276 cases, counted this in seconds: one primary mammogram reading takes 48 seconds on average (median 25), and resolving one disagreement at arbitration takes 2 minutes 12 seconds (median 1 minute 55 seconds). One arbitration review, by time, equals about five primary readings. This is a direct answer to "what does the person do after the model": they don't glance at the suggestion in passing — they carry out an operation that costs, by time, as much as five ordinary ones.

This new step comes with a measured asymmetry of trust, and it flips the usual story on

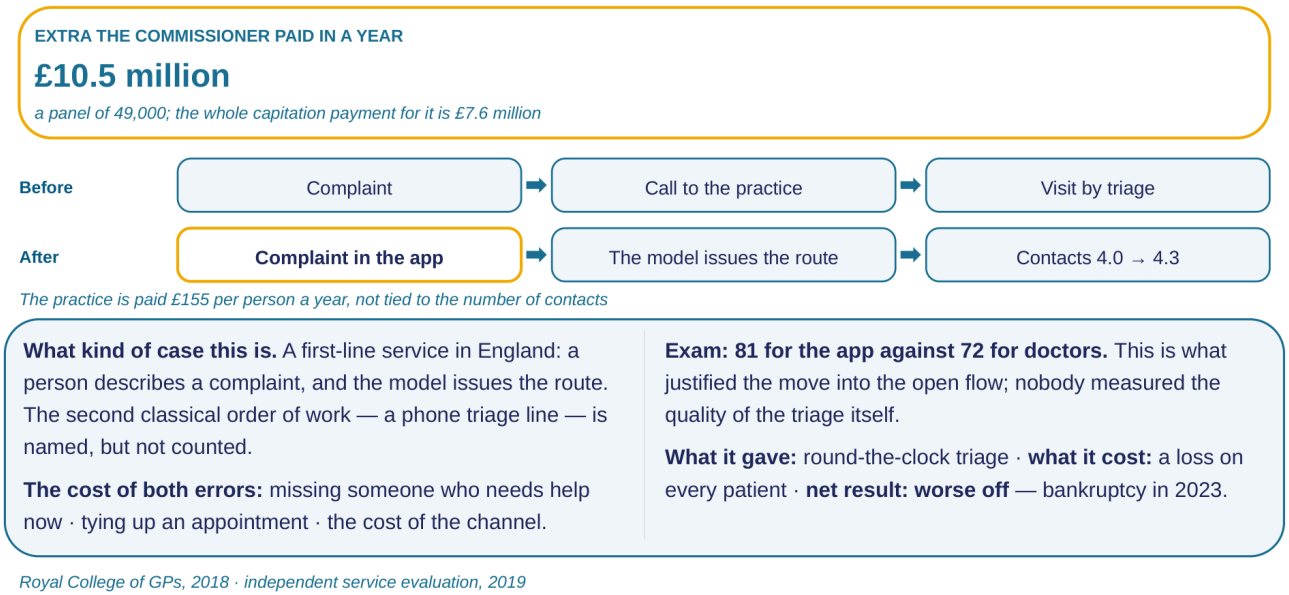
its head. A separate Swedish randomized, preregistered trial on 54,991 women (the ScreenTrustCAD study) put a finding from any of three independent reads — two radiologists and the model — through the same consensus-discussion step and measured what happens to a finding from each source. A finding from one radiologist alone went to consensus review in 14.2 percent of cases (263 of 1,858) with a confirmation accuracy of 3.4 percent (9 of 263); a finding from the model alone went to review in only 4.6 percent of cases (86 of 1,886) — but its confirmation accuracy was 22 percent (19 of 86), more than six times higher. People systematically double-check a colleague's finding more often than the model's finding, even though the model's finding turns out more accurate by a wide margin. This is the first direct measurement of exactly this side — insufficient, not excessive — of trust; the rest of the course's material on trust talks about excessive human trust in the model; here the opposite is measured, among professionals, on a sample of almost 55,000 women. And this is not the same figure as the 80.5 against 73.8 percent that measures cancer detection — that figure is about the share of cancer found out of what existed in a sample of 105,934 women; this one is about the share of findings sent to re-review, on a different sample of 54,991 women. The two figures are easy to confuse precisely because both are "about the model and trust," and they have to be kept explicitly apart.

The German program stands here for contrast. Detection rose — 6.7 against 5.7 per 1,000 among 463,094 women screened — but the construction of the work is different here, and it's named in the very first sentence of the publication's own summary: it compared double reading with model support against ordinary double reading. To put it as a figure, not a general phrase: the model replaced zero percent of human readings — both radiologists kept reading every case, with no exception. The same publication's authors separately recalculated a different, never-deployed scenario: if the cases the model classified as normal had simply not been read by a human at all, the workload would have dropped by 56.7 percent. This figure isn't an observation, it's a hypothetical recalculation on the same data, and the authors themselves call it exactly that. The conflict of interest here is named outright, not with a euphemism: the study was funded by the system's manufacturer (Vara), and some of the co-authors are its employees. The gap between 0 percent replacement (what happened) and 56.7 percent (what would have been counted under a different construction of the work), on the very same dataset, is simultaneously an engineering detail and a breakdown of an overstated claim passed off as a measurement.

And the last point — an honestly named gap, not a guessed-at one. Neither for the Swedish model (Transpara, developer ScreenPoint Medical) nor for the German one (Vara) do sources disclose the volume and composition of the training sample, the frequency of retraining after deployment, or the protocol for monitoring drift in the live flow. In both cases this is a closed-vendor commercial model, and the degree of disclosure here is lower than in the retinal-fundus case further in this section, where at least an order-of-magnitude figure for the training sample and one specific regulatory document on updates turned up. The difference between "a gap named outright" and "a gap nobody asked about" is the same one this course already requires we not lose track of.

Sources. Warren et al., "Impact of using artificial intelligence as a second reader in breast screening including arbitration," Nature Cancer 2026 · Hovda, Holen, Hofvind, European Radiology 2026 · Dembrower, Crippa, Eklund, Strand, "ScreenTrustCAD," Radiology 2025 · Eisemann et al., PRAIM, Nature Medicine 2025

Example 2: a first-line chatbot replaces a call to the practice



A first-line service in England: a person describes their complaint in an app, and the model issues the route — where to go right now. The service took the place of a call to one's own practice, and it ended in bankruptcy in August 2023. The cost of both errors in triaging contacts: missing someone who needs help right now costs an outcome; tying up an appointment for someone who didn't need it costs a doctor's time and someone else's delayed visit. There's a third cost, usually not counted at all: triage costs money for whoever funds it, and the payment structure decides whether it survives.

What was claimed. In June 2018 it was publicly announced that the app scored eighty-one percent against seventy-two on licensing exam questions for trainee doctors — against the average score of practicing doctors. The Royal College of General Practitioners publicly called these claims questionable.

The device and the service go by their own names: the app is Babylon, the first-line service built on it is "GP at Hand." The input was a complaint in free text; the output was a route — "where to go right now." What it was trained on is unknown, and that lack of knowledge is itself telling. Two people spoke about the internal construction, and they are different people with different standing to speak — folding them into one "same thing" would be exactly the mistake this section teaches you to catch. Hugh Harvey worked at the company in 2016–2017 as head of regulatory affairs for clinical AI: by his own public claim, there was nothing to train — inside sat a decision tree of "if-then" rules in an ordinary table. David Watkins is a general practitioner who never worked at the company: since February 2017 he's documented misses and written to the regulator about them, escalating the complaint to "critical" by 2018. The first is words from a person inside about the construction; the second is letters from a person outside about the system's behavior. Neither is a check, and treating either as an

established fact would be exactly the mistake this case is examined to catch. They're examined separately further on, alongside what the company itself published.

The participant's words aren't the last thing found about the chatbot's construction: the company has its own technical publication, and it can be trusted differently — I examine it separately, so the two different testimonies don't blur into one fact. The scale of the pilot, for context: with the National Health Service it started in January 2017 and was available to roughly 1.2 million residents of north London; one case in the chat took about two minutes.

Why the exam metric doesn't hold up as a foundation. An exam question arrives already formulated and contains the relevant features. A live contact arrives with no formulation: the person says "I feel awful," not "chest pain radiating to the left arm." The exam metric measures performance with a ready-made formulation — exactly the part of the task that doesn't exist in the open flow. And the right comparison existed and was ignored: a year before the demonstration, an independent comparison of three apps on the same tasks put this app in last place — it recognized typical illnesses worse than any of the others.

Now the other half of the case — the cost, and here the classical regime is counted in money. The practice is paid about one hundred fifty-five pounds per registered person a year by the National Health Service, and that amount doesn't depend on the number of contacts. A classical contact rate is four times a year for people 25 to 44 and three and a quarter for people 15 to 24. This is the base chosen, rather than the national average, because more than eighty percent of the service's panel were under forty, and the average would have understated the difference. In the service, contacts came to four point three per person a year.

The arithmetic plays out as division, and that's where all the bite is. On top — a fixed sum per person per year. On the bottom — a grown number of contacts. Every extra contact is free for the patient and gets paid for out of the same payment, divided by a bigger number of contacts. The load moved from triage to the appointment, and nobody was left to pay for it. The engine of this move is the accessibility of the channel — the model's quality has nothing to do with it. Entry became instant and free, and the number of contacts grew.

What it gave and what it cost. It gave round-the-clock triage and a low threshold for reaching out. It cost: an economics that ran at a loss on every patient, more than ten million pounds of extra expense for the commissioner in a year with a panel of about forty-nine thousand people, and a documented case where symptoms resembling a heart attack weren't recognized. Net result: a valuation at exit through a merger of more than four billion dollars, bankruptcy in August 2023, assets sold for five hundred thousand pounds.

Caveats to the slide's figures. The ages used aren't national averages — the reason is named above. The contact rate in the service is an estimate from an independent report over twelve months of data, with personal contacts counted from the single month the report was given access to count; it's an estimate with a trend extrapolation,

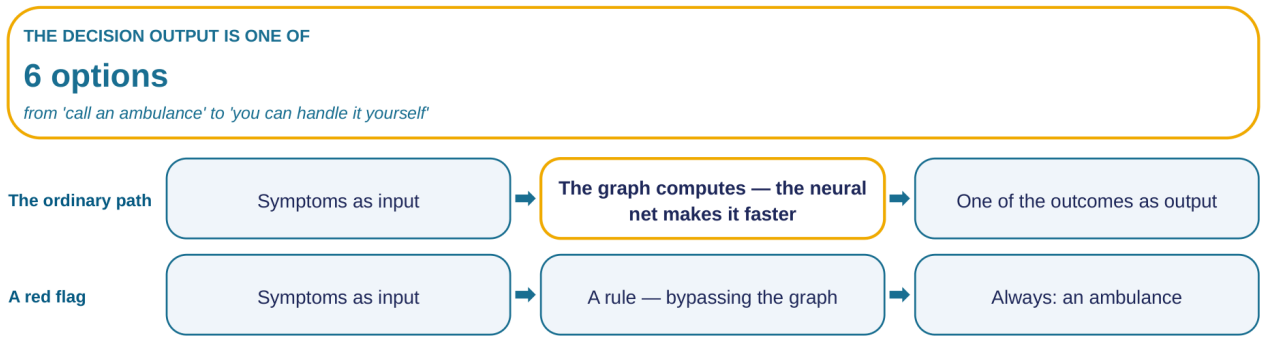
and the report says so itself. The last plate carries two different quantities in two different currencies, not one quantity at two points in time, and that's exactly why there's no arrow between them: the company's valuation was figured in dollars at the merger, the revenue from the asset sale in pounds at bankruptcy, and in some reports the same sum is given as $620,000.

What these figures don't prove: whether the rise in contacts was unmet need or demand the service itself created — the report honestly says it doesn't know. Whether the decision tree itself sorted poorly, there's no data for that either: what's measured here is a restructuring of the channel's economics, not the quality of the triage.

The case is still worth taking as it is: this is a failure of the condition, and the cost of claiming a model where there was none is its content. The lesson is a checkable criterion: a demo metric is not grounds for release into the open flow, and the question to ask is what material it was obtained on.

This regime doesn't distinguish rare from common in the open flow. Dictating a visit note into it is accepted — the record-keeping section examines that; sorting a contact, after this case, is not.

Example 2, the architecture: who said what about what's inside Babylon



Three people spoke, and that's not one source. Hugh Harvey, the company's head of regulatory affairs in 2016–2017: inside, a decision tree of "if-then" rules in a table. **David Watkins**, a doctor outside the company: he's been writing to the regulator about missed cases since 2017. **The company itself**, a 2016 publication: a probabilistic graph plus hard-coded rules for red flags on top of it — on the diagram.

Words from the inside · letters from the outside · the company's own publication. Not one independent check exists.

A trained model making the decision was never here at all.

Middleton et al., arXiv:1606.02041, 2016 — a publication by the company itself

What actually stood inside the first-line chatbot is not reliably known, and three people have spoken about it — with different standing to speak. They have to be kept apart before anything else: folding three testimonies into one "same thing" is exactly the mistake this case is examined to catch. Hugh Harvey worked at the company as head of regulatory affairs for clinical AI in 2016–2017; his public claim — that inside the "AI algorithm" sat a decision tree of "if-then" rules in an ordinary table — is words from a person inside, about the internal construction. David Watkins never worked at the company: a general practitioner, he's documented recognition misses since February 2017 and reported them to the regulator, escalating the complaint to "critical" by 2018 — that's testimony from outside, about the system's behavior, not about its internals. Folding them into one "same thing" is an attribution error, and it stood twice in drafts of this very chapter; a section that teaches people to ask where a claim came from has to be precise about who made it. The third testimony is the company's own technical publication from 2016, and it can be trusted differently: not an independent check, but also not one person's recollection. The company itself has its own technical publication from 2016, and it can be trusted differently: it's not an independent check, but it's also not one former employee's account — it's what the company itself published about its own architecture. These two testimonies must not be confused: the "if-then" decision tree is a participant's words; what's described below is a published architecture.

The publication describes the system as a probabilistic graphical model of primary care: a directed acyclic graph in which diseases and symptoms are connected by prior probabilities and conditional dependencies. The structure of the graph — which nodes connect to which — is built and checked by expert physicians, not by lone engineers. This is already not the "if-then" decision tree in a table the participant talked about; it's a probabilistic model with its own structure.

But on top of the probabilistic part sits a separate mechanism, and it must not be confused with the graph itself. For a defined list of "red flags" — specific symptom combinations — the triage decision isn't derived by probabilistic reasoning at all; it's forced by a hard rule: such a combination leads straight to "call an ambulance," regardless of what the probabilistic part is showing at that moment. And separately from both parts, the publication mentions a neural network, but its role there is just one thing: it speeds up computation on the probabilistic model — it doesn't replace it, and it doesn't make the decision itself. Of the three components — the graph, the hard rules, the neural network — a trained model making the decision isn't in any of them.

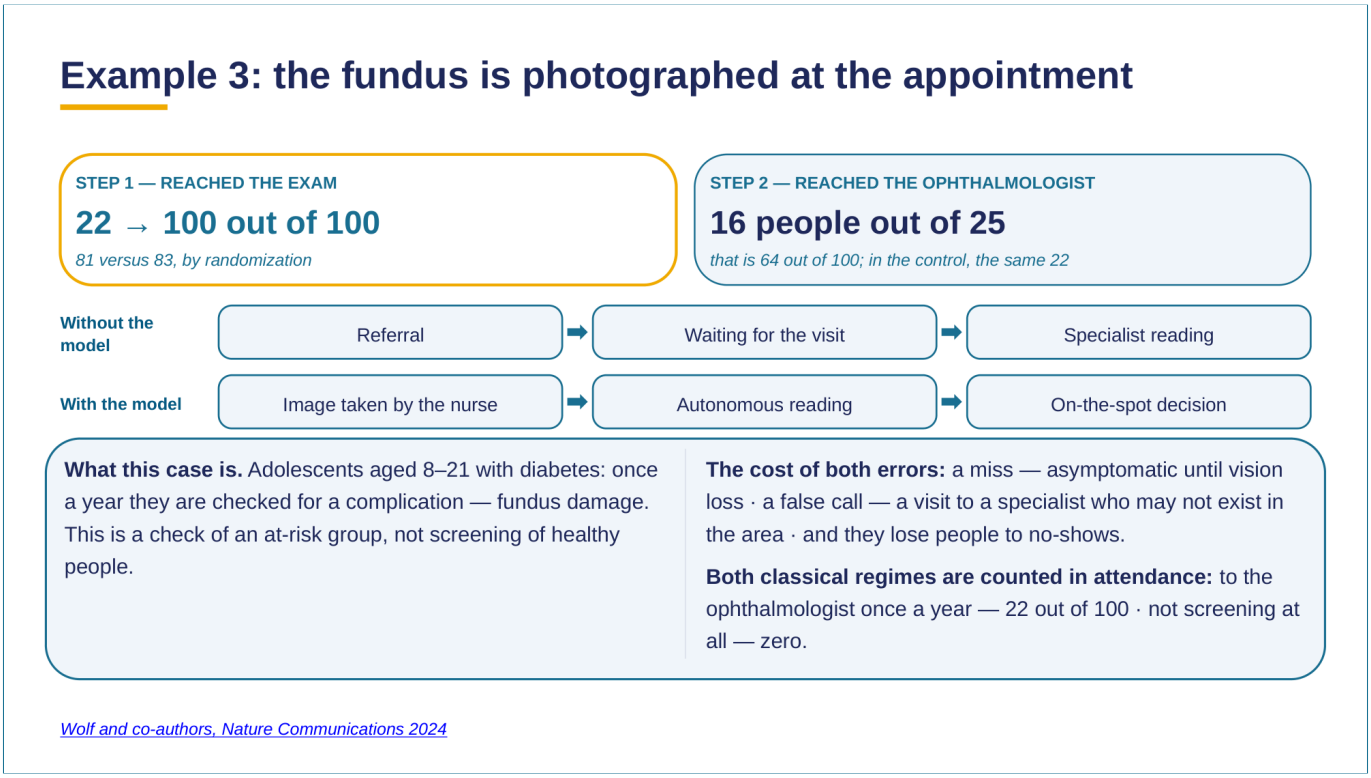
From here follows a direct conclusion, and it stands as a statement, not as a caveat about missing material: the systematic design of a trained model making the decision — the thing this lecture's other engineering cases all require — simply was not here at all. This isn't a search gap: it's the honest answer to "how was the engineering built," for this particular case.

On output, the system didn't give a binary "yes/no" — it gave one of six mutually exclusive options: call an ambulance, go to urgent care, see a doctor urgently, see a doctor without urgency, go to a pharmacy, self-care. Six outcomes, not two — and that's also part of the architecture, not a detail of presentation: the more distinguishable outcomes on the output side, the more points at which a rule or the graph can go wrong in different ways.

The feedback loop — who rewrites the graph's rules, and how often, what happens specifically after a recorded error — isn't described as a process in any source found, including this very company publication. All it has is a general phrase about "iterative expert review": no frequency, no link to specific recorded cases, no answer to whether the rule was updated right after the missed heart attack discussed on the previous slide, and if so, how fast. The difference between "a gap named outright" and "a gap nobody asked about" is the same one this course already requires we not lose track of: a blank spot that was never named is indistinguishable, to the reader, from a blank spot nobody ever looked for.

Different pieces of this breakdown deserve different amounts of trust, and that's worth saying outright, since they come from different kinds of sources too. The per-head sum and contact frequency behind the cost calculation are official statistics and an independent assessment report: these can be trusted. The exam comparison and the independent test a year earlier are taken from secondhand accounts; the primary publications aren't open. The claim about a decision tree is a participant's words, and that's the weakest item here. The graph and rules architecture is the company's own technical publication: heavier than one participant's words, but lighter than an independent check, because the company is describing its own system. The bankruptcy facts and the deal sums that close out the breakdown are public filings and the business press.

Sources. Middleton et al., "Sorting out symptoms," arXiv:1606.02041, 2016



Adolescents from eight to twenty-one years old with diabetes: once a year they are checked for a complication — fundus damage. In a randomized trial, the image is taken by a nurse or a primary-care technician right at the visit, and the decision to refer or not is issued by the authorized device on the spot, in under a minute, without an ophthalmologist. Attendance at the exam rose to one hundred people out of one hundred, against twenty-two. This is a check of an at-risk group with an existing diagnosis, not screening of healthy people for diabetes.

The cost of both errors is stated before any accuracy figure. Missing fundus damage in a person with diabetes means reaching vision loss: up to the late stages it is asymptomatic, the person feels nothing and so does not come in. An unnecessary referral costs a visit to a specialist who may not exist in the area. And a third quantity decides everything: they lose people to no-shows. A test that a person never reached has zero sensitivity, whatever it measures in the lab.

Both classical regimes are counted in one unit — attendance. The first: once a year, a person with diabetes goes to an ophthalmologist, who takes the image and reads it himself; twenty-two people out of one hundred reach the test. A separate visit to a separate specialist on a separate day, often in another city, while the person feels nothing — the regime loses people along the way. The second regime is unpleasant to name, and it is more widespread than the first: not screening at all. Its attendance is zero, and that figure comes from definition, not measurement. This is how monitoring works where there is no ophthalmologist in the area; without naming this regime, the gain in attendance reads only against a visit to the specialist, when it was also obtained against nothing at all.

The comparison base: eighty-one people against eighty-three, randomized, one device.

The device's sensitivity does not enter into this breakdown — the sources give no prevalence of the lesion in these samples, and accuracy cannot be read without the group and the prevalence.

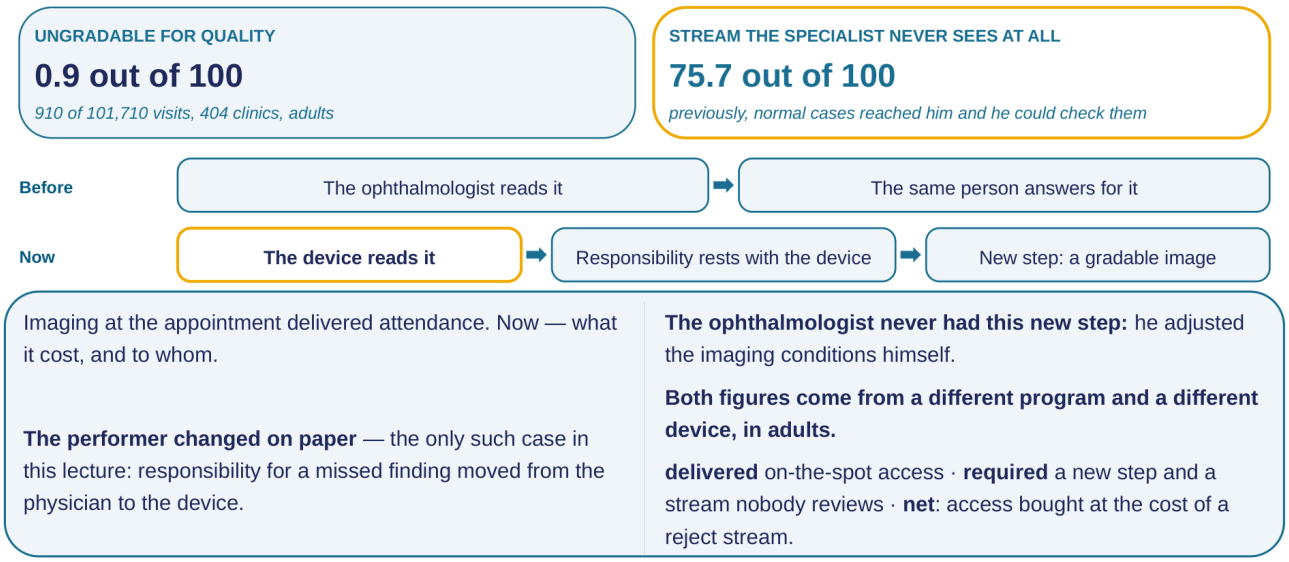
The second step of the same path is counted within the same trial: of twenty-five people whose result came back abnormal, sixteen reached the ophthalmologist — sixty-four out of one hundred, against twenty-two in the control. The criterion this breakdown leaves behind: autonomy transfers responsibility and does not eliminate work — it shifts it onto a different step and a different person. If this step was not counted, the deployment is cheaper than it actually is. Here it was counted, which is why the breakdown holds up: sixty-four percent were carried through to the specialist, not one hundred.

Not everything is known about how the device is built, and both halves are worth naming. Input — two fundus images taken by a nurse or technician in a primary-care office; output — a "refer or not" label. The system was trained on roughly 25,000 complete exams (University of Iowa and the EyeCheck project); the primary source of this figure did not come up in full text, so it is not claimed more precisely than "roughly", and it must not be rounded into a certainty when retold. The publication does not name the device that operated in this trial. The regime does have names, and they are checkable: the first authorization in history for autonomous diagnosis without physician interpretation was granted in 2018 to the IDx-DR system, which is called LumineticsCore today; the retrospective stream of one hundred and one thousand visits discussed further on was captured on an EyeArt device.

How far this evidence can be trusted: the allocation to groups is random — evidence does not get stronger than that — but the sample is small, there is one device, the age group is narrow, and there is not a single replication.

Sources. Wolf et al., Nature Communications 2024

Example 3, cost: the reject stream and a new step



The same fundus imaging at the appointment, now counted from the cost side. The decision to refer or not moved from the ophthalmologist to a device authorized by the regulator in full, and responsibility for a missed finding moved along with it: previously it rested on the physician's decision, now it rests on the device. This is the only case in the entire lecture where the performer changed on paper: the first authorization for autonomous diagnosis without physician interpretation was granted in 2018.

Hence a new step that did not exist in the old order: obtaining a gradable image. The ophthalmologist had no such operation, because he adjusted the imaging conditions himself, on the spot. Now a nurse takes the image, and its gradability has become a separate task with its own defect rate.

How much this step costs has been measured — but on different material. On a mature stream, with a different manufacturer's device and in adult patients, nine-tenths of a percent of images are ungradable for quality: 910 of 101,710 visits that passed validation, across more than 400 clinics. And more than 75 percent of the stream the specialist never sees at all. Against attendance of 22 out of 100, these figures do not add up into one ledger: two different programs, two different devices, two different age groups. Combining them into one "delivered — required" pair would be exactly the splicing of samples this lecture is dissecting — which is why the reject stream gets its own slide with an explicit caveat, instead of sitting inside the payoff count.

What happened to the ophthalmologist. He sees roughly a fifth of the stream, and it is the part the model selected. Previously it was the specialist who decided what was normal; now the normal cases never reach him — and he cannot check whether they really were normal. This is the move named precisely: the work did not disappear, it moved to a different performer, and the ability to check it moved along with it.

Seventy-five percent is exactly the size of that move, named in units of work rather than in words.

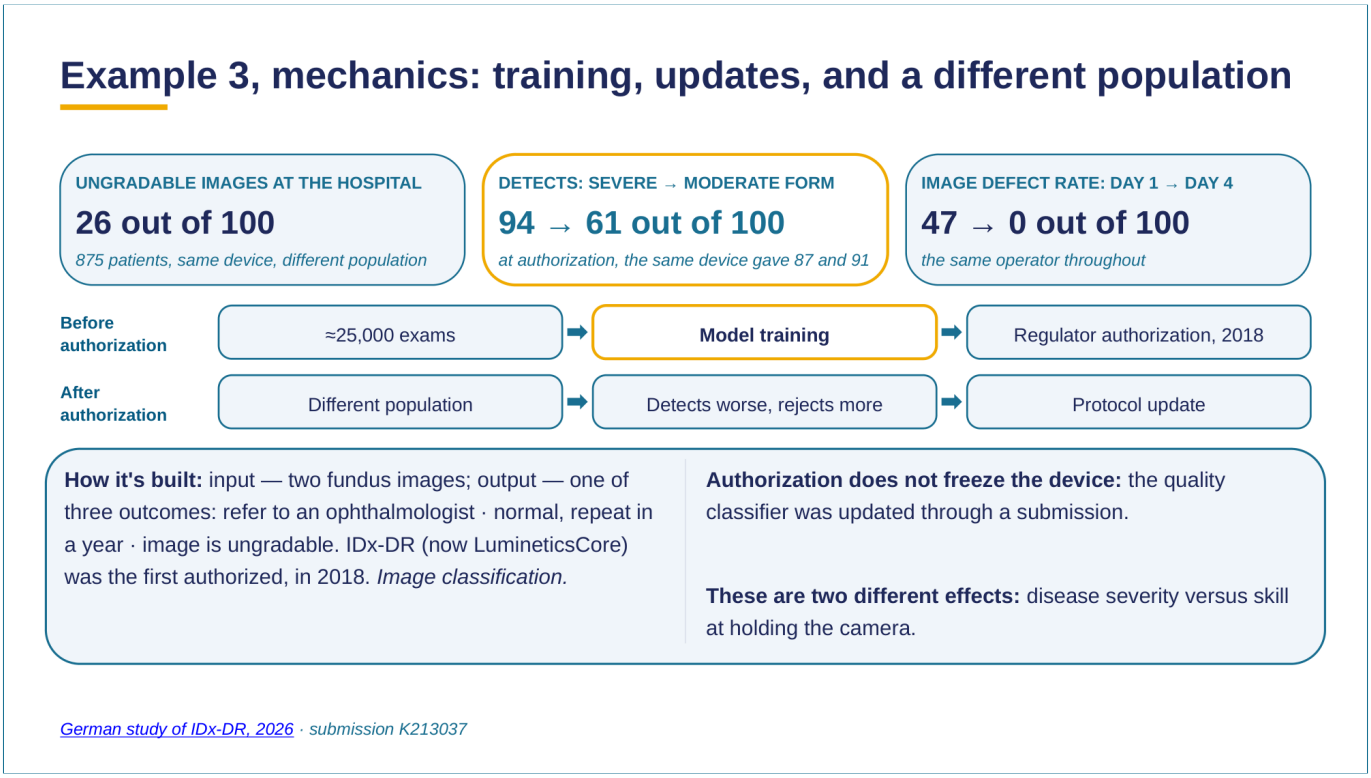
How this case differs from the Swedish screening program, where a model was "deployed" too: there, the model split the stream, but the finding for each study still came from a human — the order of work changed. Here the device issues the finding — the performer changed. The two deployments cannot be described in the same words, even though they often are.

Add the two halves together. Delivered: examination on the spot, at the same visit, without a separate trip to a specialist who may not exist in the area. Required: a new step with its own defect rate, and a stream nobody reviews. Net, this is access bought at the cost of a reject stream — and both halves have to be named together, because each one alone sounds like a conclusion on its own.

The criterion that carries forward: autonomy transfers responsibility and does not eliminate work — it shifts the work onto a different step and a different person. A deployment that did not count this step is cheaper than it actually is. For an engineer, two questions follow: who retakes an ungradable image, and on whose time; and who reviews the part of the stream the model classified as normal. Neither question is about the model's accuracy, and both are settled when designing the workflow, not when choosing the device.

The strength of evidence differs between the two halves of this breakdown, and they should not be equated. Attendance was measured by a randomized trial — the top rung of the evidence ladder, though with a small sample, one device, and a narrow age group; there is not a single replication. The figures on this slide did not come from a trial at all: they are a retrospective count from clinic records.

Sources. US regulator decision DEN180001, 2018



The EyeArt reject stream was measured on different material — a different device and a different program, named on the previous slide with an explicit caveat about non-comparability. What follows is what is known about the training of the IDx-DR/LumineticsCore device itself, how it is updated after authorization, and what a separate measurement on a new sample showed.

A German study at a tertiary diabetes hospital (Karlsburg, 2020–2021, 875 patients, average age 52, range 8 to 92) tested the same device, IDx-DR, which gave the pivotal 87.2 and 90.7 percent sensitivity and specificity on the registration sample. In this German sample, the system itself flagged 26.1 percent of images as unfit for analysis — against 96.1 percent gradable at the pivotal study. This is not proof of drift over time: the German sample consists of hospitalized, older, and more severe diabetics, with a higher likelihood of cataracts and constricted pupils, rather than the US primary-care target population for which the device was authorized in 2018. The correct reading is a measurement of how sensitive the result is to the composition of those examined, not a deterioration of the device over the years.

Sensitivity on this same German sample turned out to differ by severity of the lesion rather than being a single number: 94.4 percent for severe diabetic retinopathy — close to the pivotal figures — and only 60.9 percent for the moderate form. Exact agreement with the ophthalmologist's severity grading was only 54.2 percent of cases. The spread is wide enough that a single sensitivity number for this device simply does not exist: it is tied to exactly who is being examined.

The same study found a separate and strong effect — operator training on the camera, not a change in the algorithm: the share of ungradable images for one and the same examiner fell from 46.9 percent on the first day of working with the camera to 0 percent

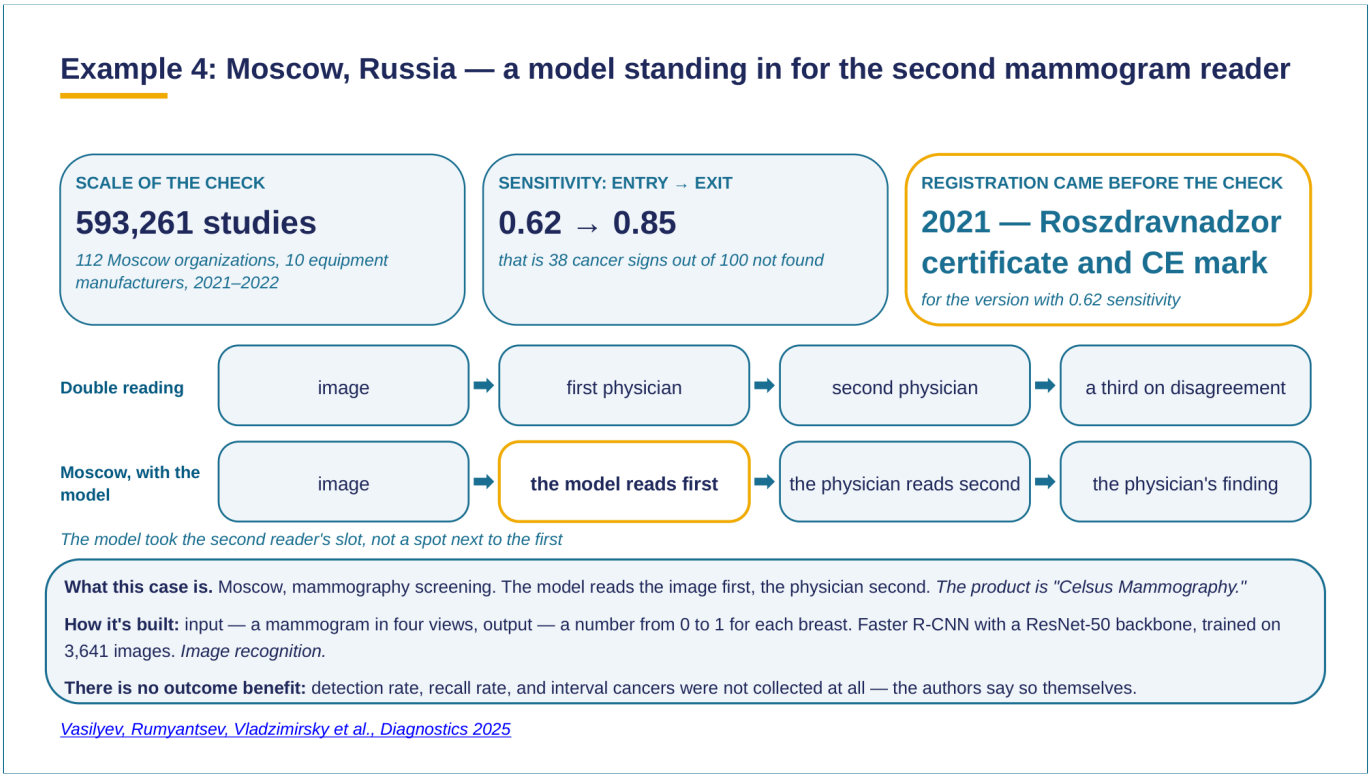
on the fourth day of practice. The model's decision did not change at all here — what changed was the hand of the person holding the camera.

The 2018 authorization does not fix the device forever. At registration, a change-control protocol was agreed: technical modifications capable of affecting the false-positive rate require a new notification to the regulator before deployment, not a quiet update. A concrete example of this mechanism in action is 510(k) submission number K213037, which updated the image-quality classifier from version 2.1.1 to 2.3.0. The full text of the submission did not come up as readable text on direct access to the document, but the fact of filing and the document number are confirmed on the regulator's site — this is exactly the second of the three states of a gap: searched in good faith, found the document, but not its contents.

The device from a different manufacturer, EyeArt — the very one that produced the reject stream and the more-than-75-percent stream on the previous slide — is built differently, judging from patent documents: two convolutional networks on shared layers, one segmenting retinal structures, the other assessing the quality of the image itself. Here, rejection for quality is not a separate filter ahead of diagnosis but is built into the same mechanism that produces the finding. The sources do not disclose the details of the training sample for the certified version of the device — there is only an adjacent validation dataset of different origin, and it cannot be equated with the training set.

The five engineering-depth questions for this case are closed unevenly, and it is worth saying so plainly: input-output and the threshold are fully closed; training only to an order of magnitude; the update is a fact of mechanism without the content of the change itself; and drift is not this same device on the same population at all, but someone else's sample of the same device. The gap between "closed" and "partially closed" here is not a reason to drop the breakdown — it is the honest answer to a question that most medical-AI material does not ask at all.

Sources. Scientific Reports / PMC12864748, "Real-world performance of the AI diagnostic system IDx-DR in the diagnosis of diabetic retinopathy and its main confounders", 2026 · FDA 510(k) K213037, accessdata.fda.gov



Moscow mammography screening: the model reads the image first, the physician second — that is, it occupies the slot where the Swedish program puts a second human. The product is called "Celsus Mammography", tested on 593,261 studies from 112 city organizations, on equipment from ten manufacturers, in 2021–2022.

This is the only breakdown where you can see how a model is authorized for a mass-scale stream and what happens to it after authorization: none of the other breakdowns have this part, and the sources on the Swedish program say nothing at all about training and drift.

I'll start with what this case is. Moscow, mammography screening, part of the city's experiment in computer vision for radiology, running since 2020. Over the course of that experiment, fifty-two different models have processed almost fourteen million studies — that is the frame our case sits in, and the case itself is a single one: the "Celsus Mammography" model on the city's mammography stream.

Now the most important thing about the mechanics, and this is exactly where this case diverges from the Swedish one. In the Swedish program, the model selects — it gives the image a risk score and uses it to decide whether to run a second human reading; the finding for each image still comes from a human either way. Here it is built differently: the model reads the image first, the physician reads second, and the intent of the city's experiment is stated directly — to replace the second radiologist with a registered medical device. It is one slot, and two different ways of filling it. One stage, different mechanics, which is why this is two breakdowns rather than one.

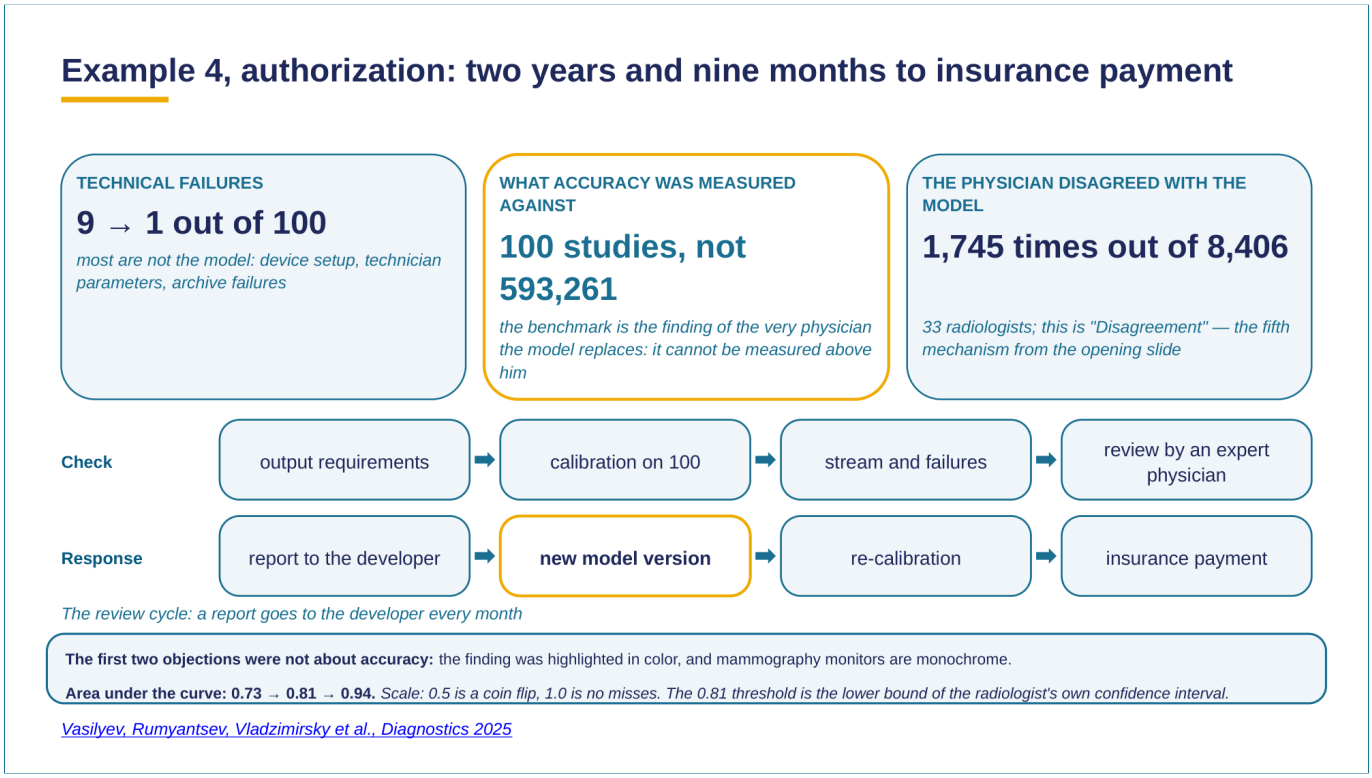
On how it is built — something the room has not gotten on a single slide of this section so far. Input — a standard mammogram in four images: two views per breast. Output — not a category, not a word, but a number from zero to one for each breast: the

probability of cancer signs. Inside — a Faster R-CNN detection network with a ResNet-50 backbone; trained on three thousand six hundred forty-one images. Pause on this pair of numbers: the training set is three and a half thousand images, the validation stream is five hundred ninety-three thousand two hundred sixty-one studies. A ratio of almost one to one hundred sixty. This is not a knock against the developer, it is a property of the task: annotating an image with a histology-confirmed outcome is expensive, and you cannot annotate as many as pass through the city in a year.

I will state the cost of both errors in the same units the Swedish program already used, because different units cannot be added here. Missing a cancer is an interval cancer: it shows up on its own, later, at a larger stage, and for the Swedes that is 1.76 cases per thousand screened. Raising a false alarm is recalling a woman for a second image, 2 out of 100 for the Swedes. Now the cost that this case specifically measures, and it is of a different kind: at entry into the check, the model had a sensitivity of 0.62. To translate: out of a hundred cancer signs in the set where they were being looked for, the model did not find thirty-eight. And with that sensitivity it already held a Roszdravnadzor registration certificate and a European CE mark — both obtained in year twenty-one, before the city check even began. This is the main lesson of the slide: registering a product and the product actually working are two different facts, and the first does not imply the second.

The two classical regimes to compare against are already counted: double reading by two physicians with a third for disagreements — the very regime the model is replacing here — and single reading by one physician, which is more widespread and does not work with a zero. And now something I have to state honestly, and which many retellings of this work do not: there is no outcomes comparison of "model instead of the second physician" against "two physicians" in it. The authors did not collect detection rate, recall rate, or interval cancers at all — and they call this their own limitation, in the text, in their own words. So this case has no measured outcome benefit, and we will not credit it to the model. What is measured in it comes on the next slide; a great deal is measured there, but it is something else.

Sources. Vasilyev, Rumyantsev, Vladzimirsky et al., Diagnostics 2025



What was done with the Moscow mammography model after registration and before the city began paying for its work out of insurance funds. The whole cycle took two years and nine months: five calibrations and four versions of the product.

The first check is functional, and it stops the model on two points, neither of which is about accuracy. First: the output document and the additional set of images lacked the "for research use only" label. Second, and worth remembering: the model highlighted the finding in color — and mammography monitors are monochrome, so the colored highlight cannot be distinguished on them. The model could have been arbitrarily accurate; its answer was physically unreadable on the equipment actually sitting in the room. This is exactly the engineering work no metric can substitute for: until someone sits down at a real monitor, you cannot see it.

The second check is the calibration check, on a hundred studies with confirmed annotation, and here the model failed on area under the curve: as a reminder of the scale, 0.5 is a coin flip, 1.0 is no misses. The threshold was set at 0.81, and it was not pulled out of thin air: it is the lower bound of the confidence interval that the radiologist himself gives on the same task. In other words, the requirement is framed as: be no worse than the least plausible human. The model scored 0.73 and was sent back.

What the developer rewrote over the following month is a list worth stating out loud, because it answers the question "what actually happens when a model fails". They expanded the training set. They reworked image preprocessing. They added a separate channel with brightness-window selection and histogram equalization. They added a head that scores the quality of the input image. They swapped the network backbone from ResNet-50 to ResNeSt. They replaced a coarse resizing of the region with a precise one. They changed the activation function. After that the area under the curve became

0.81 — exactly the threshold, not above it — and the model was let into the stream. Two more reviews followed: at month nine they added a fibrocystic-change class and annotation on the international breast-imaging reporting scale, added at the radiologists' direct request; at month fifteen, two more finding classes, an image-quality score, and a quadrant split, and the network was moved to a different architecture. By the fourth version, the area under the curve reached 0.94, and before authorization for payment there was a fifth calibration, where the figure came in slightly below the peak. Revisions do not only go up.

Technical failures are a separate story, and probably the most useful one for an engineer. They fell from nine percent to one, and that is a genuine improvement in robustness. But look at the composition: most of the failures are images the system did not process in full — cases where the model handled one, two, or three frames out of four — and the causes are named directly: incorrect mammograph setup, study parameters set by the technician, and failures in the city archive itself. In other words, a substantial share of what the report counts as a model defect is not a model defect at all. Without this breakdown, the figure "failures dropped ninefold" would read as credit to the network.

And two nuances that change the reliability of everything I have named; both are stated on the slide. First: accuracy was measured on the hundred studies of the calibration set — not on the five hundred ninety-three thousand. A stream of that size measured failures, physician agreement, and expert review, not accuracy; the authors name this as a limitation themselves. Second: in the prospective part, the benchmark was the physician's own finding — the very physician the model replaces. Measured this way, the model cannot turn out better than the human even in theory, and a shared miss by both is never visible at all. I'll add a third, smaller one: the clinical review was run by one expert physician at a time, and at month fifteen that physician changed — meaning the measuring instrument itself changed partway through the measurement, and the authors note this themselves.

What moved. The decision "is the model's output acceptable" moved from the developer to a city-level tester: output requirements are published, calibration is mandatory, and a report goes to the developer every month. And the physician gained a step that did not exist before: over the observation period, thirty-three radiologists pressed "disagree" one thousand seven hundred forty-five times out of eight thousand four hundred six responses — roughly every fifth time. This is "Disagreement", the fifth of the six mechanisms we named at the opening: a discrepancy with the model is worked through as a separate step, and that step is paid for in working time. In January of year twenty-three, the city introduced a separate paid service — mammography description and interpretation using the model. Two years and nine months from the start of testing to the tariff.

Sources. Vasilyev, Rumyantsev, Vladzimirsky et al., Diagnostics 2025

Section conclusion: a gain in access, no added accuracy from the model

What works: reaching people who otherwise would not come — **22 → 100 out of 100** · cutting the volume of reading where the procedure is already dense — **44.3 percent less second reading**.

What does not work: setting the threshold — a signature stands behind it · an open-referral stream — the one attempt that actually happened ended in bankruptcy · answering for the gradability of the image.

With the work: the workload moved from triage to the visit, and there is now nobody to pay for it · the performer changed, and a step appeared — obtaining a gradable image.

No procedure checks what the filter screened out. In screening, a human read every study; where the model decided, its "normal" never reached him.

Of the six distortions named at the opening, this section ran into one — disagreement: arbitration grew two- and five-fold, and one review takes as long as five readings.

The outcome of the selection stage: the model cuts the volume of reading and reaches people who otherwise would not come, and it cannot set a threshold, work in an open-referral stream, or answer for the gradability of an image. The stage comes together around four questions that will also be asked at every stage after it. The numbers live on the breakdowns these generalizations were drawn from.

What works. The benefit in both breakdowns is measured in access — in how many people reached the exam and how many images a second human read. It is obtained two ways: cutting the volume of human reading where the procedure is already dense, and reaching people who otherwise would not come. The second way is stronger in this section, and the reason is simple: a test a person never reached has zero sensitivity. The model added no accuracy here, in a single breakdown.

What does not work. Three boundaries, and none of them is about the quality of the model. The model does not set the threshold: which is worse — a miss or an unnecessary call — is decided by whoever signs for it. In an open-referral stream, not one breakdown produced a benefit, and the one attempt that actually happened ended in bankruptcy. And autonomous reading does not answer for the gradability of the image: obtaining it falls to a person without specialized training. And one boundary that is easy to overextend where it does not belong: coverage gets published, the mechanics of the device do not. That does happen, but it does not split by country: the domestic radiology platform publishes coverage without the mechanics, and on that very same platform, in the Moscow mammography breakdown, the device is disclosed more fully than in any other case in this section — architecture, size of the training set, origin of the threshold, composition of the failures. The Swedish and German screening programs both publish the device's mechanics, and neither says anything about training or drift. So the split is not by country, it is by who is doing the reporting: the

platform reports coverage, the device review reports mechanics.

What happened to the work. Two forms, named from the glossary. On the first-line service, the workload moved to a different step — from triage to the visit — and there turned out to be nobody to pay for the moved workload. In autonomous fundus screening, the performer changed: the decision moved entirely to the authorized device, and a derivative form appeared — a step that did not exist before. The third consequence is not a form but a property of the procedure: most of those examined no longer get a second, independent human.

What holds this up. An intervention threshold written down in a document long before any device existed. Double reading and arbitration on disagreement — where the second reader has not been replaced. And what is missing: a procedure that would check what the filter screened out. The answer to "who reviews what it filtered out" is most often "nobody", and that is not one organization's oversight — it is an empty spot in the order of work itself.

What carries forward — and this cannot be summed up in one phrase, tempting as that is. Where the model decides instead of a human, its "normal" really does not reach him: the chatbot itself answers "no need to go", and in fundus imaging the decision is made on the spot, without an ophthalmologist. But in both screening programs a human read EVERY study, and the bias there is of a different kind: he reads while seeing the model's score, and the second independent reading has been removed — so what is biased is not the composition of the stream, but the mechanics of reading. A selection error in the first case is only caught on the next cycle, by a different person on a different day; in the second, it is looked for at disagreement. That is why the next section opens with what "measured" even means.

And one mechanism out of the six named at the opening, this section genuinely ran into: "Disagreement." On the Moscow mammography stream, thirty-three radiologists pressed "disagree" one thousand seven hundred forty-five times out of eight thousand four hundred six responses — roughly every fifth time — and working through each such discrepancy is paid for in working time that, before the model, was not spent at this step.

The third consequence of the move is not a glossary form but a property of the procedure: most of those examined no longer get a second, independent human. And the answer to "who reviews what the filter screened out" is most often "nobody", and that is not one organization's oversight — it is an empty spot in the order of work itself.

What holds this workload up is an intervention threshold written down in a document long before any device existed, and double reading with arbitration on disagreement — where the second reader has not been replaced. In an open-referral stream, not one breakdown produced a benefit, and the one attempt that actually happened ended in bankruptcy.

The benefit in this section is obtained two ways, and the second is stronger: a test a person never reached has zero sensitivity, whatever it measures in the lab. So

attendance here is not an organizational footnote — it is what the model's benefit at this stage is actually measured by.

And the tally on the course's own grid, which will keep growing section by section: at this step, three cells out of forty-two are filled, and none of them is about treatment. The difference between them is worth noting right away: a mode that makes the decision is adopted slowly, while modes that formalize a decision already made spread fast. Triaging a request in words has stood still since the case we broke down; autonomous decision-making without a physician is authorized only in a handful of cases, and only where the question is binary; and the model as a second reader is spreading wider than any of them — because it does not remove a single signature.

Nine forms of the move: what each looks like in an example

	form	what it looks like in the example
1	the workload moved to a different step	triage moved to the visit — nobody to pay for it
2	the step disappeared entirely	the candidate was checked and dropped out
3	a step appeared	the decision of who needs a second reading
4	the workload grew in the same place	alarms are frequent, confirmation is rare
4a	the workload in the same place fell	the same stream, less work
5	the performer changed	the device issues "refer or not"
6	the order of work changed	rearranged, but not counted
7	the cost of an error shifted	the model's error is the signer's error
8	the performer is the same, but got worse	without the assistant, they find less
8a	*the performer is the same, but got better*	*no measurements exist*

This section has just named two of its forms — the workload moved to a different step, and the performer changed. Here is the list they were drawn from. Every breakdown from here on ends with a named form — or a named refusal to assign one, and a refusal is as much a verdict as a name. There are no numbers in this glossary on purpose: a count under each form will build up from the breakdowns, and by the end of the conversation seven of the nine forms will carry a measured value, while two will not.

This glossary of forms came from the previous lecture, and proved its worth there too. The familiar image — "the workload moved to the right" — lies in several directions at once. It describes one case out of nine possible ones and silently substitutes it for the other eight. Work can move to a different step while keeping its full volume. It can double while staying exactly where it was. It can disappear, and that is the rarest of the cases. It can appear for the first time — nobody did this work before at all. It can fall to a different hand at the same place and the same volume. It can stay exactly the same, with only the consequence of a miss changing. And it can stay exactly the same in everything except the condition of the hand doing it. Nine rows exist for exactly one reason: so these cases do not collapse into one.

The glossary splits by the question asked of it. The first five forms measure place and volume: where the work now sits and how much of it there is. Four and four-a are a pair with opposite signs at one place: the workload either grew or fell, and both halves genuinely occur, sometimes on the same measurement. The next four measure the hand, the order, the consequence of a miss, and the condition of the hand itself. This split is not a formality: the last form carries the most uncomfortable observation of the lecture — the performer stayed the same, the place stayed the same, the volume stayed the same, and the result got worse.

The eighth form was introduced by this topic, and before it there was no glossary with this axis. It is easy to confuse with its two neighbors, so the distinction is worth memorizing right away. In the fifth form the hand changed: a different person does the work. In four-a the volume fell: there is less work. In the eighth, the hand and the volume are the same, and what fell is the quality of what that hand finds. A quick check is the question of what the share is computed from. Four-a has no share at all — it counts the absolute volume of work. The eighth has a share, and it is computed from the same thing: the same cases, the same people, a different share of findings.

Row eight-a is kept empty on purpose, and it needs to be read as empty. The reverse case — the same performer getting better without the assistant present — sounds just as plausible as the eighth, and there is not a single measurement behind it. A number is reserved for it, a name is given, there is no count. This empty row cannot be papered over with a similar case: any case where a person got better while the assistant was switched on measures something else — it measures the pair, not the hand.

Now for what the glossary of forms is not. It does not answer the question of whether something is good or bad. A form only names what happened to the work; whether it is a benefit or a cost is decided separately, by a number and a baseline. A workload move to a different step can be a pure benefit, when there is less of it at the new step, or a pure cost, when there is nobody to pay for it at the new step. The same holds for a drop in volume at the same place: one and the same visual-search measurement shows both a benefit and a cost, depending on whether the cue was correct or not.

And a last point, which is why this glossary is issued right here, at the start. Alongside it works a second tagging scheme — four payment addresses: attention, skill, signature, arbitration. These two things answer different questions about the same case, and must not be confused. A form says what happened to the work: where it now sits, how much of it there is, whose hand does it, what a miss ends in. An address says at which human step this happened: does a person review it second, did they stop doing it themselves, did they sign something unread, do they work through a disagreement. Work is capable of moving in a way no human step feels at all — and a human is capable of paying at a point where the work stayed exactly in place.

And the distinction without which the glossary reads wrong: a form says what happened to the work, and a payment address says at which human step it happened. The nine names answer the first question, the four addresses answer the second, and they must not be mixed: work moves in a way no human step feels, and a human pays at a point where the work stayed in place.

Where each form was found:

- **the workload moved to a different step** — the first-line chatbot · **Selection** - **the step disappeared entirely** — the dash · **Wrap-up** - **a step appeared** — the Swedish screening program · **Selection** - **the workload grew in the same place** — early warning · **Monitoring** - **the workload in the same place fell** — the Swedish screening program · **Selection** - **the performer changed** — the fundus case · **Selection** - **the order of work changed** — breaking a promise ·

Medications - **the cost of an error shifted** — "normal" without a check · **Oversight** - **the performer is the same, but got worse** — the share fell · **Qualification** - ***the performer is the same, but got better*** — *we keep the number empty*

2. Instrumental and Laboratory Diagnostics

In screening, the decision was who to examine. Here the person is measured by a device, and their condition is turned into a value that will be read somewhere else, at another time.

In order:

Example 1: Moscow — preventive scans that no human reads

Example 2: Moscow — intracranial hemorrhage, where autonomy was withheld

Example 3: Thailand — fundus reading in a primary-care office

Example 4: skin, and the composition of the dataset

Example 5: selection into a high-attention program — a score instead of a physician's decision

Instrumental and laboratory diagnostics is the second stage in the order of work. In the first section, the decision was whom to call in for examination at all; here the person is already in view, and data is taken on them — an image is made, a sample is taken, a device is connected, a load protocol is applied, and the measured value is watched for change. The industry itself names this stage exactly this way: instrumental and laboratory diagnostics. The term splits into two halves by where the measurement comes from. Instrumental is what a device took directly off the person. Laboratory is what was measured in a sample taken from the person and carried off to a lab. Both halves do the same job: turn the person's state into a value that can be recorded, transmitted, and read somewhere else, at another time.

The boundary with the third section is worth drawing right away, because without it the two sections blur together. Here the stage produces values. A conclusion about disease from these values will be reached in the next section, and by a different person. So everything on these slides is about whether the measurement is fit for use and what it means, not about what diagnosis follows from it.

Now a point that is hard to hold onto when reading marketing materials, and it is why this section exists in the lecture at all. Any share — sensitivity, specificity, accuracy — is computed on a specific group of people, and when carried over to a different group it stops meaning the same thing. The numbers on the slide show exactly this: one system with one trigger threshold gives eight correct alarms out of a hundred in a general population stream and seventy-nine out of a hundred in a specialized department. Not one line changes inside the system — only the share of sick people in the group changes, and that share was set by whoever decided who to call in for examination. The full count, in people, comes two slides from now.

Hence the order of the section: first what AI intervenes in, then the intervention. First, what the stage's work is made of — what actions happen in it, who sets the threshold, who decides whether a measurement is usable. Then, on its own slide, the classical regime done entirely without a model: image quality control. Then a country-by-country overview — what volume of images and analyses within an ordinary referral stream passes through a trained model. And only after that, five breakdowns, the first two of them domestic.

They are worth naming right here, so the slide gets recognized later by a name already heard. Each of the five will come up on screen under exactly this heading.

Moscow, preventive scans — the only breakdown in the lecture where autonomous use of the model did not fail, and where its cost was counted as a number. Sensitivity was set to one, specificity was declared not significant, and a human stopped reading more than half the preventive stream: 315,491 of 575,549 scans. The cost: 290 clinically significant misses, and the fact that in ordinary work nobody would have found them.

Moscow, intracranial hemorrhage — the same city, the same circle of registered products, the opposite answer. On 3,409 head CT scans, the model alone raised 293 false alarms against four from the physician. This is the only breakdown in the section where the reading itself failed, rather than the setting, the dataset, or the target.

The remaining three take their place where it is possible to fail without the reading itself being wrong.

Thailand — a deployment failure. Autonomous reading of fundus images in eleven clinics rejects twenty-one images out of a hundred for quality, and both causes are about the room where the image was taken, not about the model: the room's lighting and a slow connection.

Skin, and the composition of the dataset — a failure of training-data composition. Three models lose nine to fifteen points of area under the curve on darker skin tones, which were underrepresented in the training sets. This is the only breakdown in the lecture where the subject is the training set itself, not the algorithm, the deployment, or the target. And the only one where the fix is measured: fine-tuning on a representative set closes the gap, while three algorithmic bias-correction methods did not close it.

Selection into a high-attention program — a failure of target definition. Future treatment spending was taken as the target variable, while the decision being made was about illness, and spending reflects utilization, which differs across groups with the same underlying health.

Two mentions come with a number and do not become breakdowns: a systematic review of imaging studies in COVID-19 pneumonia — sixty-two selected out of two thousand two hundred twelve, and not one fit for clinical use — and the bias in a blood-oxygen-saturation device, in which there is no trained model at all.

What this section does not have, and why. No comparison of models against one another: this is not a market survey. No laboratory accuracy figures detached from their

group — by the same rule the section follows throughout. And not a single case where the reading itself failed: there was none among those broken down, and the section's conclusion says so directly.

The image alongside is a real workstation where data is taken. The conversation about shares starts here: first, someone puts the person in front of the device.

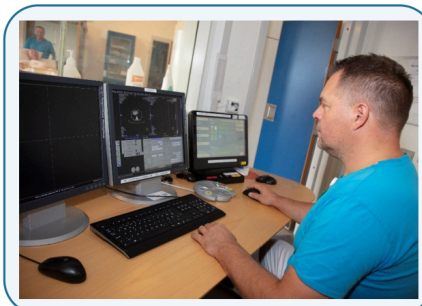
About the woman with the letter — her thread runs through the whole lecture, and in this section it lands in the most uncomfortable spot. She came in with no complaints, by age alone — that is, right in the general-population stream, where there is roughly one sick person per hundred. So every share anyone quotes her is computed on a group that is not hers, and the share of correct alarms for her own group is the worst one possible. The slide that counts the two groups leads exactly here, and her name is spoken directly there.

The evidence: six actions of the stage, and the two the model took over

Human — four: set the threshold → take the measurement → decide if it's usable → recompute for its own group

Model — two: read the measurement; report the result

Six actions of the stage; the model took over two



Four ways to take a measurement, and none can substitute for another:

- **Instrumental examination** — the device works directly on the person
- **Laboratory analysis** — the person is no longer present, a retake on the spot is impossible
- **Functional test** — a load protocol is part of the measurement conditions
- **Remote examination** — the reader never saw the patient

The model took over two of the six actions, and not the hardest two. It does not set the threshold against the cost of an error, cannot order a retake, and leaves input gradability to a human.

You're told "94 percent accuracy." On what group was it computed?

Roberts et al., Nature Machine Intelligence 2021

This stage is called instrumental and laboratory diagnostics, and that is the name the industry itself uses for it. The person is already in view by the time it starts: the previous stage decided that he gets examined. Now measurements are taken on him — an image is made, a sample is taken, a device is connected, a measurement is recorded in the chart.

The data here is of a different kind than at selection. There it was year of birth and an interval; here it is an image, a device signal, a reading from blood, urine, tissue, or a culture. And two questions get decided on this data, not one. The first is what the measured values mean. The second, which never appears in marketing materials, is whether these values are even fit to be read at all.

There are four ways to obtain such data, and all four need naming, because the breakdowns ahead will split across them. Instrumental examination gives values taken by a device: a plain film, a CT scan, an endoscopy. Laboratory analysis gives readings from blood, urine, tissue, a culture — and some of these carry their own turnaround time that will not go away: a sputum culture itself takes three weeks. A functional test takes values under load, meaning it adds to the measurement the condition under which it was taken. A remote examination is built so that the person is imaged in one place and a specialist reads the image in another; this is not exotic — it is what radiology in areas with no local radiologist runs on.

The decision this stage makes reads as: what has been measured about the person, and with what accuracy. Interpretation follows after that — reading what was measured and issuing a finding on it, and that is the work of the next stage.

Now for why this stage comes after selection and not before it. Measurement accuracy is described by two shares, and they are computed on different groups: sensitivity takes

only the sick, specificity only the healthy. The two are named as a pair and always for one threshold, and the protocol chooses the threshold against the cost of the outcome — chosen before the developer ever started work. What the physician at the visit actually gets is a third share, the share of alarms that are correct among all raised, and it depends on how many sick people are in the examined group. That group was assembled by the previous stage. So the order is forced: the denominator comes from selection, and before selection there is no readable meaning to the measurement at all. What this costs is counted in people on the next slide.

And how this decision is made without any model at all — by the classical regimes, by name. Quality control of the study: a human looks at the measurement obtained and decides whether it can be read or needs a retake, and the rule he decides by is written in the room's own instructions, not in the device. A specialist reading the measurement at a named trigger threshold: the threshold for declaring "sick" is set by protocol, and the reader works at that threshold without setting his own. The next slide is given entirely to the first of these — and it is given to it because it, and not the reading itself, is where the third of this section's five breakdowns, the Thai one, failed.

And the same field with the model, named before the breakdowns. Of the six actions, the trained model took over two: reading the measurement in full, without a specialist, and reporting the result — where instead of a value, a single score and a threshold-based cutoff land on the table. It does not set the threshold, cannot order a retake, leaves input gradability to a person without specialized training, and recomputing for its own prevalence is left to whoever applies it. It does add one step: a repeat visit for anyone whose measurement failed quality. Its benefit here is consistency: the ten-thousandth frame reads the same as the first.

A systematic review of work on COVID-19 pneumonia selected sixty-two out of two thousand two hundred twelve publications and found not one of them fit for clinical use. Every one of them computed the shares; almost none named the group they were computed on. The caveat is mandatory: this is a finding about the quality of research papers, and it says nothing about deployed systems. This mention does not become a breakdown — it has no numeric cost for either error, and no count against a classical regime.

And a question for the room, worth asking right here rather than at the end — it is also a one-move game. You're told "94 percent accuracy." I take one and the same system with one and the same trigger threshold and apply it to two groups of ten thousand people each, differing ONLY in the share of sick people among them. Not one line changes inside the system. The question: how far apart will the share of correct alarms be between these two groups? By how much — by half, by double, by tenfold? I'll ask for a number out loud, or one word in the chat. The next slide works this out, and the gap there is nearly tenfold; the room usually guesses one and a half to two. The question is asked right here, before the breakdown: a number the room names itself, it remembers and checks against later — a number it is simply handed does not stick the same way.

And a last thing worth saying before the breakdowns. The two actions the model took over are not the hardest of the six. The hard ones stayed with people: setting the threshold against the cost of an error, and deciding whether a measurement is usable. This is the overarching plot of the whole lecture, and in this section it shows most clearly of all.

Sources. Roberts et al., Nature Machine Intelligence 2021

How accuracy is counted: one trigger threshold on two groups of ten thousand

FINDS IN 100 SICK PEOPLE · FALSELY ALARMS IN 100 HEALTHY PEOPLE

94 · 11

one published setting, applied to both groups unchanged

OUT OF 100 ALARMS TURNED OUT TO BE DISEASE: STREAM → DEPARTMENT

8 → 79

94 of 1,183 versus 2,820 of 3,590; two groups of 10,000 people each

General population stream	1 sick out of 100	1,183 alarms	8 correct out of 100
Specialized department	30 sick out of 100	3,590 alarms	79 correct out of 100

One system, one trigger threshold; what differs is the share of sick people in the group

The properties of a measurement are computed on a group. One **trigger threshold** — the value beyond which the system declares "sick" — was applied to two groups of ten thousand people, differing only in the share of sick people.

The difference is nearly tenfold, and not one line changed inside the system. This share has a name — **positive predictive value**: what fraction of alarms turned out to be disease. This is what the physician at the visit actually sees, not sensitivity.

The woman with the letter stands in the general-population stream: for her it is 8 out of 100.

Rajpurkar et al., CheXNet 2017

One trigger threshold — the value beyond which the system declares "sick" — is applied to two groups of ten thousand people, which differ in exactly one thing: the share of sick people among those examined. The system catches 94 sick people out of 100 and falsely alarms 11 healthy people out of 100. The count is short. These two figures illustrate the method, taken from a published trigger threshold for a chest-image reading model; they do not serve as a characteristic of a specific product, and no procurement decision should be based on them.

Apply it to both groups with not a single change inside.

General population stream: ten thousand people in a row, disease in one out of a hundred, that is 100 of 10,000. Correct alarms 94, false alarms 1,089, total 1,183. Specialized department: the same ten thousand, but here people have already been selected by complaint or referral, disease in thirty out of a hundred, that is 3,000. Correct 2,820, false 770, total 3,590.

Now for what this count is actually for. The physician at the visit does not see sensitivity at all — he sees what share of the alarms that reached him turned out to be disease. In the first case that is 94 of 1,183, that is, 8 out of 100. In the second, 2,820 of 3,590, that is, 79 out of 100. The difference is nearly tenfold, and not one line inside the system changed: only the group changed, and the share of sick people in it was set by whoever decided who to call in for examination — not the developer. This is a selection decision, made for everyone before anybody even looked at accuracy.

This share has two names, because two professions name it differently. An engineer knows it as precision on the positive class. In clinical work it goes by the name positive predictive value, and it depends on prevalence — the share of sick people in the examined group. Prevalence is not a property of the device.

What this means for the physician. Explaining to a patient one false alarm out of ten and nine false alarms out of ten are two different jobs, and those conversations are built differently. What this means for the engineer. Carrying a system over from a specialized department to general-population screening changes the meaning of its output: the same signal of "sick" is an almost certain finding in the first case, and in the second, a reason for further examination that will end in the word "normal" nine times out of ten.

And from here, three ways to raise the share of correct alarms, none of which consists of improving the model. Raise the threshold: there will be fewer false alarms, more missed cases, and the exchange rate here is set by the cost of both errors. Change the group: examine people in whom the disease occurs more often. Add a second, confirmatory test and declare "sick" only on both — this is exactly how the two tiers of screening programs are built: the first is tuned not to miss, its output is not a diagnosis but an invitation to further examination, and its share of correct alarms is low by design. A program where the first tier declares the diagnosis loses the second tier entirely.

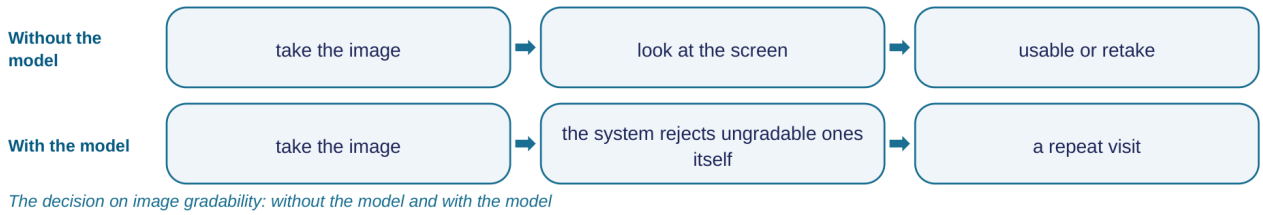
Hence the rule this section hands off to the whole lecture: a stated accuracy figure without a named group and a named prevalence does not mean anything. The right pair of questions to ask of any accuracy claim is: which two shares, and on what group; and what is the prevalence where you are planning to apply this.

One clarification of usage that this rule makes visible. "Six cases detected per thousand examined" and "prevalence six per thousand" are different statements. The first speaks to the outcome of the program's work on the examined group, the second to a property of the population. Substituting the second for the first turns up constantly in retellings and breaks the whole count, because a value nobody measured gets plugged into the denominator.

Now about the woman with the letter, because this arithmetic is about her. She came in with no complaints, by age alone, and stands in that very general-population stream, where there is roughly one sick person per hundred. So the share of correct alarms for her group is the left-hand one, eight out of a hundred, not the right-hand one. If she gets a call inviting her for further examination, there is an eight-in-a-hundred chance the finding will be confirmed, and ninety-two that it is a false alarm, and each of those ninety-two people will live their weeks with an open question. Not one line inside the system changes.

Sources. Rajpurkar et al., CheXNet 2017

The classical regime: image gradability is decided by the technician at the machine



The decision on image gradability: without the model and with the model

The classical regime of this stage — **quality control of the measurement.**

The rule the technician decides by: the room's own instructions and the visible quality of the frame. The device carries no such rule.

What happens if an ungradable image goes to reading anyway: the finding gets made on what is visible — and down the chain follow a wrong diagnosis, a repeat study, lost weeks, and treatment started too late. **Nobody notices,** until a review is needed.

With the model, the fork does not disappear — it splits in two: it rejects an ungradable image itself — and then the person gets a repeat visit; if it does not reject it, it reads whatever it was given.

The classical regime of this stage is quality control of the measurement, and it is decided by the technician at the machine. He has two rules: the room's own instructions, and the visible quality of the frame. The device itself carries no such rule, and the technician's decision rarely makes it into any document. This is work done entirely without a model, so this slide does not deal in numbers for the cost of either error or a count against the model — and it should not.

A chest-X-ray room in a district clinic. The technician positions the patient at the machine, takes the image, looks at the screen, and decides: usable, or retake. He decides based on what is visible in the frame — patient positioning, field coverage, the presence of artifacts, visibility of what is supposed to be visible. This rule lives in the room's own instructions and in the eyes of whoever took the image; it does not exist in the device at all. The diagram on the slide is exactly these three steps, and there is nothing more to the classical regime than that.

This work has its own unit of count, and it needs naming right here, while the regime is still classical: one gradability decision per image, made on the spot and almost never recorded. No record is left — which means there is no way, from the documents, to later count how many images got retaken. That is exactly why, in the breakdown that follows, the share of ungradable images came as a surprise to everyone: in the classical order nobody measured it, because there was nothing to measure it from.

If an ungradable image goes to reading, what gets read is an artifact — and the reader may never learn this, because he never saw the patient. The diagnosis gets made on what is visible, not on what was actually there. There is nobody but the technician to catch the substitution: no feedback ever comes back to him, and until a case gets reviewed, nobody usually finds out the image was ungradable at all.

Hence a general statement worth making before any breakdown at all: before any reading happens, someone decides whether the measurement is fit for use, and that someone is a human. Quality control is a procedure, and it never becomes a property of the device.

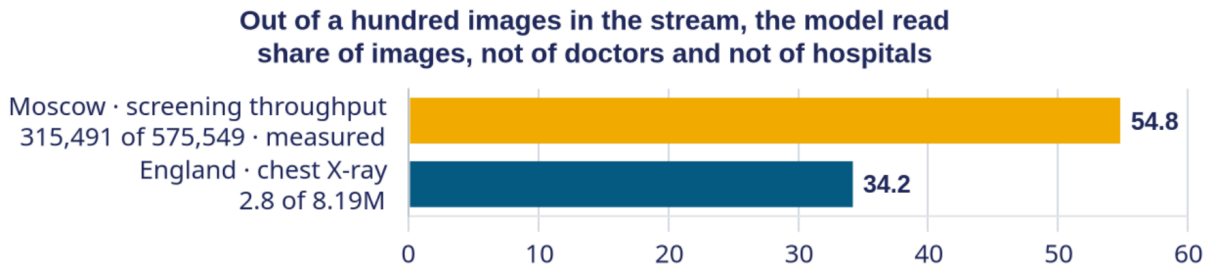
This work has a volume, and that volume is measured. The technician retakes the image, the patient changes a second time, a line forms. In the breakdown that follows, this work is counted directly: in autonomous reading of fundus images across eleven clinics, 393 of 1,838 images were rejected for quality — that is, 21 out of every hundred, over the first six months.

And here is where the scene stops being mundane. As long as a human does the reading, gradability stays an internal matter of the room: not satisfied — retake. When the reading becomes autonomous and happens on the spot, gradability turns into a condition the result depends on: there is nobody to order a retake, and nobody for a specialist to check the finding against either. Of the three reading regimes — double reading, arbitration, and quality control — this section breaks down only the third; the first two come later, together with the human who reads alongside the model, and are counted there.

It is worth saying why this slide is its own slide at all, rather than a line in the stage's description. Of the three classical reading regimes — double reading, arbitration, quality control — this section breaks down one, and it is this one because it is exactly the regime the very first breakdown will catch already changed. Double reading and arbitration arrive in the next section, together with the human who reads alongside the model, and are counted there in numbers. Here they are only named.

A question for the room fits here — a prediction on this slide's fork. With the model, the decision on image gradability splits in two: it rejects an ungradable image itself, and then the person gets a repeat visit; if it does not reject it, it reads whatever it was given. I ask: which of this fork's two errors costs the system more — rejecting a usable image, or reading an ungradable one? And I ask for one phrase of justification. There is no "universal" right answer: it depends on who pays for the repeat visit and who is held responsible for a missed disease — and that is the entire point of the question. The different answers in the room are exactly the material for the next step: the cost of an error is set before any accuracy figure, and it is not the model that sets it.

How much of the stream the model reads: one basis for three countries



Gradability of every image was decided by a human at the machine — now about the stream itself. There is one question: out of a hundred chest images taken in a country, how many did the model read. Any other comparison of countries here is a splicing of samples.

The English share has two official denominators. The numerator, 2.8 million, is stated by the developer. Health-service statistics give 8.19 million X-rays — which works out to 34 out of 100; the government's own announcement names "over 7 million" — the same 2.8 million gives 40 out of 100.

The US has no figure of this kind at all. What gets published is the share of products in the registry (76.5 of 1,430 over thirty years) and the share of physicians using the model (34 by survey). The second looks like the English figure down to matching digits, and measures something else.

Vasilyev et al., Digital Diagnostics 2025;6(1):5–22 · *NHS England, Diagnostic Imaging Dataset 2023/24* · *UK government announcement*

This slide's question is framed so it can be answered with one kind of number, and the basis was not chosen for convenience. Out of every hundred chest images taken in a country, how many did a trained model read. Any other way of comparing countries on this question turns into a splicing of samples — exactly the error this whole section teaches you to catch, and it would be odd to commit it on the slide that leads into the breakdowns.

Moscow, preventive stream: fifty-five out of a hundred. More precisely — 54.8 percent: the model received 575,549 preventive chest images over five months of 2024 and classified 315,491 of them as normal, after which not a single human read them. This is a share on a named sample, with a published method and a named trigger threshold. The breakdown of this case itself follows on the next slide, and the number comes from there.

England, chest X-ray: thirty-four out of a hundred. The numerator here is 2.8 million images a year, the same number that opened the evening; back then it stood alone, filling the whole screen, and I said it was volume and only volume. Now we are dividing it, and this is worth pausing on, because the share depends on the denominator more than it sounds like it should. That 2.8 million itself is the tool developer's own words, not the state's. The denominator comes not from them but from the health service's official statistics: in the year to March 2024, England performed 8,194,405 chest X-rays across all referral types. Division gives 34 out of 100.

But the government's own announcement about this same deployment, in the same release, names its own denominator — over seven million chest X-rays a year. With that, the same numerator gives forty out of 100. Two official denominators for one question, a gap of more than a million images, a six-point spread in the share. This is

stated here out loud, rather than one of the two being silently picked: a slide that teaches you to read the denominator has no right to hide its own.

It is worth naming separately who states which number, because these are different grades of evidence. The 2.8 million numerator comes from an interested party. The government does not give a numerator at all: it states the share of trusts — the tool runs in half of England's NHS trusts — and a count of people, four million, who got a faster diagnosis or had suspicion cleared faster. The share of trusts and the share of images do not have to match: a trust can be connected and still send through only a small part of its own stream.

Now the US, and here the slide carries not a number but a gap, named in words. The US does not publish a share of images read by a model. What gets published is something else: a cumulative 1,430 authorizations of products with a trained model, as of the end of 2025, 76.5 percent of them in radiology. This is a share of products in the authorization registry over thirty years, not a share of the annual stream.

There is a second American number, and the temptation to place it alongside is strong exactly because it looks alike: in a survey by a professional radiologists' association, thirty-four percent of physicians surveyed use a model in practice. It cannot stand next to the English thirty-four out of a hundred images — there it is a share of images, here a share of physicians. A physician using a model might run only one image in twenty through it. Putting these two numbers on one axis would mean committing a denominator swap right on the slide, and the matching first two digits would make it invisible.

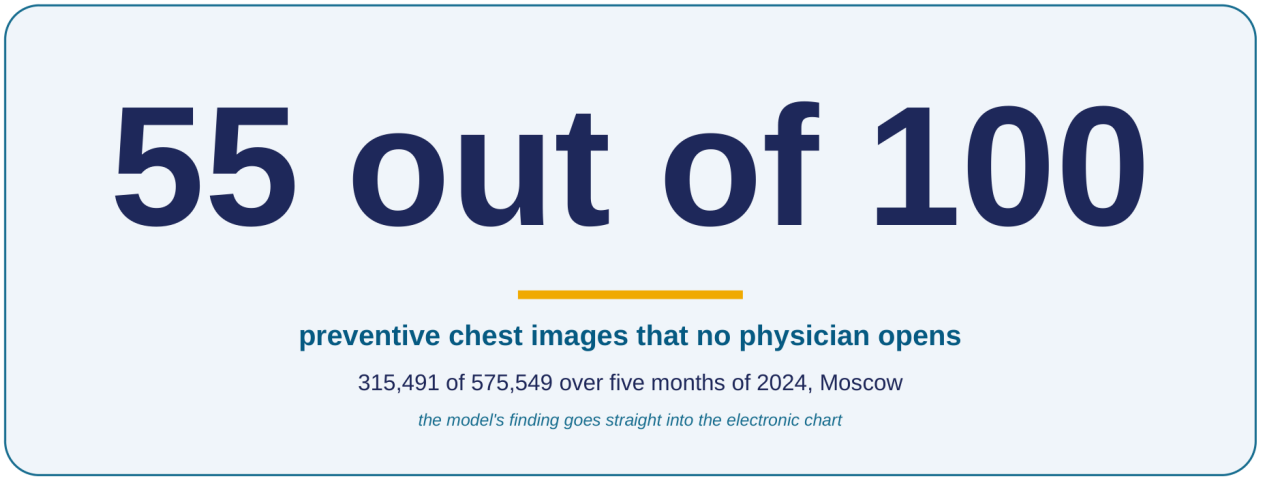
And what remains for Russia as a whole, beyond the Moscow sample. Moscow services interpret 1.4 million studies a month, about eighty percent of them coming in from the regions; cumulatively, over eighteen million studies, 75 regions, around two and a half thousand medical organizations. The mayor's office estimates this stream at a quarter of all images performed in Russia. Nobody publishes a denominator for this estimate — neither a national annual image count, nor the rule by which anything gets included in it. That is why it is not placed on the axis: the axis carries only what has a denominator named as a number.

Three grades of evidence on one slide, and they cannot be read as one row. The Moscow share is a peer-reviewed prospective study with a published sample. The English one has a numerator from an interested party, a denominator from government statistics, and the government itself provides a second one. The "quarter" estimate is an official's statement with no denominator at all.

The breakdowns follow next, and the first two of them are domestic: one about how, on this very Moscow stream, the model was sent to work without a human; the second, about a task in the same city where it was not allowed to work without one.

Sources. Vasilyev et al., *Digital Diagnostics* 2025;6(1):5–22 · NHS England, Diagnostic Imaging Dataset 2023/24, table 6 · UK government announcement, June 10, 2026

Moscow, the preventive stream: how many images no human opens



Vasilyev et al., Digital Diagnostics 2025;6(1):5–22

Fifty-five preventive images out of a hundred in Moscow go into the medical chart without ever passing through a human. The share is computed on a published sample — England has two such shares, the United States has none at all.

Fifty-five images out of a hundred. More precisely — fifty-four point eight percent: from May to September of year twenty-four, the model received five hundred seventy-five thousand five hundred forty-nine preventive chest images, fluorography and X-ray scans, and classified three hundred fifteen thousand four hundred ninety-one of them as normal. The finding on them went straight into the electronic chart. Not a single physician opened these images.

I'll state the denominator separately, because without it the number reads wrong. Five hundred seventy-five thousand is not the city's entire radiology stream, and not all chest images. It is the preventive stream: images taken on healthy people by plan, not by complaint. Six hundred forty-two thousand six hundred eighty-one studies were sent into the system; five hundred seventy-five and a half thousand went into the analysis. On a complaint-driven stream the share would be different, and the two cannot be compared.

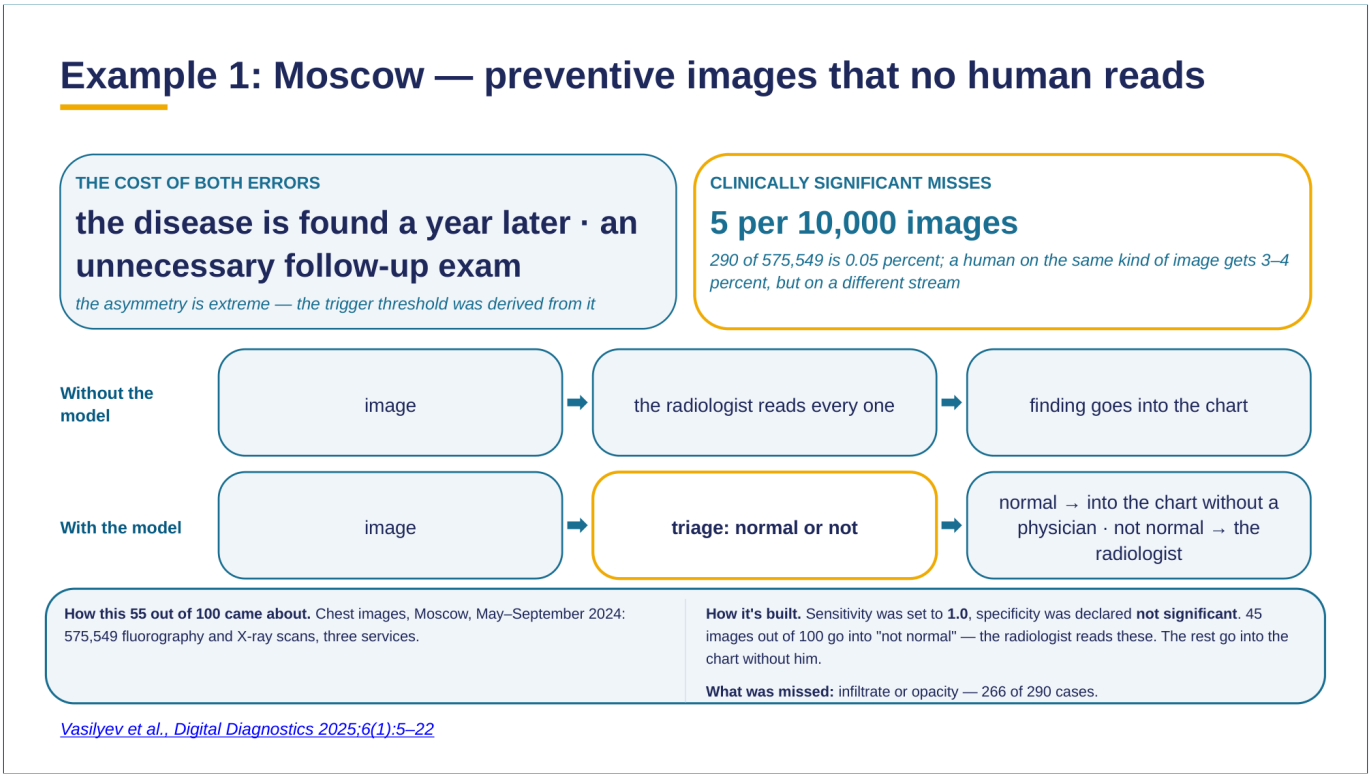
And right away, a distinction without which this hit turns into a slogan. Fifty-five out of a hundred is volume, and only volume. It says neither that this is good nor that it is bad. It says that the decision "a human will not see this image" is made in the city three hundred fifteen thousand times over five months, that is, roughly two thousand times a day, and it is not a human making it. What has been paid for this is the next slide, and the payment there is counted: two hundred ninety clinically significant misses, and minus forty-three point seven percent of tariff revenue.

Why this is a standalone hit, not a line in the breakdown. The whole evening rests on

one question: what does the human do after the model. At the third tier of participation — "looks only at what's in dispute" — the answer turns out to be literal: on half the stream, he does nothing, because that half never reaches him. The number stands as its own slide so the room hears it before it starts working through the trigger threshold and the tariffs: scale of the decision first, its mechanics second.

One note on the technique itself, in case anyone asks why it was tuned this way. Sensitivity is set to one deliberately: an alarm is raised on anything even remotely resembling an abnormality, and specificity is declared not significant. This is exactly why forty-five images out of a hundred go to a human and fifty-five do not. The breakdown of this decision is on the next slide; what matters here is only that the trigger threshold is not a default setting — it is an adopted and recorded decision.

Sources. Vasilyev et al., *Digital Diagnostics* 2025;6(1):5–22



Moscow, May–September 2024: 575,549 preventive chest fluorography and X-ray scans, three registered services. Sensitivity was set to one, specificity was declared not significant — and a human stopped reading more than half of this stream. The breakdown has to start with the cost of both errors, because the entire engineering decision for this case was derived from it.

A preventive chest image is taken on a person with no complaints. A missed infiltrate on such an image means tuberculosis or a tumor that will be caught on the next cycle, a year later, and the cost of the miss is measured by the stage at which the disease is caught. An unnecessary alarm on a preventive image is incomparably cheaper: the person just gets called in for a follow-up exam. The asymmetry here is extreme, and it is exactly what set the trigger threshold.

The classical regimes, counted in the same unit. The first: a radiologist reads every preventive image. The share of clinically significant misses for a human on a chest X-ray is three to four percent, and that figure comes from the same center that ran the study, on a sample of 8,815 images and 403 specialists. The second regime is double reading, two radiologists per preventive image. It is not used at mass scale, and not for methodological reasons: there are not enough readers even for a single reading. The first regime's cost is also named, and named by the mandatory health insurance tariff: a human reading one preventive chest X-ray costs 132 rubles 98 kopecks.

Now the model against these regimes, and the mechanics matter more than the result here. The trigger threshold was set as follows: sensitivity 1.0 with a confidence interval from one to one, specificity ruled not significant. This is an engineering decision, not a property of the model. With this setting, the system is bound to raise an alarm on anything even partly resembling an abnormality, and the price it pays for that is that

almost half the stream ends up in "not normal."

From there the stream splits in two. Images classified as "not normal" go to a reference-center radiologist — 260,058 of them, forty-five out of a hundred. Images classified as normal are read by nobody: the model's finding goes straight into the electronic chart. There are 315,491 of them, fifty-five out of a hundred, and that is the entire benefit of this case.

It was checked in a way routine practice never checks and never will. All 315,491 "normals" were re-read by radiologists from a federal academy — an organization not involved in the deployment. They raised 942 suspicions. A thoracic radiologist confirmed 411 of them, and the remaining 531 turned out to be false alarms from the checking physician himself — and that number is worth holding onto: the human doing the checking here was wrong more often than the machine being checked. Of the 411 confirmed, 290 were clinically significant, 121 were not.

The outcome in one line: 290 clinically significant misses out of 575,549 images, that is, five per ten thousand, with a confidence interval of four to six per ten thousand. The triage was correct in 99.95 percent of cases.

These five per ten thousand cannot be compared directly with the human three to four percent, tempting and devastating as the comparison looks. The human share is computed over the entire stream, including the hard images. The model's share is computed only over the subset it itself selected as easy. The two numbers do not share a denominator, and a sixtyfold difference does not follow from them.

What exactly was missed is worth naming item by item. Infiltrate or consolidation — 188 of 290 cases. An area of opacity — 78. Widened mediastinum — nine, pleural effusion — five, a break in the integrity of a bone plate — three, atelectasis — three, dissemination — two, a cavity — two. The first two items account for ninety-two percent of all clinically significant misses, and these are exactly the findings a preventive image is taken to catch.

The cost of this benefit — the money, the regulatory change, and the vanished layer of review — is broken down on the next slide.

Sources. Vasilyev et al., *Digital Diagnostics* 2025;6(1):5–22

The first thing that moved — a layer that caught the error disappeared. In routine operation, not one of the 290 misses would have been found by anybody. They were found only because the study's protocol required humans to re-read all 315,491 "normals". Once the study ends, this layer will not exist, and the share of misses will stay exactly what it is. In other words: the number we just praised the model with was obtained by means that do not exist in its routine operation.

Second: reading got more expensive per image. The tariff for reading a fluorography image classified as "not normal" is 155 rubles 62 kopecks, against 132 rubles 98 kopecks for the old blanket reading. An X-ray is 157 rubles 56 kopecks. Every image a human still reads got more expensive. The model's autonomous finding, meanwhile, costs 6 rubles 90 kopecks for a fluorography image and 8 rubles 80 kopecks for an X-ray — more than twenty times cheaper than human reading.

That is why the savings arithmetic looks exactly the way it looks. Over five months, tariff spending came to 43 million 89 thousand rubles, against a modeled 76 million 537 thousand under the old regime — minus forty-three point seven percent, a saving of 33 million 448 thousand. And it does not come from reading getting cheaper: reading got more expensive. It comes from the fact that nobody reads more than half the stream.

Third: before any work could happen, the regulatory base had to be amended. Four new service codes were added to the 2024 mandatory health-insurance program's tariff agreement. Without this amendment, the model could not operate at all — there was nothing to pay for its finding with. This is worth an engineer noting: the system's technical readiness and the ability to actually switch it on turned out to be two different things here, and the second was settled outside the technical plane entirely.

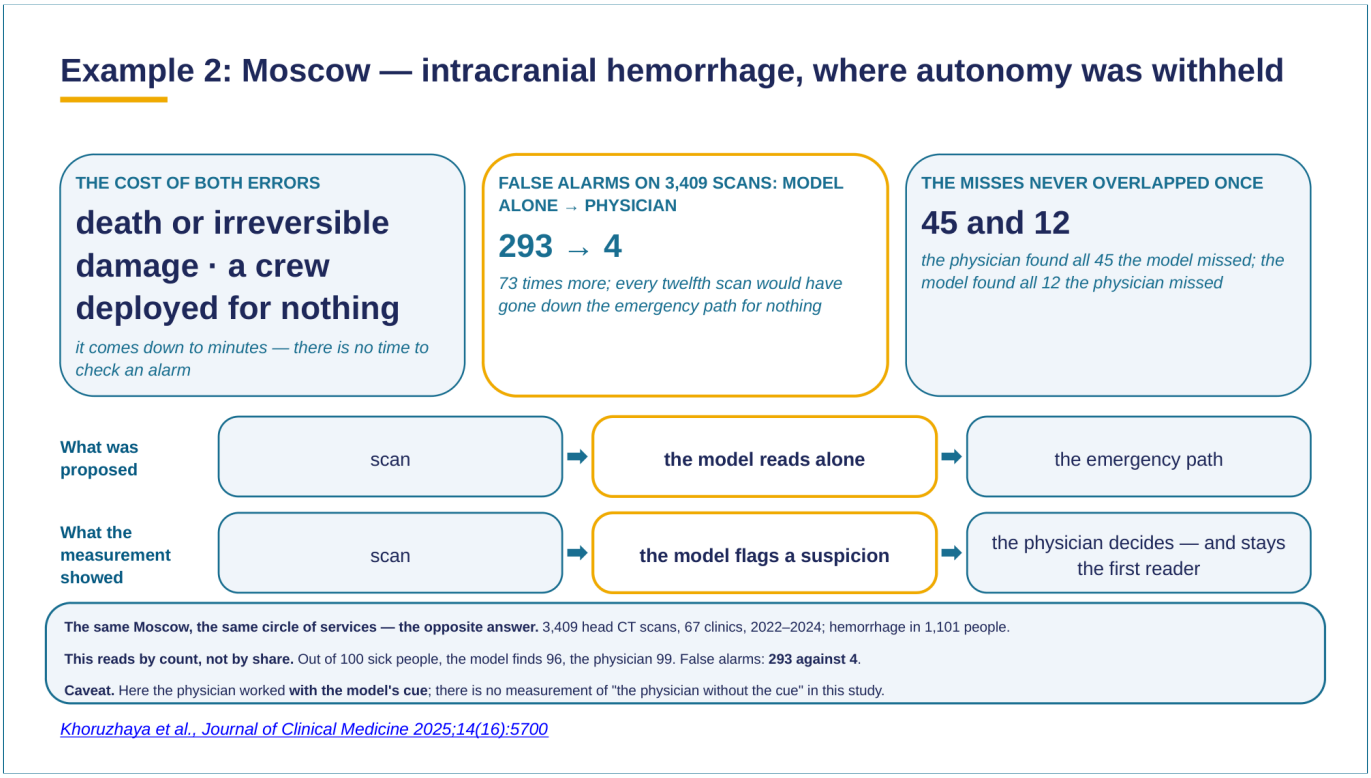
Fourth: the burden of misses fell unevenly. Half the stream is people aged eighteen to

forty-four, 155,898 of 315,491 people, and they carry the largest share of misses: twelve per ten thousand against seven in the older groups. For men, the odds ratio for a clinically significant discrepancy is 1.32, with an interval of 1.04 to 1.66. The average share across the stream — five per ten thousand — describes none of these groups.

The pair that closes this breakdown. Delivered: 33 million 448 thousand rubles over five months, and half the stream taken off the radiologist's plate. Required: 290 missed clinically significant findings, not one of which will be found under ordinary operation.

And the lesson in full, because this is the first case in the section where autonomous use of the model did not fail. It is built honestly: the task is narrow — triage into "normal" and "not normal", not a diagnosis; the trigger threshold is set against the asymmetry of cost; the benefit is counted against a named classical regime in its own rubles; the cost is named with a number and an address. And it also shows what this rests on: the share of misses is known exactly because it was measured, and in routine operation there is nothing left to measure it with.

Sources. Vasilyev et al., *Digital Diagnostics* 2025;6(1):5–22



Moscow, sixty-seven clinics, April 2022 to December 2024: 3,409 head CT scans, intracranial hemorrhage in 1,101 people. Three registered services reading the scan on their own were compared against the physician's finding. The cost of both errors here is of a different order than on the preventive stream, and everything else follows from it.

Intracranial hemorrhage is a condition where it comes down to minutes: how fast it is seen on the scan decides whether the person survives to treatment or not. A miss here is not a stage delayed by a year — it is death or irreversible damage. But a false alarm is not free here either: it is a deployed emergency crew and an operating-room slot taken up for nothing. And the key thing is that there is no time, on this path, to calmly check an alarm.

The classical regime: a radiologist reads the head scan himself. In this study he is taken as the comparison point too — but in a particular form, and that form has to be named before the numbers. The physician had access to the model's output as an assistive cue. There is no separate measurement of "the physician without the cue" in this study. So this study does not measure what the cue itself contributed; it measures something else — the model alone against the physician with a cue.

The share of sick people in the sample is thirty-two out of a hundred, and it has to be kept in mind: on a stream where hemorrhage occurs in a handful out of a hundred, these same shares would produce a completely different picture on the physician's desk.

The physician with the cue outperformed the model on every metric, and the gap in the shares looks moderate. Out of a hundred sick people, he finds ninety-nine against ninety-six. Out of a hundred healthy people, he does not alarm ninety-nine point eight against eighty-seven point four. This second pair is the entire story, but it does not read

in shares — it reads in cases. On the same sample, the model raised 293 false alarms, the physician four. Seventy-three times more.

The main observation of this breakdown, and the reason it exists at all: the misses did not overlap in a single case. All forty-five hemorrhages the model missed, the physician found. All twelve the physician missed, the model found. Agreement between the two methods: eighty-nine point six percent.

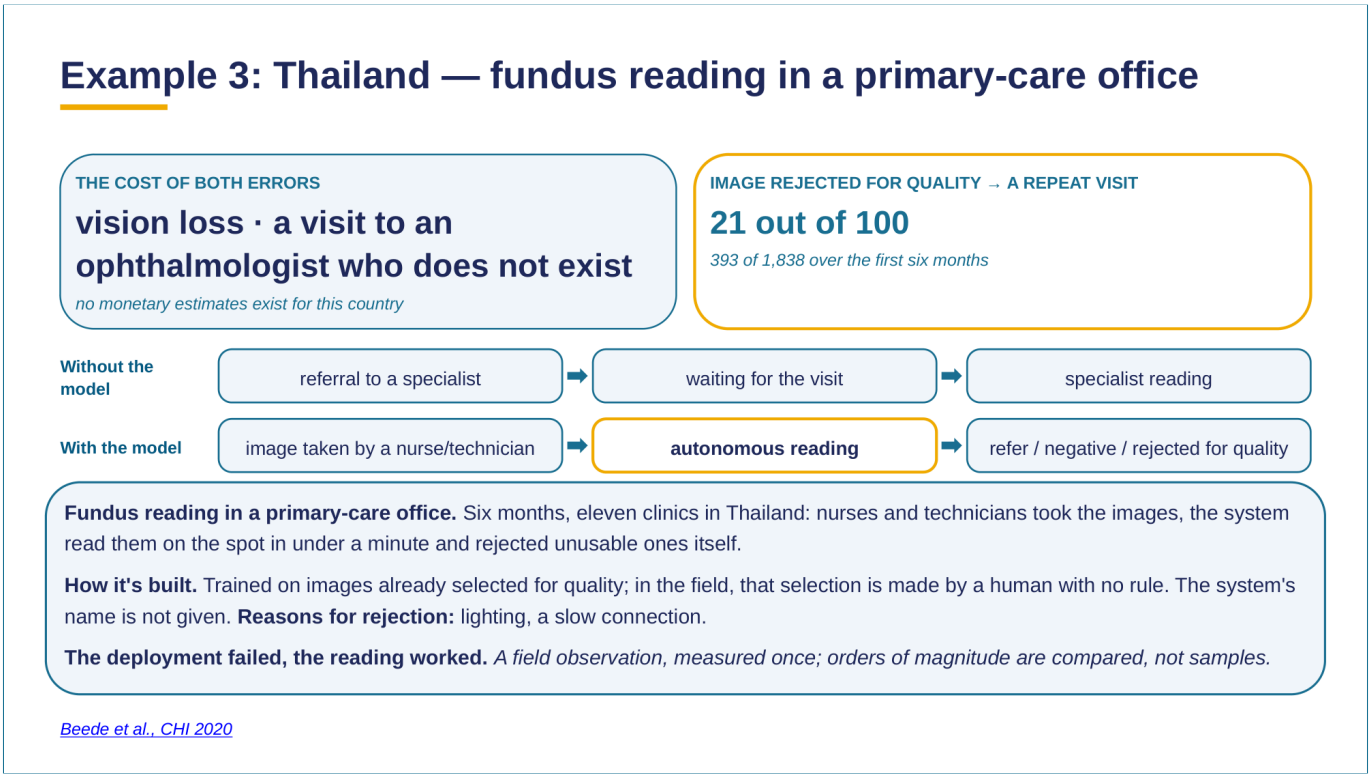
What moved here. Autonomous operation is not considered at all: 293 false alarms out of 3,409 studies mean that every twelfth scan would go down the emergency path for nothing, and on the emergency path a false call costs more than a wait in line. The complementarity of the misses moves the model from first reader to second — its value is not that it reads better, but that it errs in different places. And for that, the work of sorting through 293 false alarms arrives — done by the very physician the model is supposed to help.

The pair that closes this breakdown. Delivered: twelve hemorrhages the physician missed and the model caught. Required: two hundred eighty-nine extra reviews on the same 3,409 studies.

And the lesson these two domestic breakdowns stand next to each other for. The task is the same — reading an image with a trained model. The city is the same, the circle of products is the same, the oversight is the same. The conclusion is the opposite: there, the model was sent to work without a human, and what that cost was measured; here, the authors state directly that the model's place is second reader, not first. What tells these two cases apart is not the quality of the models, but the cost of a false alarm and whether there is time. On the preventive stream, a false alarm costs one follow-up exam, and there is time. On the emergency path, it costs a deployed crew, and there is no time. From the same property of the model — raising alarms generously — one task gets permission to work without a human, and the other gets a ban on it.

And one last thing this breakdown adds to the section: it is the only one of the five where the reading itself failed. Not the setting, not the composition of the training set, not the definition of the target — the reading.

Sources. Khoruzhaya et al., *Journal of Clinical Medicine* 2025;14(16):5700



Eleven clinics in Thailand, the first six months of deployment: primary-care nurses and technicians take the fundus image, the system reads it on the spot in under a minute and rejects an unusable one itself. It rejected twenty-one images out of a hundred — 393 of 1,838. The cost of both errors here is counted in outcomes and in visits; there are no monetary estimates for this country.

A missed fundus lesion in a person with diabetes means irreversible vision loss: up to the late stages it is asymptomatic, the person feels nothing and does not come in on his own. A false alarm is a referral to an ophthalmologist who may not exist in the area at all, and a line where those who actually need the referral have to wait. There are no monetary estimates for this country in open sources, and American ones do not carry over here, so the cost is named in outcomes and visits.

The classical regime is named in words, with no number: once a year, a person with diabetes visits an ophthalmologist, who takes and reads the image himself. There is no count for this regime on the same Thai sample in the source, and plugging in attendance from a different trial — a different country, a different age group — would mean splicing two samples into one. There is no second classical regime for this task at all: double reading of fundus images never happens in primary care — there is nobody there to do it.

The model against this regime: autonomous reading of two fundus images right in the primary-care office, a result in under a minute, no ophthalmologist. Input — two images, output — the same label a specialist would assign. It was trained on a lab dataset of images that had already been selected for quality, and this is exactly where the whole breakdown lies: in the field, this selection is made by a human given no rule for it. The study names no name for the system — there is nothing to name it with, and

this is a refusal, not an omission.

The source is silent on the size and composition of the training set, on how often it is retrained after deployment, and on any drift-monitoring protocol: this is an observational deployment study, describing the setting and its problems, not the model's engineering.

And here is what the field showed. Over the first six months of deployment across eleven clinics, 21 images out of a hundred were rejected for quality — 393 of 1,838 taken. Both named causes are about the setting, not the model: the lighting in the room where the image was taken, and a slow connection that kept the image from making it to processing in time. Neither cause is visible on the lab dataset.

Twenty-one out of a hundred is the share of ungradable images, and it must not be confused with a no-show rate: attendance was measured by a different breakdown, on a different sample, in a different country.

The lab accuracy of the predecessor model is published, but it does not enter this breakdown: it has no named group and no named prevalence. The section's rule applies even to a number that would flatter the breakdown.

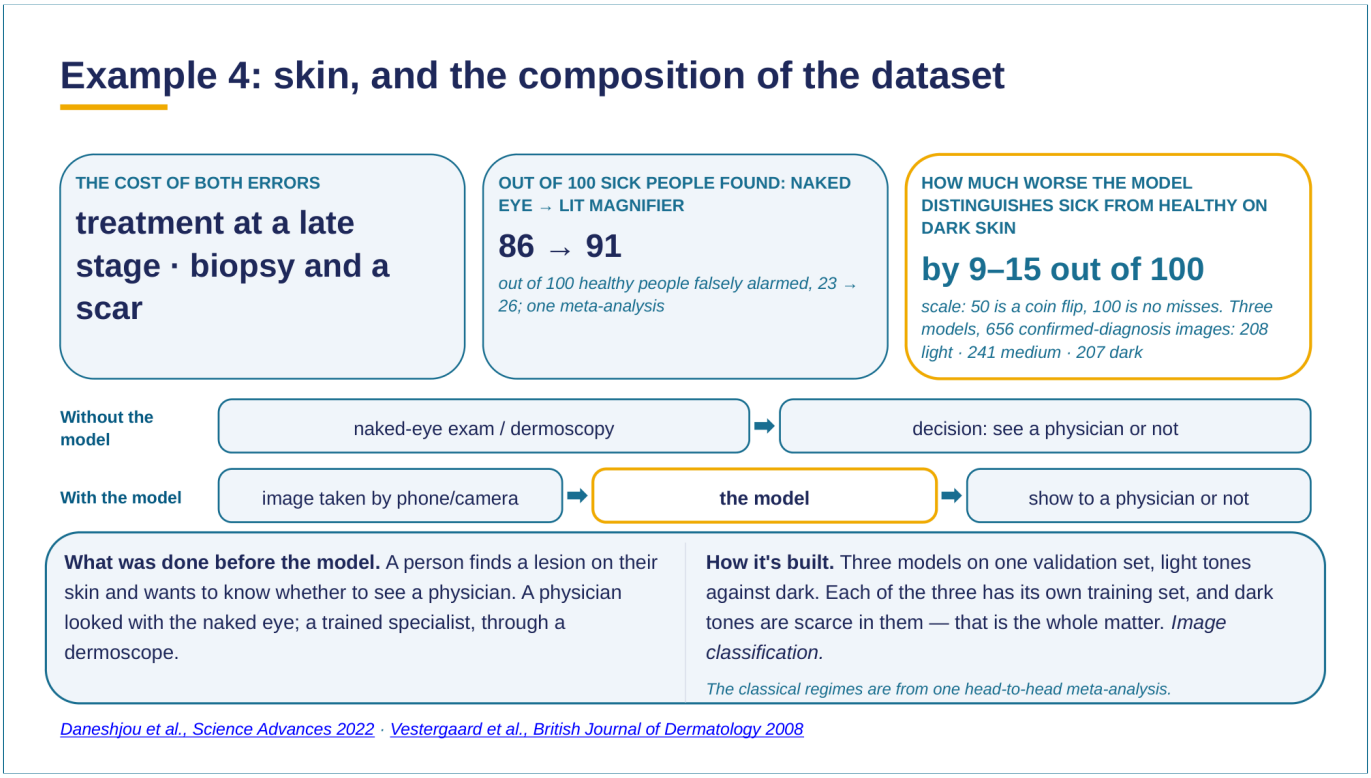
Now for what moved in the work, and this is the case's leading change. The workload grew in the same place. The place and the hand are the same — the primary-care technician, the same camera — but the volume of work is larger: he has gained the job of obtaining a gradable image. The ophthalmologist never had this operation at all, because he compensated for the imaging conditions himself: adjusted the position, retook the shot, looked at the result with his own eyes. Alongside it, a step appeared that did not exist in the classical order: a repeat visit for anyone whose image did not pass. That is 21 people out of every hundred examined, and each of them has to come back a second time.

The pair that closes this breakdown. Delivered: a result right in the primary-care office, with no referral and no waiting for a specialist. Required: a gradable image under conditions the system was never tested against. Net: at the start of deployment, the result came in worse than the lab measurement, and the gap is explained by the mechanics of the setting, not the mechanics of the model.

The lesson worth taking from this breakdown as a whole. The deployment failed — the reading worked. The model arrived ready; the setting for it was not, and nobody measured that very setting before deployment. Neither of the two named changes is about the model: both are about the setting.

And an honest caveat on the strength of the evidence. This is a field observational deployment study: eleven clinics, the first six months. Measured once, with no direct replication.

Sources. Beede et al., CHI 2020 · illustration — Skevas et al., BMC Ophthalmology 2024, fig. 3



A person finds a lesion on their skin and wants to know whether to see a physician. Before the model, this question was settled two ways: a physician looked with the naked eye; a trained specialist, through a dermoscope, a lit magnifier. Here the same question is answered by a model from an image, and three such models were tested on one set of six hundred fifty-six images with confirmed biopsies. On darker skin tones, all three distinguish sick from healthy nine to fifteen points worse than on lighter tones.

The cost of both errors is named by stage. A missed malignant skin lesion means treatment at a late stage rather than an early one; there is no single survival figure by group, computed on one cohort over one period, in the open summaries, so the cost is named by stage rather than by percentage. An unnecessary alarm means a biopsy, a scar, and months of waiting for the result.

Both classical regimes come from one study and one methodology: a head-to-head meta-analysis showed that a naked-eye exam finds eighty-six sick people out of a hundred at a specificity of seventy-seven, and dermoscopy by a trained specialist finds ninety-one at a specificity of seventy-four. A more recent review of the same task gives different numbers for the naked-eye exam, and gives no pooled estimate for dermoscopy at all, due to the heterogeneity of the studies; the two methodologies are not placed side by side, or the comparison would run between counting methods rather than between regimes.

The composition of the validation set is named item by item: two hundred eight light tones, two hundred forty-one medium, two hundred seven of the darkest. The denominator is respectable, and there is no basis here for talking about "a handful of examples."

Here is how it's built: input is an image of a skin lesion, output is a "show to a physician

or not" score. The three models each have their own training set, and dark tones are scarce in them — that is the entire breakdown, and it is the only case in the lecture where the subject is the composition of the training data, rather than the algorithm, the deployment, or the choice of target. The publication gives no names for the models themselves; the validation sample is named — DDI — and the study can be found by it.

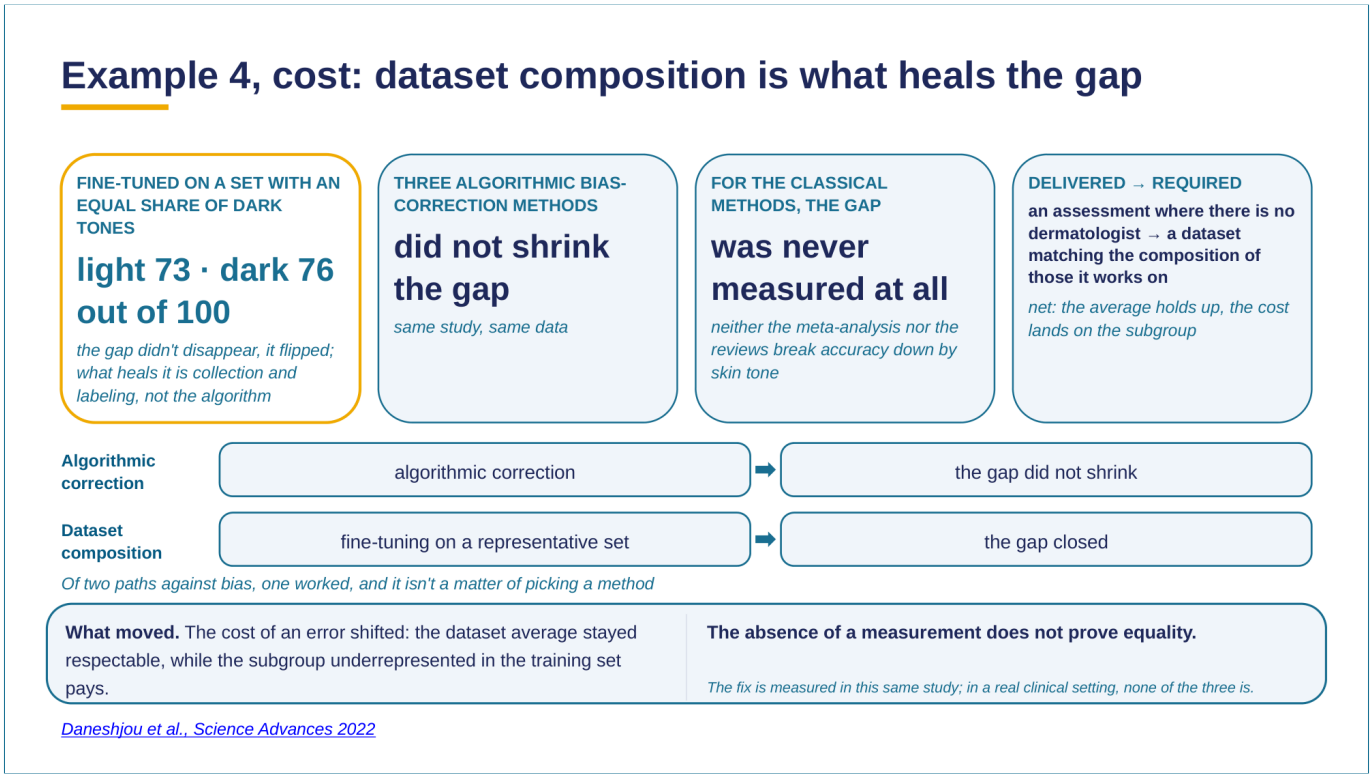
This breakdown does not take one commonly circulated figure, and the refusal is worth stating directly. The phrasing "sensitivity drops twenty to thirty percent on dark skin" was not found in the publication, and its source cannot be identified. The nearest actually measured figure looks different and belongs to one of the three models: sixty-nine sick people out of a hundred on light tones against twenty-three on dark. The section's rule is applied to a number that would have flattered this breakdown — and that is exactly why it is applied.

The unit of measurement is named precisely: these are points of area under the curve, a measure of distinguishing sick from healthy, not a share of sick people found. One cannot be substituted for the other — they are quantities of a different nature, and a gap in area-under-the-curve points does not translate into a percentage of misses. If anyone in the room converts it into a percentage, they will get a number nobody measured.

The strength of the evidence. This is a test on a curated, biopsy-confirmed corpus, not a field observation of deployment: the study says nothing about how these models behave in an actual clinical setting. The denominator is named, the composition is named, the unit is named — and that is where its value ends. The finding cannot be carried over to a specific product: there are three models, and none of them is named.

What to take from this case. The model works, and it works unevenly for different people, and the difference lies in whose images it was trained on. The task here is a familiar one — image classification; the subject of this breakdown became the training set, and this is the only place in the lecture that asks not about the algorithm and not about the deployment, but about the composition of the data.

Sources. Daneshjou et al., Science Advances 2022 · Vestergaard et al., British Journal of Dermatology 2008



There is one form here, and it is named directly: the cost of an error shifted. The place is the same and the performer is the same, but what the error actually costs got distributed differently — differently for different people. Average accuracy on the dataset stays respectable, and a deployment report with one number will show everything is fine. The subgroup whose images were underrepresented in the training set is the one that pays, and its contribution to the average sinks by construction: it is the smaller part of the dataset.

The fix in this breakdown is actually measured, and that is a rare thing — bias usually gets discussed without the remedy ever being checked. Fine-tuning on a representative set closes the gap, and even flips it. Three algorithmic bias-correction methods did not shrink it significantly. The conclusion here is uncomfortable for engineering taste: what fixes the gap is collection and labeling — the part of the work most often outsourced and treated as grunt work. The model does not fix it.

What an engineer should take from this. The composition of the training set is not grunt work and not something to hand off to an intern: here it turned out to be the only thing that fixes the measured gap, while three methods built specifically to fight bias did not fix it. This is an uncomfortable conclusion, and it is checkable — the check sits in the same study, right next to the gap itself.

The pair that closes this breakdown. Delivered: a standing assessment of a skin lesion where there is no dermatologist — access, not accuracy. Required: a dataset that mirrors the composition of those the system will actually work on, and knowing that composition in advance. Net: the average holds up, the cost of the error moved onto a subgroup, and you cannot see this from the average.

And the line that carries the whole breakdown. The gap by skin tone is measured for

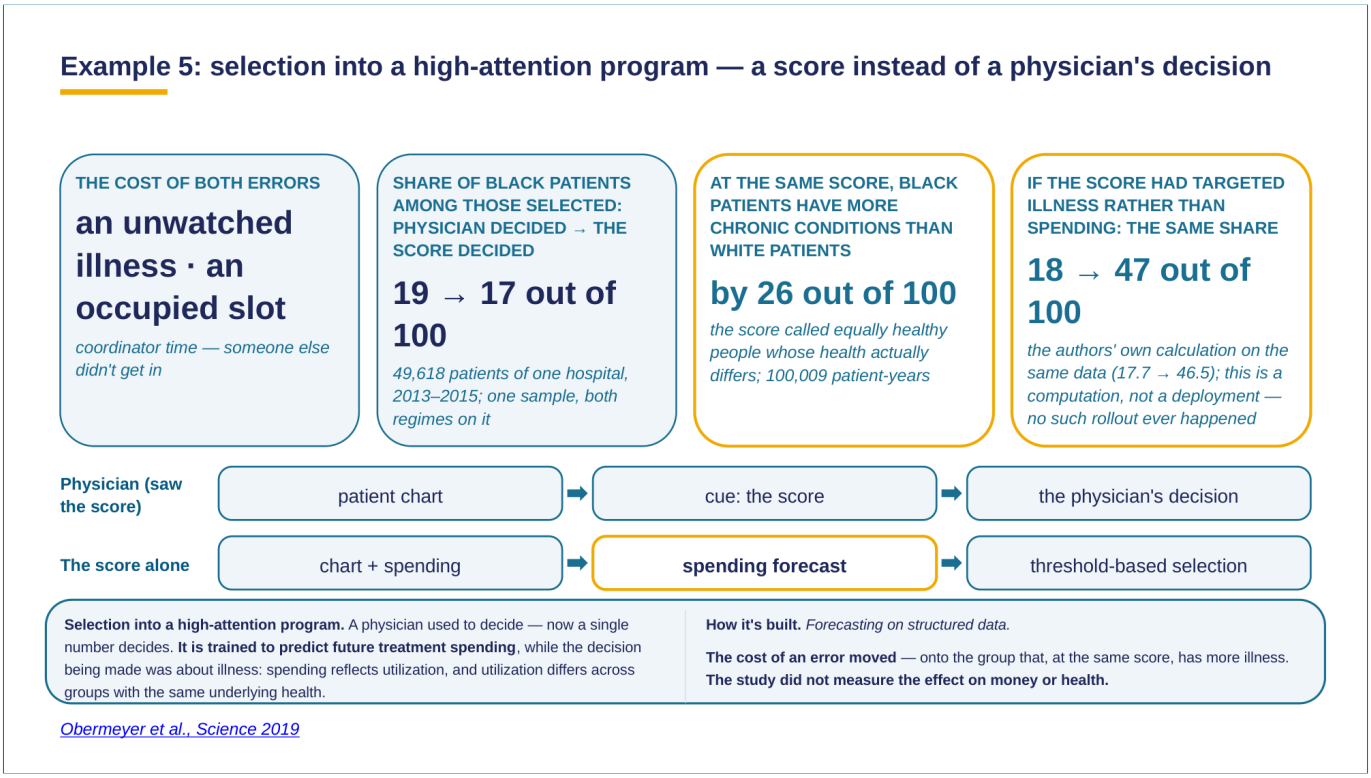
the model, and never measured at all for the classical methods: neither the direct-comparison meta-analysis nor the reviews break accuracy down by skin tone. So there is nothing to say "the model is worse than the human on this subgroup" with, and claiming it would be exactly the mistake this breakdown exists to catch. The correct statement is different, and checkable: for the model the gap is measured and named; for the human it is measured by nobody, and the absence of a measurement is not evidence of equality.

This is worth holding onto for a second, because the opposite gets said more often. From the fact that the gap by skin tone is unmeasured for classical examination, it does not follow that classical examination has no gap: it only follows that nobody looked. The difference between "measured and equal to zero" and "unmeasured" is exactly the difference between evidence and its absence, and confusing the two is equally wrong in either direction.

Hence a practical question for any vendor: on whom was your average computed, and which subgroups dissolved into it. This question is worth asking before purchase, not after deployment, because after deployment the average will be the only number in the report — and it will look good.

And about how this cost looks in a report. If a system like this were deployed and measured with one number — average accuracy across all images — the report would show smooth, honest well-being: the average really does hold up. The shifted cost can only be seen by breaking the measurement down by subgroup, which means knowing in advance which subgroups to break it down by. This study broke it down by skin tone; which subgroup will turn out to matter in the next task does not follow from the model itself in any way, and someone who knows the subject has to name it.

Sources. Daneshjou et al., Science Advances 2022



This is about a high-attention program: a hospital sets aside a limited number of slots for people who need closer-than-usual monitoring. The cost of both errors is counted in slots. Someone who needed the program and didn't get in — the illness progresses unwatched, and it is found out at hospitalization. Someone who didn't need it and got in anyway — a slot and a coordinator's time are spent, meaning someone else, a third person, didn't get in.

Both classical regimes are counted on the same sample. With selection by the treating physician's decision, the share of the group in question among those actually selected is 19.2 out of 100. With selection by a single spending-forecast score, it is 17.2 out of 100. The two numbers are close, differing only by the label of the regime, which is why they are not said back to back.

A limitation of this count has to be named out loud, and it matters: the physician selecting patients had already seen the score — above the threshold, he received a cue. So what is counted is selection by a physician who has the score at hand; there is no selection without the score anywhere in the publication. The comparison therefore runs not between "human" and "model", but between two regimes, in one of which the human works with a cue.

The model against these regimes. The system computes one score per patient, selection runs on a threshold, and the target variable taken is future treatment spending. The result: at the same score, one group turned out to have 26 percent more illness. In other words, to get into the program, they had to be sicker at the same assessed risk. The base is named in full: 49,618 patients of one academic hospital, 100,009 patient-years, 2013–2015. The denominator belongs to one hospital, and conclusions about the size of the skew belong to it, not to the industry.

Here is how it's built: input is chart records and spending data, output is one score per person and threshold-based selection. The publication gives no name for the commercial algorithm — and this is a refusal to name it, not an omission: the authors are analyzing a system they are not permitted to name.

The mechanism is worth unpacking slowly: the choice of target variable was engineering-sound, and it is instructive exactly because of that. Spending exists for every patient, it is numeric, requires no labeling, and really does correlate with illness. But spending reflects utilization — how much a person has used medicine — and utilization differs across population groups with the same state of health. The error is not in the model, and not in the data, taken separately: nobody wrote down what the decision was supposed to target, and what the data happened to contain became the target instead.

The authors calculated what would happen if the health gap at the selection boundary were removed: the share of this patient group among those selected would rise from an observed 17.7 to 46.5 out of 100. This has to be stated in the conditional, or the exercise gets passed off as an implemented fix. The calculation works like this: at the boundary, a healthier patient is swapped for a sicker one until the gap closes, and the resulting composition is observed. There is no deployment behind this, and no retraining either.

The authors also actually retrained a model, and the result is weaker than the calculation. Their own predictor, based on the number of active chronic conditions, gives 26.7 out of 100 in the highest-risk group, against 14.1 for their own spending predictor trained on the same data. And the observed share of 17.7 belongs to the commercial algorithm the authors are analyzing — a third model — and one cannot be subtracted from the other: no such difference exists in the original source. The more precise lesson: changing the target variable moves the composition of those selected, and still falls short of 46.5.

What moved. The performer changed: selection used to be done by the treating physician, now a single score does it. And the cost of an error shifted — onto the group that, at the same score, has more illness. The pair that closes this breakdown: delivered consistent selection where manual selection is inconsistent; required a match between spending and illness that does not exist; net, for one group selection got worse than before, and the fix is, to this day, only calculated.

On scale and its denominator — separately, because these two numbers are easy to combine wrongly. Models of this class are applied, by industry estimate, to roughly 200 million people a year. This is a coverage estimate, not a measurement sample: the skew was measured on 49,618 patients of one hospital, and conclusions about its size belong to that denominator, not to the two hundred million. These numbers are in different units and cannot be added together — which is why only the second one is on the slide.

And something worth saying to an engineer directly. The target variable in this task was

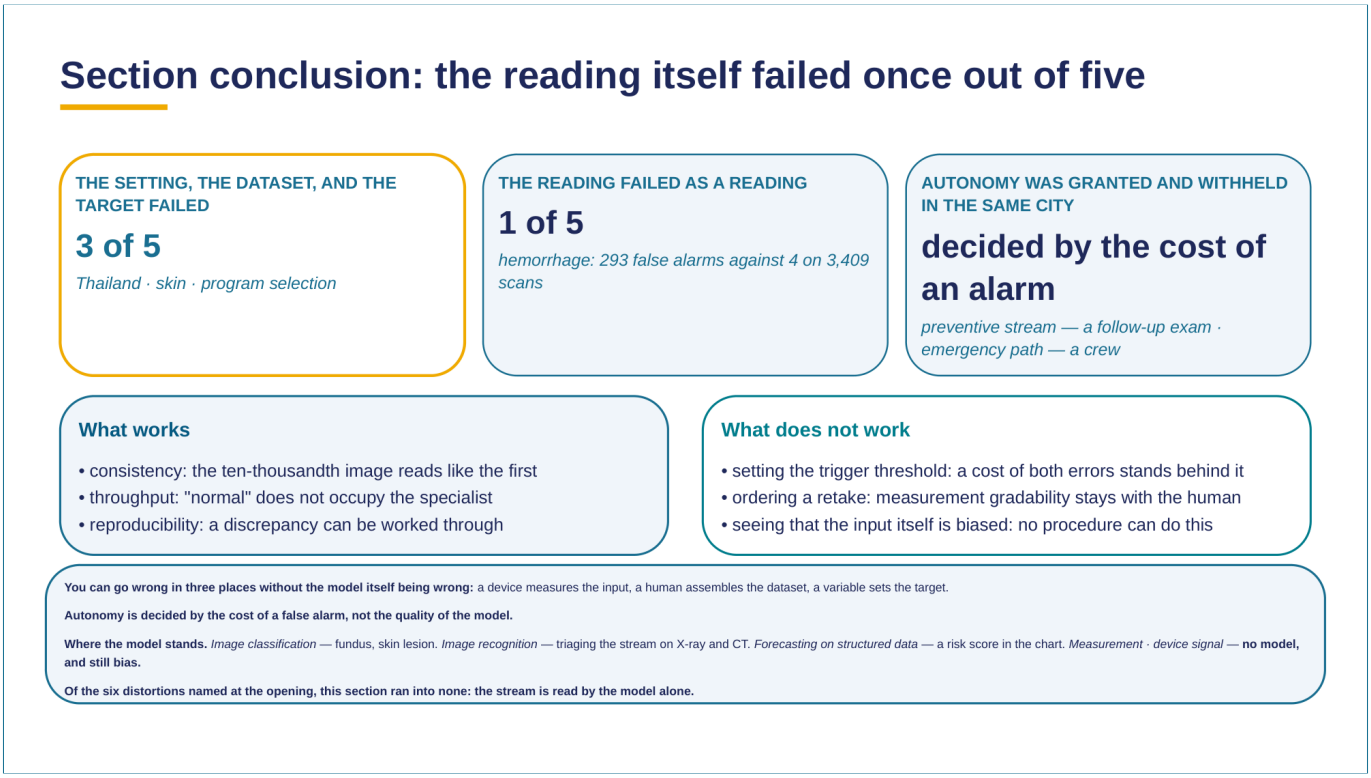
not chosen out of laziness: spending exists for every patient, it is numeric, requires no labeling, and really does correlate with illness. Anyone sitting in this room would have chosen the same one, given the same data and the same deadline. The error is not in the choice — it is that nobody wrote down what the system treats as its target, and so nobody checked whether that target matches the decision's actual purpose. There is no procedure for that check in the order of work, and the section's conclusion says so.

And a last point, on the limits of this lesson. This was measured at one academic hospital over three years, on 49,618 patients. The conclusion about the mechanism — that choosing the target variable moves the composition of those selected — carries over; the specific size of the skew carries over nowhere, and it must not be stated as an industry figure.

Sources. Obermeyer et al., Science 2019

What exactly was being decided: who to call in for intensified monitoring.

And about the tool itself: the publication gives no name for the algorithm.



The outcome of this stage: the reading itself failed once out of five. In three breakdowns, what let things down was the setting, the composition of the training set, and the definition of the target — that is, it was possible to go wrong without the model itself being wrong. The benefit here is real and measurable, and a specialist will take it: consistency of reading and relief for the stream. The system reads the ten-thousandth image like the first, and resolving a discrepancy gains a reproducible side — you can ask the machine again, you cannot ask yesterday's version of yourself.

There is one boundary, and it is the same one the section opened with. Shares are computed on a group, and the meaning of the output changes along with the group. The same "sick" signal means different things in a general-population stream and in a specialized department, and the gap between them is nearly tenfold — eight out of a hundred against seventy-nine.

Let's state the risk so it sticks. A device measures the input. A human assembles the dataset. A variable sets the target. You can go wrong in three places without the model itself being wrong — and three breakdowns in this section are exactly that: in fundus reading, the deployment failed; in skin assessment, the composition of the training set; in program selection, the definition of the target. In none of these three did the reading itself fail, and what got confirmed instead was that nobody measured the setting for the model, the data for it, or its target before deployment.

And the two domestic breakdowns add something the first three do not have. In the intracranial-hemorrhage task, the model alone produced 293 false alarms against four from the physician on the same 3,409 scans, and missed forty-five hemorrhages: here the reading itself failed. And right next to it — the same city, the same circle of products, the preventive stream, where that same generosity with alarms turned out to

be a benefit: a human stopped reading more than half the images, and that saved 33 million rubles over five months. From the same property of the model, one task got permission to work without a human, and the other got a ban on it. What tells them apart is the cost of a false alarm and whether there is time: on the preventive stream, a false alarm costs one follow-up exam; on the emergency path, it costs a deployed crew, and there is no time to check it. This is exactly what an engineer should take from this section, instead of the question "is the model good."

What happened to the work. The workload grew in the same place: the primary-care technician gained the job of obtaining a gradable image, which the ophthalmologist never had, because he compensated for the imaging conditions himself. A step appeared: a repeat visit for anyone whose image did not pass. The performer changed: program selection used to be done by the treating physician, now a single score does it. And the cost of an error shifted — twice, by two different mechanisms: in the skin breakdown it was shifted by the composition of the training data, in the selection breakdown by the choice of the target variable.

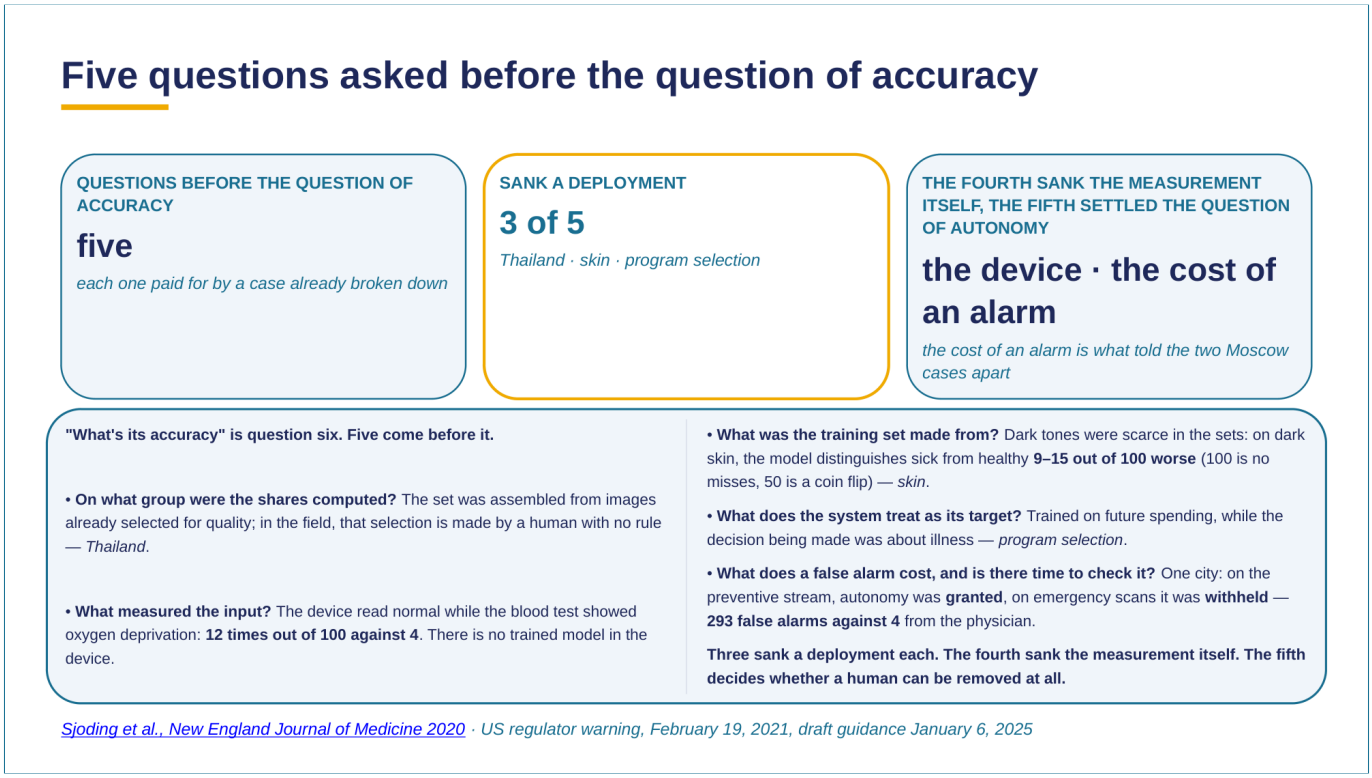
What holds this workload up. A trigger threshold, set against the cost of both errors by the protocol, not by the developer. Measurement quality control — a procedure written in the room's own instructions. And recomputing a share for one's own prevalence before applying someone else's number at home. There is one dash here, and it is set in bold not for decoration: nobody checks the system's target. None of the three procedures asks what exactly the algorithm treats as its target — and in the third breakdown that is exactly what let things down.

And one encouraging observation worth stating, because otherwise this section reads as hopeless. The one fix actually measured in this section came from data composition: fine-tuning on a representative set closed the skin-tone gap, and even flipped it, while three named algorithmic bias-correction methods did not shrink it significantly. This is the part of the work that genuinely sits in an engineer's hands — not the choice of method, but collection and labeling.

Now about the cells of the course's own grid occupied by this stage — there are four, and the fourth is the most interesting. Image classification: autonomous fundus reading and skin-lesion assessment, and both classes of devices bypass the specialist and both only produce a "show to a physician or not" triage. Image recognition: detection and triage of the stream on X-ray and CT — this is what the platforms from the country-by-country overview are doing. Forecasting on structured data: a risk score inside the medical chart, bought by an administrator or a payer, meaning the decision of who gets taken under watch is made wherever the money is counted. And the fourth cell — measurement and device signal — is occupied with no trained model at all: the blood-saturation device's bias arrives before any model does. The full grid will come together in the summary, across all sections at once.

Further down the cycle, everything this section read lands with a human who will make the decision and sign for it. What happens to that human is the subject of the next section.

So three out of five failed, and in the fourth place — the reading itself — the fifth one failed.



Five questions get asked before the question of accuracy: on what group were the shares computed, what measured the input, what was the training set made from, what does the system treat as its target, and what does a false alarm cost. "What's its accuracy" is question six. Each of the five is already paid for by a case already broken down, so none of them hangs there as generic advice.

The first question is about the group, and it runs through the entire section. We started with the fact that one and the same system gives eight correct alarms out of a hundred and seventy-nine out of a hundred, and the difference is set not by the device but by whoever decided who gets examined. But this question has a second, less obvious half, and Thailand is what sank it. The lab set used to measure reading accuracy was assembled from images that had already passed a quality selection. In the field, there is a selection too — made by a human nobody told the rule for. So the group the shares were computed on and the group the system actually works on differed not by diagnosis but by input quality, and nobody measured this before deployment.

The second is what measured the input. This is the cheapest of the four questions and the least often asked. "What data was it trained on" gets asked often; "what obtained every value in that data" — almost never, even though the device's own documentation answers it, not a study. A device that reads blood oxygen saturation from light absorption in tissue stands here as proof that bias can exist before any training at all: a discrepancy with direct blood testing in 11.7 percent of measurements in patients with dark skin against 3.6 percent in light-skinned patients. Without the denominators, this number does not read: eighty-eight discrepancies out of seven hundred forty-nine measurements against ninety-nine out of two thousand seven hundred seventy-eight. This is one named cohort from one study; the multicenter cohort in the same paper gives different numbers, and the two are not stated side by side — two sets from one

study read as a single sample and get spliced together.

The direction of the bias matters more than its size here. The device reads HIGHER than the blood test does — meaning it misses dangerous drops in saturation more often, that is, the error leans toward "everything is fine." And this is not researchers' speculation but a recognized measurement problem, recorded in two documents: a US regulator warning from February 19, 2021, and a draft guidance from January 6, 2025, requiring these devices to be tested across different skin tones. There is no trained model in the device at all — it is optics and light-absorption arithmetic devised long before any machine learning. A system that takes blood saturation as an input — and early-warning systems do, we will meet one in the next section — inherits the bias: it does not create it, does not amplify it through any fault of its own, but receives an already-biased number and works with it as if it were a measurement. Strength of evidence: a prospective comparison against direct blood testing, with the direction confirmed independently and recorded in a regulator's document — high.

Hence a conclusion that is easy to miss while arguing about algorithmic fairness: what needs checking is not only the model, but also whatever measured its input.

The third is what the training set was made from. The skin breakdown: three models lose nine to fifteen points of area under the curve on dark tones, which were scarce in the sets. And this is also the section's one piece of good news: this gets fixed by collection and labeling, not by choice of method. Fine-tuning on a representative set closed the gap; three algorithmic bias-correction methods did not.

The fourth is what the system treats as its target. Selection into a high-attention program: the target variable chosen was future treatment spending, because it exists for everyone, it is numeric, it needs no labeling, and it correlates with illness. And what it reflects is utilization, which differs across groups with the same underlying health. Nobody wrote down what the decision was actually supposed to target, and what the data happened to contain became the target instead.

Now about the count, and this slide's count is honest rather than rounded. There are four questions, and three deployments in this section — and the temptation to say "each question sank a deployment" has to be put down. Three questions sank one deployment each: group — Thailand, dataset — skin, target — program selection. The fourth, about the input, sank none, because the pulse oximeter has no trained model at all: there was nothing there to deploy. And that is exactly what makes it the most useful of the four — it is the only one checking something that sits upstream of the model and will outlive any replacement of it.

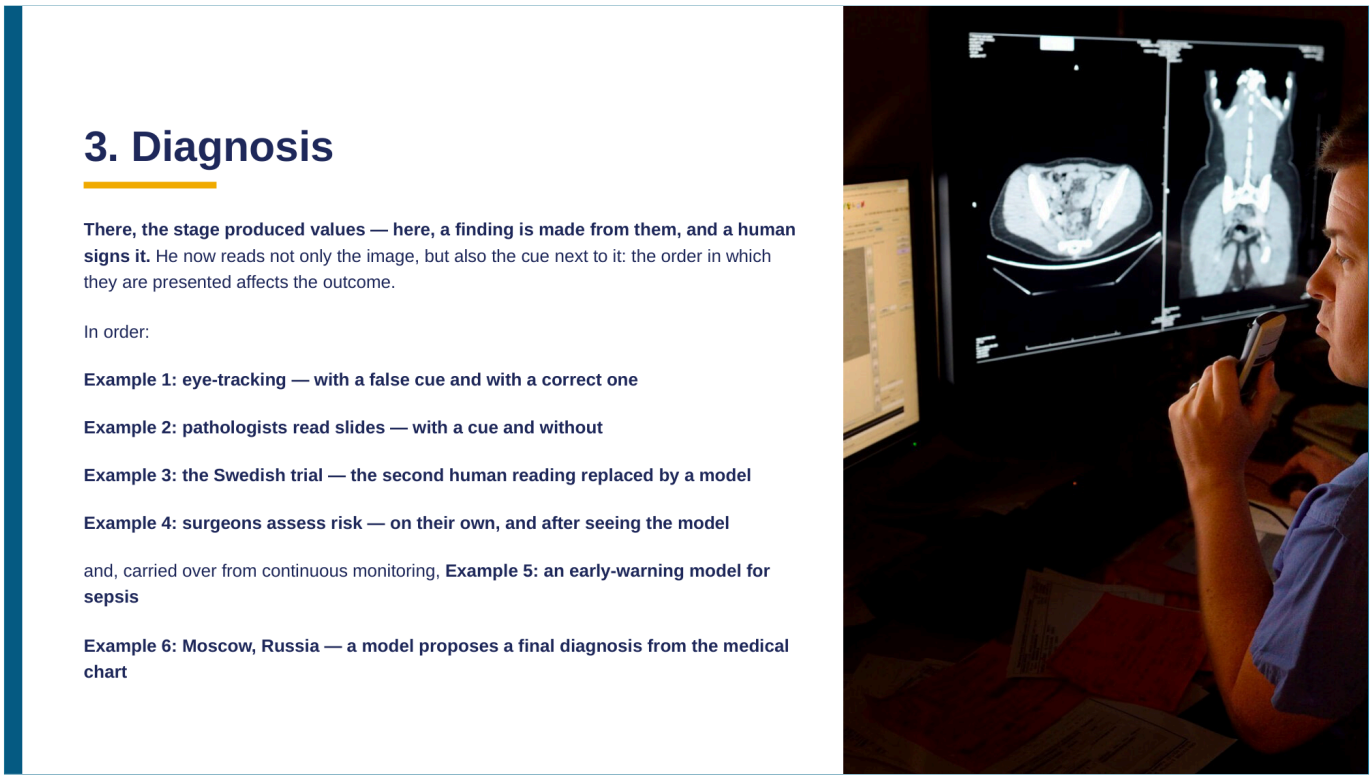
If asked why the accuracy question is placed fifth, and whether it is even needed. It is needed, but it is derivative: the answer to it only reads correctly after the first two. A stated accuracy figure without a named group and a named prevalence does not mean anything — that is the rule this section hands off to the whole lecture, and it is also what explains why the order of questions runs this way and not in reverse.

The fifth is what a false alarm costs and whether there is time to check it. This question

appeared last in the section, and turned out to be the one that settles the question of autonomy. The two Moscow cases stand side by side not for balance: one city, one oversight body, one circle of registered products. On the preventive stream, the model was tuned to raise alarms generously — and sent to work without a human, because a false alarm there costs one follow-up exam, and there is time. On emergency head scans, the same generosity produced 293 false alarms against four from the physician, and the authors state directly: the model's place is second reader, not first. A false alarm there costs a deployed crew, and there is no time to check it. The quality of the models does not tell these two cases apart.

Sources. Sjoding et al., New England Journal of Medicine 2020 — the discrepancy between the blood-saturation device and direct testing. The addresses for the other four questions are on the slides of their own breakdowns.

And the question for the room on this slide is a prediction. Five questions are in front of you, and behind each stands a case already broken down today. I ask you to name which of the five, in your view, sinks a deployment most often — and to say why in one phrase. The answer we'll arrive at: the first three each sank a deployment, the fourth broke the measurement itself, the fifth decides whether a human can be removed at all — meaning the costliest thing is not accuracy, but the trigger threshold and who works through the false alarms. The room usually names the composition of the training set; it is third in order, and not first in cost.



This section crosses the same boundary that separated the second section from the first, only from the other side: the data has already been measured and read by the system, but a human signs the finding. The cue sits next to the image, and this section's question is what the human actually looks at after that — the image itself, the cue, or by now a habit of trusting the screen. The industry calls this stage diagnosis, and it settles one question: what explains the finding. Its decision is to name the disease and sign for that name.

Without a trained model, this stage is held up by three regimes, and all three get their own slides further on. Reading by a single specialist, with a signature. Double independent reading, where two people look separately and do not know each other's opinion. And arbitration — a third person who works through a disagreement. Independence here is not politeness, it is a condition: the moment the two share a common source of a cue, the second opinion stops being second, and the measured discrepancy rates show it.

The six breakdowns run in increasing order of exactly what is measured, and they are worth naming here, so the slide gets recognized later by a name already heard.

The first is eye-tracking. The same reader, on the same mammograms, misses twice as many cancers where the cue stayed silent; this shows up in eye movement — a false cue narrowed the visual search. The same slide carries the same quantity with the opposite sign: in a different program, where the reader requests the annotation himself, missed cancers went down. Both halves sit on one slide deliberately: without the second, the first breakdown reads as a failure rather than a measurement of a mechanism, and the sign of the effect is set not by the cue itself but by whether it is correct.

The second is pathologists reading slides. The same mechanics of the effect, carried over from the eye to the decision: the same cue raises the odds of a correct answer fourfold when the model is right, and drops them fourfold when it is wrong, while the gain across the group as a whole is not significant. A report with one number would show "no effect" here — and that is the most useful thing this section can say about averaging.

The third is the Swedish trial. The only randomized trial in this section with preregistered outcomes, and the only breakdown where the pair came out stronger than either side alone: one human reading on top of the model, with arbitration on disagreements, against double human reading — with half the human reading.

The fourth is surgeons assessing risk. The breakdown that was promised in the previous lecture. On three outcomes out of four, the pair is more accurate than the surgeon alone, and on septic shock — the one outcome where the human had his own edge — it dropped below both the human and the model.

The third and fourth stand next to each other deliberately: the tool is the same, and the results are opposite, and what diverges between them is not the model but the mechanics of the procedure — whether the independence of the two opinions is preserved, and in what order the cue is presented.

The fifth is an early-warning model for sepsis, and it is the only one in this section where the decision is made continuously, rather than on a single case. It comes from the continuous-monitoring stage, which does not run as its own section in this lecture, so on the divider it is named by its own job, not by its place in the cycle. This breakdown rests on two bases that must not be confused: an alarm is raised for eighteen hospitalizations out of a hundred, and twelve out of a hundred raised alarms get confirmed. To find one case, a shift has to work through eight alarms. The classical regime here is not zero — it is a point scale computed by hand, and it was confirmed eight out of a hundred against twelve for the model — the model never left that band.

Between the breakdowns sit two unnumbered slides, and they have no number for a reason, not by oversight. A cardiogram with an incorrect machine readout answers the question of what trust is calibrated against, and does not assign a cost to either error. A review of a hundred and six experiments draws a rule from other people's measurements: where the pair beats the human and where it falls below the model. Neither of the two fits the form of a detailed breakdown, and passing them off as breakdowns would double-count.

This section holds one through-line, and it is worth naming before the breakdowns. The "human plus model" pair is not equal to either the human or the model alone, and which side will tip the balance cannot be said in advance — it has to be measured. Usually, though, anything but this gets measured: the model's accuracy separately, the physician's satisfaction separately, and the third quantity, the one the pair was actually assembled for, is counted by nobody.

The "task × data type" grid is not drawn section by section — it is assembled once, in

the wrap-up, across all sections at once.

The image alongside is a real reading room: a physician at two monitors, both showing CT slices, a voice recorder in hand.

Diagnosis: the stage's seven steps, and the three where the model operates

FREQUENCY OF DIAGNOSTIC ERROR: ADULT OUTPATIENT VISITS

5.08 out of a hundred visits

identified via a trigger — an unusual patient return, an abnormal result with no follow-up workup; so only part of the errors is counted; the share itself was extrapolated to the US adult population; the authors call this an underestimate: the true frequency may be noticeably higher than five percent

The workflow	gather the data →	list candidate diseases →	order a differentiating test →	cross out what's ruled out
Occupied by the model	composing the list	the second reading	the sign-off	

How the list narrows: the stage right after the workup

One visit: a cough for three weeks, an evening fever, a four-kilogram weight loss. Of four candidate diseases, the X-ray and blood test ruled out two — two remained, indistinguishable on the image.

The model occupies three of the seven steps: composing the list, the second reading, the sign-off. **And it added one — deciding whether to trust the suggestion.**

Singh et al., BMJ Quality & Safety 2014

The stage has been named, and now step by step. It has seven steps: gather the data, list the diseases that could explain it, order a differentiating test, cross out what's been ruled out, have a second independent reading, resolve any disagreement with a third reader, sign the report. The diagram shows the first four — the ones that narrow the list; the fifth, sixth, and seventh come after it and are covered on the next slide. Here is what matters about each of the seven.

Gather the data. Complaints, the physical exam, whatever was already measured at the previous stage. That is everything the physician has going in.

List the candidates. Diseases that could explain what's observed: prolonged bronchitis comes first as the common cause, tuberculosis because the picture is classic and missing it is costly, lymphoma because the same two signs can be read differently, an autoimmune condition less often, but it still belongs on the list.

Order a differentiating test. The X-ray and the blood test come first for a reason: start with what's cheap and rules out a lot. The clinical version of hypothesis elimination is harsher than the engineering one in one respect — some tests are themselves risky or expensive, so the order of testing becomes a separate decision that has to be justified.

Cross out what's ruled out. Bronchitis is ruled out by the image: bronchitis doesn't produce changes in the upper lobe. The autoimmune condition is ruled out by calm inflammatory markers. Two diagnoses remain that the image can't distinguish, and a sputum culture decides between them — three weeks of waiting, part of the price the patient pays.

Have a second, independent reading. A second opinion is a report on the same case made by someone who hasn't seen the first one; the whole point of that definition rests

on the second half.

Resolve the disagreement. A third reader sees both reports and steps in only where the two disagreed.

Sign the report. The signature is the point where responsibility becomes attached to a name.

Picking the wrong item off the list is an error that gets noticed: a test result comes back and doesn't match expectations. Failing to put on the list something that should have been there is an error that leaves no trace: no one orders a test for it, the case leaves with a different diagnosis, and it comes back a year later.

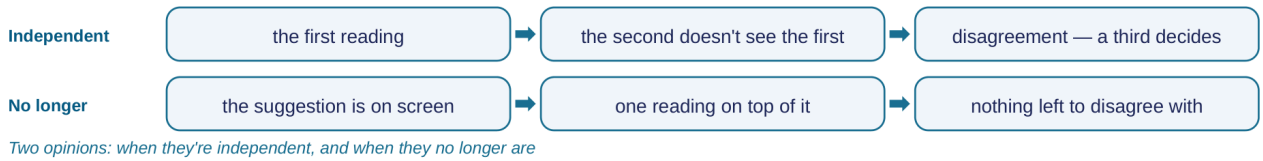
And the same field with the model, named before the case studies. Of the seven steps, the trained model occupies three. The second — the suggestion edits the composition of the list, adding a line a tired physician wouldn't have come up with, or dropping one that should have been there. The fifth — it becomes the second, independent reader. The seventh — the sign-off, and at the fourth rung of involvement it's given without looking. One step gets added — deciding whether to trust the suggestion — and the physician never had that operation before at all. Three places where it doesn't work are named for what breaks: independence — the suggestion is visible before the physician's own conclusion forms, and the model's error is the same for every reader, so it doubles instead of averaging out; selectivity — not a single group managed to disbelieve the suggestion exactly where the model was wrong; explainability — a simplified explanation of an incorrect recommendation drops the result below working without a model at all.

A question for the room, while the list is on screen. Which of your own hypotheses have you failed to write down — and how did you find out it was missing? We're after a short answer, one sentence: most often you find out the same way as in the clinic — a result comes back that no one expected, because you don't order a test for something that was never written down. Someone will mention an incident review, someone else a debugging session where the cause turned out to be outside the list under consideration. It's the same mechanism, and it's needed here so the second row of the diagram reads not as an improvement but as a shift in where the error sits: the model edits the composition of the list — that is, it intervenes in exactly the step whose failure leaves no trace.

And one last thing worth pausing on in the second row of the diagram. Of the seven steps, the trained model occupies three — composing the list, the second reading, the sign-off — and it added a fourth on its own: deciding whether to trust the suggestion. The physician never had that operation before, no one is trained for it, and the visit norms set aside no time for it. The rest of the section measures exactly this: how a person's attempt to tell a correct suggestion from an incorrect one turns out. Three places where it doesn't work are named for what breaks. Independence — the suggestion is visible before the physician's own conclusion forms, and the model's error is the same for every reader, so it doubles instead of averaging out. Selectivity — not a single group we'll examine managed to disbelieve the suggestion exactly where the

model was wrong. Explainability — a simplified explanation of an incorrect recommendation drops the result below working without a model at all.

A second opinion: what holds it up, and when it stops being second



Two opinions: when they're independent, and when they no longer are

Held up by independence: the second reader is denied access to the first report, and a third reader resolves any disagreement.

A second opinion from the model that suggested the first one is no longer a second opinion: the error is shared, and it doubles instead of averaging out.

The order of presentation has been measured. A suggestion shown before the person formed their own opinion, versus a suggestion shown after — **8.9 points** in decisions to act. *70 physicians, 254 cases.*

Which of you has asked a model for a second opinion on a decision you'd already made yourself? What did you do when it disagreed?

Goh et al., a continuation of the same trial

A second opinion is a report on the same case made by someone who hasn't seen the first one. The whole weight of that definition rests on the second half.

The mechanism exists because a single reading misses things, and it misses them for any reader: attention is limited, the image is large, a sign can be small and sit where it wasn't expected. What makes a second opinion useful is a single observation: the two readers' misses don't coincide. The first missed a density because they were distracted by calcifications in the center; the second was distracted by something else. If both systematically failed to see the same thing, a second reader would add nothing but cost. The gain from double reading comes from the mismatch between errors and nothing else — it doesn't double the strength of the first reader.

Independence is built by the organization's own hands: a separate reading queue, a hidden report field, sometimes a separate shift or institution. It doesn't rely on a reader's diligence. There's exactly one checkable sign — could the second reader physically have seen the first report at the moment of reading; access rights give the answer, and a written policy is not proof here. The partition costs money, and it's the first thing cut when someone looks for savings. Formal independence looks the same and works differently: the rate of disagreement falls, and management reads that as growing consistency.

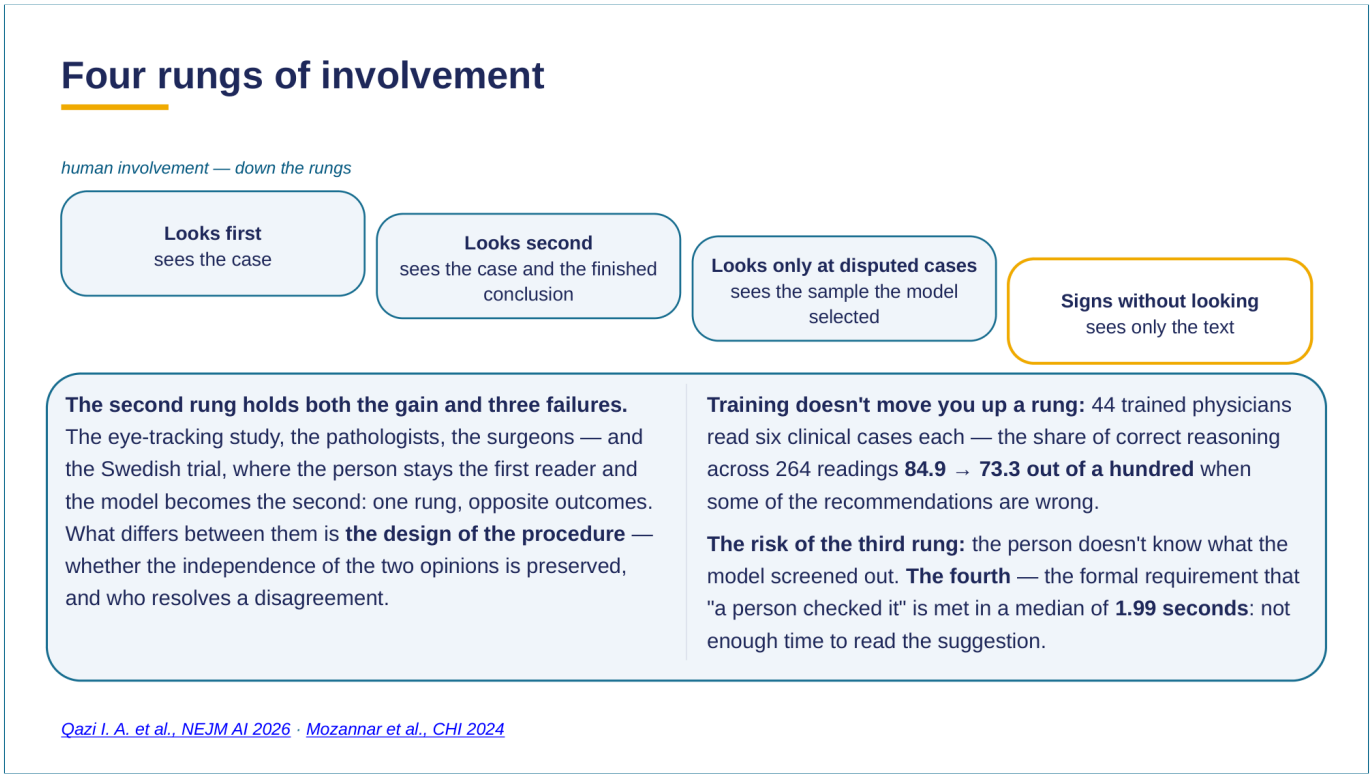
Arbitration is a third reader who sees both reports and steps in where the first two didn't match. Their workload is set by the number of disagreements, meaning it can be estimated in advance and is governed by a threshold. A case conference works differently: several specialists discuss a case while seeing each other's opinions, and independence is deliberately given up there — the value lies in exchanging arguments. That's why the screening stream is never read by case conference.

The model's suggestion breaks independence from two sides. It's visible before the person's own opinion has formed. And it's the same for every reader: the trained system's miss is identical for everyone, so its error doubles across two readers rather than averaging out; the averaging that a second reader was set up to achieve never happens. From this comes the conclusion the section rests on: "add a second opinion from the model" and "add a second physician" are different changes to the process, even though they look like the same block on a diagram.

And one measurement the lecture doesn't take on board — named out loud, because otherwise you can't see how its own rule works. On anchoring in the reverse direction — that is, on how a physician's text moves the machine — there's a figure: the model's agreement with the physician's reasoning fed into it rises from 65.8 to 93.5 percent. On its subject, this figure is precious — it strikes right at the independence of the second opinion from the machine's side — but it's measured in a preprint where there are zero live physicians: the physician reasoning in it is simulated by other models. A lecture that treats strength of evidence as its own rule doesn't say that number from the stage: it's more honest here to say there's no measurement.

For the woman with the letter, the partition between the two readers is exactly what her program rests on: the gain from double reading comes from the mismatch between two people's errors, and that mismatch only occurs between people who haven't seen each other's work. Both readers see the model's suggestion, and the partition stops being a property of the procedure: both readers are now answering the question "do I agree," and that's a different task with a different error rate. Arbitration by a third reader still happens, but now it has to resolve a disagreement between two opinions that shared a common source.

And one check on this diagram. The partition in it isn't a rule of conduct but a mechanism: there's exactly one checkable sign — could the second reader physically have seen the first report at the moment of reading. Access rights give the answer, and a written policy is not proof here. That's why "add a second opinion from the model" and "add a second physician" are two different changes to the process, even though they're drawn as one block on the diagram.



The four rungs of human involvement describe what a person manages to see before a decision moves on. At the fourth rung, where a person only signs off, the median review time is 1.99 seconds, and that's the time BEFORE the decision is made; the second median in the same study is 6.48 seconds.

The formula "human in the loop" answers a question nobody asked: it says the person is present, and says nothing about what they do. The four rungs separate this out by what a person manages to see. Reading first, they see the case and reach their own conclusion before learning the machine's; this is the most expensive rung in time, and it preserves independence. Reading second, they see the case together with the finished conclusion — independence is lost by construction, and there's a gain in speed. Looking only at disputed cases, they see the sample the model selected, and whatever was selected incorrectly never reaches them at all. Signing without looking, they see only the text of the report.

The fourth rung has a measured figure, and it was found outside medicine: twenty-one programmers, 1,096 processed suggestions from a code assistant, behavioral telemetry. The median time spent over a suggestion before deciding to accept it is 1.99 seconds. The same study has a second median, and the two must not be confused: if you count the review that continues after accepting the suggestion, you get 6.48 seconds — meaning more than half of all review time falls after the decision has already been made. The first figure answers the question "does a person have time to read before signing off." More important is this: a suggestion accepted without prior review later needed a fix with probability 0.53; one reviewed before acceptance, 0.18. There's no medicine in this sample, and carrying it over to a physician rests on an analogy.

The medical measurement alongside it is about training not raising you a rung. The

baseline for comparison here is working with correct model recommendations; there's no condition at all without a suggestion in the trial: the same 44 physicians — twenty-two per arm, six clinical cases each, meaning 264 readings, not 264 cases — give 84.9 out of a hundred with correct model recommendations and 73.3 when some recommendations are deliberately wrong; the wrong ones were given in three of the six cases. The adjusted difference is minus 14.0 points, with an interval from −18.9 to −9.1: the point estimate's sign matches the interval's, and it has to be said out loud, or the number reads as a gain. The strength of evidence is high — a peer-reviewed randomized trial, not a preprint — the sample is small, it's a single country, and neither fact cancels the other. We leave the −18.9 to −9.1 interval as is and don't correct it: the two published versions of this work give different bounds, both centered on minus 14.0, and we have no right to change the bounds without reading the journal version. A physician who completed a twenty-hour course stayed on the rung where they started: the course doesn't turn them into a reader who can selectively disbelieve.

And one last thing about the fourth rung, without which it reads as a reproach. Signing without looking is behavior with a measured payoff, and that payoff isn't created by the interface but by the order in which a case is reviewed after the outcome. A physician who accepted a machine suggestion in favor of standard treatment is judged, on review, as more reasonable than one who rejected the suggestion; for a non-standard recommendation there's no difference. So the rung doesn't rest on laziness: disagreeing and being wrong costs more than signing off. The measurement itself and its caveats are presented in the section on licensing, oversight, and liability — where signature, procedure, and regulation are examined; it's named here so the ladder doesn't read as a scale of diligence.

And right away, a caveat without which the ladder lies. A rung names what a person manages to see, and nothing more: it can't rank case studies by quality, and it can't tell them apart either. Later in this section you'll meet two case studies on the exact same rung, "looks first," with opposite results — in one, the pair didn't outperform a human without the model and stayed below the model itself; in the other, it came out stronger than both sides. What tells them apart is the design of the procedure: whether the independence of the two opinions is preserved, and who's assigned to arbitrate a disagreement. Their rung is the same. And the case study where the pair came out below both the model and the human individually sits on a different rung — "looks second": there, the suggestion was shown before the person had formed their own opinion.

Sources. Qazi I. A., Ali A., Khawaja A. U. et al., Automation Bias in Large Language Model–Assisted Diagnostic Reasoning among Physicians Trained in AI Literacy — A Randomized Clinical Trial, NEJM AI 2026;3(5), 23.04.2026, DOI 10.1056/AIoa2501001; medRxiv preprint 10.1101/2025.08.23.25334280 · Mozannar et al., Reading Between the Lines, CHI 2024

Example 1: eye-tracking — with a false suggestion and with a correct one

MISSED CANCERS WITH A FALSE SUGGESTION: WITHOUT → WITH

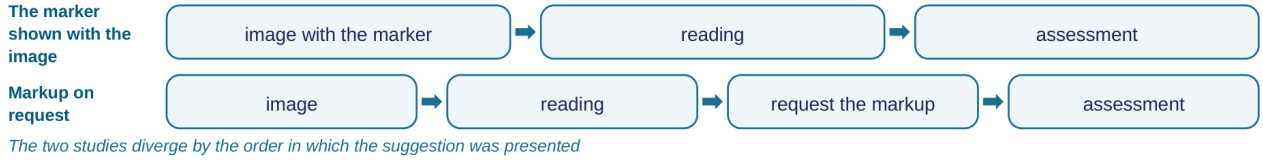
29 → 61 out of a hundred present

14 cases where the suggestion was silent; significant. The cost is later, worse treatment

MISSED CANCERS WHEN READERS REQUESTED THE MARKUP THEMSELVES: WITHOUT → WITH

165 → 115 out of 900 readings

75 cancer cases per condition; a 5.6-point gain with an interval of −0.3… +11.4 — not significant. Reading time didn't change



The two studies diverge by the order in which the suggestion was presented

Two measurements of one quantity. 10 radiologists read 60 mammograms twice: on their own, and with a suggestion deliberately wrong in both directions. 12 radiologists in a different program: 150 exams, also read twice, with an approved system.

A different order of presentation — the opposite sign.

How it's built. Input — a mammogram, output — a suspicion score for the region. The second study names the product: **Transpara 2.1.0, ScreenPoint Medical. Focus:** the area of the image scanned fell from 11.1 to 9.5 out of a hundred, with longer dwell time in the right spot — 4.4 → 5.4 seconds.

This narrowing is the second of the six distortions from the opening of the evening: where the model stays silent, the gaze doesn't linger, and the model's miss becomes the reader's miss.

Taib et al., Radiology 2026 · *Gommers et al., Radiology 2025*

Ten radiologists read the same set of sixty mammograms twice: first on their own, then six weeks later with a suggestion that was deliberately wrong in both directions. Both the result and the eye movements were measured, so here you can see what mechanism changed the result. On cases where the suggestion stayed silent, missed cancers rose to 61 out of a hundred against 29 for the same readers — twice as many. That's what narrowing looks like, the second of the six distortions presented at the opening of the evening.

The set is enriched with cancers to roughly half, and that has to be kept in mind: in a live program, cancer occurs in a handful per thousand.

The baseline is the same reader's median sensitivity on the same cases without a suggestion: 71 out of a hundred, meaning 29 missed cancers out of a hundred present. There were fourteen silent cases, and the difference on them is robust. The cost of this figure isn't abstract: a missed cancer means later, worse treatment.

The second half of the same measurement shows the sign flipping within a single study: on the fourteen cases with a false-positive suggestion, the reader's specificity rose — from 21 to 39 out of a hundred. In other words, a wrong suggestion isn't simply harmful; the sign of the effect is set by which direction it was wrong in.

Exactly how this happened was measured, not guessed. On lesions the model missed, the reader's gaze lingered noticeably less often. The false suggestion narrowed the visual search. There was no verbal persuasion involved: no one argued to the reader that the lesion wasn't there — their eye simply stopped less often where no marker had been placed. And that's the most uncomfortable part of this case study: the effect sits below the level at which a person can control themselves by force of will.

The design of the second study, and why it sits on the same slide. Twelve radiologists from a screening program across ten institutions, experience ranging from 4 to 32 years with a median of 12. One hundred fifty exams, also read twice, but with an approved system — Transpara 2.1.0 made by ScreenPoint Medical. Here the result is the opposite: missed cancers went down, 115 out of 900 readings versus 165. A 5.6-point gain with an interval from minus 0.3 to plus 11.4 — not significant; reading time didn't change.

The area under the curve rose from 0.93 to 0.97: plus 0.03, with an interval from 0.02 to 0.05, and it rose for all twelve readers without exception.

Separately — a figure that sits on the lecture's own axis. The model alone scored 0.99, with an interval from 0.97 to 1.00, meaning higher than the pair of "reader with the system." The authors say this outright, and we won't hide it: in this study, the person next to the model didn't improve it — they made it worse.

Focus was measured here too: the share of the image's area covered by gaze fixations fell from 11.1 to 9.5 out of a hundred, while dwell time on the lesion area rose from 4.4 to 5.4 seconds. The same narrowing of the search as in the first study — but with a different outcome.

Why the signs diverged, and this is the only substantive explanation available. In the first study, the marker is displayed together with the image — that is, before the reader has formed their own opinion. In the second, the reader requests the markup themselves, after already having looked. The order of presentation isn't a matter of formatting — it's a controllable parameter with a measured price.

On the vocabulary of forms. The load at the same spot went down: same place, same performer, less work. But the sign of this form is set by whether the suggestion was right — with a false suggestion, the same form turns out to be a cost rather than a gain, because the model's miss becomes the reader's miss.

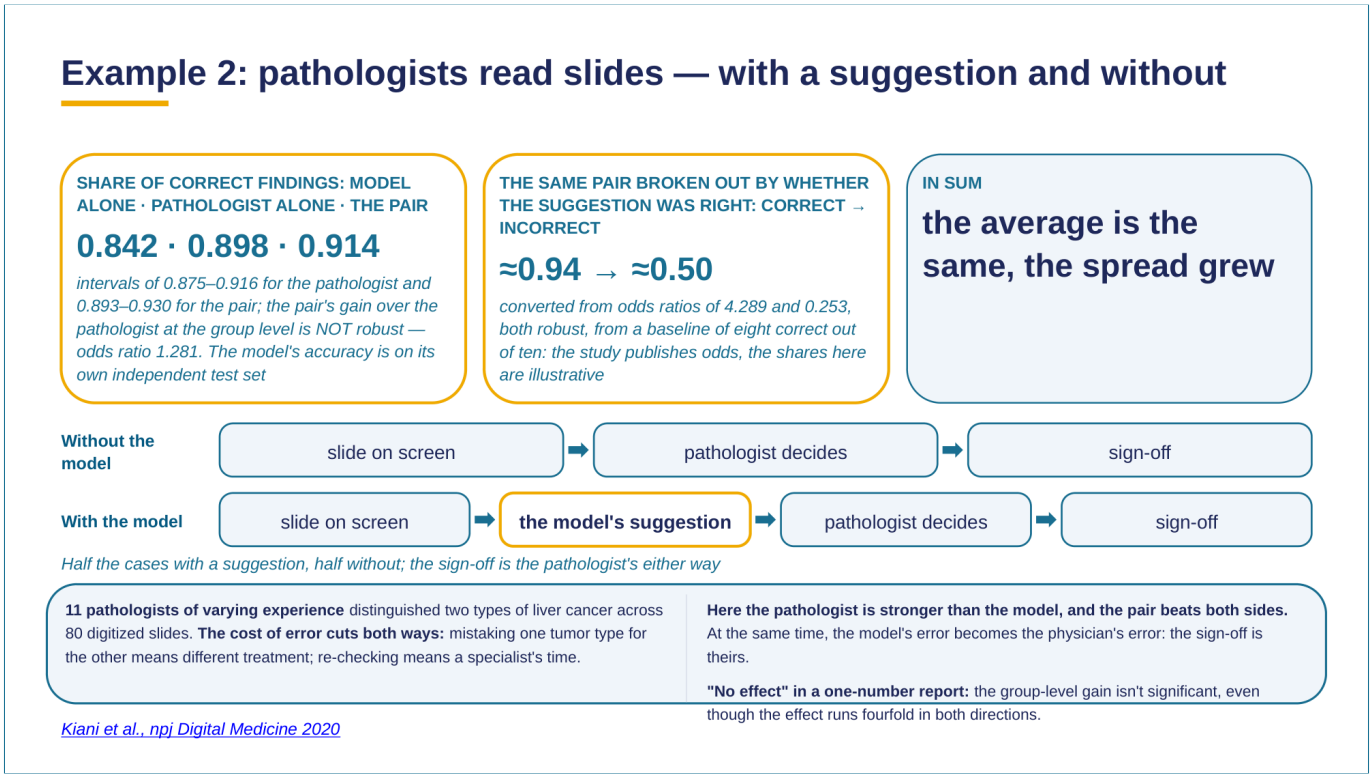
The strength of evidence is named alongside the figures. The first study — a reader compared against themselves — is strong in mechanism, medium in reader sample size: ten of them, and no generalization to the whole profession follows from it. The second — twelve readers from ten institutions, so wider in locations, but also not population-level. Both sets are enriched with cancers; ScreenPoint Medical provided the tested system, and that's disclosed in the publication itself.

The rung of involvement in the first study is "looks second": the suggestion appears on screen together with the case, that is, before the reader has formed their own opinion. In the second, they look first and request the suggestion themselves. These are different rungs, and it's them — not the model — that drive the results apart.

A short question for the room, and it's a prediction based on this very measurement. The reader's gaze was measured twice: on their own and with a suggestion. The area of the image scanned with a suggestion fell from eleven and one tenth to nine and a half out of a hundred, while in the marked spot the gaze lingered longer — from four and four tenths to five and four tenths seconds. I ask: what happened, then, to the cases

where the suggestion stayed SILENT — did misses stay the same, go down, or go up? And by how much? I ask you to name it before I show it. The figure sits a line below on this same slide, and it's worth covering it with your hand for a second: the prediction "the same" is the one heard most often in the room, and missed cancers there became twice as many.

Sources. Taib A., Partridge G., Phillips P., Chen Y. et al., Automation Bias in Action: Eye Tracking of Humans Reading Screening Mammograms with and without AI Prompts, 2024 · Gommers J. J. J., Sechopoulos I., Broeders M. J. M. et al., Radiologists' Performance and Visual Search Patterns When Reading Screening Mammograms with Artificial Intelligence Support, 2025



Pathologists read specimens with a single suggestion from the model, and in half the cases the suggestion was wrong. What's measured here is the same thing as in the eye-tracking studies, but at the level of the decision rather than eye movement: the same tool shifts the result fourfold in both directions — up with a correct suggestion and down with an incorrect one.

Significance levels for these figures. For the group as a whole, the pair's gain over the pathologist didn't reach significance: odds ratio 1.281 at p equal to 0.184 — that's exactly what produces "no effect" in a one-number report. But both fourfold shifts are robustly significant: odds ratio 4.289 with a correct suggestion (interval from 2.360 to 7.794), and 0.253 with an incorrect one, p less than one thousandth in both directions. The pair reads together: no effect at the group level, and fourfold in each direction individually.

The design of the study, which determines how the rung is marked. Eleven pathologists of varying experience — three gastrointestinal subspecialists, three specialists in other fields, three residents, and two whose specialty wasn't specified — distinguish two types of liver tumor from digitized slides. The set is eighty slides; the model's own accuracy on an independent test set is zero point eight hundred forty-two. The same pathologist read the same set twice, in separate sessions: half the cases with a suggestion, half without, with the assignment flipped in the second session and a washout period of at least two weeks. The comparison runs between sessions: a first and second look at the same case are never compared against each other here.

The baseline is reading a slide without a suggestion. When the suggestion is correct, the odds of a correct answer rise roughly fourfold; when it's wrong, they fall by roughly the same factor. This held at every level of experience and across cases of every

difficulty alike: a subspecialist's own field gave them no protection, and that matters more than the size of the effect itself.

How to read "fourfold" without getting the direction wrong. Odds are the ratio of the probability of a correct answer to the probability of an incorrect one. If, without a suggestion, the pathologist answered correctly in eight cases out of ten, odds grown fourfold come out to about ninety-four out of a hundred, while odds fallen fourfold bring it down to about fifty. Symmetry in the odds produces asymmetry in the shares, because high accuracy has a long way to fall and almost nowhere to rise.

The second classical mode — a subspecialist second opinion — is named and not counted: it wasn't measured on this set, and plugging in a number from a different study would be exactly the kind of pooling of mismatched samples this course warns against.

The average gain and the sign-flip are one observation read two ways, and they come from the same sample. At the group level, without splitting cases into those where the model is right and those where it's wrong, the gain doesn't reach significance. That's the price of this case study, and it's organizational in nature, with nothing clinical about it: a deployment report carrying a single number would show no effect exactly where the effect was fourfold in both directions. There's exactly one decision such a report leads to — "works about the same as before, keep it" — and it gets made blind.

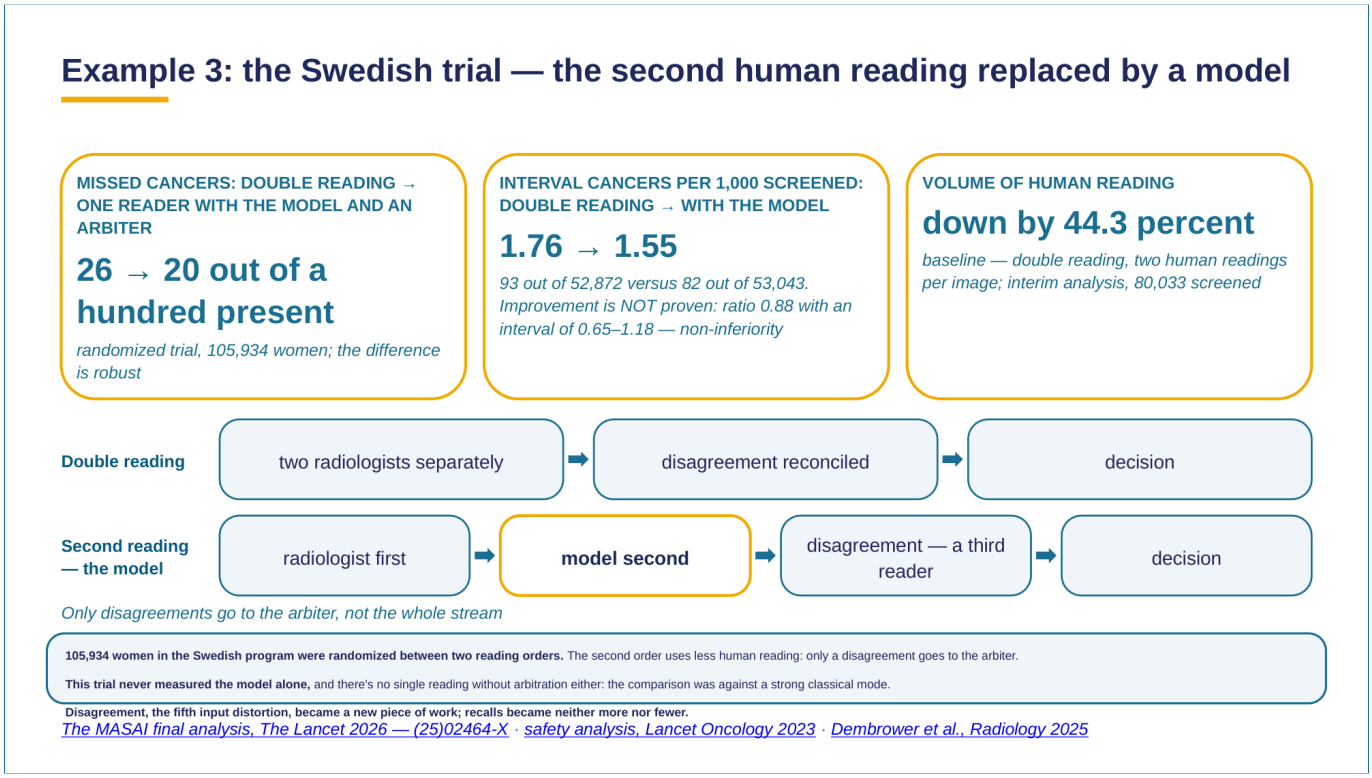
What shifted. One form: the cost of error moved. The steps, the amount of work, and the performer stayed the same — the pathologist looks at the same slide for the same amount of time. What changed is the consequence: the model's error became the physician's error, because the sign-off on the report is theirs. The shift is measured in both directions, and that's rare: usually only the loss or only the gain gets measured.

The pair that closes out this case study reads straight off the plates. Gave — a gain with a correct suggestion. Cost — an equal loss with an incorrect one. In sum — the average is the same, the spread grew. For the procedure, that's the worst possible outcome: the average gives no reason to step in, while the spread means some patients got a worse decision than they would have before.

Alongside this, in a different genre and weaker in strength of evidence, sits a review of clinical decision-support systems: a wrong suggestion raises the risk of a wrong decision by twenty-six percent, drawn from a review covering seventy-four studies out of thirteen thousand eight hundred twenty-one screened. A caveat on the review is mandatory: that body of work includes many rule-based systems, with trained models making up only part of it. So this figure stands as a mention; it doesn't work as an argument here.

Sources. The study doesn't publish its training set either. Kiani A. et al., Impact of a deep learning assistant on the histopathologic classification of liver cancer, npj Digital Medicine 2020, PMC7044422. Study design: eleven pathologists of varying experience — three gastrointestinal subspecialists, three specialists in other fields, three residents, and two with unspecified specialty; eighty slides, two readings, suggestion assignment

flipped in the second session, a washout period of at least two weeks. A mention removed from the slide as the weakest in strength of evidence: Goddard K., Roudsari A., Wyatt J., Automation bias: a systematic review of frequency, effect mediators, and mitigators, JAMIA 2012;19(1):121–127.



A Swedish randomized trial of the national screening program: 105,934 women randomized between two reading orders. In the control arm, classical double reading runs as usual — two independent radiologists with a third arbitrating any disagreement; in the second arm, the second human reading is replaced by the model, and the radiologist reads first. A disagreement is still resolved separately, and this is disagreement, the fifth of the six input distortions, appearing a second time this evening — in a different country and a different trial.

Strength of evidence: four sites, enrollment from April 2021 to December 2022, outcomes preregistered. Nineteen people were excluded from the analysis; 53,043 entered the model arm and 52,872 the control. The design is single-blind, for non-inferiority. This is the only measurement in the section where a patient outcome is actually measured; the rest measured reading accuracy, gaze, or a reasoning score.

What was compared. The control arm ran classical double reading: two independent radiologists, a third arbitrating any disagreement, sensitivity 73.8 out of a hundred with an interval from 68.9 to 78.3 — this is the baseline; the model arm got 80.5 with an interval from 76.4 to 84.2, and the difference is robust, p equal to 0.031. This is the only one of the trial's three figures where a difference is actually proven: for interval cancers, only non-inferiority is shown, and reading volume was counted on a different sample. The model arm: the model reads the stream, one radiologist reads on their own, and an arbiter resolves any disagreement between them; sensitivity 80.5 with an interval from 76.4 to 84.2, and the gain is significant. Specificity is the same in both arms — 98.5 out of a hundred healthy women. In other words, the second reader was replaced, while independence and arbitration were preserved and rebuilt around the new participant.

Arbitration and the threshold. The model issues a suspicion score, the radiologist gives their own report, and on disagreement the case goes to an arbiter who sees both. The arbiter's workload is set by the rate of disagreement, and that in turn by the threshold: shifting the threshold toward suspicion raises both the number of disagreements and the number of recalls for further workup. Classical double reading had a threshold too, but it was set by each reader's own practice, where it's unrecorded and untunable; moving to the model turned it into an explicit parameter that someone now has to maintain. The second classical mode — a single reading without arbitration, which programs with no resource for a second reader run on — wasn't reproduced in this trial: the comparison was against the strong mode.

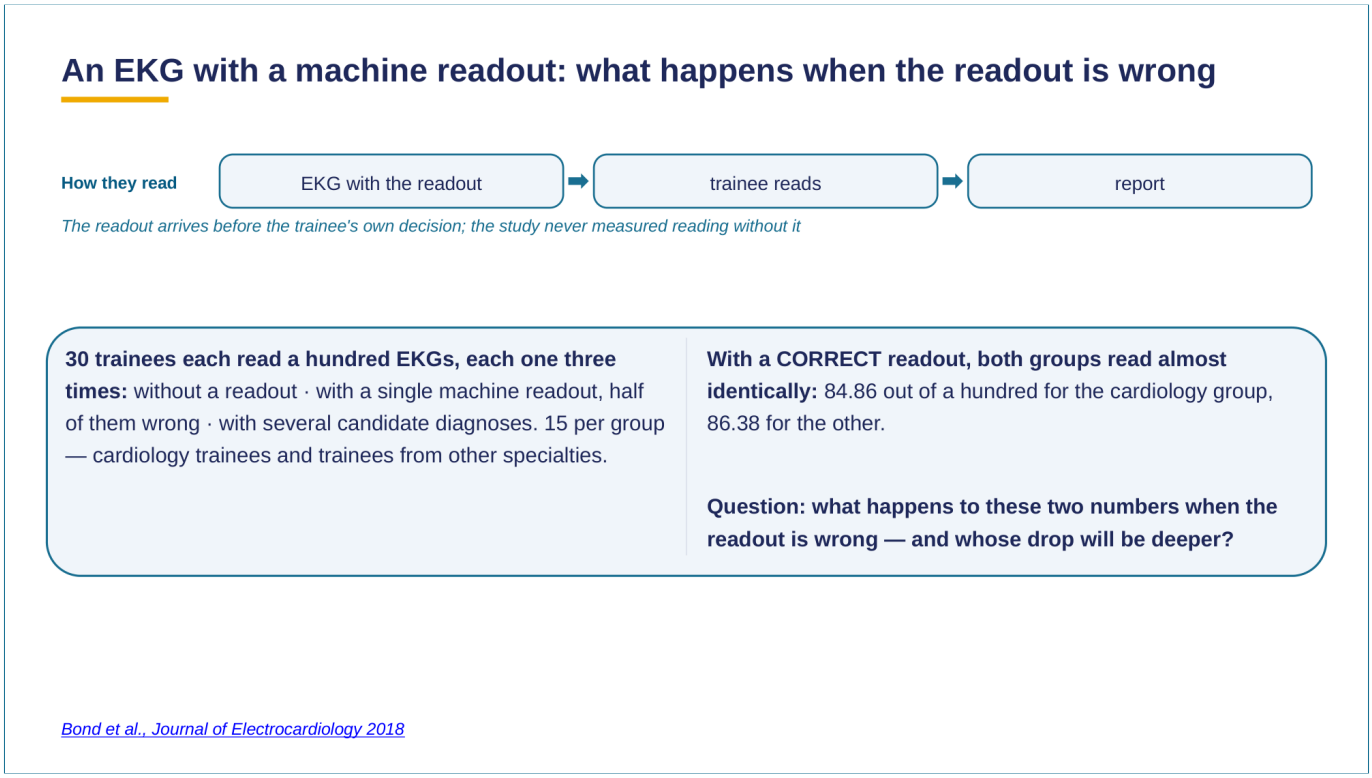
The tool's design is named just as precisely, and just as incompletely, as in the last case study: input — a mammogram, output — a suspicion score against which a threshold is set. What the system was trained on and how often it's updated after deployment, the publication doesn't disclose — and that's not a gap in the research, it's the nature of a trial report.

The reading physician's rung is "looks first": they see every exam on their own and produce their own report, with no one else's shown to them. The arbiter looks only at disputed cases, and that's set by the program's own design. The task type is image recognition. What the system was trained on, the program doesn't publish: there's no way here to check a vendor's claim against the product's name, and either way the gain in this case study belongs not to the device but to the design of the procedure around it.

On interval cancers — disease found between screening rounds — precision matters. In the double-reading arm it's 1.76 per 1,000 screened, with the model 1.55: that's 93 cases out of 52,872 versus 82 out of 53,043. The rate ratio is 0.88, with an interval from 0.65 to 1.18 — the interval crosses one, there's no significance, and the trial was designed to claim non-inferiority. Reading 1.55 against 1.76 as a proven improvement is not allowed. What is proven, and proven strongly, is something else: sensitivity is significantly higher, the outcome didn't worsen, and the composition of what was missed shifted toward the less dangerous side — 16 percent fewer invasive cancers (75 versus 89), 21 percent fewer large ones at stage T2 or above (38 versus 48), and 27 percent fewer aggressive subtypes (43 versus 59).

And why this measurement turns the whole section around. The other measurements all showed the same error mechanism: a person trusts the suggestion more than they should, and pays for it in attentiveness, in odds of a correct answer, or in access to the tool wasted for nothing. The asymmetry of trust, examined earlier in this section on a Swedish program with three independent readings, is broken in the opposite direction here: a source that was confirmed as cancer more than six times as often was recalled a third as often. So the rule of this section isn't "people trust the machine too much" — that's only half of what's been measured. Trust in a finding's source is set by the design of the procedure, not by the source's accuracy, and it's calibrated independently of that accuracy — in either direction. Don't soften the conclusion: the model and its accuracy are sound in this measurement; what broke is what was built around it.

Sources. MASAI, final analysis, The Lancet 2026 · Dembrower et al., Radiology 2025, ScreenTrustCAD



Thirty trainees read EKGs: fifteen cardiology trainees and fifteen from other specialties. Each one went through a hundred EKGs three times: without a readout, with a single machine readout half of which were wrong, and with several candidate diagnoses. Nine thousand readings in total. The order of reading decides everything here, and it's shown in the diagram: the readout arrives BEFORE the reader's own decision, not after — meaning the reader sees someone else's answer before forming their own.

Nine thousand readings are repeated measurements on thirty readers; they aren't independent observations, and the authors themselves name this limitation. The third condition, with several candidate diagnoses, doesn't take part in the rest of the case study: the slide is carried by the first two, where there's a single suggestion that's either right or wrong.

The strong side of this design is that a reader is compared against themselves. The same thirty people, the same hundred EKGs, the only difference being what sits next to the tracing. So the result can't be explained by different training between the groups, or by different difficulty in the material: the material is the same, and the reader is the same.

And one thing this study doesn't measure, even though it looks like it does. Half of the readouts are wrong by the authors' own design — a hand-made substitution, not the behavior of a real device. So what's in front of us is a measurement of the human, not a measurement of the quality of a machine readout: the question isn't "how good is the readout" but "what does a readout the reader has come to trust do to them."

The baseline. With a correct machine readout, both groups read almost identically — eighty-four and eighty-six hundredths out of a hundred for the cardiology group, eighty-six and thirty-eight hundredths for the other. A point and a half of difference.

Everything that follows is counted from this point, and it has to be read precisely: the baseline for the drop is the CORRECT readout, not reading with no suggestion at all. The study doesn't publish a "with no readout at all" column by group, there's nothing to reconstruct it from, and there's no comparison here against reading on its own. This narrows the conclusion: what's measured is what an incorrect suggestion does to a reader against a correct one, not what a suggestion does against having none.

Now the question, and it gets asked twice, because it has two halves. First: how far will these numbers fall when the readout is wrong? Name the order of magnitude — five points, fifteen, by half? Second, and this one's more interesting: whose drop will be deeper — the cardiology trainees' or the other specialties' trainees'? You can answer out loud or in one word in the chat: "cardiology" or "the others."

The common assumption here is that the suggestion's winners and losers split roughly evenly, and that if someone falls deeper, it's the person with less training; the second half of this assumption happens to be true, and the room usually guesses it. But the first half — the depth — the room almost always underestimates: people name five to ten points. The real number is on the next slide, and the gap between what's named and what's real is exactly why it's worth pausing here.

If the room stays silent, both halves get named as options and the room votes by show of hands: "a drop of less than fifteen points" against "a drop by half." A closed question with two options almost always gets a show of hands; an open one almost never does.

One caveat before the reveal, so it doesn't later look like an excuse. Experience is NOT the same between the groups — thirty-six plus or minus eleven months against twenty-eight plus or minus thirteen. What actually distinguishes them is specialty, and the authors build their conclusion on exactly that, but saying "their experience is the same" would be saying something false.

Sources. Bond et al., Journal of Electrocardiology 2018

The reveal: both groups fall by half, the less expert one falls deeper

ACCURACY LOST ON AN INCORRECT MACHINE READOUT: CARDIOLOGY TRAINEES → OTHER SPECIALTIES

−43.20 → −58.95 points

the difference is robust in both groups; the less expert group loses 15.75 points more — the authors' own conclusion, stated twice

AGREED WITH THE INCORRECT READOUT

30.77 versus 42.31 out of a hundred

medians for the same two groups, in the same order: the less expert group agrees more often

The ones who fell deeper were trainees from other specialties — those with less training in this kind of reading.

report accuracy, out of a hundred	correct	incorrect
cardiology trainees	84.86	**41.66**
other specialties	86.38	**27.43**

Anchoring, the first of the six input distortions. Specialty protects, but it doesn't save you: both groups fall, and both by half.

Bond et al., Journal of Electrocardiology 2018 · Buçinca et al., CSCW 2021 · Yu et al., Nature Medicine 2024

First, the answer to both halves of the question, in the order they were asked. Depth: the fall isn't five points, or fifteen — it's by half. For cardiology trainees, accuracy falls from eighty-four and eighty-six hundredths to forty-one and sixty-six hundredths out of a hundred — forty-three and twenty hundredths points. For trainees in other specialties, from eighty-six and thirty-eight hundredths to twenty-seven and forty-three hundredths — fifty-eight and ninety-five hundredths points. Both drops are significant, p less than one thousandth. Who falls deeper: the less expert group, and the room usually guesses this correctly.

The authors state the load-bearing conclusion of this study themselves, and they state it twice: less expert readers are affected MORE STRONGLY. The gap between the losses is fifteen and seventy-five hundredths points in favor of the expert group, and that group agrees with the incorrect readout less often: thirty and seventy-seven hundredths versus forty-two and thirty-one hundredths out of a hundred, by median. In other words, specialty protects, and protects measurably — it just doesn't save you: both groups fall, and both by half. This has to be said in exactly this order, because the reverse order gives the reverse conclusion, and that's exactly the mistake this slide carried before the fix: a caption reading "specialty tells them apart" erased the expertise axis the authors themselves call load-bearing.

Let me name what's happening to the person here: it's a cognitive bias. The first of the six presented at the opening of the evening, and it's called anchoring — the report shifts toward the suggestion, even one that's knowingly wrong. The opening slide showed it on chest X-rays: twenty-three and six tenths versus ninety-two and eight tenths out of a hundred across two hundred twenty readers; here the same mechanism is captured on an EKG.

Alongside this, not instead of it: the link between a reader's confidence and being right weakens for both groups. You can get out of a fall in accuracy through a recheck procedure; you can't get out of broken calibration, because the person doesn't know which cases to send for a recheck. This shouldn't be ranked above the between-group difference: calibration is the study's second finding, not its first.

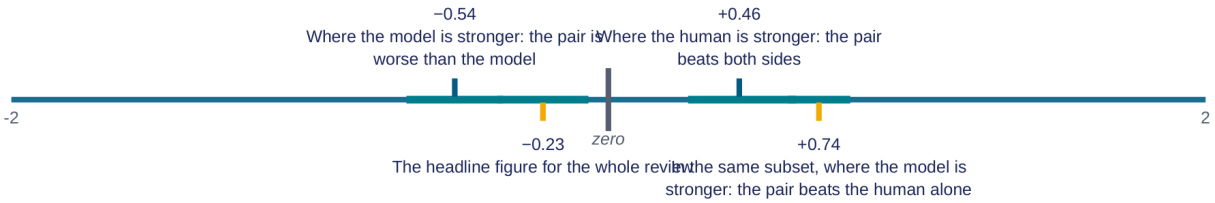
What does NOT save you. One hundred forty radiologists, fifteen chest X-ray tasks: experience, subspecialty, and familiarity with the tools don't predict who the model's help will benefit and who it will hurt; the common assumption that weaker readers come out ahead is refuted. What does predict it is the presence of model errors. There's deliberately no figure for this quantity: its whole content is the absence of a link. But that study measured what predicts the BENEFIT from a suggestion, while the EKG study measured the DEPTH OF THE FAILURE on an error — these are two different questions, and there's no conflict between the two sources. Collapsing them into one conclusion, "experience doesn't help," would give exactly the opposite conclusion from the one the EKG study's authors actually print.

Where's the ceiling, if someone asks "can this be fixed by changing how it's presented." One hundred ninety-nine crowd-platform participants, a nutrition-composition task — non-clinical and with no stakes; the suggestion's accuracy was simulated at a seventy-five percent level, no model was trained. Forced presentation techniques — not showing the suggestion until requested; the person's own decision first, suggestion second; a delay before display — cut agreement with the incorrect suggestion from sixty-four to forty-eight out of a hundred. This pair is counted on one task in the set; on the aggregate metric across all tasks the shift didn't reach significance. Both halves hold together: sixteen points is a real improvement, and even with it, close to half of the agreements remain. The cost is measured by the same study: the interfaces that cut reliance the most were rated worse by users. There's no breakdown by expertise in this study at all, so its figures don't fit under the matrix above, and they can't be placed there.

And the overall point of this pair of slides, the reason it sits right here, after three case studies on the suggestion. In all three, the person either trusted the suggestion more than they should have, or got nothing from it at all — and in none of the cases was the reason that they knew too little. Expertise does enter the picture, and measurably so: it reduces both the depth of the failure and how often people agree with an incorrect suggestion. You can't say "experience doesn't help"; the true statement is "experience isn't enough." Hence the ceiling: redesigning the presentation is possible, and it buys a significant sixteen points, but close to half of the agreements with an incorrect suggestion remain even under the strongest measure tried. So "we'll teach people not to trust blindly" isn't a process measure, it's a wish; the only thing that counts as a measure is something that changes the order of presentation and the distribution of responsibility.

Sources. Bond et al., Journal of Electrocardiology 2018 · Yu et al., Nature Medicine 2024 · Buçinca et al., To Trust or to Think, CSCW 2021 · Dietvorst et al., Algorithm Aversion, 2015

A review of 106 studies: where the pair beats the human, and where it falls short of the model



effect size, in units of spread · 74 studies · 106 experiments · 370 effect sizes out of 5,126 screened

none of the four intervals crosses zero

A preregistered review with a meta-analysis. Only studies measuring all three quantities were included: the human alone, the model alone, and the pair.

The headline figure for the whole review is minus 0.23: on average, the pair is worse than the stronger of the two sides. From this figure, both halves of the review read as parts, not as the final verdict.

Where the model is stronger than the human, both figures are measured on the exact same studies: the pair is worse than the model by 0.54, while also beating the human alone by 0.74. Both together are the rule: the pair outperforms the weaker side and doesn't catch up with the stronger one.

Across the whole review, the pair beats the human alone by 0.64 — but that's a different base and different studies, and this number doesn't go next to minus 0.54.

Vaccaro, Almaatouq, Malone, Nature Human Behaviour 2024

This section's rule is built from a review of a hundred six comparisons. The three case studies before it gave three different outcomes: a drop in sensitivity in the eye-tracking study, both signs at once in pathology, and a gain on the control line. A single trial, however vividly measured, remains a single case, and three single cases don't add up to a rule either. To talk about a rule, you need a review, and one exists: a preregistered review with a meta-analysis, 74 studies out of 5,126 screened, 106 unique experiments, 370 effect sizes, publications from January 2020 through June 2023. The inclusion criterion decides everything here: a study was only included if all three quantities were measured — the human alone, the model alone, and the pair. Without the third quantity there's nothing to compare the pair against, and that's exactly why most deployment reports never answer this question.

Effect size is a measure of "how far it shifted," expressed in units of the data's own spread. It's needed where you have to combine a hundred six studies on different scales: one measured the share of correct answers, another an expert score, a third, time. Numbers like this are read by two features: the sign, and whether the interval sits entirely on one side of zero. An interval from −0.71 to −0.37 is negative throughout — meaning the effect will stay negative if the studies are repeated. A third decimal digit carries no real meaning here: in substance it's "about half a spread's worth downward" against "about half a spread's worth upward."

The caveat travels with the figure every time. The review's headline figure of −0.23, with an interval from −0.39 to −0.07, is computed against the better of the two sides; the human isn't the baseline here. It's precisely against this figure that minus 0.54 reads as a subset of the studies, not as the final verdict.

Next comes a distinction that's easy to slip on. The study has TWO "pair versus human

alone" figures, and they're computed on different bases. The first — across the whole review: 0.64, with an interval from 0.53 to 0.74, a statistic with ninety-eight degrees of freedom. The second — on the same subset of studies where the model is stronger than the human: 0.74, with an interval from 0.63 to 0.85, a hundred four degrees of freedom, meaning exactly the same studies used to compute minus 0.54. This slide's point — "the pair beats the human and falls short of the stronger side" — is stated with the SECOND pair, minus 0.54 and 0.74, because both halves of it stand on the same subset of studies and on comparable degrees of freedom. Putting 0.64 next to minus 0.54 would collapse two different questions on two different subsets of studies into one conclusion, and the room would read them as two halves of the same measurement.

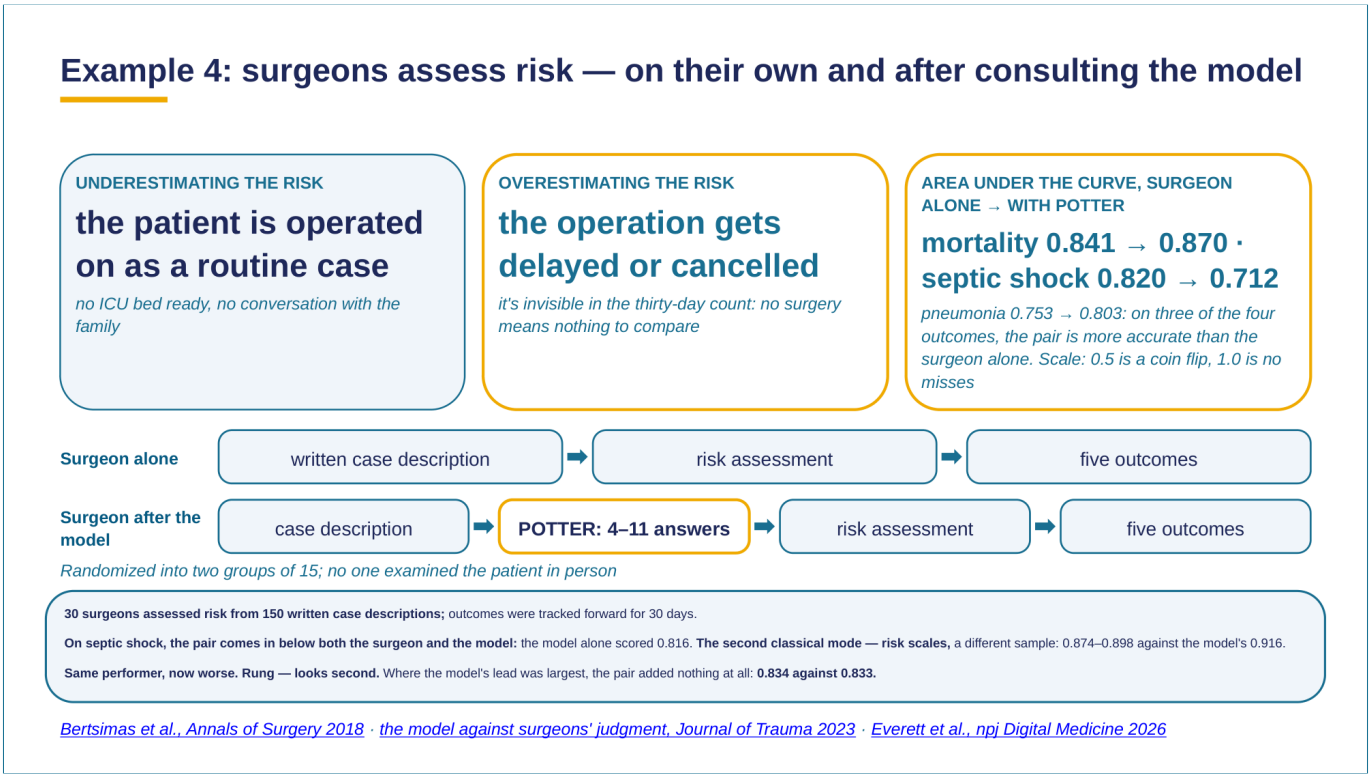
The opposite subset is named too: where the human is stronger, the pair beats both sides — 0.46, with an interval from 0.28 to 0.66. The reading "the human-and-model pair works poorly" isn't supported by this data. There's one correct formulation: the pair is usually better than the human and worse than whichever of the two is stronger.

What the review doesn't say. It isn't about patient outcomes, heterogeneity within it is high, and medical studies make up only part of it. No form from the vocabulary applies here: behind this review there's no deployed process at all — no location, no workload, no performer — so there's nothing to fit to the nearest form. The rule that follows is short: first measure who's stronger on their own, then decide who decides.

Sources. Vaccaro, Almaatouq, Malone, Nature Human Behaviour 2024 · a preprint of the same study, open access

What of this gets said aloud, while being removed from the slide itself. First: a rule isn't built from a single trial — here it is, assembled from a hundred six studies by a preregistered meta-analysis, and that's a different strength of evidence than any of the four case studies before it. Second: after this slide, the question "does AI help" reads differently — as a question about who in this pair is stronger on this particular task, and that question is checkable, unlike the original one. Third: the rule isn't tied to a modality — it's assembled from studies with text, images, and tabular data all at once, and that's exactly why it can be carried out of the lecture and applied to your own work.

And one last thing, without which this slide reads as a verdict against the pair. The pair doesn't inherit the strength of the stronger side — but it does beat the human alone, and beats it robustly: 0.74 on the same subset of studies, with an interval from 0.63 to 0.85; 0.64 across the whole review, with an interval from 0.53 to 0.74 — both entirely above zero. So the question isn't whether to bring the model into the work, but which side you measure the gain against. Comparing against a human without the model gives one picture; comparing against whichever side is stronger gives another, and almost any deployment report knows how to swap the second for the first.



The task we're entering is built differently from reading a scan. The patient arrives on an emergency basis, and within the next few hours someone has to decide: take them to the operating room now, or not. The surgeon makes that decision, and makes it from what they can see — the patient's appearance, their age, what they've had before, labs drawn an hour ago. There's no time for a case conference, and often no second opinion either.

The cost of both errors is named here before any accuracy figure, and the two halves aren't comparable in how measurable they are — that has to be said before any number. Underestimating the risk means taking the patient to the operating room as a routine case: no ICU bed prepared, no transfusion on standby, no conversation with the family about what might come next. A complication in that case is counted on day thirty, and counted well: the national surgical quality database collects exactly these outcomes. Overestimating the risk means delaying the operation or not performing it at all — and the person dies from what wasn't done. This half is completely invisible in the thirty-day count: the patient wasn't operated on, there's nothing to compare, and there's no way to learn that the surgery would have saved them, because the attempt never happened. The half that has no number is named outright and isn't replaced by an estimate.

At this point there isn't one classical mode but two, and both are counted. The first — formal risk scales used long before any machine learning: a physiologic-status scale, an emergency-surgery scale, an open online calculator. All three are computed on the same validation sample as the model, by the same measure, and that's rare: usually a new system is compared against someone else's published number, or not compared at all. The three scales give between 0.8743 and 0.8975; the model, on the same sample, gives 0.9162. The gain — that is, 0.019 to 0.042 — is noticeable but not dramatic, and it

has to be stated exactly that way, not as "the model beats the scales."

The second kind of classical mode is a surgeon's judgment with no scale at all, and this study is built as follows. Thirty emergency surgeons were randomized into two groups of fifteen: one group assesses risk from a written case description on their own, the other after consulting the POTTER calculator. One hundred fifty cases, outcomes tracked forward thirty days. None of them examined the patient in person: they worked from the description, and that limitation has to be stated out loud, not hidden. The calculator itself is a decision tree, not a neural network, trained on 382,960 cases; input is four to eleven answers about the patient, output is the probabilities of five outcomes over thirty days. This measurement has never been repeated.

A caveat about the samples is needed here, or the two rows of numbers will blur into one. The three scales and the model are computed on a 2014 validation sample — tens of thousands of cases from the same national database. The surgeons were measured on a different one hundred fifty cases, tracked forward thirty days. The two rows sit side by side as two classical modes for one task, and they don't add up into one ledger: comparing 0.9162 to 0.820 directly is not allowed.

The result on three of the five outcomes landed exactly as deployment promised: on mortality, on bleeding, and on pneumonia, surgeons with the suggestion became more accurate than surgeons without it. Most deployment reports stop right there, which is exactly why the other two are worth naming.

The fourth outcome — ventilator dependence. This is the spot where the model's lead over the human was the largest of the five: 0.928 against 0.833. The rule the lecture just assembled from a hundred six studies predicts a gain for the pair here, and there isn't one: 0.834 against 0.833 — that is, exactly zero. Dropping this row and keeping the four convenient ones would have turned this case study into a pure demonstration of the rule.

The fifth outcome — septic shock — and this is the lecture's single most important number. Here the human was more accurate than the model: 0.820 against 0.816. The difference is tiny, and what matters isn't its size but its direction — this is the one place in the task where the surgeon had their own advantage. The surgeon who saw the suggestion scored 0.712. They didn't just fall below the model — they fell below themselves, on the one outcome where they didn't need any help. The pair came out worse than both the model alone and the human alone, and that has to be stated exactly that way, not as "the pair didn't improve the result."

What shifted here, by the vocabulary. The eighth form: same performer, now worse. Its checks pass literally. Same place — the risk assessment happens in the same spot. Same workload — the same one hundred fifty cases. Same hand — the surgeon gives the assessment, and the order of work hasn't changed. What changed is the state of the hand: the surgeon stopped relying on their own advantage exactly where they had it. Sitting alongside this is the fifth form — part of the risk assessment moved from the surgeon's own judgment onto the screen — but the load-bearing one here is the eighth, because what's measured isn't where the work happens but the quality of what the

hand finds.

Two caveats on this measurement: no one examined the patient — the surgeons worked from written case descriptions; and this measurement has never been repeated.

The answer, assembled: four cases, four effects, one conclusion

case	effect	conclusion
Surgeons and POTTER. 30 surgeons, 150 cases, randomized	on three of four outcomes the pair is more accurate than the surgeon; on septic shock **0.712** against 0.820 and 0.816	lost exactly where the human was stronger
Physicians with a language model. 50 physicians, randomized	reasoning score **76 versus 74**: a 2-point difference, interval −4…8	indistinguishable from a reference manual
A review with a meta-analysis. 74 studies, 106 experiments	where the model is stronger — the pair is worse than the model by **0.54**, better than the human by 0.74; across the whole review, −0.23	the scoring runs against the stronger side
The eye-tracking study. 10 readers, against themselves	where the suggestion stayed silent (14 cases) — missed cancers **29** → **61** out of a hundred	the sign is set by whether the suggestion was right

In none of the four did the human's decision actually reach a patient. The outcomes serve as the reference standard for scoring accuracy, not as a consequence of the suggestion.

surgeons: Journal of Trauma 2023 · *physicians with a language model: JAMA Network Open 2024* · *the review: Nature Human Behaviour 2024* · *eye-tracking: Radiology 2026*

The four measurements that came up in this section answer one and the same question, and they have to be assembled out loud, or they'll stay four different numbers in four different places. The question was asked earlier: can adding a human to a machine make the result worse, and does this happen with professional clinicians. The answer is yes, and it's been measured four times, from four different angles.

The first measurement is the strongest of the four by study design. Thirty emergency surgeons were randomized into groups, one hundred fifty cases were tracked forward thirty days, and both human groups and the model were measured on the same cases. On septic shock — the one outcome where the human was more accurate than the machine — the pair scored 0.712 against 0.816 for the model alone and 0.820 for the surgeon alone. This is a direct measurement of what was promised: the pair falls below each side on its own.

The second measurement is the trio the lecture opened with. Fifty practicing physicians, randomized into two human conditions: a physician with ordinary reference materials scored 74 out of a hundred, a physician on top of the model scored 76, and that difference isn't significant. The model-alone run, 92, wasn't an equal randomized arm and ranks lower in strength of evidence than the rest. The tasks here are constructed case descriptions, there's no physical exam, and no one checked patient outcomes. So this measurement is weaker than the first on two counts at once, and both have to be named.

The third measurement isn't a single study but a review of a hundred six. A preregistered review with a meta-analysis pulled together seventy-four studies, and the inclusion criterion decides everything there: a study was only included if all three quantities were measured — the human alone, the model alone, and the pair. Where

the model is stronger than the human, the pair is worse than the model: an effect size of minus 0.54, with an interval from minus 0.71 to minus 0.37 — negative throughout — and on that same subset of studies the pair beats the human alone by 0.74, with an interval from 0.63 to 0.85. Both halves have to be named together, or the table row reads as a verdict. The headline figure for the whole review is minus 0.23, with an interval from minus 0.39 to minus 0.07, and it's against this figure that minus 0.54 reads as a subset of studies. The review is stricter than any single case by design, and broader in scope: medical studies make up only part of it, it doesn't touch patient outcomes, and the studies are of different kinds.

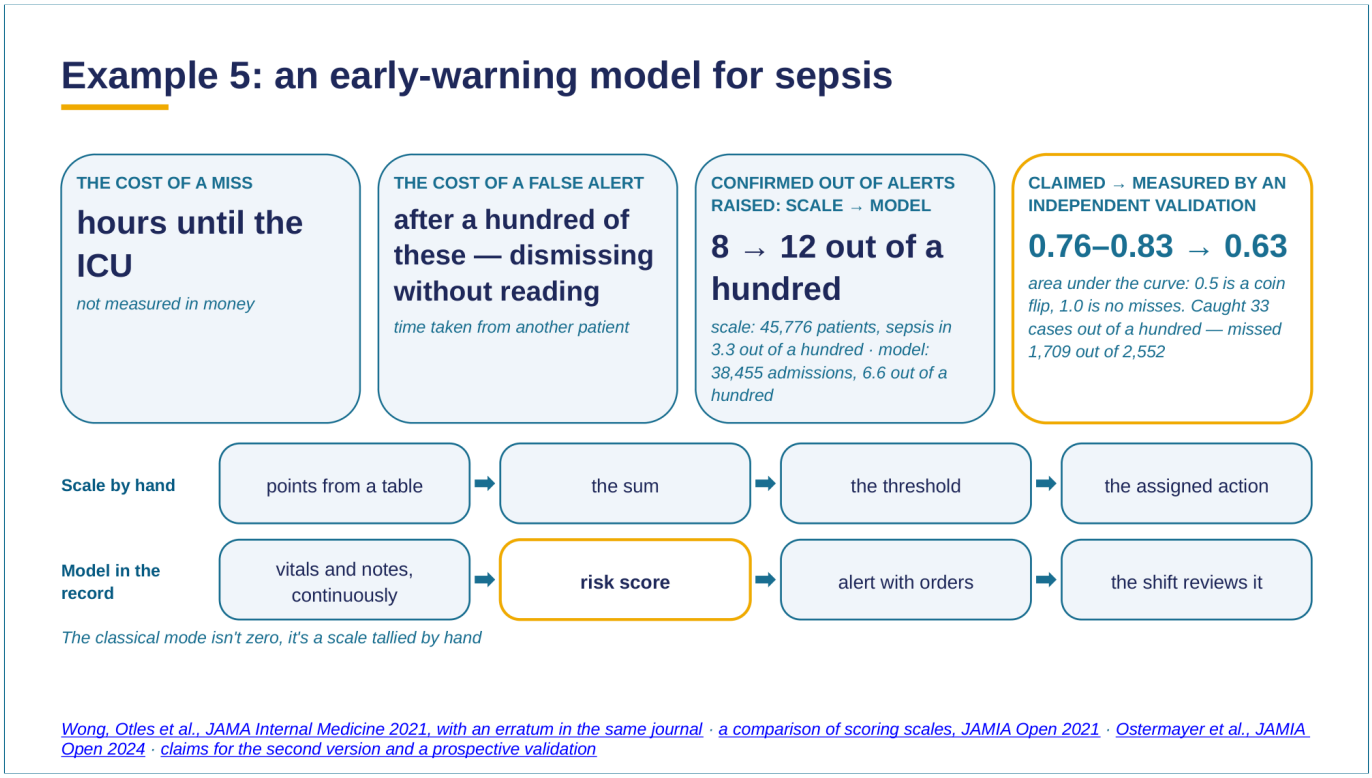
The fourth measurement answers not "how much" but "through what mechanism." Ten readers from a national screening program read the same set of sixty cases twice — first without suggestions, then six weeks later with them. On cases where the suggestion missed the finding, the same reader's sensitivity fell from 71 out of a hundred to 39. The comparison is a person against themselves, which removes the between-person variability that's large in reading scans. The weakness is named in the same breath: there are ten readers, and you can't generalize to the profession from ten.

Now the conclusion, one for all four. The human-and-model pair doesn't inherit the strength of the stronger side. That isn't the same as "the pair performs poorly": the pair usually beats the human without the model, and that has its own figure — an effect size of 0.64, with an interval from 0.53 to 0.74 across the whole review (0.74 on the subset where the model is stronger). The formulation the data supports reads: the pair is usually better than the human and worse than whichever of the two is stronger. From this comes a practical rule: before deciding who decides, you have to measure which of the two is stronger on this specific task.

And an honest caveat that applies to all four at once; it has to be formulated precisely. You cannot say "a patient outcome isn't measured in any of the four": for the surgeons, a primary outcome is stated outright — one hundred fifty patients, thirty days of prospective follow-up — and the surgeons' own slide says that a complication is counted on day thirty. This was checked against the stated primary outcome of each of the four studies, the same way any borrowed figure gets checked.

The correct formulation holds, and the point stays the same: in none of the four did the human's decision actually reach a patient. For the surgeons, the case descriptions are built from admissions that had already happened, the participants' assessments never fed into treatment, and the outcomes serve as the reference standard against which the accuracy of those assessments is scored. For the physicians with the language model, there are no patients in the study at all — the tasks are constructed. The review, by its very subject, isn't about outcomes. The eye-tracking study measured accuracy, reading time, and gaze behavior. In other words, what's measured is how a person looks and how they trade a correct decision for an incorrect one — and what's not measured is what happened to the patient because the person looked worse. The difference between "the outcome isn't measured" and "the decision never reached a patient" isn't nitpicking: the first is false, and demonstrably false; the second is true,

and is in fact the conclusion itself.



An early-warning model for sepsis, built into the medical record, replaced a scoring scale that had been tallied by hand before it. You have to start with how the scale is built: without it, there's nothing to compare against. An early-warning scoring scale is a table where every value measured at the bedside gets points for deviating from normal: a respiratory rate from twelve to twenty scores zero, twenty-one to twenty-four scores two, twenty-five and above scores three. The points are summed, the sum falls into a band, and the band assigns an action: continue observing, increase the frequency of measurements and call the on-call physician, or call the resuscitation team immediately. The weights in this table were set by a consensus committee of a professional society and printed on the chart that sits on the ward. Hence a property the model doesn't have: an error in the scale is handled as an error in a document — a committee convenes, revises the table, issues a new edition, and anyone who disagrees knows exactly who to object to and on which point.

I'm naming the cost of both errors before any numbers, and on the slide it stands as two plates. Missing a deterioration: sepsis is a leading cause of in-hospital death, and the clock here runs in hours until the ICU, not in money. Raising a false alert: an antibiotic given "just in case," time taken from another patient, and — after a hundred such alerts — the habit of dismissing a notification without reading it. The same shift pays for both errors, and the choice between them gets made many times a night.

Now the numbers, starting with what the model is compared against. For the scale, eight alerts out of a hundred are confirmed: the remaining ninety-two were raised for nothing, and that's the cost of the classical mode, before any machine learning at all. The measurement was taken on forty-five thousand seven hundred seventy-six patients, sepsis in three and three tenths out of a hundred, one sample covering all three scales at once; the other two scales in the same measurement came in at twelve

and eighteen. A figure that circulates alongside this in the literature is eighty-eight and eight tenths — the false-alarm rate of a bedside monitor — and it can't be added to ours: that's a device at the bedside, with a threshold set by hand, with no trained model inside it at all. It's a different task and a different baseline; I'm saying this here so the slide doesn't carry two shares the room will add together.

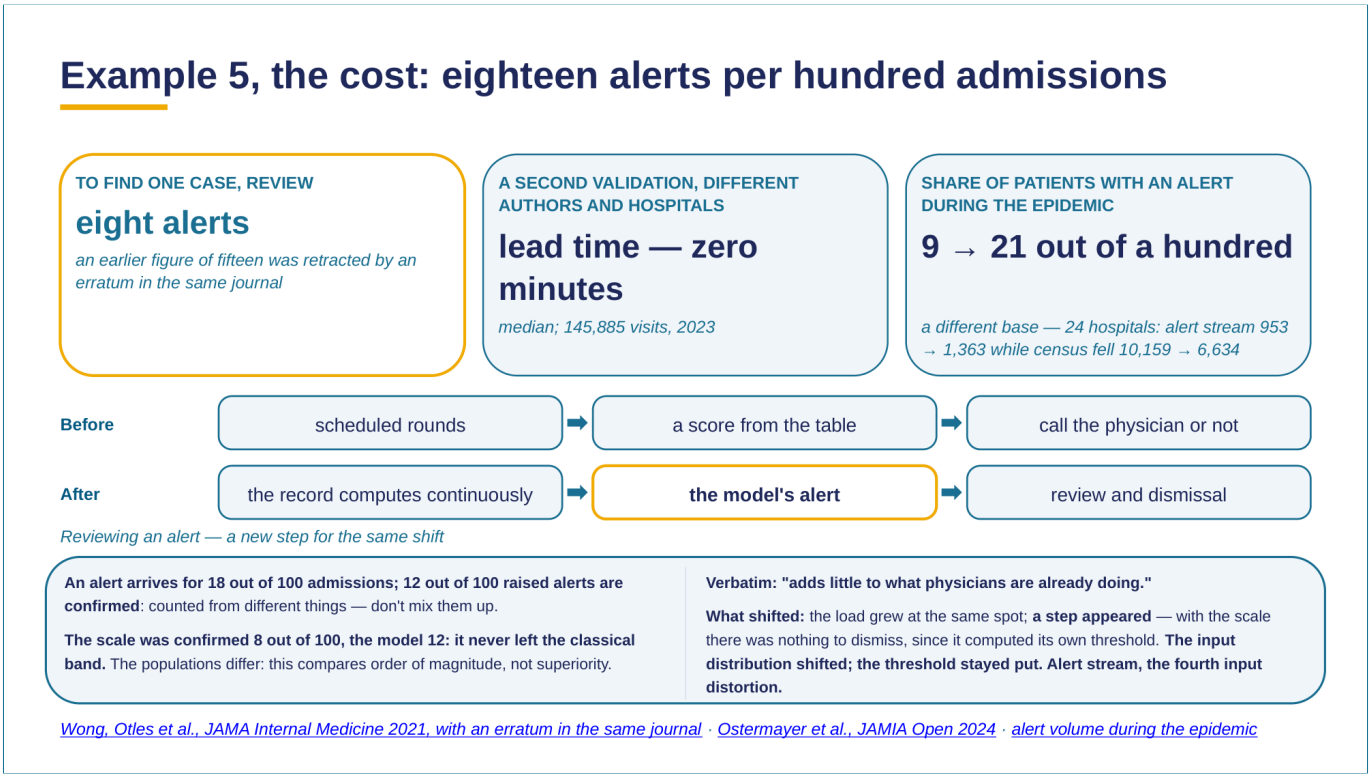
What the developer claimed, and what an independent validation measured. For the first version of the model, an area under the curve of 0.76 to 0.83 was claimed. An external validation across 38,455 admissions measured 0.63 — discrimination only a bit better than a coin flip. The model caught 33 sepsis cases out of a hundred: it missed 1,709 out of 2,552. The developer released a new version after this, and what matters here isn't that the first fell short of what was claimed, but that this was learned from the outside: there was no internal validation that would have shown the same thing.

The two denominators on the slide are different, and confusing them is the main place where case studies like this lie. Eighteen out of a hundred is the share of admissions that received at least one alert: six thousand nine hundred seventy-one alerts across thirty-eight thousand four hundred fifty-five admissions. Twelve out of a hundred is the share of already-raised alerts that were confirmed. They're counted from different things. From this comes an arithmetic worth saying slowly: to find one case of sepsis, the shift has to review eight alerts. An earlier version of this case study said fifteen — that figure was retracted by an erratum in the same journal, and I'm naming this out loud, because this lecture is about how numbers live longer in retellings than in the original sources.

A second validation, by different authors and in different hospitals, adds something the first didn't measure — time. In the emergency department, it's the physician who recognizes sepsis: they examine the patient, order labs, start treatment. The validation compared the timing of the alert against the timing of that decision, across one hundred forty-five thousand eight hundred eighty-five visits to two departments in 2023. The median lead time was zero minutes; in roughly half the cases, physicians were already managing sepsis before the model spoke up. The authors' conclusion, verbatim: "adds little to what physicians are already doing." For the current version, validated prospectively across two hundred twenty-seven thousand ninety-one admissions in four systems, discrimination rose to a band of zero point eighty-two to zero point ninety-two, while confirmation stayed low — thirteen to twenty-six out of a hundred.

What shifted. The alert now exists and runs without a human — around the clock, across every patient at once, from data that already sits in the record in machine-readable form. That's a real gain, and a specialist will take it. But reviewing and dismissing an alert became a separate line item of working time that didn't exist before this was deployed, and at some hospitals this step has its own staff now. The vocabulary form here is the fourth — the load grew at the same spot: same place, same hand, more reading and dismissing. And the form "a step appeared" — about the review of the alert itself: with the scale, there was nothing to dismiss, since it computed its own threshold.

Sources. Wong, Otles et al., JAMA Internal Medicine 2021, with erratum · a comparison of scoring scales, JAMIA Open 2021 · Ostermayer et al., JAMIA Open 2024 · a prospective validation of the current version, JAMA Network Open 2026 · alert volume during the epidemic, JAMA Network Open 2021



The cost of the early-warning sepsis model, counted from the alert stream hitting the same shift. An alert arrived for eighteen out of a hundred admissions — six thousand nine hundred seventy-one alerts across thirty-eight thousand four hundred fifty-five admissions; twelve out of a hundred already-raised alerts were confirmed. The two denominators are different, and confusing them is the main place where case studies like this lie: collapsing these two numbers into one thought means saying something false.

That's how the alert stream works, the fourth of the evening's six input distortions: the model sends them one after another, and the physician dismisses almost all of them, including the ones that shouldn't be dismissed. The dismissal rate is measured across a whole class of such systems — a 2024 systematic review of drug-interaction alerts gives an average of 90 out of a hundred, with an interval from 85 to 95. This is worth reading both ways: most alerts really are excessive, and a high dismissal rate is in part a reasonable adaptation, not only fatigue.

From this comes the arithmetic: to find one case of sepsis, the shift has to review eight alerts. Per thousand admissions, that's roughly a hundred eighty-one alerts and twenty-two confirmed cases. The figure of fifteen alerts per case, which circulates in retellings of this study, was retracted by an erratum in the same journal: numbers live longer in retellings than in the original sources.

The comparison everything was counted for. The scale is confirmed eight out of a hundred, the model twelve. The model never left the band of the classical mode. The populations in the two studies are different, so this compares order of magnitude, not superiority; claiming superiority isn't allowed. And the reverse isn't true either: the model didn't fail — it landed roughly where the scale, which cost incomparably less,

already stood. That "roughly where it already stood" is the actual result of this case study.

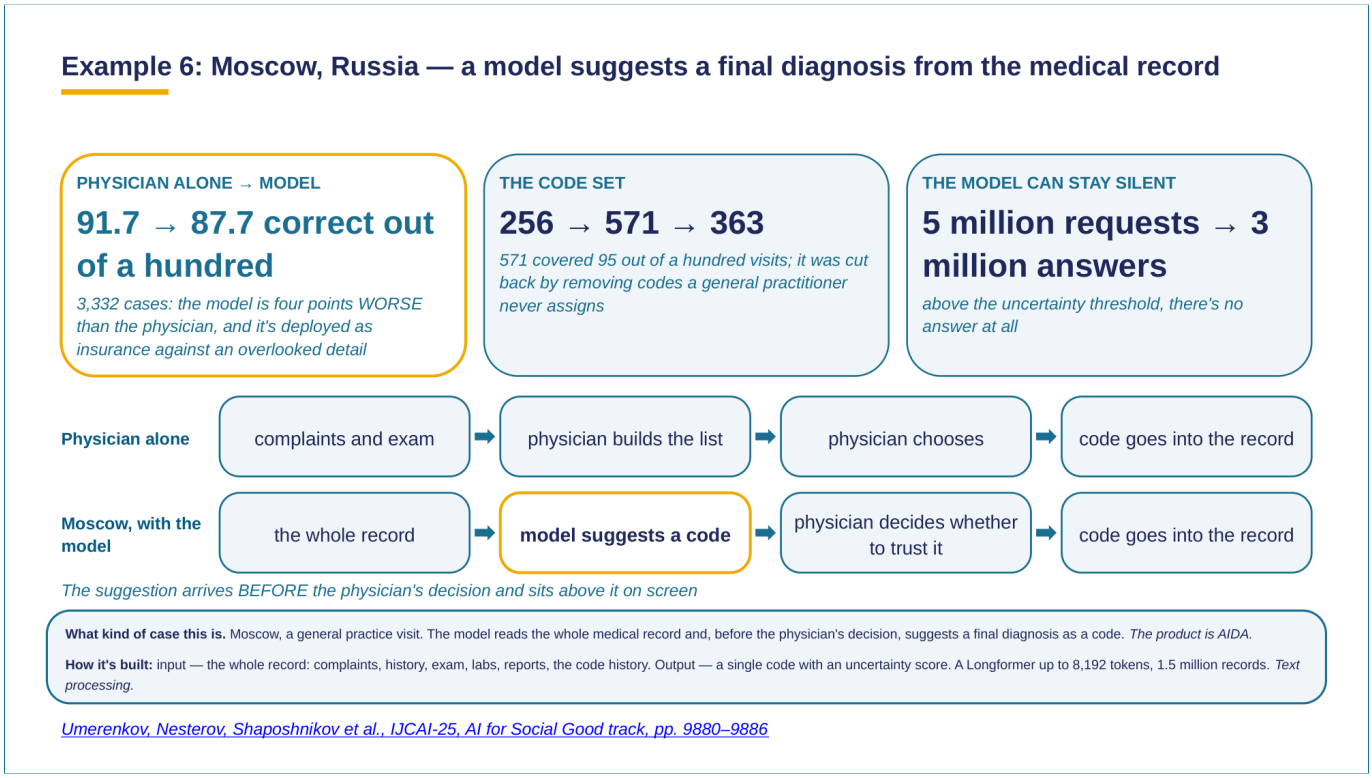
A second validation, by different authors and in different hospitals, adds something the first didn't measure — time. In the emergency department, it's the physician who recognizes sepsis: they examine the patient, order labs, start treatment. The validation compared the timing of the alert against the timing of that decision, across one hundred forty-five thousand eight hundred eighty-five visits to two departments in 2023. The median lead time was zero minutes; in roughly half the cases, physicians were already managing sepsis before the model spoke up. The authors' conclusion, verbatim: "adds little to what physicians are already doing." For the current version, validated prospectively across two hundred twenty-seven thousand ninety-one admissions in four systems, discrimination rose to a band of zero point eighty-two to zero point ninety-two, while confirmation stayed low — thirteen to twenty-six out of a hundred. The strength of evidence differs between these two validations, and that has to be stated: the first is retrospective, the second is prospective and multi-site.

What shifted. The alert now exists and runs without a human — around the clock, across every patient at once, from data that already sits in the record in machine-readable form. That's a real gain, and a specialist will take it: no scoring scale gets recalculated across the whole hospital every fifteen minutes. But reviewing and dismissing an alert became a separate line item of working time that didn't exist before this was deployed, and at some hospitals this step has its own staff now. The vocabulary form here is the fourth — the load grew at the same spot: same place, same hand, more reading and dismissing. The form "a step appeared" is about the review of the alert itself: with the scale, there was nothing to dismiss, since it computed its own threshold. And the form "same performer, now worse" doesn't apply to this case study, and this has been checked: no one measured the quality of the shift's work before and after the deployment, and such a measurement can't simply be invented, any more than a number can.

And one last figure that only reads correctly with its baseline; it sits on the slide as a pair, and the pair has to be said in full. Across twenty-four hospitals during a viral epidemic, the daily alert stream rose from nine hundred fifty-three to one thousand three hundred sixty-three — up forty-three percent. On its own, that means nothing: more patients, more alerts. But census over the same period fell from ten thousand one hundred fifty-nine to six thousand six hundred thirty-four patients — down a third. So alerts per patient didn't rise by forty-three percent — they rose by more than double: the share of patients with an alert rose from nine to twenty-one out of a hundred.

Here's what happened: the input distribution shifted — different patients, different illness — while the threshold stayed put, because no one recalculated it. The model didn't break and wasn't retrained; it kept computing the same thing it always had, on a flow that had changed. This carries over outside medicine almost without edits: at the exact moment the flow changes the most, the tool is wrong the most, and that's exactly the moment the organization has the fewest hands free to recalculate the threshold.

Sources. Wong, Otles et al., JAMA Internal Medicine 2021, with erratum · Ostermayer et al., JAMIA Open 2024 · alert volume during the epidemic, JAMA Network Open 2021



Moscow, a general practice visit: AIDA, a clinical decision-support system, reads the whole medical record — complaints, history, exam, labs, reports, the code history — and, before the physician's decision, suggests a final diagnosis as an ICD-10 code. The suggestion sits at the top of the working screen. The system was trained on a million and a half city records; in its first year it handled five million requests and three million recommendations.

The other case studies at this stage measured the suggestion against a single piece of data: an image, a slide, an EKG, a risk score. Here the model reads the whole record and gives a final diagnosis, not a preliminary one.

What kind of case this is. Moscow, a visit to a general practitioner or family physician at a city polyclinic. The system is called AIDA; it's built into the city's clinical decision-support system and runs on top of the same electronic medical record the physician is already working in. One important clarification, so as not to confuse two products: Moscow previously ran a service that suggested the three most likely preliminary diagnoses from the complaints recorded at intake. AIDA is a later product in the same line, and its subject is different: not a preliminary diagnosis from a complaint, but a final diagnosis from the whole history. For our stage, that's exactly the case we need: this section is about how a conclusion is drawn from everything that's been gathered.

How it's built. The input isn't a single field but the entire record in text form: complaints, history, exam findings, lab results, reports and conclusions from imaging and other studies, the history of previously assigned codes. The output is a single code from the International Classification of Diseases plus a self-reported uncertainty score. Inside is a Longformer language model — that is, a transformer with sparse attention that can hold

a sequence of up to eight thousand one hundred ninety-two tokens; it was first trained to predict masked words in unstructured clinical notes, then fine-tuned to predict the code. The training set is a million and a half de-identified records from the city system, of which one million three hundred fifty thousand went to training and one hundred thirty-three thousand to validation. The labels were diagnoses confirmed by laboratory or imaging findings and flagged in the record as final.

Two engineering details from the same source, both instructive. First: when full lab results were fed into the text, accuracy DROPPED — a long printout of lab values crowded the old history out of the window, and the model lost the long-term trends. So now only values outside normal range are fed in. Second: adding the time gaps between visits, which looks obviously useful, produced no measurable improvement at all; nor did height and weight. Feature selection here was measured, not guessed, and the heaviest-weighted features turned out to be complaints and history, not lab values.

The code set is a separate story about how a task gets narrowed to fit practice. They started with the two hundred fifty-six most frequent diagnoses. They expanded it to five hundred seventy-one, and that set covered ninety-five out of a hundred visits, excluding obstetrics, oncology, and routine checkups. Then they CUT it back to three hundred sixty-three, dropping codes a general practitioner never assigns, codes absent from clinical guidelines, or codes unsuitable as a final diagnosis. In other words, the set was edited not for the metric's sake but to match what can actually be a final diagnosis at this point in care.

Now the classical mode this is compared against. There's one here, and it's measured: the physician decides alone. On the set of labeled cases, the physician scores ninety-one and seven tenths correct out of a hundred. There simply isn't a second mode here — a colleague's second opinion — at a polyclinic visit: in the case study on second opinions we already covered what holds that mechanism up, and a fifteen-minute visit has nothing to hold it up with. This isn't a gap in the measurement, it's a property of the stage, and I'm naming it outright.

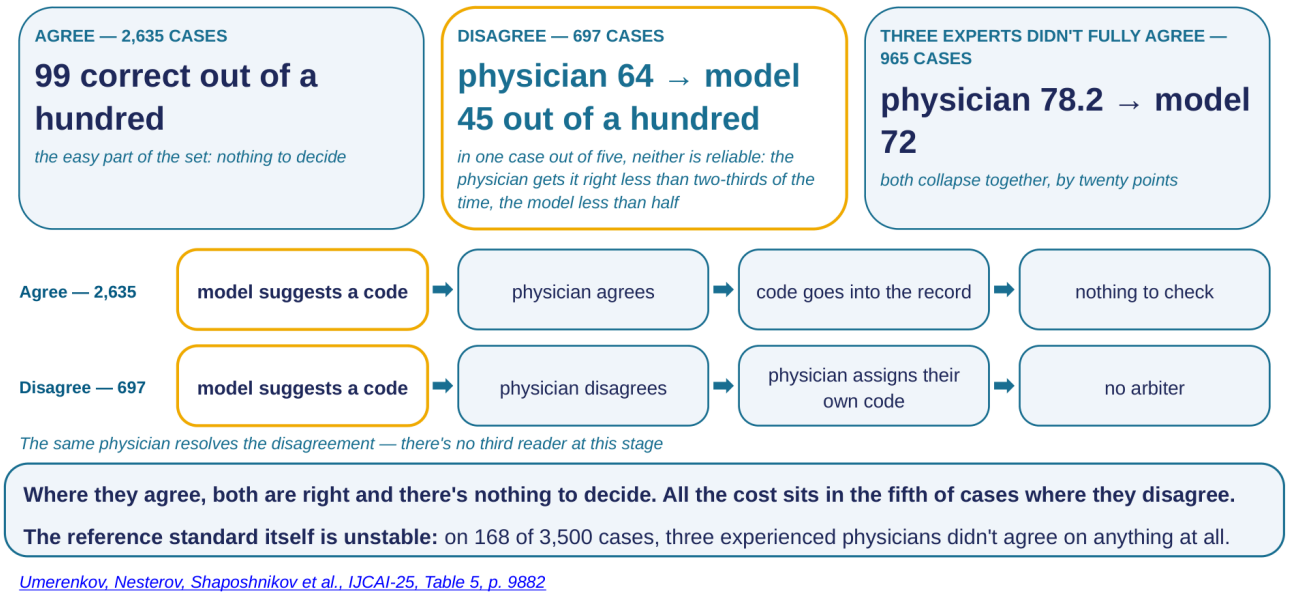
And the key figure of this slide, the reason this case study is worth doing. The model scores eighty-seven and seven tenths against ninety-one and seven tenths for the physician. The model is FOUR POINTS WORSE than the physician — and it's deployed anyway. The authors don't hide this or make excuses for it: they write that the model works as a check and a safeguard, so that no significant detail in the record goes unnoticed. This has to be kept separate: the argument "the model is more accurate than the human" never comes up here at all, and yet three million recommendations got made in the first year regardless. The basis for deploying it is something else — not accuracy, but thoroughness of review.

And one last thing on this slide: the model can stay silent. It outputs a code with an uncertainty score — at first this was simply the maximum of the output distribution, then a separate uncertainty-estimation model was added — and if the uncertainty is above a threshold, there's no answer at all; instead the screen shows a message that the record's data is insufficient, and separately there's a warning for an "especially

complex case." Hence a gap in the numbers that would otherwise look like data loss: five million requests in the first year, and three million recommendations. Two million times, the model declined to answer. The authors write that this very refusal is what raises physicians' trust in the tool — meaning the ability to stay silent is part of the product here, not a shortcoming of it.

Sources. Umerenkov, Nesterov, Shaposhnikov et al., IJCAI-25, pp. 9880–9886

Example 6, the cost: where the physician and the model disagree



All the cost of this sixth case study sits in a fifth of the cases. Where the physician and the model agree — 2,635 of 3,500 cases — both are right almost always, ninety-nine out of a hundred, and there's nothing to decide there. Where they disagree — 697 cases, nineteen and nine tenths percent — the physician is right in sixty-four cases out of a hundred, and the model in forty-five. This is exactly the spot that case studies on AI suggestions almost always skip, and it's the only one where the suggestion makes any difference at all.

The set everything was counted on: three thousand five hundred de-identified cases, pulled from the same city system, but AFTER the training set was frozen — so training cases wouldn't leak into the validation ones. Each case was reviewed independently by three experienced physicians, each seeing the whole history; the reference standard was the diagnosis at least two of them agreed on.

And here's the first thing worth saying slowly, because it's about the measure itself, not about the model. All three agreed in sixty-seven and a half cases out of a hundred. Two of the three agreed in twenty-seven and a half. And in five cases out of a hundred — that's a hundred sixty-eight cases out of three thousand five hundred — three experienced physicians, looking at one person's full history, didn't agree AT ALL: three different diagnoses for one case. Remember this the next time you hear about the accuracy of a diagnostic model: the very line it's measured against is itself shaking. Where the reference standard is assembled by a two-out-of-three vote, "accuracy" means "agreement with the majority," not "correctness."

Now the key figure of the evening for this section, and I'll break it into three subsets of one set. First: cases where the physician and the model gave the same answer — two thousand six hundred thirty-five, or three-quarters of the set. There, both are right

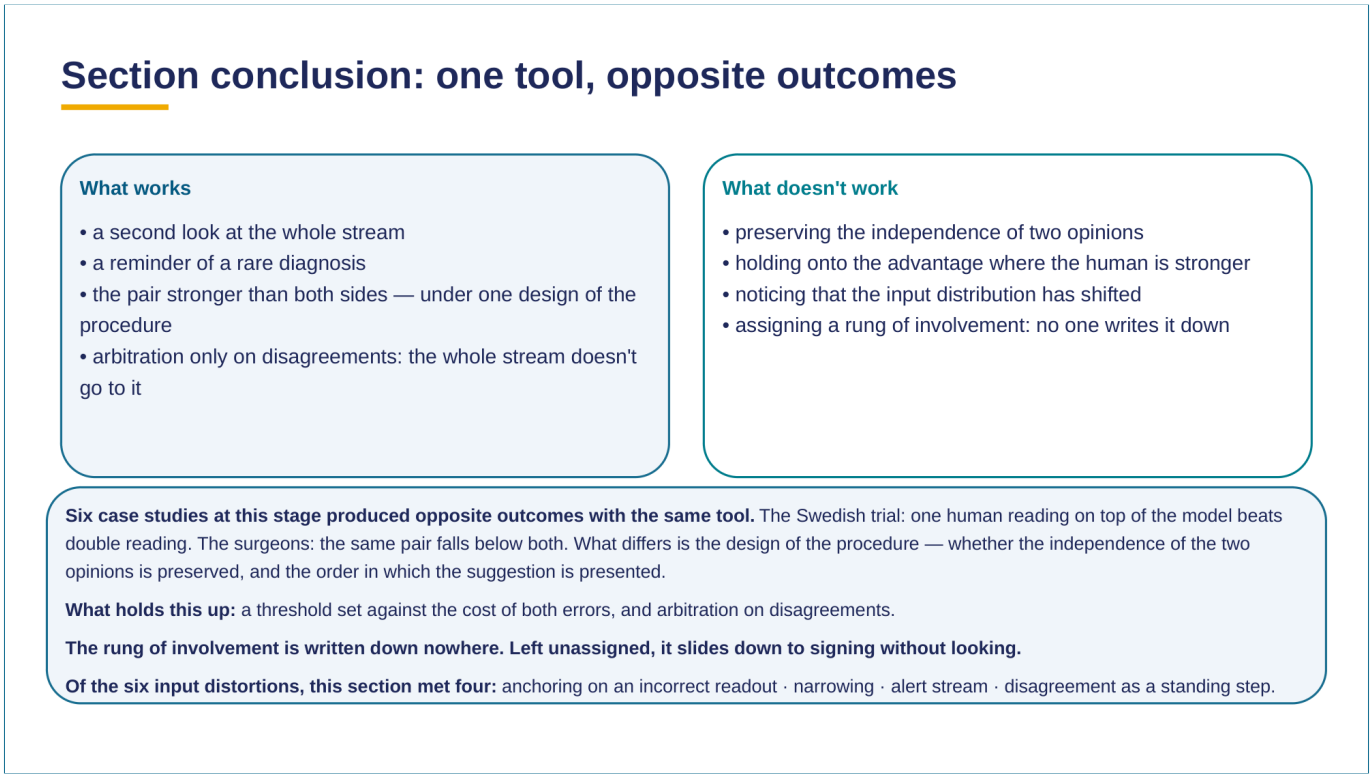
almost always: ninety-nine out of a hundred. There's nothing to decide there, no need for a suggestion, and these cases are exactly what drive the bulk of the attractive headline number. Second: cases where the physician and the model DISAGREED — six hundred ninety-seven, almost exactly a fifth. And here it is: the physician is right in sixty-four cases out of a hundred, the model in forty-five. Let me state the conclusion plainly, because it's worth stating: in that fifth of cases, the only place where the suggestion could possibly have changed anything, NEITHER of the two is reliable. The physician is right less than two-thirds of the time, the model less than half. "Listen to the physician" gives you a third wrong there; "listen to the model" gives you more than half wrong.

The third subset finishes the picture. Take the nine hundred sixty-five cases where the three experts didn't fully agree — the objectively hard ones. There, both collapse together: the physician from ninety-seven down to seventy-eight, the model from ninety-four down to seventy-two. They fall together, and by roughly the same amount. So the model isn't hard where it finds things hard on its own terms — it's hard exactly where the human finds things hard, and their errors are correlated. For a pair, that's the worst possible arrangement: insurance that fails at the same moment as the thing it's insuring isn't insurance at all.

And a nuance about the percentages themselves, which the slide carries. Eighty-seven and seven tenths and ninety-one and seven tenths were obtained by a scoring method where a synonymous classification code is counted as correct — meaning the same disease, recorded under a different code, counts as a hit. By strict scoring, code for code: the physician scores exactly ninety, the model eighty-six and one tenth. The difference is small, but the direction matters: the headline figure is always the more generous of the two, and which one was actually used is worth asking before you repeat it.

What shifted. Three things, and none of them is about accuracy. First: the suggestion arrives BEFORE the physician's decision and sits above it on the screen — not alongside it, not on request, but at the top. We named this mechanism earlier, at the opening, and it has a name: anchoring. The conclusion shifts toward the suggestion even when the suggestion is wrong, and the order of presentation is a controllable parameter that's set here to the riskiest possible position. Second: the physician gained a step that wasn't part of their work before — deciding whether to trust the suggested code; in a fifth of the cases, this step is actually carried out, and we've just seen how it turns out. Third, and this comes from the authors' own surveys: the tool proved most useful to younger specialists — they feel more confident with it. We've already met the experience gap on the EKG with the incorrect machine readout; here it shows up from the other side, and honesty requires saying that these surveys are qualitative — there's no figure attached to them.

Sources. Umerenkov, Nesterov, Shaposhnikov et al., IJCAI-25, Table 5, p. 9882



Six case studies at this stage, using the same tool, produced opposite outcomes, and what differs between them is the design of the procedure — whether the independence of two opinions is preserved, and the order in which the suggestion is presented.

What works. A second look at a stream no human could ever read in full: the model reads everything and doesn't lose sensitivity on a Friday evening. The reminder of a rare diagnosis works by the exact same mechanism through which a suggestion also does harm — it adds to the composition of the list, and a line a tired physician wouldn't have come up with gets added. And one design of the procedure where the pair is measurably stronger than both sides: sensitivity of eighty and a half against seventy-three and eight, at the same specificity.

What doesn't work. Independence is lost because of how the interface is built; the physician's attentiveness isn't the cause. Selectivity is a procedure the system lacks: no group got a tool that let them tell a correct suggestion from an incorrect one, and none of them managed the task. Explainability has a measured cost: a simplified explanation of an incorrect recommendation drops the choice below the line the same physician gives with no model at all.

What happened to the work. The load fell at the same spot — in the eye-tracking study, that was a cost; with the sign reversed, a gain; so the form itself carries no sign — the sign is set by whether the suggestion was right. The cost of error shifted: the model's miss became the reader's miss, and the human signs off on it. The performer changed: the model became the second reader, and the drop in human reading volume is a consequence of that change; it isn't an achievement on its own.

What holds this up. The order of presenting the suggestion is a controllable parameter with a measured cost: an interface that shows the marker alongside the image buys

speed at a high price. Arbitration by a third reader, where the human and the model disagreed. And one thing written down nowhere: the human's rung of involvement. Left unassigned, under load, it slides down to signing without looking.

And the input mechanism this section ran into harder than any other: anchoring. The sixth case study, from Moscow, puts the model's suggestion at the TOP of the working screen before the physician has decided — setting the order of presentation to the riskiest position this stage allows. The same study measures how this turns out in the one spot where the suggestion makes any difference at all: in that fifth of cases where the physician and the model disagree, neither is right — the physician in sixty-four cases out of a hundred, the model in forty-five.

This link has to be said out loud: in the case study where the result fell, and in the control line where it rose, the exact same system from the same vendor was at work.

The figure behind the line on explainability is worth naming, because it runs against intuition. In a vignette study on antidepressant selection, an incorrect recommendation delivered with a simplified explanation dropped the share of correct choices from 35.7 to 26.2 out of a hundred — below what was achieved with no model at all. The explanation didn't help anyone tell a bad piece of advice from a good one; it helped the bad advice get through.

Two things hold this load up: the order of presenting the suggestion, which has a measured cost, and arbitration by a third reader — but only where the human and the model disagreed. We don't repeat the numbers behind the four vocabulary forms: each one already stood on its own case study, and repeating them here would turn the conclusion into a summary of the section. The same tool dragged the result down in one case study and raised it in the control line — what differed between them was the design of the procedure, and that's the conclusion of this stage.

How to say the four vocabulary forms that stand in a row. The load fell at the same spot — in the eye-tracking study — and the sign of this form is set by whether the suggestion was right: the same drop in scan coverage turned into a gain in the mirrored study. The cost of error shifted — the model's miss became the reader's miss, signed off by a human. The performer changed — the model became the second reader, and human reading volume fell as a consequence of that change, not as a goal in its own right. Same performer, now worse — the surgeon's own advantage on septic shock collapsed under the same workload; this form didn't exist in the vocabulary before this section, and it had to be added.

And about the bolded line. The human's rung of involvement is written down nowhere: not in the program's regulations, not in the product manual, not in the reader's job description. An unrecorded rung doesn't stay neutral — it slides downward, toward signing without looking, because that's the cheapest way to satisfy the requirement "a human makes the decision." The bill for this comes due in the section on licensing, oversight, and liability, and it's presented not to the interface, but to whoever failed to assign the rung.

Observation looked for deterioration in numbers; this work is built differently. The decision here isn't made by a device or a shift on duty — it's made by a person who reads the document later, sometimes months later, and never saw the patient in person. For them, the quality of the record isn't a matter of style; it's the only channel through which someone else's decision reaches them.

This work has four classical modes: the physician types the text by hand right after the visit; carries a previous note forward into a new one, editing what's changed; dictates to a live scribe; or edits a resident's draft under their own signature. Two of the four are counted, and in different units — the physician's own writing in minutes per note, carrying forward in a share of the chart's text; for the live scribe and the resident's draft, there's no figure in our sources, and the reason for each is named on the spot. The model enters this work from two directions. First — recording the visit itself: the tool listens to the conversation and hands back a draft note, which the physician proofreads and signs. Second — recording the outcome of the whole hospitalization: the model assembles the discharge document from the case history, the physician again proofreads and signs, and treatment is then ordered elsewhere from this document.

Both case studies rest on the same form. The classical mode is counted separately from the model's claimed effect; the measured effect comes from a randomized trial or a blinded comparison, not a self-report from people who chose the tool themselves; and each one names a pair — what became possible, and what it required.

The case study on the visit note shows how the claimed time savings and the measured result are two different quantities, counted in different units. The case study on the discharge summary goes further: the error count for the model and for the human is comparable, but the error is built differently — not a typo, but a plausible-sounding

detail that was never in the history. The old proofreading habit of "reading it the way you'd read a colleague's note" doesn't carry over to this document.

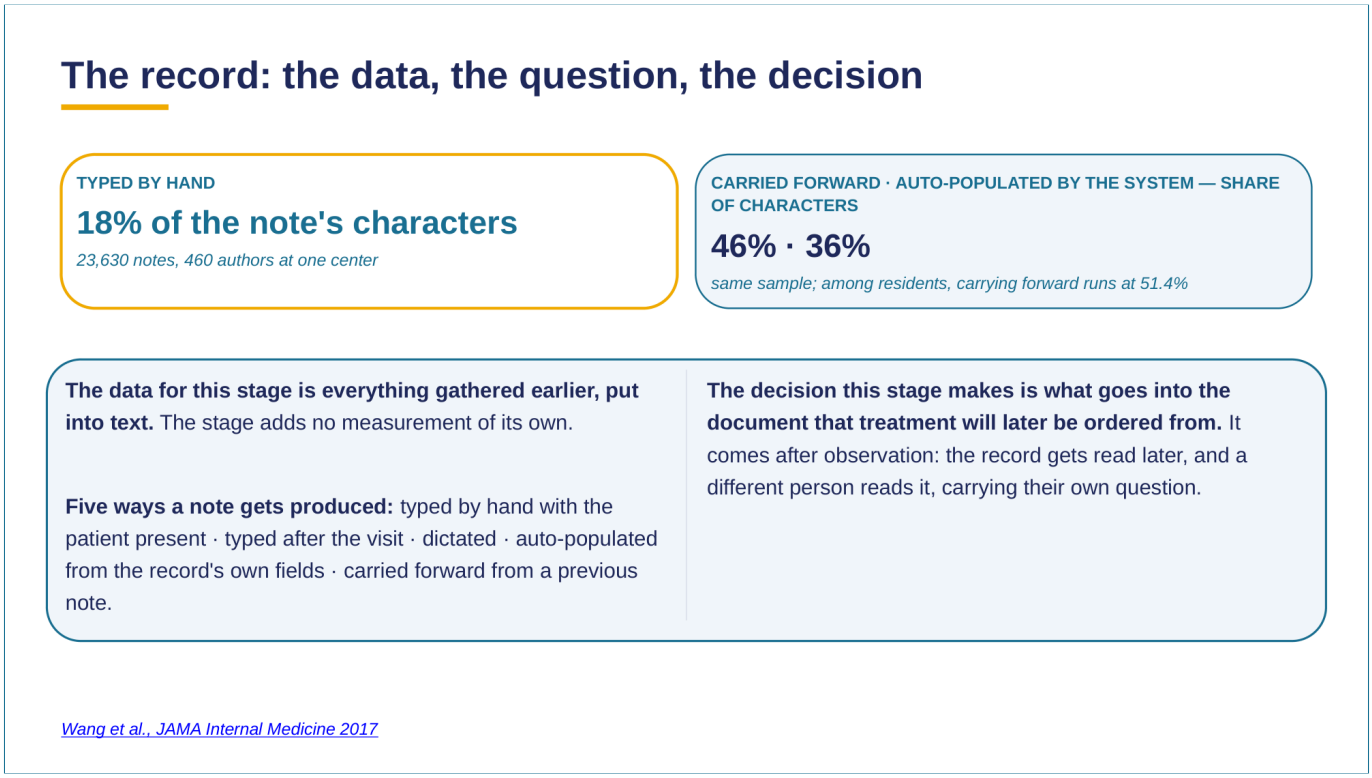
This section ends with a document, and in medicine a document almost always sits under someone's signature. That's the bridge to the next section: what actually holds the cost of an error in check once a decision has already been made and recorded.

Let me say right away how this section is built differently from the last one. In observation, the subject was numerical: a trail of values, a score, a threshold. Here, the subject is text, and there's nothing to check it with except reading. A score has a threshold; a record doesn't: you can't say "this record is seventy percent complete" and make a decision on that number.

This section has two different readers, and they face a different cost of error. The visit note is read by the patient themselves and by the next physician — an error there costs a repeated workup or a mistaken continuation of treatment. The discharge summary is read by the local physician, the next hospital, and the insurer — an error there costs a decision made at a different institution, where there's no one left to ask. That's why this section has two case studies, not one: one tool, two stages, two different costs.

And one property worth holding in mind for the whole section, because it explains both costs. A gap in the record and the absence of an event look the same to a reader. If the document doesn't say a medication was given, the reader can't tell "it wasn't given" apart from "it was given and not recorded." No amount of careful reading fixes this: the distinction simply isn't in the document. From this follows something that isn't obvious to an engineer: completeness of the record matters more here than brevity, and any tool that makes a document shorter has to be checked against exactly that.

One last thing about the order of the two case studies. The visit note comes first because it has real randomization behind it — the strongest evidence in the whole lecture, and this stage deserves to open with it. The discharge summary comes second because it's heavier: treatment gets ordered from it at a different institution, and the error there is built differently — not a typo, but a plausible-sounding detail that was never in the case history. The old proofreading habit of "reading it the way you'd read a colleague's note" doesn't carry over to a document like this, and that will be this section's main finding.



Keeping medical documentation is a separate piece of a physician's work, with its own form, its own deadlines, and its own accountability for content. The stage adds no measurement of its own: it puts into text everything the preceding stages gathered, and decides what goes into the document a decision will later be made from. The previous stage left behind a trail of measurements, alerts, and what was done about them; from here on, the document does the work. A completed record serves three purposes at once: a source of information for the next physician, the subject of a quality-of-care review, and the primary evidence when a case gets reviewed.

The data for this stage is everything the earlier stages gathered, put into text. The complaints and history from the visit, exam findings, test figures, the diagnosis as formulated, the treatment plan, the trail of repeated measurements. The stage adds not a single measurement of its own; it converts what's been gathered into a form that can be handed off.

There's one question being settled with this data, and it isn't about archiving: what does the next reader, who wasn't at the visit, need to learn. The record isn't made for storage and isn't made for an inspector — it's made for a person who will act on it without having seen the patient.

Now about the ways a note actually gets produced, and this is worth pausing on, because it's built the opposite way from how the stage is usually pictured. Five ways are listed on the slide; I'll go through each. In a modern record they all work together, inside one document. This list is about where the characters in a note come from; the modes of writing that come later in the section are about who puts the text into the document. Two different axes. Typing by hand during the visit — the physician types while talking with the patient. Typing after the visit — between patients, or at the end

of the day, that same take-home work with the chart. Dictation — speech transcribed into text, either by a person or by speech recognition. Auto-population — content the system pulls into the note on its own, from structured fields: the medication list, lab results, monitoring readings. And carrying forward a previous note — an earlier note moves into the new one, whole or in blocks, with edits for what's changed.

The proportion between them was measured keystroke by keystroke, and it's on screen as plates. The electronic record system at a large medical center tracks the origin of every character in a signed note: typed by hand, auto-populated by the system, or carried forward. Across twenty-three and a half thousand notes from four hundred sixty authors, it came out like this: eighteen percent of characters typed by hand, forty-six carried forward, thirty-six auto-populated by the system. Among residents, the share typed by hand is even lower, and the carried-forward share higher — eleven and eight tenths against fifty-one and four tenths.

The unit here is characters, and that has to be held firmly: later on, a carry-forward share of fifty and one tenth percent comes up, and that's a share of WORDS, from a different study with a different sample. Forty-six and fifty-one aren't the same number measured twice.

This proportion has to be read plainly: the physician types a smaller part of the note by hand. Four characters out of five come from previous text or from the record's own fields. And from this comes the first thing to know about the ceiling of any tool that promises to take typing off someone's hands: it's going after exactly that eighteen percent. The strength of this evidence: a single-center cross-sectional measurement from the record system's own log — a behavioral measurement captured by the machine itself — and generalizing it to other institutions or other record systems hasn't been checked.

The document forms at this stage differ, and the work inside them isn't the same: a visit note in the chart, a progress note in the hospital, a discharge summary — the final document on the hospitalization, which the next physician then works from — and a referral to a specialist. Method and form are different things: the same form can be typed by hand, dictated, or carried forward.

Now the readers, and I'll name them out loud: there's no room for them on the slide. The record has five, and each one acts on it. The local physician, a month later: the record tells them what's already been checked, what's been ruled out, and what treatment was given. The next hospital, on readmission: the discharge document is the main source of information about last time. Another specialist at the same institution the patient was referred to. A review committee, when a case gets examined: it reads the record and decides whether the physician's actions matched approved protocols, and the record itself is the evidence. And the patient themselves, who's entitled to a copy.

From this comes a property of the stage that sets the cost of error: whoever decides based on the record never saw the patient. A physician reading someone else's note can't tell "this didn't happen" apart from "this wasn't recorded." A gap in the record and

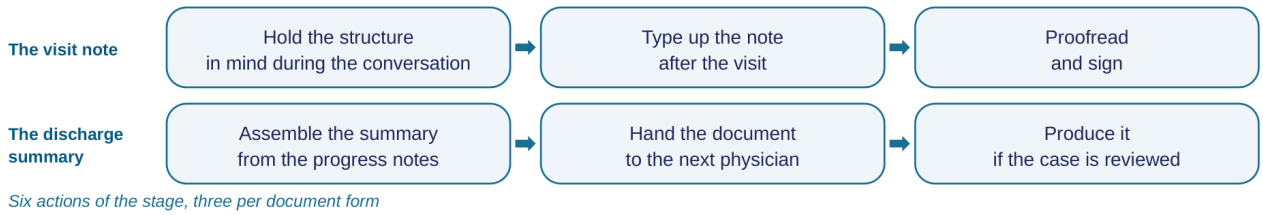
the absence of an event look identical to them, and they have nothing to tell the two apart with.

The decision this stage makes sounds simple and carries a lot of weight: what goes into the document a decision will later be made from. The document triggers actions — treatment gets ordered and cancelled based on the note, admission at the next institution is based on the summary, a place in the queue is based on the referral, and a review assesses compliance with protocol based on the contents of the chart.

And why this stage comes right after observation. The direct answer: the record gets read after it's made. But the second half matters more — it's read by a different person, carrying their own question. Observation decided "intervene now, or wait for rounds" from a trail it kept itself. Here, the decision is made by someone who has no trail — only text — and the quality of that text, for them, isn't a matter of style; it's the only channel through which someone else's decision reaches them.

Source. Wang, Khanna, Najafi, JAMA Internal Medicine 2017

What happens at this stage: the visit note and the discharge summary



Six actions of the stage, three per document form

The visit note. They listen to the conversation; the text holds the complaints, exam findings, diagnosis, and treatment plan. It's read by the patient themselves and the next physician.

The discharge summary. There's nothing to listen to — it's assembled from the progress notes covering the whole hospitalization. It's read by the local physician, the next hospital, and the insurer.

The two document forms this whole section moves between are broken down here step by step: the visit note and the discharge summary. They have to be broken down before any talk of tools — without that, it's unclear exactly what the model later intervenes in.

The visit note is made like this. While the patient is talking, the physician holds the structure of the future text in mind: what goes into complaints, what into history, what into exam findings. That's the first action, and it runs in parallel with the live conversation — that's where its whole weight lies. The second action is typing up the note; more often after the visit, between patients, or by the end of the day, and whatever didn't get done goes home with them. The third is proofreading and signing. The signature isn't a formality here: from that minute on, the text becomes what others will act on, and what gets produced if the case is reviewed.

The discharge summary is built differently, and the difference deserves to be stated clearly. There's nothing to listen to there: the patient has been on the ward for a while, and the source is the progress notes for the whole hospitalization, the test results, the medication list. The first action is to assemble a summary from this: what happened, what was done, how it ended, what condition they were discharged in, and what to do next. The second is handing the document to a reader who was nowhere near the hospitalization. The third is producing it if the case is ever reviewed.

Six actions, three per form. Remember this split: later in the section it turns out that the model didn't land in arbitrary spots — it landed in the exact same spot in both chains: the one where the text gets typed up or assembled. Holding the structure in mind, proofreading, and signing all stay with the human in both cases, and so does producing it if the case is reviewed.

And one thing without which what follows will sound like nitpicking. These two forms have different readers and a different cost of error. The visit note is read by the patient themselves and by the next physician at the same clinic — an error there costs a repeated workup or a mistaken continuation of treatment, and the author can still be asked. The discharge summary is read at a different institution, where the author isn't present: treatment gets ordered from it, and there's no one to ask. That's why this section has two case studies, not one, and the second is heavier than the first.

Now something the diagram doesn't draw, but that has to be said. These six actions aren't equal in weight, and the first is the heaviest of all. Holding the structure of the future text in mind during a live conversation is work no one sees and that isn't counted as part of the visit. The patient is talking, the physician is listening, and at the same time sorting what they hear into future fields: this goes under complaints, this under history, this under exam findings. If it doesn't get sorted now, it has to be recalled in the evening, and recall is worse than hearing.

The second thing worth saying about the visit note: it's more often typed up away from the patient, and this has a cost. Time passes between the conversation and the typing — sometimes a whole day of visits — and whatever got forgotten in that gap never makes it into the document. That, incidentally, is where the temptation to carry an earlier note forward wholesale comes from: carrying it forward doesn't require remembering.

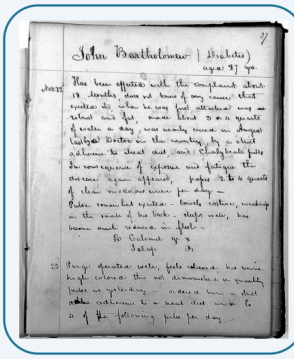
One more thing about the discharge summary. It's assembled not from conversation but from what's already written: progress notes, orders, test results, procedure reports. In other words, this is work with someone else's text, long before any model enters — the physician reads what was written over the whole hospitalization, including what colleagues wrote, and merges it into one document. Remember this: when we later say the model assembles the summary from the record, it's doing exactly the work that was already work with text, not with the patient.

And one last, administrative but important point for the accounting. The sixth action — producing it if the case is reviewed — doesn't happen every time, and that's exactly why it gets forgotten when people estimate what documentation actually costs. Its price only becomes visible there: a document that doesn't record an action means, at review, that the action never happened.

Why keep medical documentation at all

A decision about a person outlives the person who made it. The physician will go on leave, change jobs, forget; the patient will turn up a year later at a different institution. The only thing that crosses between them is the text.

A case book, 1819. Name, age, diagnosis, how long the complaint has lasted, what was prescribed, an entry for the next day.



Two hundred years later, these are the same fields — only now their composition and the order they're kept in are set by a government order, not by a physician's habit.

The case book of the Philadelphia Almshouse infirmary, 1819–1820, Wellcome Collection L0016766 · Russian Ministry of Health order No. 530n of 05.08.2022

Why keep medical documentation at all. A decision about a person outlives the person who made it: the physician will go on leave, change jobs, forget, while the patient will turn up a year later at a different institution — and the only thing that crosses between them is the text. The question sounds naive, and the answer usually given is wrong: "for reporting," "for the inspector," "because that's how it's done." If that were really the reason, this section wouldn't be worth five minutes.

On screen is a page from the case book of the Philadelphia Almshouse infirmary, eighteen nineteen. John Bartholomew, twenty-seven years old, diabetes. Then, in the physician's own hand: has suffered about eighteen months, doesn't know the cause, was strong and full-bodied before the illness, passes three to four quarts of water a day, was nearly cured in August by a country doctor with a strict meat diet and iron pills, the illness returned after exposure to cold and fatigue. Pulse somewhat excited, weakness in the loins, sleeps well, has lost a great deal of flesh. Prescribed. The next entry, the following day: it has acted, feels relief, urine deeply colored, the quantity hasn't decreased, pulse as yesterday.

Look at what's here. Name and age. Diagnosis. Duration. What came before, and how it was treated. What the exam found. What was prescribed. And, under a separate date, what came of it. These are exactly the fields that sit in today's standardized form, the same ones we just broke down step by step. Two hundred years, a different country, a different medicine, a pen instead of a keyboard — the design of the record is the same.

Why is it the same? Because the record has one job, and it hasn't changed. One person makes a decision about someone, and another will act on it — a week from now, a year from now, somewhere else. The physician will go on leave, change jobs, forget this visit among a thousand others. The patient will come back, and the person sitting across

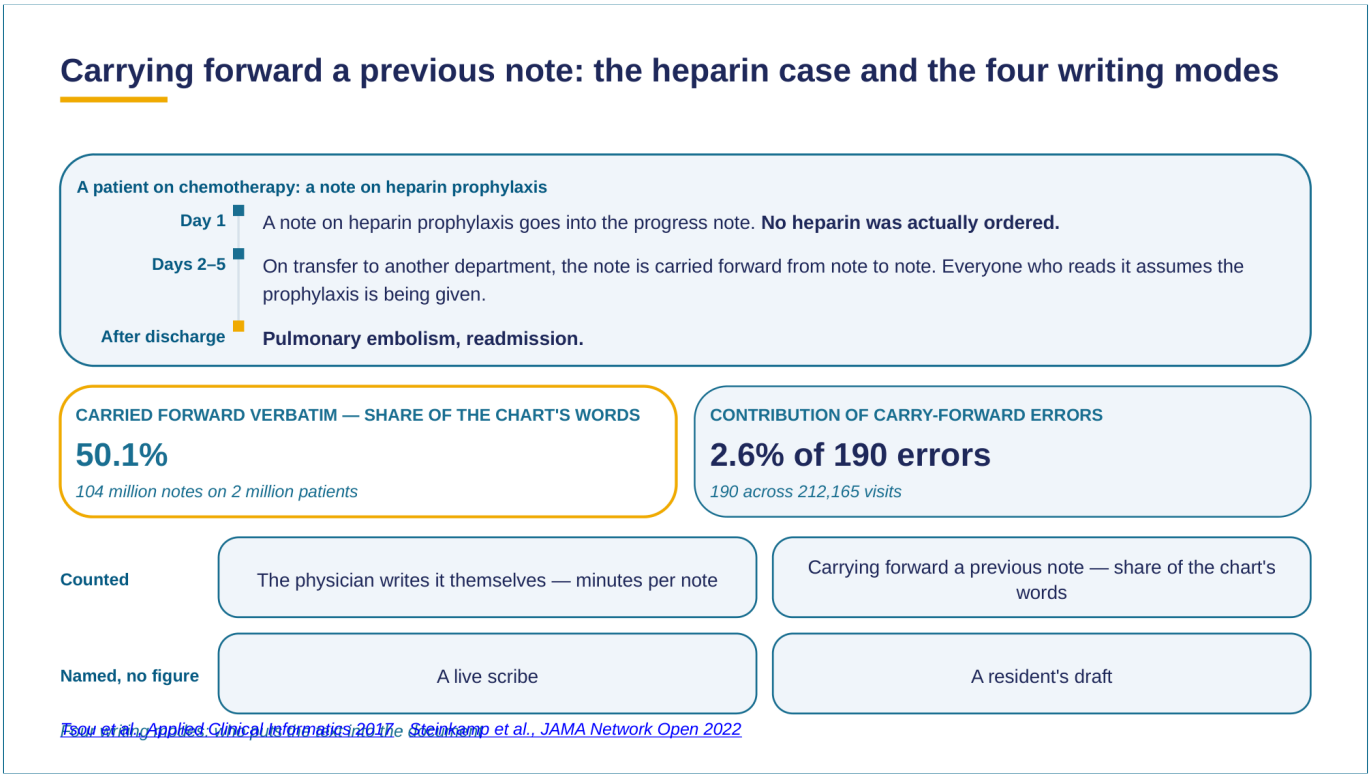
from them won't have seen them before. The only thing that crosses between these two people is the text. Not memory, not an impression, not a conversation. The text.

Everything else in this section follows from here. If the record is the only channel, then its completeness matters more than its brevity, and any tool that makes the document shorter has to be checked against exactly that. If decisions get acted on from the record, then an error in it costs not a reprimand but an action. And if the text is now written by someone other than the person who spoke with the patient — that's exactly the change this section exists to examine.

One clarification, so it doesn't sound like nothing has changed. Something has: in eighteen nineteen, the composition of the record rested on a physician's own habit and the school they trained in. Today, standardized forms and the order they're kept in are set by a Ministry of Health order, and this is no longer a matter of the writer's own taste. Later in this section we'll see what follows from this for the tool: the model has somewhere to put the text, but it isn't the model that assigns where it goes.

I'll also say what's missing from this page, which is no less interesting. There are no units of measurement beyond quarts of water a day. No scales, no scores, no threshold. This physician's decision is recorded in words, and the only way to check it is to read it. This feature of the stage stays with us for the rest of the section: a score has a threshold; a record doesn't. You can't say "this record is seventy percent complete" and make a decision on that number. This directly explains why scoring a machine-produced document is so hard: there's nothing to measure it by except a human reading it — and we'll see later that every measurement in this section runs up against exactly that.

And one note for anyone from IT who's used to a different order of things. The record here isn't an event log. A log is written for whoever will debug a failure, and it's written mechanically, completely, with no selection at all. A medical record is written by a person, for a person who will act on it, and so it holds not everything, but what's been selected — what the writer judged to matter. Selection is the substance of the work. A tool that takes typing the text off the human's hands doesn't take selection off them.



For a patient on chemotherapy with a previous pulmonary embolism, a note about ordering heparin for prophylaxis made it into the progress note. Heparin itself was never ordered. On transfer to another department, this note was carried forward five days running, and everyone who read it assumed the prophylaxis was being given — the note was the only thing anyone checked against. Shortly after discharge, the patient developed a pulmonary embolism and was readmitted.

The strength of this evidence: a single case report, the weakest kind there is. It shows how the error is built and says nothing about how often it happens — frequency is carried by the two figures below it, half the words in the chart, and two and six tenths percent of diagnostic errors found. The order matters: first you see how the error is built, then how often it occurs. The reverse order gives you shares behind which the room doesn't see a single person.

This is the second of the four classical writing modes — carrying forward a previous note. Before comparing the model against them, all four have to be presented and counted at the same cost: two of them are counted, in different units, and the other two have no named figure at all. Otherwise the model gets compared against an empty space, a comparison it always wins and that means nothing. The units of the two counted modes don't add up into one ledger either: on time, the model is compared against the physician's own writing; with carrying forward, it converges on a different question.

The modes are about who puts the text into the document. The five methods from the previous slide are about something else: where the characters in the note come from. Two different axes, and carrying forward a previous note belongs legitimately on both.

The first mode — the physician writes it themselves. It's counted inside the randomized

trial itself: the trial has a control group that kept writing notes the old way, and their time per note is four minutes twenty-two seconds. That's the baseline for every time comparison in this section. The mode has its own drift too, and the plate carries it as a pair: over the observation period, time in the control group fell on its own to four minutes four seconds. Eighteen seconds is a small figure with a large meaning: every claim of "got faster" has to be measured against this drift, or those eighteen seconds of time simply passing get credited to the tool. The strength of this figure: a preprint, never peer-reviewed.

The second mode — carrying forward a previous note: the earlier note moves into the new one, whole or in blocks, with edits for what's changed. For a long time this was treated as a bad habit of individual physicians; measurement showed it's the main way text actually gets into the chart.

A full review of the records at one large health system over six years — a hundred four million notes on nearly two million patients — found that half of all the text in the chart, fifty and one tenth percent of the words, verbatim repeats something already written about the same patient before. The unit here has to be stated plainly: this is a share of words, not a share of notes — not "half the notes are copied," but "half of what's written had already been written." The share rises year over year: thirty-three percent for notes from year fifteen, fifty-four for notes from year twenty.

And one line without which the room will hear a contradiction where there isn't one: fifty and one tenth is a share of the chart's WORDS, forty-six a share of a note's CHARACTERS. Different units, different samples: the numbers don't contradict each other, and they don't add together.

The strength of this figure: a peer-reviewed measurement of a complete sample from one system, with a count reproducible from the published method. The weak side is named by the authors themselves: the signal is a verbatim match, so carrying forward with rewording is invisible to the count, and the share is more likely understated than overstated.

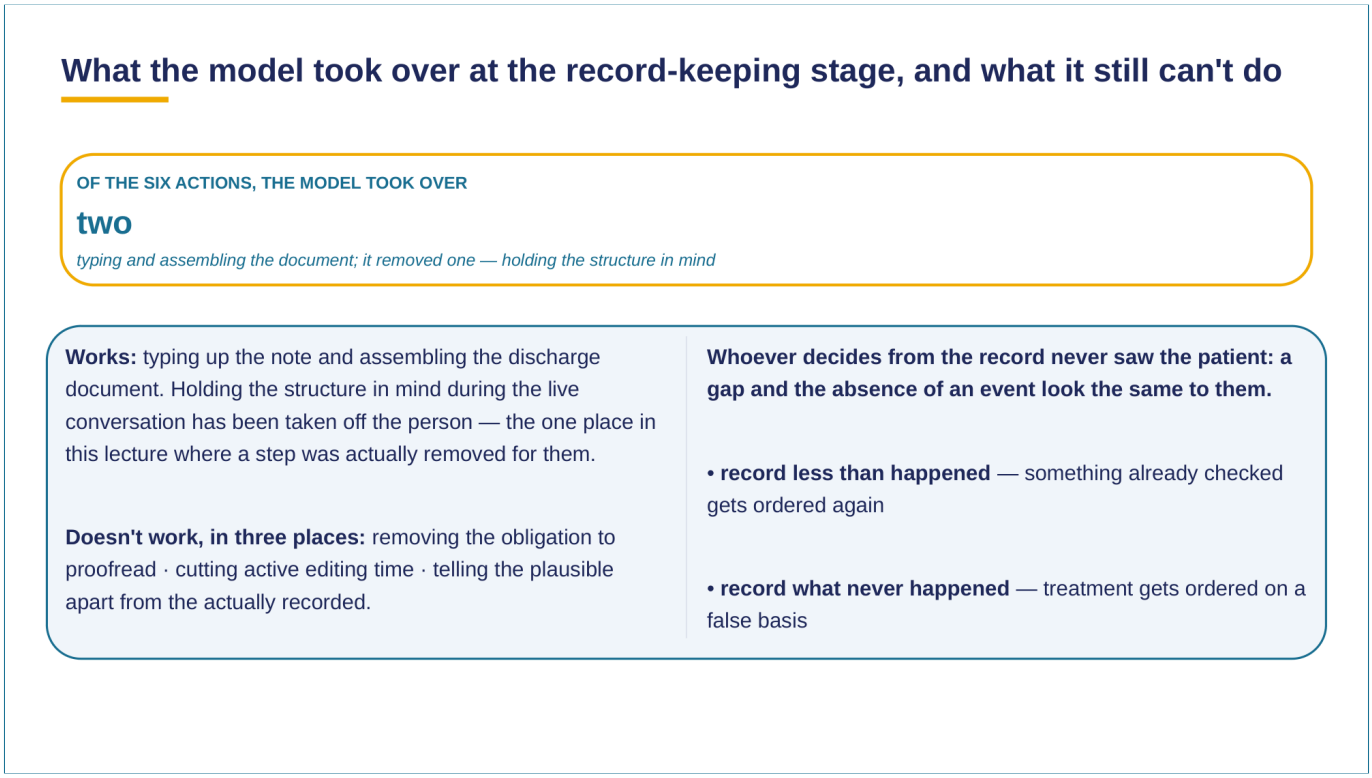
This mode's cost of error has been measured too. On diagnostic errors in primary care: out of two hundred twelve thousand visits over a year at two large urban centers, researchers pulled cases where unplanned care followed a visit within two weeks, and physician reviewers found a hundred ninety diagnostic errors. Carry-forward errors contributed two and six tenths percent of all those found. A caveat on attribution is needed here, and it's mandatory: the full text of the original study couldn't be accessed, and these figures were read from a systematic review that cites it; until checked against the original, the attribution runs explicitly through the review.

The third mode — a live medical scribe: a person at the visit, in the room or remote by connection, writing the note on the physician's behalf while they talk with the patient. This isn't exotic: this mode has its own market, its own cost per visit, and its own literature on saving physician time. It's named and not counted: it was never an arm of the randomized trial, and pulling its cost from a different study would mean combining two unrelated samples.

The fourth mode only applies to the discharge document: a resident writes the draft, the attending physician signs it. It's also named and also without a figure, and the reason is its own, named explicitly: the blinded comparison of a hundred hospitalizations, which this section comes to as its second case study, used the attending physician's own document, not the resident's. So the blinded comparison says nothing about this mode, and its numbers can't be substituted in here.

And the last thing to carry away from this slide. The model doesn't converge on the second mode by time — it converges on a question that turns out to be load-bearing for this whole section: what happens to the reliability of the record when the document contains plausible-sounding text that no one actually said at this visit.

Sources. Tsou et al., Applied Clinical Informatics 2017 — the review the heparin case and the figures 7.4 · 35.7 · 2.6 are drawn from · Steinkamp et al., JAMA Network Open 2022 — the share of the chart's words carried forward verbatim



Let's break down the work of this stage before talking about tools. A twelve-minute visit: the patient talks, the physician asks, examines, prescribes — and in parallel holds the structure of the future note in mind. After the visit, or even by the end of the day, that structure gets typed into the chart; whatever didn't get done becomes homework with the chart in the evening.

The six actions of this stage are listed on the slide, and I'll add what each one actually involves. Holding the structure in mind — during the live conversation. Typing up the note — after the visit, or even by the end of the day. Proofreading and signing. Assembling the discharge document for the whole hospitalization. Handing it to the reader. Producing it if the case is reviewed.

The record isn't made for an archive, and that's the first thing worth saying. It has readers, and each one acts on it: the local physician a month later, the next hospital on readmission, another specialist at the same institution, a review committee when a case gets examined, and the patient themselves, who's entitled to a copy.

And right away, a measurement that shows how strongly the reader shapes this work. Three physicians who had implemented the same electronic record system in hospitals inside and outside the US took data from the vendor itself and counted the length of an outpatient note. The American one came out four times longer: over six thousand characters against roughly twelve hundred for specialists outside the US — with the same record vendor, the same fields, the same insertion tools. The difference can't be a difference in the product, since the product is the same; the authors name the cause outright — documentation outside the US is shorter because only what's clinically relevant goes into it, while the American note is bloated by also serving as a billing and audit document. I'll name the strength of this honestly: it's an editorial, not a study with

a method, the figures were supplied by the vendor, and the sample and counting method aren't disclosed — it shows a direction and claims a magnitude. A second figure comes from the same source, with a separate sample: in the US in year twenty-one, the rules for billing a visit were revised specifically to remove documentation needed only for billing from the physician's plate; three years later, across one billion seven hundred million notes, average length didn't fall — it rose by eight and one tenth percent, while time per note fell by eleven and one tenth. People are writing faster, not shorter; length only fell at the institutions that cut back on auto-population and copying. These two series don't add up into one ledger — different samples, different years. And Russia: here, the composition of the record isn't left to an institution's discretion — standardized forms and the order they're kept in are approved by a Ministry of Health order, and the record is checked by a medical-care quality review, which looks at compliance with protocols, not at billing by volume of text. That's a different mechanism of pressure on length than the American one. There's no comparable measurement of note length for Russia: no publication with a character count, no departmental statistics — and I'm saying this outright, so that the next person who opens this question doesn't plug in a similar American number here.

Question for the room — about your own work: who here has a document you write not for a colleague but for an inspector, and how much longer is it than the substance actually requires?

From this comes a property of the stage that sets the cost of error: whoever decides from the record never saw the patient. A physician reading someone else's note can't tell "this didn't happen" apart from "this wasn't recorded." A gap and the absence of an event look identical to them, and no amount of careful reading fixes this — the distinction simply isn't in the document.

The cost of both errors is named before any figures. Recording less than happened — the next reader orders something already checked all over again, and at review, an action the physician actually took turns out to be unproven. Recording what never happened — treatment gets ordered on a false basis. Both errors are paid for by a person who wasn't part of this visit.

I've already presented the classical writing modes, and I won't recount them here. Briefly, as one package, so the tally doesn't drift: there are four modes; two are counted — the physician's own writing in minutes per note, and carrying forward a previous note in a share of the chart's text, fifty and one tenth percent of words; the live scribe and the resident's draft under the attending's signature are named, have no figures of their own, and the reason for each is named in the same place. The units of the two counted modes don't add up into one ledger: on time, the model is compared against the physician's own writing; with carrying forward, it converges on a different question — what happens to the reliability of the record when the document contains plausible-sounding text that no one actually said at this visit.

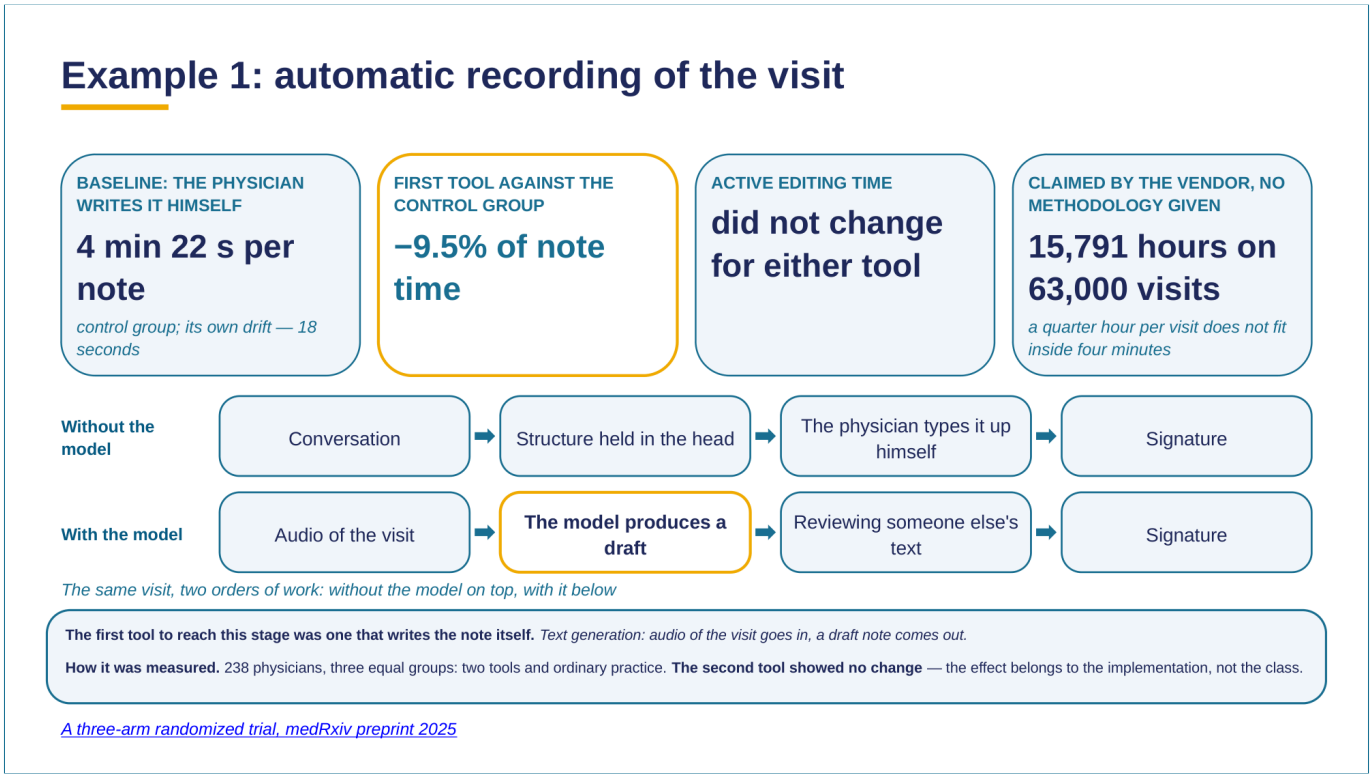
And a figure to hold onto whenever someone says "it got faster": in the control group, time per note fell on its own from four minutes twenty-two seconds to four minutes

four. Those eighteen seconds belong to the passage of time and to practice, not to the tool.

Now the same field with the model. Of the six actions, the trained model took over two: typing up the note — a draft is ready by the end of the visit — and assembling the discharge document from the whole record. The first action was actually taken off the human: an instrument now holds the structure of the future note during the live conversation, and this is the one place in the whole lecture where a step was genuinely removed for the person. It wasn't removed for free — a new one appeared in its place, and it's heavier than it looks: proofreading someone else's text and finding in it what was never there. Three places where the model doesn't work: it can't remove the obligation to proofread — the signature stays human; it doesn't cut active editing time — typing fell, but editing stayed the same; and it can't tell the plausible apart from what was actually recorded, without tying every line back to its place in the source record.

The note on the woman with the letter is the report on her scan, plus the outcome line that will go out to her by mail. She isn't the one who reads that report: a different person reads it, two years later, when she's called back in, and they'll carry their own question — has anything changed. That's why completeness of the record is worth more than brevity here.

Sources. a three-arm randomized trial, medRxiv preprint 2025.07.10.25331333, not peer-reviewed · trial registration NCT06792890 · Downing, Bates, Longhurst, Annals of Internal Medicine 2018 — note length inside and outside the US · Epic Research, 1.7 billion notes, 2023 — length after the billing-rule change · Russian Ministry of Health order No. 530n of 05.08.2022



The automatic visit-recording tool: audio of the visit goes in, a draft note comes out. It was the first tool to reach this stage. And the first thing said about it was a claim: a large health system reported saving 15,791 hours on 63,000 visits over ten weeks of work. The counting methodology was not disclosed.

Let's look for the base ourselves, out loud. 15,791 hours on 63,000 visits is about a quarter of an hour per visit. Now recall what unit the note's time is measured in: minutes and seconds, on the order of four minutes. Fifteen minutes of savings per visit do not fit inside four minutes of work on the note — you cannot save more than the whole operation cost. So the claim is counted in a different quantity, most likely total time spent in the chart altogether, and it cannot be set against the measured result.

From this comes a formulation worth carrying out of the lecture whole: a number with no stated counting methodology can neither be verified nor refuted — it is not a statement about the world. This is not nitpicking over the magnitude. Fifteen thousand hours might be true; the trouble is there is nothing to prove or disprove it with, and in the vendor's presentation it sits next to measured numbers in the same type size.

Now the measured result, and I'll name the strength of the evidence before the number: this is a preprint, it has not been through peer review, trial registration NCT06792890. A trial with genuine randomization and three equal groups covers 238 outpatient physicians across fourteen specialties: two commercial tools and ordinary practice. For the first tool, note time dropped by 9.5 percent relative to the control group; for the second, there was no significant change. Both tools affected the well-being scales — the overall professional well-being score rose by 2.76 on a scale from ten to fifty — and there was no difference between the two tools on these scales.

The control group's own baseline matters too: there, note time over the observation

period dropped on its own from 4 minutes 22 seconds to 4 minutes 4 seconds. Eighteen seconds with no tool at all — that is practice and the passage of time, and every "it got faster" is counted against this figure, not against zero.

In a multicenter observation of emergency departments, the note was finished earlier, patients more often noted that they were being listened to, and the strain on attention fell — but the active time spent editing the note did not change. Typing shrank; editing stayed the same. This is what stands on the slide as the gold plate: of all the numbers in this breakdown, this is the one least convenient for selling and most useful for planning.

From this comes a conclusion stronger than either number: one design, one population of physicians, two products of the same mode — and one result significant, the other null. The measured effect belongs to the implementation, not to the tool's class, and the phrase "we deployed automatic visit recording" carries no information about the result.

About the mechanics, one also has to say what is missing from the trial. Input is the audio of the visit, output is a draft note that the physician reviews and signs; what either tool is trained on, the preprint does not disclose. They have no names in the trial either: they are called "the first" and "the second," and that is a limitation of the work, not of our retelling — there is no way to verify from this publication which product gave the significant result. The names of this mode are known, and I'll name them so the mode can be recognized: Abridge, Dragon Copilot, Nabla, Heidi. By task type this is text generation.

The question for the room is about their own work: who reads an automatic meeting note line by line before sending it out. For many in the room such a note is already written automatically, and the answer "I skim it" is exactly the review whose cost is being measured here.

Sources. a three-arm randomized trial, medRxiv preprint 2025.07.10.25331333 · trial registration NCT06792890

The same task — automatic recording of the visit — here at home. It should be read as a pair with the foreign measurement, which had randomization, a control group, and minus 9.5 percent of note time: the same tool, but a different deployment and evidence of a different strength.

What was done. The Moscow voice assistant runs in the background during the visit: it transcribes the conversation between physician and patient and enters the complaints, history, diagnoses, and recommendations — whatever was said out loud — into the note itself. At the end of the consultation the physician sees the finished document, checks it, corrects it if needed, and signs it. It is trained on de-identified data from real visits so that it can recognize medical terminology; the data and transcripts travel over the city medical system's secured channels.

On scale. It started with neurologists, and there the service is already in every adult city polyclinic. Since September 2026 it has covered every primary care physician — more than 3,000 specialists. A primary care visit is noticeably harder for this task than a neurology visit, and the city says so itself: people come in with all kinds of complaints, often with several conditions and a long history, and the service has to do more than recognize speech — it has to sort what it hears into the note's sections. The refinement work went on over the summer, with feedback from the physicians themselves.

Now to the numbers, and here one has to be careful, because the temptation to read them as a measurement is strong. Experts rated the quality of the prepared notes at 4.8 points out of 5 — and the first publication says something important: the same as if a person had filled out the note. This is the only number in the section where a domestic result is directly set against the human mode, and it is worth remembering next to the foreign breakdown of the discharge summary, where a blinded comparison

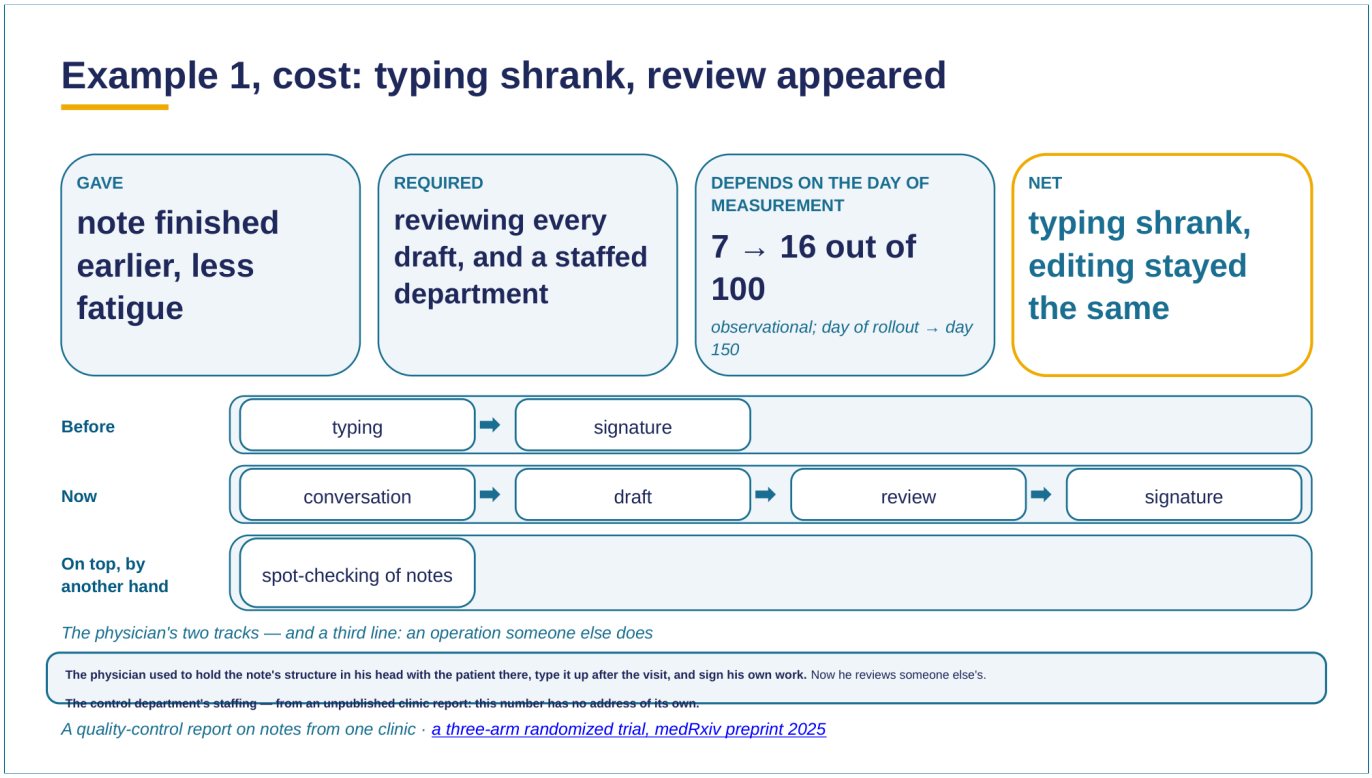
gave, for completeness, not "the same" but worse: 3.72 against 4.13. The two results diverge, and they diverge not by country but by how they were measured: there, 22 reviewers read the documents blind, not knowing which text was machine-written; here, an expert rating with no declared blinding and no published protocol.

Next come two numbers that have no base, and I'll say so directly. First: the physician's time of focused attention on the patient more than doubled. Doubled from what — not stated: there is no starting figure in minutes in the publications. Second: filling out notes takes physicians anywhere from a third to half of the visit's time. That is an estimate from the head of a vendor company's division, not a stopwatch measurement, and it describes the "before," not the result. Compare it with the domestic breakdown that comes later: there, a timing study across seven polyclinics gave 10 minutes 15 seconds by keyboard against 8 minutes 2 seconds by speech recognition — both figures in the same unit. That is what a number with a base looks like, and that is what a number without one looks like.

What I looked for and did not find. Publications with a minute-level measurement of this Moscow service — neither in peer-reviewed journals nor in agency reports; as of today the sources on it are statements from the city health department. I record this as a fact of the section: the gap belongs to the measurement, not to the deployment, and minutes from the foreign trial cannot be substituted in here — that trial is in a different language, with a different visit note, and a different tool.

And the conclusion this slide stands next to the previous one for. Scale of deployment and strength of evidence are different things, and they grow apart. The foreign trial covered 238 physicians and delivered genuine randomization; the Moscow deployment covered more than 3,000 and delivered an expert rating. An engineer choosing such a tool will have to decide what weighs more for them: that the whole city already uses it — or that its effect has been measured by someone against a control group.

Question for the room. The notes were rated by experts, with no blinded comparison and no control group. Which way would such a rating skew the result — and why? Type one word in the chat.



The cost of automatic visit recording, counted along the physician's two tracks of work: typing shrank, and review appeared as a separate, mandatory operation. Before, the physician held the document's structure in his head during the conversation and typed it up after the visit, signing his own work; now he speaks freely and then reviews someone else's text. The volume of work is comparable, the location changed — this is the form "the load moved to another step," and here it really is a move, not a disappearance.

A fifth operation is not done by the physician, and that is exactly why it stands as a separate third line on the diagram. In one clinic's report, a standing note-quality control department read more than 3,600 physician responses and collected 1,252 surveys — with its own permanent staff. The operation "spot-read machine drafts and collect feedback" did not exist in the previous process simply because there were no machine drafts. This is the form "a step appeared," and it has a measurable cost: staff. The engineer should pause here: this cost does not show up in any comparison of note time, because a different department pays for it.

One observation looks like a contradiction and is not: fatigue is lower, but active editing time is the same. What was removed was not time — it was the need to hold the document's structure in your head while talking to a person. This is a real gain, and it is measured; it is just measured in a different quantity than the one it is sold in. And this is exactly the case where the honest answer "the tool helped" and the honest answer "it did not save time" are both true at once.

The size of the effect also depends on when it's measured. A longitudinal before-and-after comparison with a control gives a reduction in note time of 7 out of 100 on the day of rollout and up to 16 out of 100 by day 150. From this come two

cautions at once, pointing in opposite directions. A quick pilot underestimates: measuring the effect in the first week may miss what shows up by the fifth month, and reject a tool that actually works. And the reverse: part of the gain by day 150 is the person getting used to it, and an observational comparison cannot separate habituation from a property of the tool. So the question "how much does the tool give" has no single answer without naming the day of measurement, and a vendor who quotes one number is naming their favorite day.

For the second tool in the same trial, there was no change in time at all, under the same design and on the same population of physicians. So the measured effect belongs not to the mode of work as such, but to a specific implementation.

Let me read the diagram line by line, because the whole point of the slide is in it. The top line is how it was: type the text, sign. Two steps, both done by the physician. The middle line is how it is now: talk, get a draft, review it, sign. Four steps, all four also done by the physician. From this alone it is already clear that there are more operations, not fewer — even though one of them, typing, disappeared.

The third line stands apart, and it is the most important one here. Spot-checking notes is not done by the physician. It is a new department with new staff, and its work is not counted at all in note-time comparisons — because what gets counted is the physician's time, while the organization pays for the control. The engineer should carry this away as a rule: when an operation moves from one pair of hands to another, the savings are counted against the first pair, and the cost appears at the second, and in one report these two figures almost never meet.

The "gave — required — net" set reads off the plates left to right. Gave: note finished earlier, less fatigue. Required: reviewing every draft, and a staffed department. Net: typing shrank, editing stayed the same. None of the three plates cancels the other two, and the honest answer to "did the tool help" consists of all three at once.

Two forms from the glossary are named here explicitly. "The load moved to another step" — for typing and review. "A step appeared" — for spot-checking. The third form, "the performer is the same, but got worse," is not in this breakdown, and I will not invent it: nobody measured the quality of review before and after the rollout.

Sources. a three-arm randomized trial, medRxiv preprint 2025.07.10.25331333

Example 2: the discharge summary

AT LEAST ONE ERROR IN THE DOCUMENT
9 of 11 machine-written summaries
of 19 fabrications, 10 were major

COMPLETENESS: HUMAN → MODEL
4.13 → 3.72
a five-point reviewer scale

UNIQUE ERRORS PER DOCUMENT
1.82 → 2.91
human → model

POTENTIAL HARM OF A SINGLE ERROR
1.34 → 1.35
out of seven; not distinguishable

Without the model	Chart and progress notes	The physician assembles it himself	Reviewing his own work	Attending physician's signature
With the model	Chart and progress notes	**The model assembles it**	Reviewing someone else's work	Attending physician's signature

Assembling the discharge document, two orders of work

Text generation: the medical record goes in, a draft of the narrative section comes out.

How it was measured. 100 hospitalizations, 22 reviewers: each document was read blind by two of them.

Shorter and more coherent, but less complete. The study names neither the model nor its training data.

The cost of error shifted: the same person writes it, but the reader pays.

A blinded comparison of 100 hospitalizations, JAMA Internal Medicine 2025

A discharge summary written by the model: the medical record and the hospitalization's progress notes go in, a draft of the document's narrative section comes out. 100 hospitalizations, 22 reviewers, each document read blind by two of them. The tool is the same one as in the previous breakdown, but the document carries more weight: treatment elsewhere gets prescribed off the discharge summary.

I'll name the classical modes before the measured result. Of the stage's four modes, two apply to this document. The physician writes the summary himself — this mode is counted, and it is the baseline for every comparison of the work. A resident's draft signed off by the attending physician — named but not counted: the blinded comparison took the attending physician's document, not the resident's, and a number from a different study cannot be substituted in here. The other two modes of the stage do not apply here: the live scribe writes the visit note, not the discharge summary, and carrying forward a previous note does not transfer to a hospitalization's final document.

How it was measured is the main point of the breakdown, and the design of the measurement is worth stating slowly. 100 hospitalizations lasting three to six days. 22 attending-physician reviewers. Each document was examined independently by two of them, and none of the reviewers knew which text was machine-written and which human. Blinding here is not a formality: a self-report from physicians who chose the tool themselves is evidence of a quite different strength, and most of the enthusiastic publications about this stage rest on exactly that.

Now the numbers. Overall quality is comparable: 3.67 for physicians against 3.77 for the model — slightly higher for the model. The model's document is shorter and more coherent. And yet it is less complete: completeness is 4.13 for the human against 3.72 for the model. Unique errors per document: 1.82 for the human against 2.91 for the

model — the model has more errors. And the potential harm of a single error is 1.34 against 1.35 on a scale from zero to seven: both ratings are low and not distinguishable from each other.

These four numbers have to be read as a whole, or you end up with one of two falsehoods. You can say "the model writes better" by leaning on overall quality. You can say "the model writes dangerously" by leaning on the error count. The true statement is the third one: the model's document is ready sooner, reads more easily, contains less of what is needed and more of what isn't, and none of its individual errors is scarier than a human one.

How it works: input is the medical record and the hospitalization's progress notes, output is a draft of the document's narrative section. The study names neither the model nor the composition of its training data, and that is a limitation of the publication.

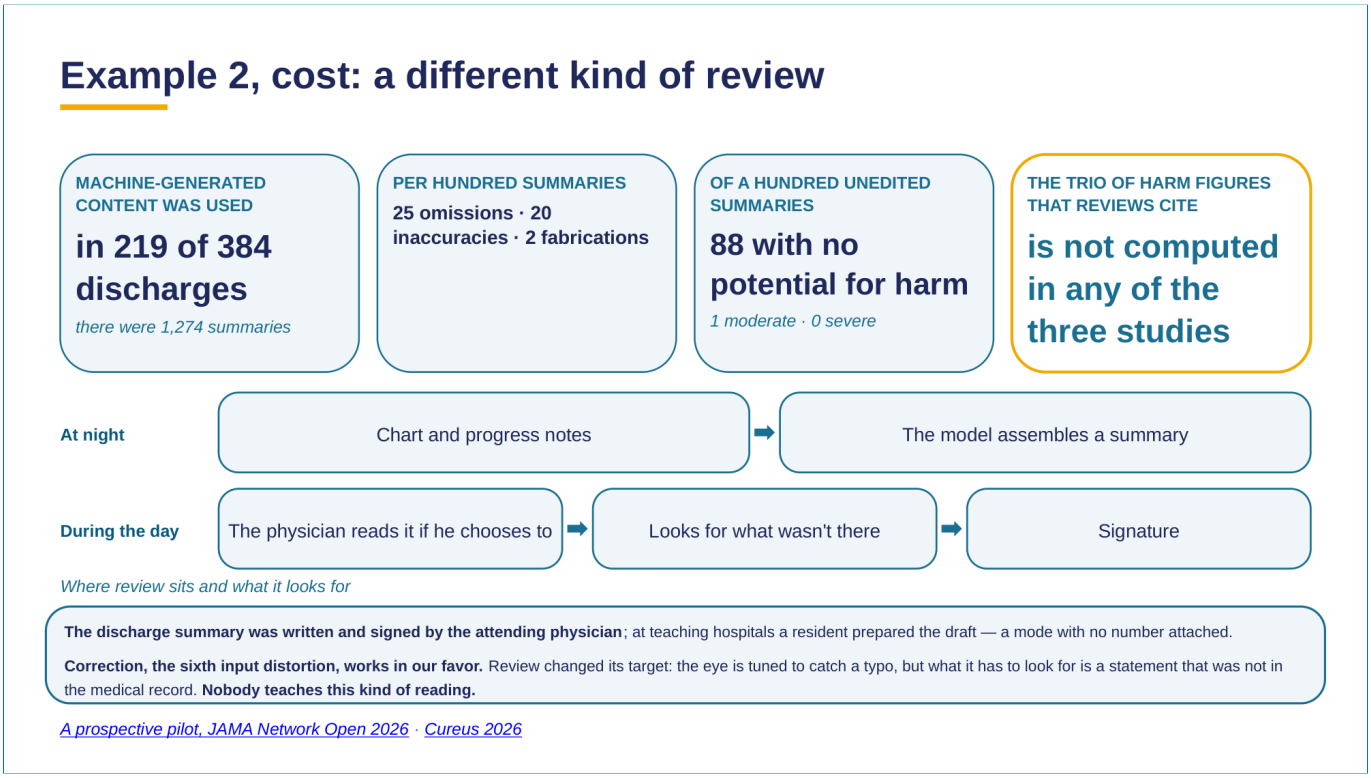
And what moved. The steps and performers on the slide are the same: assembled, reviewed, signed. The cost of error shifted. One person writes the document, but a missed line costs another person — the one who will prescribe treatment off this document a month later, at a different institution. This is the glossary form named here explicitly, and it is the most unpleasant of all: none of the participants feels the deterioration, because the deterioration lands on someone else.

Why blinding in the measurement matters deserves its own word. Almost all the enthusiastic literature about this stage rests on self-reports from physicians who chose the tool themselves and rated the result themselves. Such a report measures satisfaction with the choice, not the quality of the document. Here the design is different: documents are de-identified, the reviewer does not know which one is machine-written, and each document is examined independently by two people. This is more expensive and slower, and that is exactly why there are so few such studies.

Separately, about what the study does not say — and this has to be stated, not skirted. The publication names neither the model nor the composition of its training data. So you cannot repeat the measurement on your own document, and you cannot carry the result over to another product of the same mode either: we saw in the previous breakdown that two products of the same mode gave different results under one trial design.

And last, what carries on down the line. The document is assembled, reviewed, and signed. The signature on it is human, but the basis of the decision came from outside — from text that the person did not write. What that signature means, and who checks that the signer is capable of pushing back against the machine, is the whole of the next section.

Sources. a blinded comparison of 100 hospitalizations, JAMA Internal Medicine 2025, DOI 10.1001/jamainternmed.2025.0821



Review of a machine-written discharge summary did not disappear — it changed its target: the eye is tuned to catch a typo, but what it has to look for is a plausible detail that was not part of this hospitalization. This is how correction works, the sixth of the evening's six input distortions and the only one of them that works in our favor: the physician corrects the model's conclusion when he finds an error in it. Correction has a measured magnitude, and it was taken in a different field, on ECGs: out of 159,630 real records from one American clinic for 2011–2023, the physician corrected the machine's conclusion in 31.3 cases out of 100. That is, almost one in three — and this is the most direct answer available to the question of whether the human in the loop really works. Alongside it, from the same group of studies, stands the flip side: a suggestion accepted without checking is later corrected in 53 cases out of 100, against 18 if it was checked right away. Correction exists; it simply happens later than it should.

The fourth step of the breakdown is to name what moved. What moved here is review, and it moved not to a different place but to a different target.

The old review skill was built for human error. The eye is tuned to a typo, a cut-off sentence, a skipped section — to what a tired person leaves behind at the end of a shift. A machine error is built differently: it is a coherent, stylistically appropriate statement that was not in the medical record. The old skill cannot find it, and nobody yet teaches this new kind of reading — not in residency, not in continuing-education courses, not in the tool's own instructions.

How much of this review actually gets done was measured prospectively. For 384 discharges the system produced 1,274 summaries; physicians used the machine-generated content in 219 cases — 57 out of 100. The unit here has to be held firmly, and the plate names it: 57 out of 100 is the share of DISCHARGES, not the share

of summaries produced. There were three times as many summaries as discharges, and if you take summaries as the denominator, the same event gives 17 out of 100. One and the same observation, two legitimate numbers, different denominators — and both can be printed without lying.

Per hundred summaries, omissions were found in 25, inaccuracies in 20, fabrications in 2. And of a hundred unedited summaries, 88 were judged to carry no potential for harm, one capable of causing moderate harm, with no cases of severe harm. The definitions here differ from those in the blinded comparison, and the two sets cannot be combined into one list.

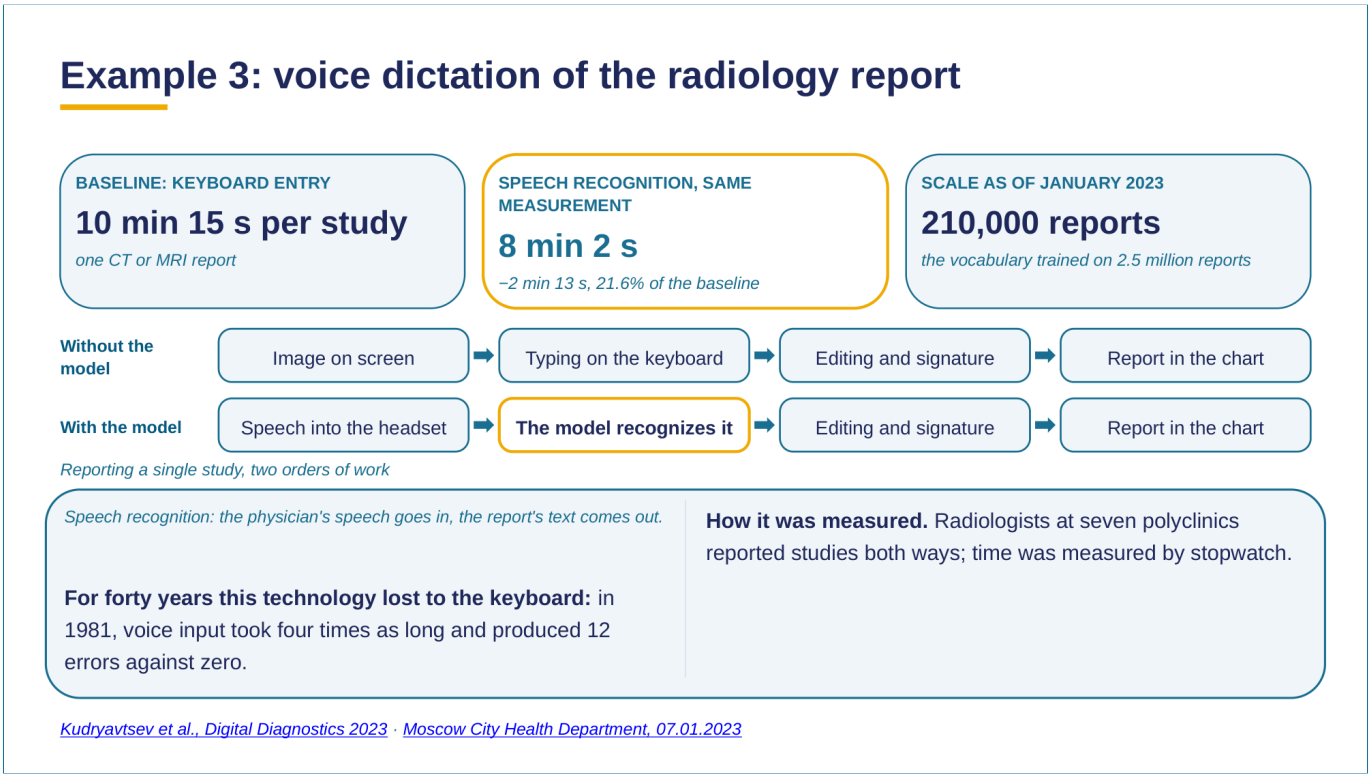
Now about the trio, and this is the most transferable part of the section. A tidy trio of numbers circulates in reviews about this case, citing a "study of a hundred discharge summaries." A line-by-line reading of the three available primary sources does not produce it: no study counts in the definitions in which it is retold. Numbers that sound convenient live longer in retellings than in primary sources.

The practical conclusion for an engineer sounds the same in any industry, and it stands on the slide as its own line. When a review carries a number with a citation to a study, the citation has to be checked not for whether the study exists, but for whether the study actually computes this exact quantity under these exact definitions. Here it does not, and the three numbers held up only because nobody opened the primary source. A retelling is weaker evidence than a study even when their numbers happen to match.

And last, about the third source. A single-center before-and-after comparison in a rehabilitation unit was published in a journal with author fees and minimal peer review. I name it rather than hide it: the evidence is weak, and conclusions should not be built on it alone — but it helped check the trio exactly as much as the two strong studies did.

On conflict of interest. It is disclosed in the publications themselves: some of the authors work for the company that makes the model, or hold a patent on it. A disclosed conflict does not make the study worthless — it makes it a study that is read more carefully, and the first thing to check is who decided what counts as success. An undisclosed conflict is worse than a disclosed one for exactly this reason: it cannot be accounted for while reading.

Sources. a prospective pilot, JAMA Network Open 2026 · a blinded comparison of 100 hospitalizations, JAMA Internal Medicine 2025 · a single-center comparison, Cureus 2026 — a journal with author fees and minimal peer review



Radiologists at seven polyclinics reported radiology studies both ways, and time was measured by stopwatch. Keyboard entry — 10 minutes 15 seconds per report; speech recognition — 8 minutes 2 seconds. Minus 2 minutes 13 seconds, 21.6 percent of the baseline.

This is the only case in the whole section where the classical mode and the mode with the model are counted in the same unit, on the same physicians, and on the same work. And it is the only case all evening where you can see how one and the same technology failed to work for forty years, and then started to.

I'll start with the failure, not the win, and that is a deliberate reversal of the form. In 1981, keyboard and voice input were compared on sixty radiology reports. Report length did not depend on the method of entry — so the complexity of the studies described was similar too. And the time it took to fill out by voice exceeded the keyboard time fourfold, and the reports filled out by voice contained an average of twelve recognition errors, while the ones typed on a keyboard contained none. The systems of that time did not suppress outside noise, and physicians' speech was recognized poorly. Remember this pair of numbers: four times slower, and twelve errors per document. The same task, the same field, the same type of tool — and the opposite answer to today's.

Now the current measurement. The first study of speech-recognition effectiveness in radiology departments was carried out in 2020 at seven city polyclinics of the Moscow City Health Department, on the Voice2Med voice-input system from the TsRT group of companies — at that time still the version with 93 percent recognition accuracy. The study compared the speed of filling out documentation by keyboard versus speech recognition; radiologists filled out reports for CT and MRI studies. The timing study gave

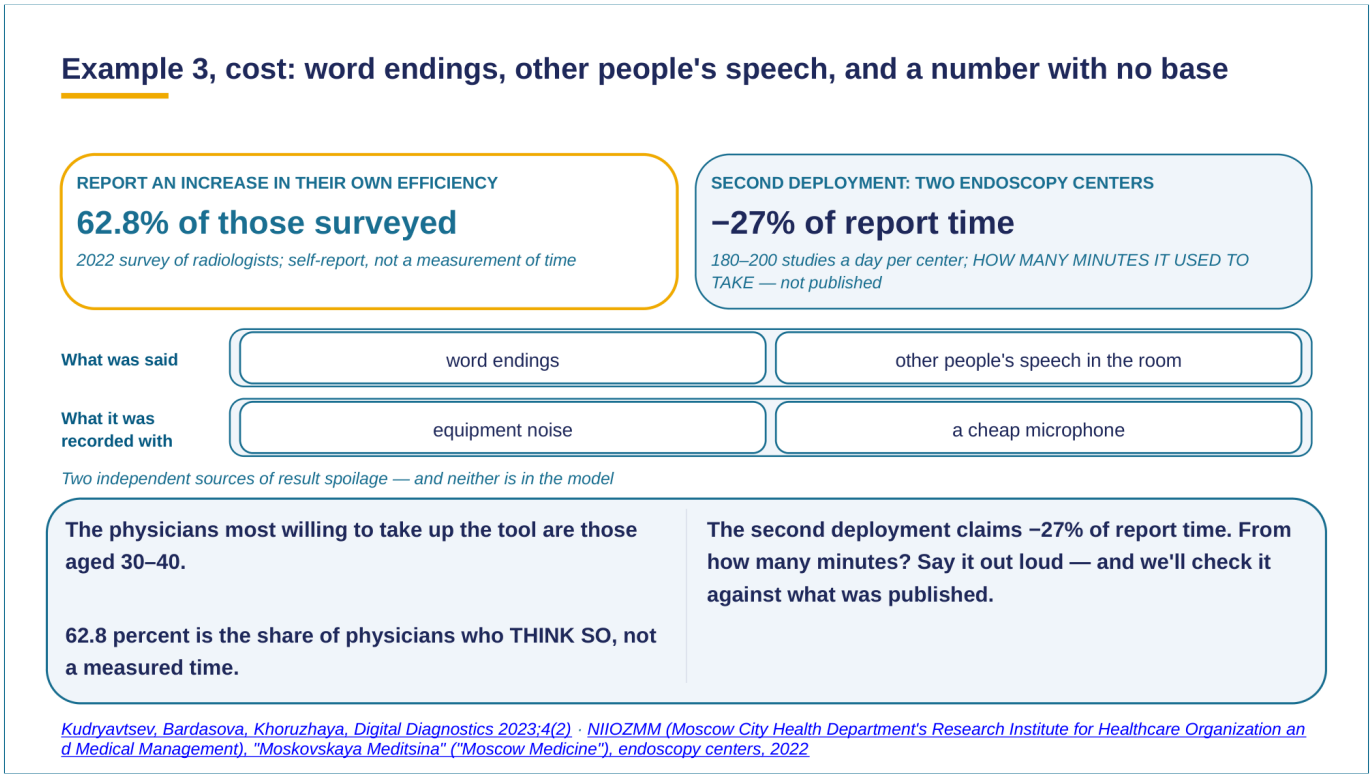
the following: average time to report one study by keyboard — 10 minutes 15 seconds; by speech recognition — 8 minutes 2 seconds.

Let's subtract it ourselves, out loud: two minutes thirteen seconds, that is, 21.6 percent of the baseline. Keep in mind that the baseline is named here, and it could be named — unlike half the numbers shown to us about this stage. By January 2023 the Moscow City Health Department reported more than 210,000 radiology reports filled out by voice, and a reduction in preparation time of "more than 20 percent." Notice something rare: the agency's figure and the independent timing study agree. That is not always the case — in this very section we saw a claim of fifteen thousand hours that agrees with nothing.

On the mechanics. Input is the physician's speech through a headset, during the study or right after it. Output is text that is automatically carried into the open report in the medical information system; the physician edits and signs it. What it is trained on: the medical-terms vocabulary was built on 2.5 million radiology reports, and the recognition accuracy of medical terminology in Russian rose from 93 percent to 98. Task type is recognition, modality is audio. This is a different cell of the course's grid than the two previous breakdowns: those were text generation, this is audio recognition. The model does not decide what goes into the document and proposes nothing — it turns speech into characters.

Now about the classical mode that our section was missing, and the domestic material closes that gap. The live scribe — the third writing mode — was named in our breakdown and never counted. In domestic healthcare, its closest relative, dictation transcription centers, was tried, but there are no publicly available scientific studies of its effectiveness, and no such centers remain in the current domestic system. The review's authors draw a cautious conclusion from this: the application turned out to be impractical. This is an honest refusal to deploy, and it is worth more than one more case of success: the mode existed, it was tried and dropped, and nobody left us a measurement after that.

Sources. Kudryavtsev N.D., Bardasova K.A., Khoruzhaya A.N. Speech recognition technology in radiology diagnostics // Digital Diagnostics. 2023;4(2):185–195 · statistics from the Moscow City Health Department, January 7, 2023



The fourth and fifth steps of the breakdown: what moved, and what the win was paid for. I'll start with what is usually treated as a trifle, and for an engineer this is exactly the substance.

Word endings. This is not nitpicking about product quality, but a property of the Russian language, named by the developers as a separate challenge: the system has to choose case and number where English speech requires no such choice. For a document this matters more than it looks: "a lesion in the right lung" and "lesions in the right lungs" differ by exactly their endings, and an error here does not look like an error — it looks like text.

Other people's speech. Physicians reported situations where the system recognized someone other than them. Three people speak in a radiology room: the physician, the patient, and staff. A tool that puts what was said by someone other than the author into the document creates exactly the defect the whole section is concerned with — plausible text that nobody meant to put into this particular report.

Noise and microphone. Recognition quality is negatively affected by the operation of diagnostic equipment and by medical staff talking with the patient or with colleagues, as well as by low-quality audio recording devices. Remember this as a rule of transfer: for a tool that listens, part of the result lies not in the model but in three thousand rubles' worth of hardware and in the room's acoustics. Neither of these is usually described in a publication about the model.

Age and adoption. Younger specialists are more open to the technology, and physicians aged thirty to forty take it up more readily than others. This too is a cost: a tool whose effect depends on who is using it cannot be rolled out by directive and expected to deliver the publication's average result.

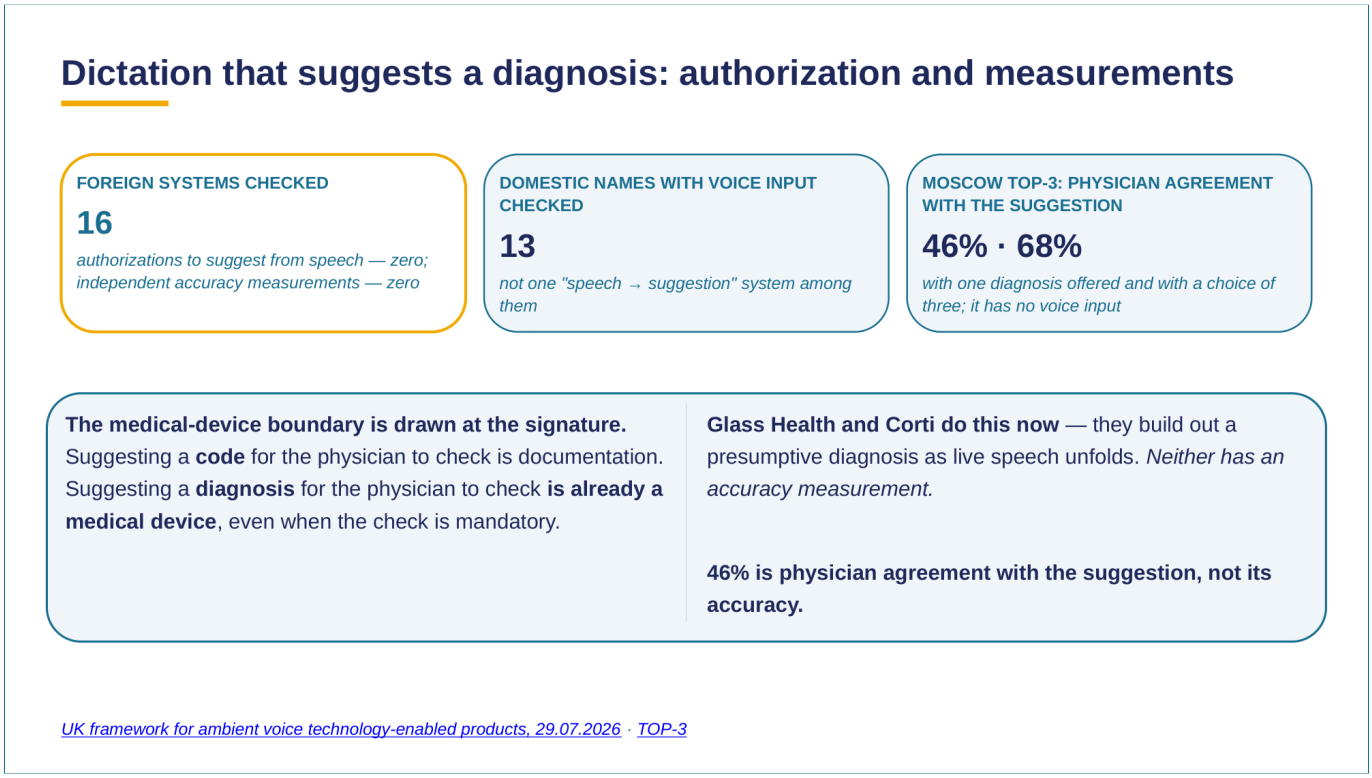
Now about a number that is easy to mistake for a measurement. A 2022 survey of radiologists found that 62.8 percent of respondents report an increase in the efficiency of their own work when using speech recognition. I'll say this slowly: 62 percent is the share of those who think so. Not the share of time saved, and not the share of reports filled out faster. Agreeing with a tool and benefiting from a tool are two different quantities, and the first is always higher than the second, because it includes the relief a person feels even when a stopwatch does not register it. We have a separate time measurement — the very timing study — and it gives 21 percent, not 62.

And the second domestic deployment, worth pausing on for a minute. In 2022, the same technology for reporting endoscopic studies was deployed at two Moscow endoscopy centers — at Botkin Hospital and Buyanov Hospital, each performing between 180 and more than 200 studies a day. A timing study of the average report-filling time before and after rollout was carried out, giving an average reduction of 27 percent, with recognition quality above 97 percent and full physician satisfaction.

And now what is missing from this publication, and this is exactly our course's business. How many minutes filling out the report used to take BEFORE the rollout — not published. Only the share is named. Twenty-seven percent of ten minutes and twenty-seven percent of two minutes are different things for planning, and there is no way to choose between them from this publication. Compare it with the previous slide: there, both 10 minutes 15 seconds and 8 minutes 2 seconds are named, and the share can be calculated yourself. Here there is nothing to calculate. This is not a reason to dismiss the rollout — it is real, on a real patient flow; it is a reason to know that a share with no base of its own does not transfer to another hospital.

The question for the room here is set up as a game, and the answer to it is a gap, not a number. I state the claim: minus twenty-seven percent of report time in the second deployment, two endoscopy centers, 180 to 200 studies a day per center. And I ask the room to say out loud how many minutes this is counted from: ten? two? Let at least one guess be voiced. After that I reveal it: the starting time was not published at all. Only the share is named. Twenty-seven percent of ten minutes and twenty-seven percent of two are different things for scheduling a visit, and there is no way to choose between them from the publication. By this point the room has usually already named its own number — and that is exactly what every reader does with a statement like this: fills in the base themselves, without noticing that they did.

Sources. Kudryavtsev N.D., Bardasova K.A., Khoruzhaya A.N. Speech recognition technology in radiology diagnostics // Digital Diagnostics. 2023;4(2):185–195 · NIIOZMM of the Moscow City Health Department, "Moskovskaya Meditsina" ("Moscow Medicine"): speech recognition in endoscopy centers, 2022



Systems that turn the visit's speech into a clinical suggestion — a recommendation or a diagnosis, not just a note — exist abroad; there are none in Russia, and nowhere in the world has the accuracy of the suggestion itself been independently measured. The UK regulator drew the medical-device boundary at the signature: a code for the physician to check remains documentation; a diagnosis for the physician to check is already a medical device.

First, about where such a system even sits. The six actions of the documentation stage that we have examined do not contain such a system at any point. Its input sits here, in documentation: it listens to the live visit. But its output sits at the diagnosis stage, that is, somewhere the lecture has already visited. This system sits between two stages at once. A tool that does not fit into the order of stages is an ordinary occurrence: a tool shows up later than the work has already taken shape, and the only honest reaction is to say which two stages it landed between.

Now the main thing, the reason this slide stands in a lecture about the signature. In July 2026 the UK regulator drew the medical-device boundary exactly at the spot this whole lecture is built around — at the signature. Transcribing the conversation, a retelling, a draft letter, suggesting a code for the physician to check — that is documentation, and it does not turn the product into a device. But suggesting a diagnosis or a treatment does, and it does so even when the suggestion goes out for mandatory physician review. Read that again: a physician's review does not soften the category. The regulator is saying that a diagnosis suggestion signed off by a physician is a device that has to be registered, with everything registration entails.

For an engineer, there is a transferable lesson here, and it is not about medicine. The line of responsibility runs not along the technology — the model is the same in both

cases — but along exactly what the product puts under a person's signature. It is not the system's architecture that separates a code under signature from a diagnosis under signature, but the cost of error behind it. This is exactly what the lecture keeps repeating all evening: the model's class decides nothing; the procedure around the person does.

About a separate caveat that appears in no source, but has to be spoken aloud. The only domestic system close to the subject at hand is Moscow's TOP-3: it suggests three preliminary diagnoses and is registered as a medical device. Two things to note. First: the physician types the complaints into it by hand; it has no voice input, so it has nothing to do with the "by dictation" question. Second, and more important: its public figure — 46 percent with one diagnosis suggested and 68 percent with a choice of three — is the share of cases where the physician **agreed** with the suggestion. Agreement and accuracy are different quantities, and the difference here is not academic: if the physician and the model make the same mistake, agreement will be high and both will be wrong. The metric will not show this at all. There is no verified reference standard in the publications against which accuracy was computed; there is only the overlap of two opinions, one of which this very system also shaped.

And the count that stands as plates. Sixteen foreign systems were checked by name; fourteen of them claim to deliver more than a note — codes, orders, cues, a presumptive diagnosis. Not one has a regulator's authorization specifically to deliver a clinical suggestion from speech. Independent, peer-reviewed accuracy measurements of this suggestion: zero; what gets passed off as measurements measures something else — for one system, the trial counts billing codes as a consequence of a better note; for two others, time spent on the note, and that is the very trial on 238 physicians the section opened with. Thirteen domestic names with voice input were checked, and not one system of the "speech straight to clinical suggestion" class is among them: here, voice gives only text, and a diagnosis is suggested from the text of the chart that the physician typed in by hand. The two lines exist, and they are not connected.

What follows from this in practice. If you are shown a system that listens to the visit and delivers a presumption, the first question is not about accuracy, but about what it puts under a signature and whether it is registered for that. The second is what it was measured by: agreement with the physician, or accuracy against a verified diagnosis. The difference between these two questions is the difference between documentation and a medical device.

Sources. UK framework for ambient voice technology-enabled products, 29 July 2026, sections 5.1–5.2 · TOP-3, manufacturer's page — the figures 46% and 68% were published in a statement by the Moscow City Government on July 14, 2021.

Section conclusion: what works, what doesn't, and where the load moved

THE ENTIRE MEASURED SAVINGS OF THE STAGE
9.5% of note time
against the control group, 4 min 22 s; absent for the second tool

ACTIVE EDITING TIME
did not shift for either tool

DOCUMENT COMPLETENESS: HUMAN → MODEL
4.13 → 3.72
a blinded comparison of 100 hospitalizations

DOMESTIC TIMING STUDY OF THE REPORT
10 min 15 s → 8 min 2 s
keyboard versus speech

What works. The draft is ready by the end of the visit; the structure no longer has to be held in your head while a living person is in front of you — the one step that was actually removed.

What doesn't. The model does not write more completely than a human. The claimed 15,791 hours have nothing to confirm them.

The load moved from typing to reviewing someone else's text; spot-checking of notes appeared, with its own staff. Next in the cycle is what holds that review in place: **the signature.**

Nobody teaches this other kind of review anywhere.

Of the six input distortions, the section met one — correction: the physician corrects the machine's work.

The conclusion of the documentation stage rests on one number, not two columns: after eight summaries of the same type in a row the room starts predicting the screen, and this number cannot be predicted.

And the stage's tally, so it lines up with what I've been saying along the way: there were four classical writing modes, two of them counted, and in different units — the physician's writing in minutes per note, carrying forward a prior note in a share of the chart's text. Of the stage's six actions, the model took over two and removed one. There are three breakdowns in the section: recording the visit, the discharge summary, and domestic voice dictation of the report. Nine and a half percent is about the first of the four modes and no other, and it is counted against the control group, not against zero.

Nine and a half percent is the entire measured savings of the stage for the foreign automatic visit-recording tool. Not "up to thirty percent," as people say at conferences, not fifteen thousand hours, as the health system reported: 9.5 percent of note time, in a randomized trial, for one of two tools. For the second tool, of the same mode and the same trial design, there are no savings at all — and that is the first thing to say right after the number, or the number will read as a property of the mode.

Active editing time did not shift for either of the two tools. The note is ready by the end of the visit, not by the end of the day, and the document's structure no longer has to be held in your head while a living person is there — this is a real gain, measured in a different quantity than the one it is sold in.

What happened to the work. The load moved from typing to reviewing someone else's text — the glossary form is named here and fits literally. A step appeared: spot-checking of notes, which has its own staff. And the cost of error shifted: a typo is

visible, a plausible detail is not visible at all.

About the claimed savings. 15,791 hours on 63,000 visits — with no disclosed counting methodology. It cannot be confirmed, it cannot be refuted; a quarter of an hour per visit does not fit inside four minutes of work on the note, so a different quantity was being counted. A number like this does not belong next to measured ones — it is of a different nature.

The document does not write more completely than a human, and this is measured by the blinded comparison: completeness 3.72 against 4.13, unique errors 2.91 against 1.82. At the same time the potential harm of a single error is the same. The document comes out faster, reads more easily, and contains less of what is needed.

What holds this in place. Review and the physician's signature — nothing more, and that matters: no technical check mechanism appeared here. And right alongside it, stated plainly: nobody teaches this other kind of review, the kind that looks for a coherent statement with no basis in the chart. The procedure relies on a skill that is not in the training curriculum.

There are two task types in this stage, and they are worth telling apart: text generation — the visit's speech and the chart's fields go in, a draft note comes out; and audio recognition — domestic voice dictation, where the output is the text of what the physician said, and nothing beyond it. A draft awaiting review is exactly where the first mode operates, and it lands inside that same eighteen percent of characters the physician types by hand. The decision of what goes into the document, and the check that what went in was actually there, are not handed over to the machine.

Treatment gets prescribed off this document down the line — and when something goes wrong, it is the signature that gets examined. This is the bridge to the next section: it is about the signature itself, what it means, and who checks that the signer is capable of pushing back against the machine.

And one remark about the form of this slide itself, because it is transferable. Every other summary in the lecture is built on four headers: what works, what doesn't, what happened to the work, what holds it in place. Here, in their place, is one number set in large type. This is not for variety's sake: when the same form repeats six times, the room starts predicting the screen and stops reading. Breaking the rhythm brings attention back exactly at the point where this section has one number worth remembering in place of everything else.

Why nine and a half percent specifically, and not any other number in the section. It is the only quantity obtained by genuine randomization: three equal groups, random assignment, a control. Every other number in the section is either observational, or claimed, or measured in a different quantity. The next time you hear a figure of tens of percent about this stage, the first question will not be "where from," but "was there a control."

And what is deliberately absent from this conclusion. There is no conclusion "the tool is unnecessary": it removed a real load, and that is measured on the well-being scales.

Nor is there a conclusion "the tool works": active editing time did not shift, and the document became less complete. The section ends with both of these things being true at the same time, and it is the one who signs who will have to choose between them.

Professional qualification: maintenance and erosion. This stage measures not the work but the person who remains the decision-maker inside it, and three measurements speak to that person — retention among developers, the rate of procedures with a finding for the same physicians after the assistant is switched off, and the same quantity in a Russian trial. Each has its own weakness.

The previous section ended with a signature that may have nothing behind it: the cost of error moves onto someone who did not sign it. Here the same thought turns to face the person as a whole: there is one more step that nobody has touched — the person himself who decides. The work has been measured all evening: the stream of alerts, the document, the scan. None of these measurements says anything about what happens to the one who remains the decisive link inside that work.

This section differs from the general conversation about technology's effect on people in that it carries numbers taken from real work, not speculation about the future or general anxieties. The numbers are few, and that is exactly why they are precious: a direct measurement of deskilling on real clinical work happened once, and it has not been repeated. The first breakdown is taken on developers, not in a clinic, and it is randomized — stronger by design, but its magnitude cannot be carried over to physicians. The section states both of these limitations directly, without hiding or softening them.

There are three breakdowns in the section, and each does work the other two don't. First comes the randomized experiment on developers: it measures exactly the second half of the pair — the result without the assistant — and measures it by lottery, a method none of the section's clinical breakdowns has. It stands first for two reasons: it gives a yardstick for the grade of evidence before that yardstick gets used, and it is

taken on the very work the room itself does. It is not clinical work, and the two grades must not be confused. Second comes the detection rate, measured on the same physicians before the assistant arrived and after a period of working with it: the only direct evidence of erosion on real clinical work. Third comes the Russian ArtInCol trial: it measures the same quantity on the same procedure with a stronger grade of evidence, but it has no observation period after the assistant is removed. A plane of "what about trust has not been measured" and a diagram with empty axes once stood here as separate slides and have been dropped by the owner; it is still worth saying that trust is not measured — but by voice, in the section's summary, not as its own slide. The new roles and steps that appeared once part of the decision came to be trusted to the model are gathered in the summary together with the rest.

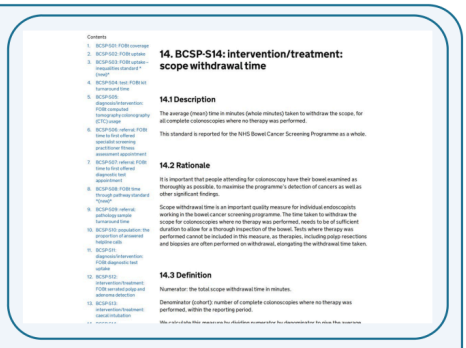
That is why this stage comes last among the substantive ones: it carries through to the end the very thought the evening began with, and it does so by the same method the lecture uses to read other people's numbers — naming what is measured, and what is not.

Qualification: what is measured, and how

Here, what gets measured is the performer himself: is he in form. Skill is spent in silence.

An adenoma is a precancerous thickening of the mucosa: removed today, it will not become cancer years from now.

Data is gathered four ways, and all four run on the ordinary flow of work: an audit of the decisions made on the job — the rate of procedures with a finding · timing by stopwatch · a count of procedures for the year · periodic re-certification. **Case review comes after the outcome and is not part of the count.**



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14.1 Description

The average (mean) time in minutes (whole minutes) taken to withdraw the scope, for all complete colonoscopies where no therapy was performed.

This standard is reported for the NHS Bowel Cancer Screening Programme as a whole.

14.2 Rationale

It is important that people attending for colonoscopy have their bowel examined as thoroughly as possible, to maximise the programme's detection of cancers as well as other significant findings.

Scope withdrawal time is an important quality measure for individual endoscopists working in the bowel cancer screening programme. The time taken to withdraw the scope for colonoscopies where no therapy was performed, needs to be of sufficient duration to allow for a thorough inspection of the bowel. Tests where therapy was performed cannot be included in this measure, as therapies, including polyp resections and biopsies are often performed on withdrawal, elongating the withdrawal time taken.

14.3 Definition

Numerator: the total scope withdrawal time in minutes.

Denominator (cohort): number of complete colonoscopies where no therapy was performed, within the reporting period.

The finding rate cannot be raised by trying harder: an adenoma lands in the denominator even when the physician did not look behind the fold. Withdrawal time is a discipline — easier to keep up.

The decision: authorize, narrow the authorization, suspend it, or not renew it.

Why this stage stands here: erosion accumulates across all stages at once — it can only be gathered once all of them have passed.

Of the four methods, one measures the result; the rest measure behavior and volume. Which one — and why is it the easiest to substitute with the other two?

The image shows BCSP-S14 standard "Scope withdrawal time" from the national bowel cancer screening programme, GOV.UK · colonoscopy quality indicators, Gastrointestinal Endoscopy 2015

This stage is built differently from all the previous ones, and that distinction has to come first. Every previous section measured the work: the stream of alerts, the document, the scan, the procedure around the signature. Here the object being measured is the performer himself, and there is one question for him: is he staying in form. An engineer is used to thinking of qualification as something issued once: a diploma earned, an authorization completed, a certificate in hand. In medicine it works differently, and this carries the whole section: a physician's skill is a consumable.

Now, what data about the person does this stage gather, and how. There are four methods, and all four run on the ordinary flow of work, with no separate trial.

An audit of the decisions made on completed work is the detection rate, and more on that separately below. The volume-of-practice count comes from the log of completed work: skill at a manual procedure is maintained by performing it, and from the outside this is checked by one quantity — how many times the person performed it. Professional societies set the minimum number per year, and below it a specialist is considered to have fallen out of form.

The detection rate is counted from ordinary reports, with no separate trial at all — and that is a rare property: the profession gets a check on an individual performer's condition straight out of his everyday work, with no exam set up for him. The rate is built cleverly. It is the share of procedures in which the physician found at least one precancerous growth, out of all his screening procedures, and it cannot be gotten by trying hard, by willpower, or by careful wording in the report: a growth that requires looking behind a fold lands in the denominator regardless of whether the physician looked, and lands in the numerator only if he did. So the detection rate measures behavior, not reporting about it.

Withdrawal time is timed with a stopwatch: the examination is done during withdrawal of the endoscope, and the slower the withdrawal, the more gets seen in time, with the lower bound set by guidelines. The fourth method is periodic re-certification: it gathers both accumulated study and accumulated practice, because between two re-certifications a specialist's condition changes in both directions. Case review is not among these four: it comes AFTER the outcome, that is, it checks an error that has already happened, and the silent drain of skill never reaches it. There is no fifth method — a trial with the assistant switched off — in any jurisdiction.

What decision this stage makes. Authorize · narrow the authorization to part of the procedures · suspend it · not renew it. Behind this decision stands everything that came before: the person either continues to decide at the earlier stages — calling patients in for examination, reading a measurement, making a diagnosis, signing — or he does not. And this is the one stage all evening whose decision is not about the patient.

And why this stage's measurement is taken off the person as a whole, across all stages at once, rather than being set as a step inside the cycle. The previous section ended with a signature that may have nothing behind it: the cost of error moves onto someone who did not sign it. Here the same thought turns to face the person as a whole. The one who signs is the one person all evening that no measurement has measured: the work has been measured the whole lecture, and nobody has measured the one who remains the decisive link inside that work. This cannot be placed as a step alongside selection, diagnostics, and observation: a work-step does not show a measurement like this at all, and lining up nine stages in a row would mean pretending they are uniform.

One caveat about the order. This stage stands last among the substantive ones all evening not because it is less important, but because a person's erosion can only be measured after it has been shown what he was doing and what held him in place. Put it earlier, and there would be nothing to measure.

Sources. colonoscopy quality indicators, Gastrointestinal Endoscopy 2015

And the same field with the model, named before the breakdown. Of the eight actions, the trained model took over one — the withdrawal examination, where a box appears in real time on the relevant part of the image. One step was added: deciding whether to trust that box. Preparation, inserting the scope, the fold, removing what's found, histology, the report, and tallying the physician's own metrics stayed with people and the profession. And the three places where the model does not work are named by what is missing there: it cannot maintain skill — the direct opposite is measured; it has nothing to show the physician that he would not have seen without highlighting inside the procedure itself; and it cannot measure his own result without the assistant, and neither can any procedure in any jurisdiction. The direction the tool is developing in is worth noting right away: from detection it is moving toward characterizing what is found, and further toward automatically tallying the physician's own metrics — that is, it is also becoming a measuring instrument for the person who uses it.

The stage's eight actions in order: prepare the patient · insert the scope to the cecum · examine during withdrawal · check behind the fold · remove what's found · send it for histology · write up the report · tally the physician's own metrics. The physician does the first seven by hand; the profession does the eighth for him, per published guidelines — and this is the only place in the stage where someone other than the physician himself measures his form.

The two metrics of this outside check are built differently. The detection rate — the share of procedures in which at least one growth was found — is a quantity taken straight from the flow of work, and there is no way to raise it directly by trying: it is either there in the work, or it isn't. Withdrawal time is a disciplinary quantity: its lower bound is set by guidelines, and compliance is checked by a stopwatch. The first measures the result, the second measures behavior, and one must not be substituted for the other — though in practice it is: time is easier to hold to.

The physician came back after a year: what keeps him in form, and where the model landed

The physician came back after a year with no procedures — an administrative post, illness, parental leave. Same diploma, authorization never revoked: there was nothing to revoke it for.

The first procedure goes differently: the scope moves slower, the hand takes longer to find the angle, he passes a fold by. No complications, the report is written, **and he doesn't know it.** A bad examination looks exactly like a good examination in which nothing was found.

That is why three pillars keep him in form, and none of them asks the person himself: volume of practice for the year, authorization from the organization, periodic re-certification.

Russia: periodic accreditation once every five years — 144 hours of study plus a portfolio. **The specialist writes the work report himself. There is no numeric threshold on the result — neither a finding rate nor a withdrawal time — in the requirements.**

The model took over one action out of eight — the withdrawal examination. **One was added: whether to trust the box.**

Professional society guidelines, Gastrointestinal Endoscopy 2015 · Order No. 709n of the Russian Ministry of Health, 28.10.2022, "On approval of the Regulation on the Accreditation of Specialists"

The physician came back after a year with no procedures: an administrative post, a long illness, parental leave, a transfer to another department. Same diploma, certificate still valid, authorization never revoked — there was nothing to revoke it for. The first procedure goes differently. The scope advances more slowly. The hand takes longer to find the angle for passing a bend. A fold that needs checking, the physician passes by.

And here is the reason this picture stands here: he himself does not know it. The procedure went without complications, the report is written, the patient is sent home. There is no feedback and there won't be: a growth that went unseen stays in the person until the next examination, and that is years away. A bad examination looks exactly like a good examination in which nothing was found. That is why the three pillars are built the way they are: a specialist's own sense of well-being is a useless instrument for measuring his form.

The room knows this picture from the other side. A skill not used for a year is lost — everyone who has gone back to a language or a subject area after a break knows this. There is one difference. In engineering work, the nearest build, test, or code review reports the loss of skill the same day; here the nearest instrument that reports it is the same patient's next visit, years later, and to a different physician.

Now the pillars — how this decision is made with no model at all. There are three of them, they work at the same time, and none of them asks the person himself.

Volume of practice: professional societies set a minimum number of procedures per year, below which a specialist is considered to have fallen out of form. Accreditation: authorization is issued for one kind of activity, issued by the organization that is responsible for that activity, and tied to the workplace and the procedure — it can be narrowed, suspended, not renewed. Re-certification: qualification is confirmed

periodically, and confirmation presupposes both accumulated study and accumulated practice, because between two confirmations a specialist's condition changes, and in both directions.

The three pillars answer one question — is the person staying in form — and none of them turns to the person himself. The profession does not ask the physician whether he is in form: it counts on his behalf, by the detection rate and the withdrawal time.

Now how this is set up in Russia, because that is closer to the room than any foreign guidelines. A specialist's periodic accreditation takes place once every five years; the procedure is set by Order No. 709n of the Russian Ministry of Health, dated October 28, 2022. It requires accumulating 144 hours of continuing-education training — in one course or several — or at least 72 hours of continuing-education training plus informal education recorded on the Continuing Medical Education portal — and submitting a portfolio: a report on professional activity over those five years with information on education and length of service.

And here is what follows from this, and it has to be phrased precisely. Saying "the result of the work is not looked at in Russia" would be stronger than true: the portfolio does require a report on professional activity, and it does contain the results of the work performed in the job function. What is true is something else: the specialist draws up this report himself. There is no numeric threshold on the result — one that would be counted on his behalf and that would block renewal of authorization if not met — in the requirements: not a finding rate, not a withdrawal time. That is, a Russian endoscopist has no mandatory external numeric check on his own form — the very thing the whole section is built around.

An earlier version of this slide said "the result of the work — none at all": a statement stronger than the source supports — exactly the class of error we've been examining in vendors. Here, the detection rate is counted where a center decides on its own to count it, not because accreditation asks for it. So a Russian endoscopist is missing, as a mandatory matter, exactly the instrument the physician from the opening picture lacked: there is nowhere for him to learn of his own decline.

Now about the field of work itself, so that it's clear what the model intervenes in. The examination is made up of eight actions. Prepare the patient. Insert the scope to the cecum. Examine during withdrawal. Check behind the fold. Remove what's found. Send it for histology. Write up the report. Tally the physician's own metrics. The person does the first seven by hand; the profession does the eighth for him, per published guidelines, and he himself takes no part in that count.

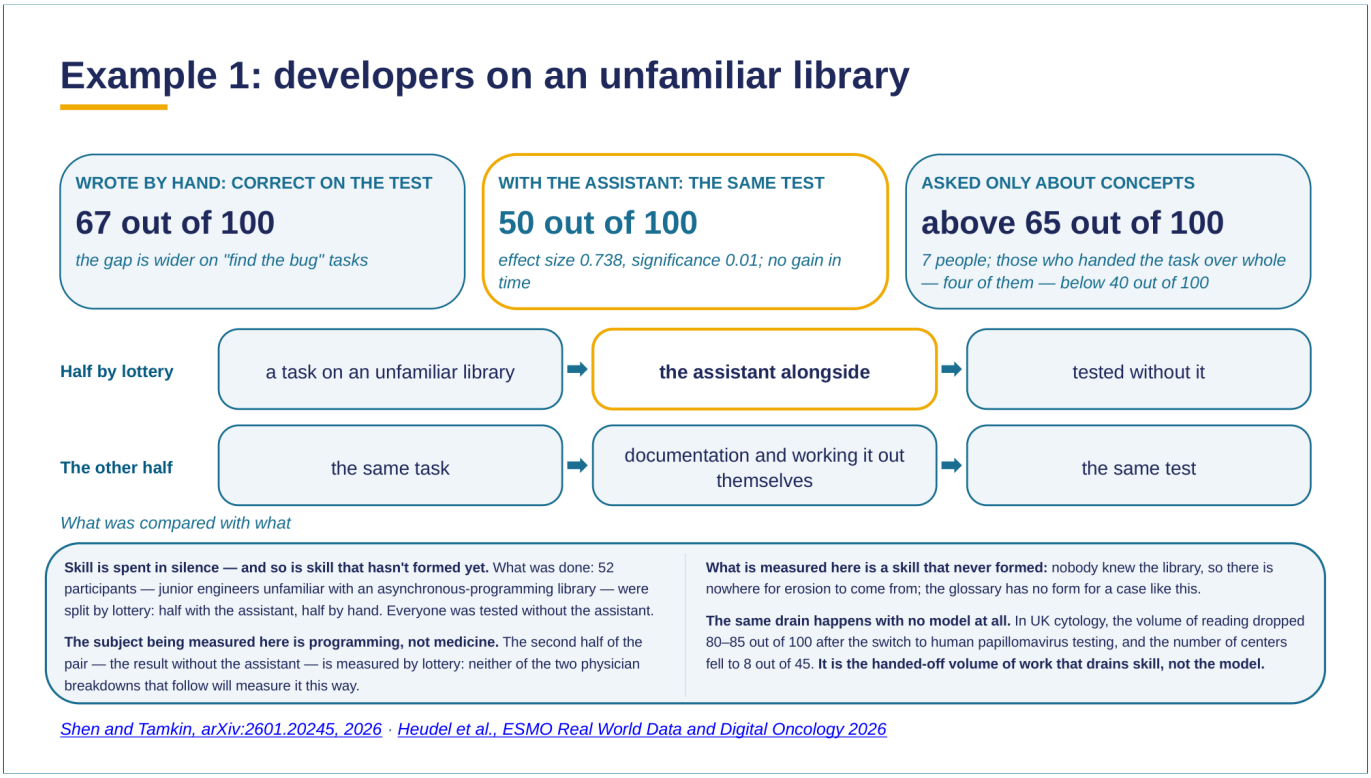
Of these eight, the trained model took over one — the withdrawal examination, where a box appears in real time on the relevant part of the image. And one step was added: deciding whether to trust that box. Preparation, inserting the scope, the fold, removing what's found, histology, the report, and tallying metrics stayed with people and the profession.

And the three places where the model does not work are named by what is missing

there. It cannot maintain skill — the direct opposite is measured, and that is the next slide's breakdown. It has nothing to show the physician that he would not have seen without highlighting, inside the procedure itself. It cannot measure his own result without the assistant, and neither can any procedure in any jurisdiction. The direction the tool is developing in is worth noting right away: from detection it is moving toward characterizing what is found, and further toward automatically tallying the physician's own metrics — that is, it is also becoming a measuring instrument for the person who uses it.

Only one question for the room is asked in this stage, and it is a prediction, not a poll: of the four measurement methods named in the previous slide, one measures the result of the work, and three measure behavior and volume. I ask the room to name which one measures the result, and to say why it is easiest to substitute with withdrawal time. The answer: the detection rate; it gets substituted because time is easier to hold to, and the rate cannot be raised by trying harder. There is deliberately no second question in this stage: two questions in a row read to the room as rhetoric, and they answer neither.

Sources. colonoscopy quality indicators, Gastrointestinal Endoscopy 2015



Fifty-two junior engineers, unfamiliar with an asynchronous-programming library, were split by lottery: half wrote code with an AI assistant, half by hand. Right after the work, everyone was tested with no assistant. Those who had written with the assistant got 50 correct tasks out of 100; those who wrote by hand got 67. Effect size 0.738 at significance 0.01; there is no gain in time at all, and the gap is wider on "find the bug" tasks. The subject being measured here is programming, not medicine.

Behind this breakdown stands a yardstick for the two clinical ones that follow, and each of them has its own hole. The Polish study measures exactly what's needed — the result for the same physicians without the assistant — but it is observational, looking backward through an archive, and the authors make no claim of causality. The Russian trial is stronger in grade of evidence, it is randomized, but it has no observation period after the assistant is removed at all, and the authors say so themselves. One study measures the right thing by a weak method, the other measures the wrong thing by a strong method. Here the second half of the pair is measured by lottery, and this is the only place in the section where that happens.

The second reason to put it first, and it is the more important one. It is taken on the very work the room itself does. About physicians, you can always say: they have their own specifics, it's not like that for us. About junior engineers writing against an unfamiliar library, you cannot say that — this is exactly what half the audience was doing this week. The section is about what happened to the person who decides, and it begins with the person sitting in the room.

And right away, a distinction that cannot be glossed over. The Polish study, which comes next, measured EROSION: the same nineteen physicians were in form and got worse. Here what's measured is FORMATION: nobody knew the library, and the question

was whether the skill formed at all. The eighth glossary form, "the performer is the same, but got worse," cannot be assigned here: the same performer is not being compared against anything here. There is no form in the glossary for a case like this, and it is more honest to call it a refusal than to force a fit.

The design of the experiment. Fifty-two participants, mostly junior engineers; each has written in this language for at least a year, is familiar with AI assistance, but is not familiar with the specific asynchronous-programming library — and this matters, because what is being measured is exactly the formation of a new skill, not work with one already held. The task: write two features using this library. Split by lottery: half were given an AI assistant, half were not. Right after the work, everyone was tested with no assistant, with four types of tasks: find a bug in code, read someone else's code, write code, explain a concept.

The result. The group that wrote by hand scored 67 correct out of 100; the group with the assistant scored 50. Cohen's effect size of 0.738 at significance 0.01. That's about a two-grade difference — the kind you'd see on an exam. The gap is wider on "find the bug" tasks — and this is worth pausing on: finding a bug in someone else's code is exactly the work the assistant takes over first, and afterward there is nothing left to train that skill on. There is no gain in time here: the group with the assistant finished about two minutes earlier, a difference that is not significant. So they paid in understanding and didn't buy speed.

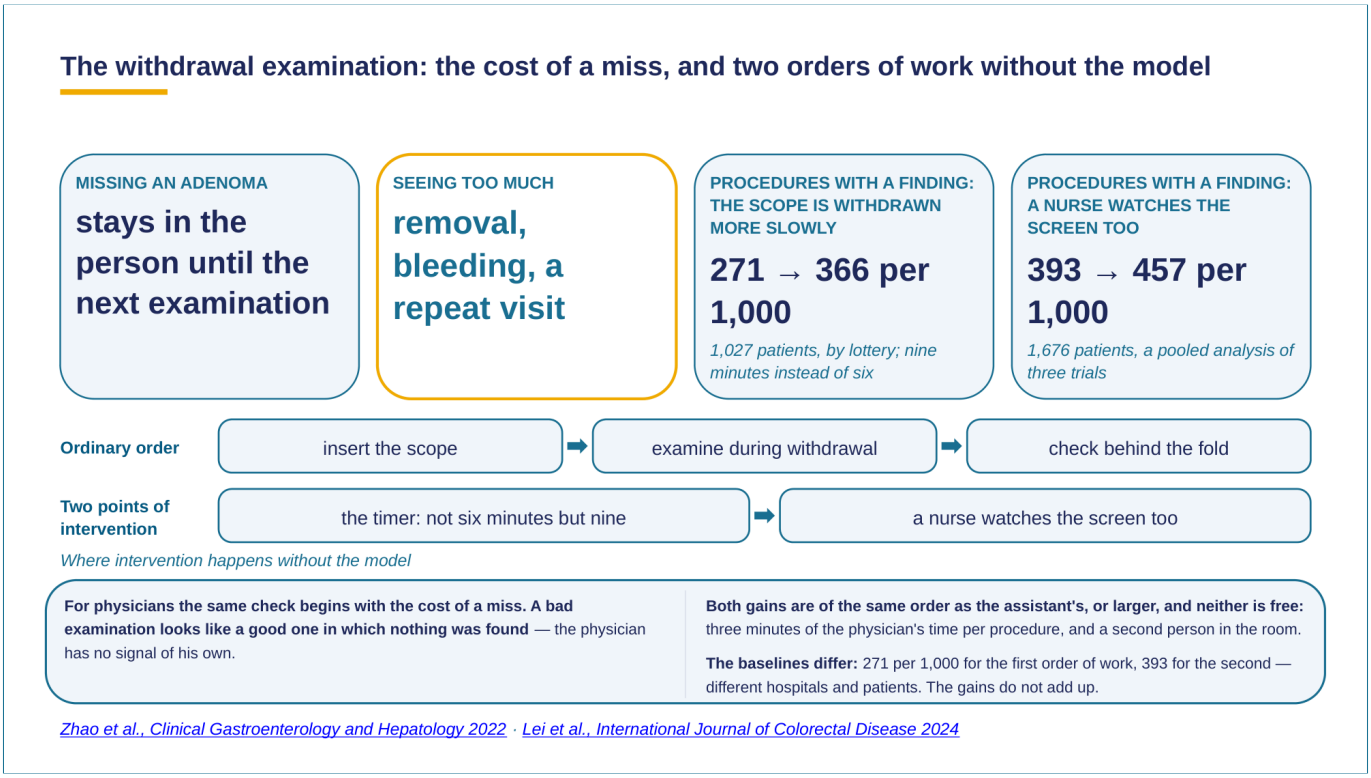
Now the reason this case is more valuable than the two that follow: it names not only the drain, but also a way not to pay it. The authors broke down exactly how people dealt with the assistant, and identified six approaches. Three of them preserve retention. Those who asked only about concepts and worked out the bugs themselves — seven people — stayed above 65 out of 100. Those who asked for code together with an explanation — three of them — did too. And those who handed the task over to the assistant whole — four of them — were the fastest of all and below 40 out of 100. These numbers are small, and they cannot be presented as a measured regularity; they can be presented as a direction, because it matches what we already saw with schoolchildren: there, guardrails removed the effect; here, the way the assistant is used removes it. In both cases the cure is the design of access, not the amount of practice.

I name the limitations as the authors themselves do. The sample is small. What was tested was immediate understanding right after the work, not skill over months — that is, this measurement has no long horizon, unlike the Polish study, where three months passed between windows. The work is code only. And there is no comparison against help from a live person — a mentor, a colleague — at all, and that absence matters: we don't know whether any kind of help produces the same effect, or specifically help from a model.

And the slide's last line, which frames the whole section if said properly. The same loss of skill happens with no trained model at all. In the UK, cytology switched to primary human papillomavirus (HPV) testing — that is a laboratory assay, with no model in it at all. The volume of cases for reading dropped 80 to 85 out of 100, the labs were cut

down from 45 centers to 8, and the review itself names the consequence directly: there was nothing left to train on. The conclusion from this carries weight, and it runs exactly against an easy reading of the whole section, which has only just begun. It is the handed-off volume of work that drains skill, not the trained model as such. Blaming AI specifically for deskilling would be an error of the same class as crediting the model with the classical mode's win — and this is the error the lecture has spent all evening catching others in. So the right question sounds like this: how much work have we taken away from the person, and where is he getting it now. And it's with that question that we'll go to the physicians.

The grade of evidence for the cytology mention has to be named: this is a consolidated review of deskilling, not the primary source of the program itself. In that same review sit two more numbers worth saying out loud: in web-reading experiments, more than thirty out of a hundred participants abandoned their own correct conclusion under an incorrect model cue, under time pressure; and in an experiment with twenty-seven radiologists, incorrect cues raised the false-call rate by up to plus twelve percentage points, including among experienced ones. Both numbers are taken from the review, not from the full texts of the experiments themselves — and that is exactly how they should be cited. And that same review has its own error, exactly on our subject: it calls the Polish study, which comes next, a randomized multicenter trial, when it is observational. A peer-reviewed review got the grade of evidence wrong. We will not get a better illustration today of why we spend the whole evening looking at how a measurement is built, rather than at the word "measured."



The withdrawal examination is the procedure step where a finding either gets seen, or doesn't. The cost of a miss is named here before any accuracy, and the first thing to understand about this stage is: feedback arrives years later. A missed growth stays in the person until the next examination — this is exactly how an interval cancer arises, found between examinations. Seeing too much costs removal, sometimes bleeding, histology, and a repeat visit. Neither the physician nor the model set the asymmetry between these two costs: a health authority set it, when it decided who to call in for examination, and how often.

About the unit the whole section counts in, and this has to be said once, here. The quantity the profession uses to measure an endoscopist is called the adenoma detection rate (ADR): the share of screening procedures in which the physician found at least one precancerous growth, out of all his screening procedures. On the section's slides it is converted into a number of procedures per thousand — that way it can be added and compared aloud without confusing it with a share of patients. Two hundred seventy-one procedures with a finding out of a thousand means: nothing was found in the remaining seven hundred twenty-nine, and part of those "nothings" is not the absence of an adenoma but an adenoma that was missed. The rate is built cleverly, and that is its strength: a growth that requires looking behind a fold lands in the denominator regardless of whether the physician looked, and lands in the numerator only if he did. So it measures behavior, not reporting about it, and it cannot be raised by trying harder in how you write it up.

Why this spot is built more dangerously than the others. In engineering work, the nearest build or check reports a miss the same day. Here, the nearest measuring instrument is the same patient's next visit, years later, and to a different physician. A poorly performed examination does not feel bad in that minute, or a month later, and

the person has no signal of his own about a decline at all. That is why the procedure counts the discipline on his behalf — he himself cannot find out.

Now the second step of the breakdown's form, the one whose omission is where breakdowns go astray. A model compared against nothing always wins, and that win means nothing: wherever the assistant arrives, two orders of work were already operating there, and both are measured in the same unit.

The first mode — discipline of withdrawal time. Withdrawal time was extended from six minutes to nine. A randomized trial, 1,027 patients, 514 against 513, time timed by stopwatch: procedures with a finding became 366 per 1,000 against 271, that is, the detection rate rose from 27.1 to 36.6 percent at a significance level of 0.001. A separate observation also gives a lever on the mode: each additional minute of withdrawal adds about six percent to the odds of finding an adenoma, measured in a window from six to thirteen minutes.

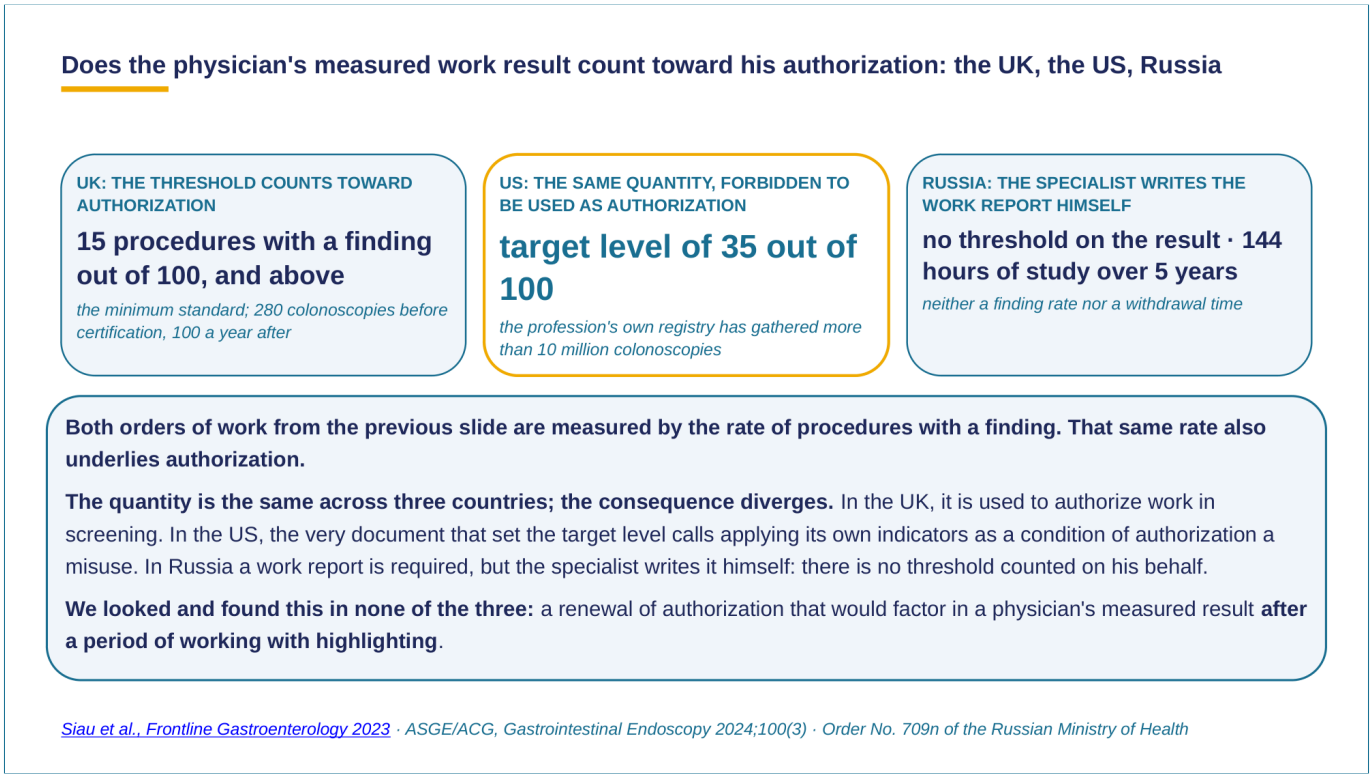
The second mode — a second observer in the room: alongside the physician, a nurse also watches the same screen. A pooled analysis of three randomized trials on 1,676 patients: from 393 procedures per 1,000 up to 457, that is, from 39.3 to 45.7 percent, relative risk 1.16 at an interval of 1.04 to 1.3. A second pooled analysis, across eleven trials and 8,278 patients, gives 1.15, from 1.05 to 1.26. The reviewers' own caveat is named together with the number: all the included trials carry a high risk of systematic bias, and there is no gain at all in this same review for polyp detection or for advanced findings. And here is my own caveat: part of the effect may be the effect of being observed while observing, and these trials could not separate one from the other.

A caveat about the baselines. The starting rates for the two modes differ — 271 procedures per 1,000 for the first and 393 for the second, different populations and different centers — so the two gains are not compared against each other. Each is compared with the assistant's gain in its own body of work, not with the other; adding plus ninety-five and plus sixty-four into "plus one hundred fifty-nine" is not valid in any form.

The conclusion this step stands apart for: both classical modes deliver a gain of the same order as the assistant, or larger, and neither is free — the first costs three minutes of the physician's time per procedure, the second costs the presence of a second person.

And nobody has measured a drain on skill for these two modes: there is no measurement. To say "they don't drain skill" would be claiming more than has been measured — exactly what this lecture examines vendors for.

Sources. Zhao et al., Clinical Gastroenterology and Hepatology 2022 · Xu et al., Gastroenterology Research and Practice 2016 · Lei et al., International Journal of Colorectal Disease 2024



The rate of procedures with a finding — the very quantity used to measure both orders of work without the model — also underlies a physician's authorization to work. And it enters that authorization differently in three countries. In the UK it is a threshold: below fifteen procedures with a finding out of one hundred, authorization for work in the screening program is not granted, and 280 colonoscopies are required before certification, and 100 a year after. In the US, the same quantity is a target level of thirty-five out of one hundred, and the profession's own document explicitly calls applying its own indicators as a condition of authorization a misuse. In Russia it is not counted on the specialist's behalf at all: a work report is required, but he writes it himself.

The question this slide answers follows right from the previous one: who counts this rate, and what happens to a person whose rate is low. The answer diverges across countries so much that you could not guess one jurisdiction from another — and what diverges here is the order of work itself, not the product that carries it out.

Let's start with where the quantity counts toward authorization literally. In the UK, the professional society's guidelines on training and certification in colonoscopy set thresholds on completed work, averaged over three months: reaching the cecum without assistance above ninety out of one hundred procedures, examining the retroflexed view in the rectum above ninety, retrieving what's found above ninety, procedure dissatisfaction below ten out of one hundred — and a rate of procedures with a finding of fifteen out of one hundred and above as the minimum standard for a trainee. Alongside it stands volume: a minimum of two hundred eighty colonoscopies before certification, or two hundred if there is certification in sigmoidoscopy, and no fewer than fifteen procedures over the last three months. After certification the guideline is at least one hundred colonoscopies a year, with a desirable level of one

hundred fifty. And the most important point: without certification, a person is not authorized to work in the screening program. That is, the result of his work, counted on his behalf from his own reports, directly decides whether he is let into that work.

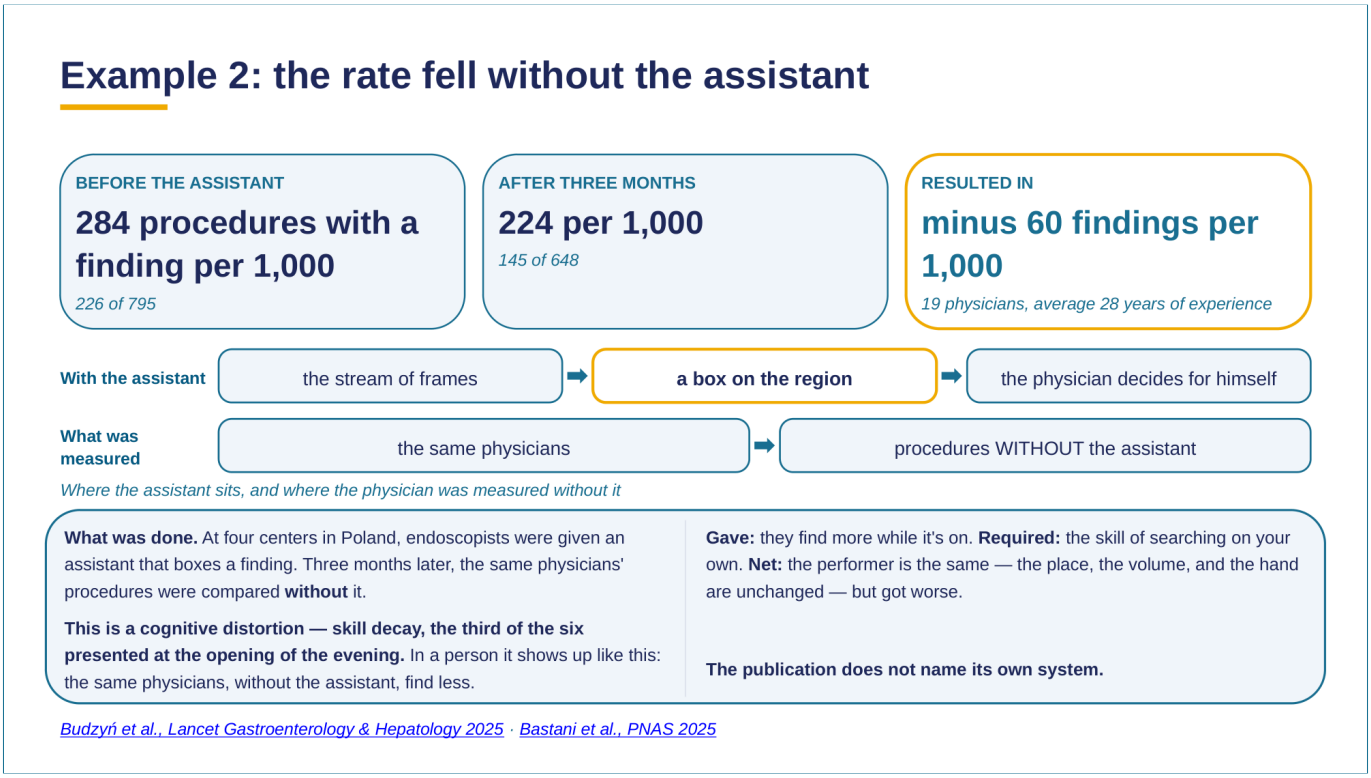
Now the US, and here is a finding worth saying slowly. They have the same indicator, and it is set: the target level of the adenoma detection rate (ADR) is thirty-five out of one hundred, and in its own document the profession marks it as an outcome indicator rather than a process one, that is, the strongest grade there is. It is gathered by an industry registry that has accumulated more than ten million colonoscopies. It would seem the basis for authorization is in place. And the very same document states, word for word, that the included indicators and target levels do not reflect a standard of care, authorization requirements, or training standards, and that applying any of this document's indicators that way is a misuse of it. The profession itself forbade turning its own measured indicator into a condition of authorization. Authorization is held up by something else: one hundred points of Maintenance of Certification (MOC) over five years plus a knowledge exam. And the same document calls stripping someone of the right to perform the procedure an extreme measure, not an ordinary outcome of a measurement.

Why this matters for the whole lecture. All evening we've been having a conversation about how a measured number is stronger than an opinion. Here we see the reverse case, and it is not foolishness: a profession that has ten million measured procedures refuses to turn that number into an administrative decision — because the indicator was counted to improve the work; there was never an intent to punish with it, and it has known distortions that, once turned into an authorization criterion, would start driving behavior. This is exactly the kind of judgment the course teaches: a number is fit for the decision it was collected for, and no other.

And Russia. A specialist's periodic accreditation takes place once every five years; the procedure is set by Order No. 709n of the Russian Ministry of Health, dated October 28, 2022. It requires a total of one hundred forty-four hours of continuing-education training — in one course or several, or at least seventy-two hours of courses plus recording on the Continuing Medical Education portal. The second section of the portfolio is a report on professional activity, and it does contain the results of the work performed in the job function. This has to be stated precisely, because the easy and wrong formulation "the result of the work is not looked at" suggests itself here on its own. The correct one sounds different: the specialist draws up the work report himself. There is no numeric threshold on the result — one that would be counted on his behalf and that would block renewal of authorization if not met — in the requirements: not a finding rate, not a withdrawal time. Here, the detection rate is counted only where a center decides on its own to count it; accreditation does not ask for it.

And the slide's last line, one of the reasons it exists at all. We deliberately looked for a jurisdiction where a physician's measured result, after a period of working with highlighting, would factor into renewal of authorization. We looked through the UK and US societies' guidelines, Russia's accreditation regulation, and a consolidated deskilling review. We did not find one in any of the three. This is not a gap in our search that

someone will close by tonight: the profession has an indicator for such a measurement, but there is no procedure for it anywhere. So when, two slides from now, we see the measured drop in the rate for the same physicians, keep this in mind: there is nowhere for a physician to learn of that drop in ordinary work, and no order of authorization in these three countries will catch it.



Four centers in Poland: endoscopists were given an assistant that boxes a finding on video, right during the procedure. Three months later the same nineteen physicians' procedures were examined without it — findings dropped from 28.4 to 22.4 out of 100 procedures, minus six percentage points. This is what skill decay looks like, the third of the six distortions presented at the opening of the evening: the assistant lifts the result while it's on, and that state is not free — the skill of searching on your own gets drained.

The strength of the evidence and the full shape of the dispute. The study is retrospective and observational, not randomized; it does not prove causality, and the authors make no such claim. The windows are short, there is one region, one assistant system, and not a single replication. A correction, three letters to the editor, and the authors' reply have been published around this work — the dispute is open, and it cannot be hidden behind the word "measured." There are three substantive objections: secondary outcomes did not drop (mean adenomas per procedure 0.54 against 0.43, advanced ones 0.062 against 0.063, both not significant); withdrawal time was not measured in the study at all, meaning the finding lacks the behavioral link that would make it checkable; and workload over the same period nearly doubled — from 795 procedures to 1,382.

Anchoring — the first of the six — is absent here entirely: what gets looked at are procedures WITHOUT the assistant, where there is no cue and nothing to anchor on. Anchoring works while the model is there; erosion is visible once it's removed.

And it was not the procedures with the assistant that were examined, but the procedures without it: the drain on skill is visible only where the tool isn't.

Composition. Nineteen endoscopists — sixteen gastroenterologists and three surgeons,

each with more than two thousand colonoscopies performed, average 28 years of experience. Two three-month windows, with the assistant deployed between them. The main comparison included 1,443 procedures without the assistant: 795 in the early window and 648 in the later one. These are not residents or novices, and that is half the strength of the observation: the drop happened among people who had held their form for two decades.

The result. The detection rate in procedures without the assistant was 28.4 percent before deployment, that is 226 findings out of 795 procedures, and 22.4 percent after, 145 findings out of 648. The absolute difference is minus 6.0 percentage points, at an interval of minus 10.5 to minus 1.6, and significance 0.0089. In a multivariable model, having worked alongside the assistant gives an odds ratio of 0.69 at an interval of 0.53 to 0.89; patient age sixty and above gives 3.60 in the same model — that is, it predicts detection noticeably more strongly. The ratio between these two figures is not stated: dividing a ratio above one by a ratio below one compares nothing. The six points lie in the same range where effects from patient mix live, and that sets the scale of the finding without canceling it out.

A caveat about the denominator, without which the number 795 reads as something it is not. In the early window, the assistant was not yet at the centers, so 795 means "procedures without the assistant" and "procedures in total" at the same time; in the later window they diverge — 648 without the assistant, 1,382 in total. The comparison of 226 of 795 against 145 of 648 is done honestly, by one column, and the rise in overall workload explains part of the drop just as well as skill decay does. This is exactly what the three letters to the editor take aim at.

The mechanics of the assistant: input is the stream of frames during the procedure, output is a box on the suspicious area in the same second. The publication does not name its own system, and manufacturers do not publish what systems of this class are trained on. The mode has names, and they are verifiable: GI Genius by Medtronic, CAD EYE by Fujifilm, ENDO-AID CADe by Olympus; the difference between them in pooled comparisons is small, and that matters more than the names.

The pair that closes the breakdown. Gave: with the assistant on, physicians find more, and the gain in detection rate in randomized trials is measured and replicated many times over. Required: the skill of searching on your own. The drain is visible within three months, even among physicians with an average of twenty-eight years of experience: this is a trained skill that is held up by volume of performance, and the drop ran exactly along centers, not individual physicians. Net: the assistant raises the result while it's on, and that state is not free. Nobody has done a count in one currency — how much the assistant on added against how much was lost while off — and it must not be presented as measured.

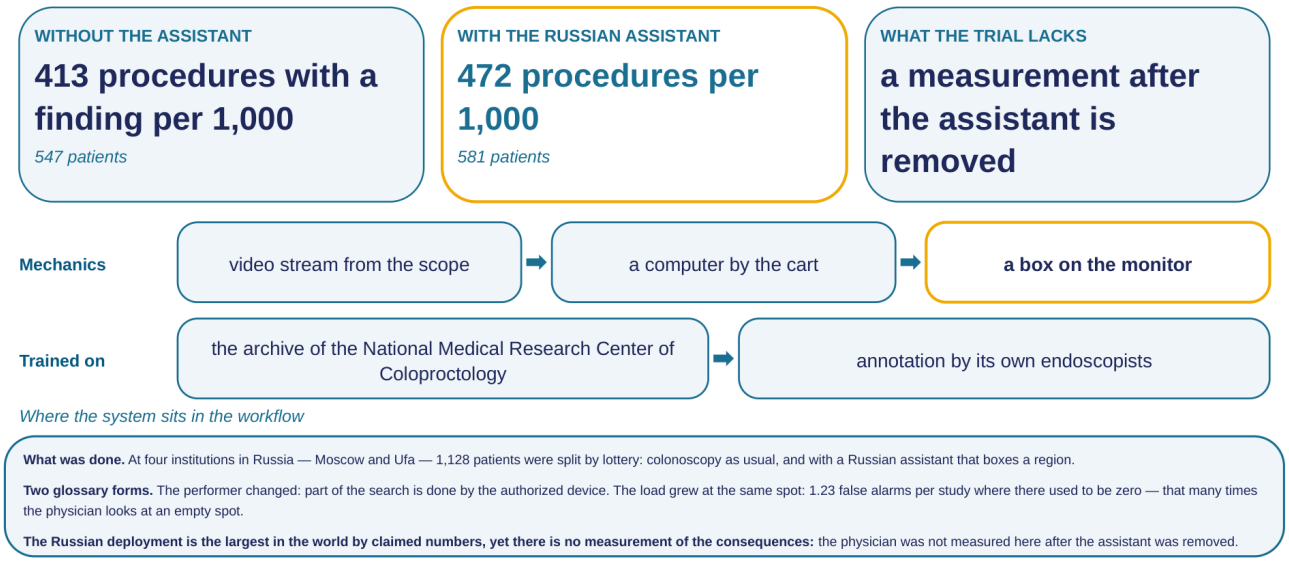
The eighth glossary form: the performer is the same, but got worse. The place of work, the volume, the order of steps, and the hand are unchanged; what changed is the condition of the hand itself. The neighboring forms don't check out: the fifth requires the hand to have changed, and the comparison here is built inside the same nineteen

physicians; the form about operation volume misses too — procedure volume is unchanged, it's the finding rate that fell. The reverse case stands in the glossary as an empty slot: it has a name, and not a single measurement.

The same phenomenon is measured by randomization outside the clinic — among high-school students, in math: an exam with no model available turned out 17 percent worse than the control for the group with free access, while the group with teacher-set guardrails came out about at the control's level. A remedy against erosion was found there, and it consists in the design of access, not the amount of practice. This is not a replication of the clinical finding: a different study, different people, a different task.

Sources. Budzyń et al., Lancet Gastroenterology & Hepatology 2025 · a published correction to the same work, 2025;10(11):e12, and three letters with the authors' reply, 2025;10(12):1060–1062 · Bastani et al., PNAS 2025 · the original article: 2025;10(10):896–903

Example 3: ArtInCol, Russia — an assistant on the same procedure



Four institutions in Russia — Moscow and Ufa: 1,128 patients were split by lottery between ordinary colonoscopy and colonoscopy with the Russian assistant ArtInCol, which boxes a region on video. Procedures with a finding became 472 per 1,000, against 413.

This trial measures the same quantity on the same procedure as the Polish study, and is built more strongly: multicenter, randomized. Its result does not cancel out or replicate the Polish finding: they measure different things.

Composition. August–December 2024, four medical institutions: the Ryzhikh National Medical Research Center of Coloproctology, Botkin Hospital, Pirogov Russian National Research Medical University in Moscow, and the Republican Clinical Oncology Dispensary in Ufa. 1,128 patients, split by lottery: 547 for colonoscopy without the assistant, and 581 with it.

The result, in the same unit the whole section counts in. Procedures with a finding became 472 per 1,000 against 413 — that is, the adenoma detection rate was 47.2 percent against 41.3, effect size 5.9 percent at significance 0.048. The mean number of adenomas found per procedure was 0.97 against 0.79, and this difference is more robust than the main one: significance 0.01.

Now something the Polish breakdown lacked entirely, and the reason this slide is especially needed for an engineering audience: the mechanics is named in full. The system is vendor-neutral, carries a Roszdravnadzor registration certificate, and connects to standard endoscopic equipment. Input is the video stream during the procedure, which goes to a compact industrial computer next to the cart. Output is a detection box on the monitor, in real time, on areas where the probability of a growth is high. It is trained on the coloproctology center's own archives, annotated by the

endoscopists of its own endoscopic diagnostics and surgery department. Before refinement, the prototype gave an accuracy of 83.2 percent and a sensitivity of 77.2 on the test set.

And the cost of attention, stated as a number — something most publications don't even look for. The mean number of false alarms came to 1.23 per study. This was measured on thirty-five randomly selected colonoscopies, and an alarm was counted as a box appearing for no less than three seconds. More than once per procedure, the physician shifts attention to a spot where there's nothing. This is the answer to the question the room asks about its own work: a model deployed in an acceptance loop always has a figure like this, and if the vendor doesn't name it, it wasn't measured.

The limitations are named from the article itself, not invented. There was no blinding in the study: the endoscopists knew they were working with the assistant and that their results were being compared, and the authors explicitly name this a possible source of bias. One center and five endoscopists from different institutions were excluded from the final analysis for protocol violations — inflating the inclusion criteria and, as a result, abnormally high detection rates. The equipment was expert-grade, the staff were trained, and the authors separately note that in resource-limited regions, deployment will run into a shortage of technical support and trained staff.

And for this section's axis — a gap, named exactly where it is. The trial has no observation period: the authors say so themselves, in a separate line in the limitations. That is, the Russian study measured exactly half the pair — what the assistant gives while it's on — and did not measure the other half: what becomes of the same physicians when it's switched off. The Polish study measured exactly that second half, and was weaker in strength of evidence. There is no clinical study anywhere, in Russia or elsewhere, that has measured both halves on one sample; the second half has only been measured by lottery in the developer study the section opened with — and that measurement is not clinical. This is the honest answer to the question "what about us": the gain is measured better than theirs, and the drain is not measured at all.

And the slide's line about the country has to be said in both halves, or it comes out as either boasting or reproach. First: the radiological AI project deployed in Russia is the largest in the world by claimed numbers. Second: we looked for a measurement of what this deployment did to physicians' skill — through deployment-program publications, specialty journals, and operational reports — and found not a single one. Scale of deployment and scale of measuring its consequences are two different things, and the first says nothing about the second.

And one remark about withdrawal time, because it closes the second objection to the Polish study. Here this quantity was measured: a median of nine minutes against eight, a difference that is not significant. That is, in the Russian trial the gain is not explained by withdrawing the scope more slowly with the assistant. The Polish study had no such measurement, and that is exactly why its finding cannot be checked all the way through.

And the last step of the breakdown — what moved. There are two forms here, and both

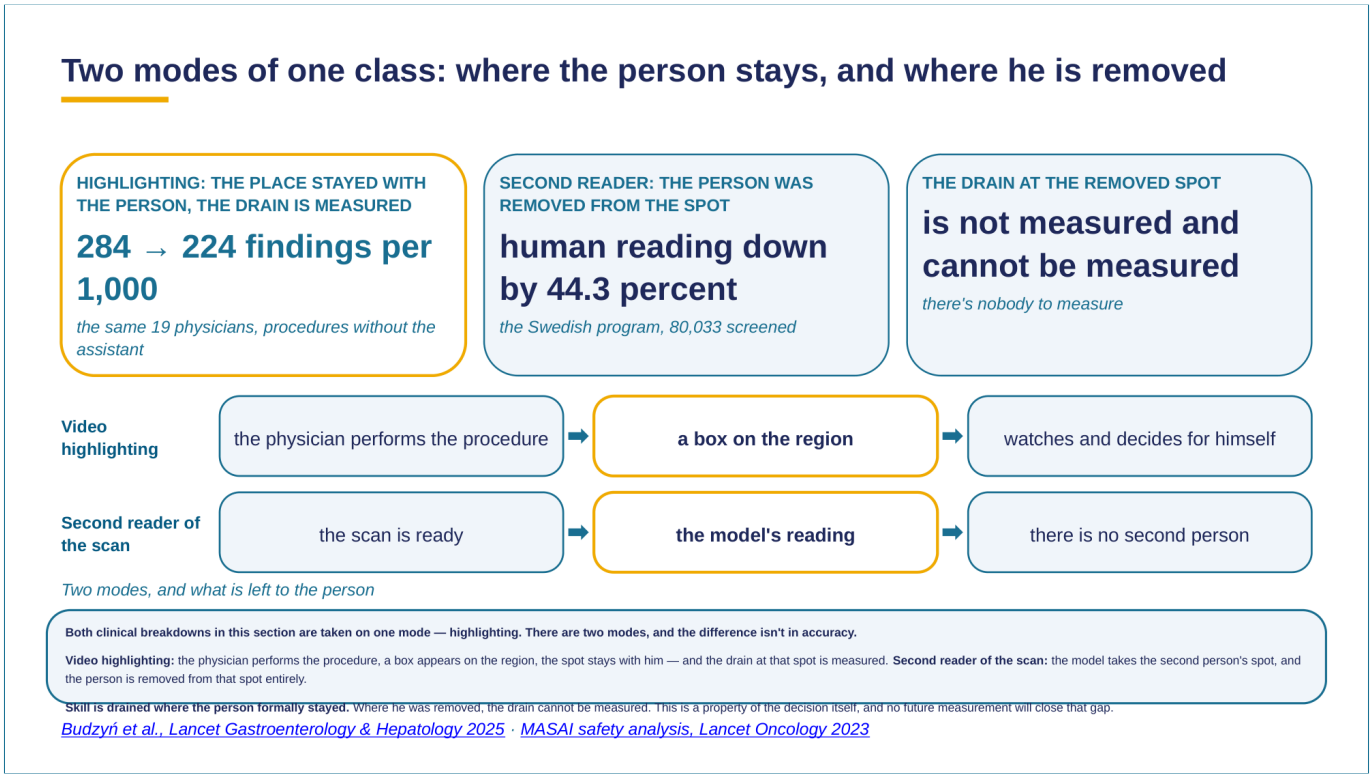
have to be stated.

The performer changed: part of the search at this spot is now done by an authorized device — a Roszdravnadzor certificate, a video stream in, the system itself marking areas out. This is the case the glossary sets apart from the Polish study and keeps for the loop's switched-on state. The form's tally: 413 procedures with a finding per 1,000 against 472.

And the load grew at the same spot: every cue costs seconds of attention to the same physician at the same workplace, and the seconds are counted — 1.23 false alarms per study where there are zero without the system.

Why it's not "a step appeared" here, even though that form suggests itself first. A new step would have had its own cost in time: where that cost has been counted in the lecture, arbitration cost two minutes twelve seconds against forty-eight seconds of reading. Here, examination time was measured and did not change — a median of nine minutes against eight, no significance. So the box lands inside a step that already exists, without creating a new operation. The form was rejected by the measurement, and it's worth saying that to the room directly.

Sources. Achkasov, Shelygin et al., Koloproktologiya 2025; 24(3):12–21



One class of tool works in two modes, and the difference between them is what's left to the person. Both clinical breakdowns in this section, the Polish and the Russian, are taken on one of the two: video highlighting during a procedure, where the physician performs the procedure himself and a box appears on the region. The second mode removes the person from that spot. The consequences for the person differ between the two modes. If this isn't separated out now, the room will walk away from the lecture with the takeaway "AI drains skill," and the correct takeaway is different, and more complicated.

The first mode — video highlighting. The physician performs the procedure himself, a box appears in real time on the relevant part of the image, and he decides whether to trust it or not. The place of work stays with the person entirely: he inserts the scope, withdraws it, checks behind the fold, removes what's found. The tool did not take over a single one of these actions; it added a cue to one of them. And it is exactly at this spot that the drain on skill is measured: for the same nineteen physicians, in procedures without the assistant, the rate dropped from 284 findings per 1,000 procedures to 224, over three months. This is the breakdown we just went through.

The second mode — the second reader. The scan is already taken, a person reads it, and the program requires two independent readers. The model takes the second reader's spot and delivers its own interpretation. Here the person is removed from the spot entirely, and that is not a figure of speech: in the Swedish program, the volume of human reading turned out to be lower by 44.3 percent — this is an interim safety analysis, 80,033 screened. More than forty percent of the work people used to do, people stopped doing.

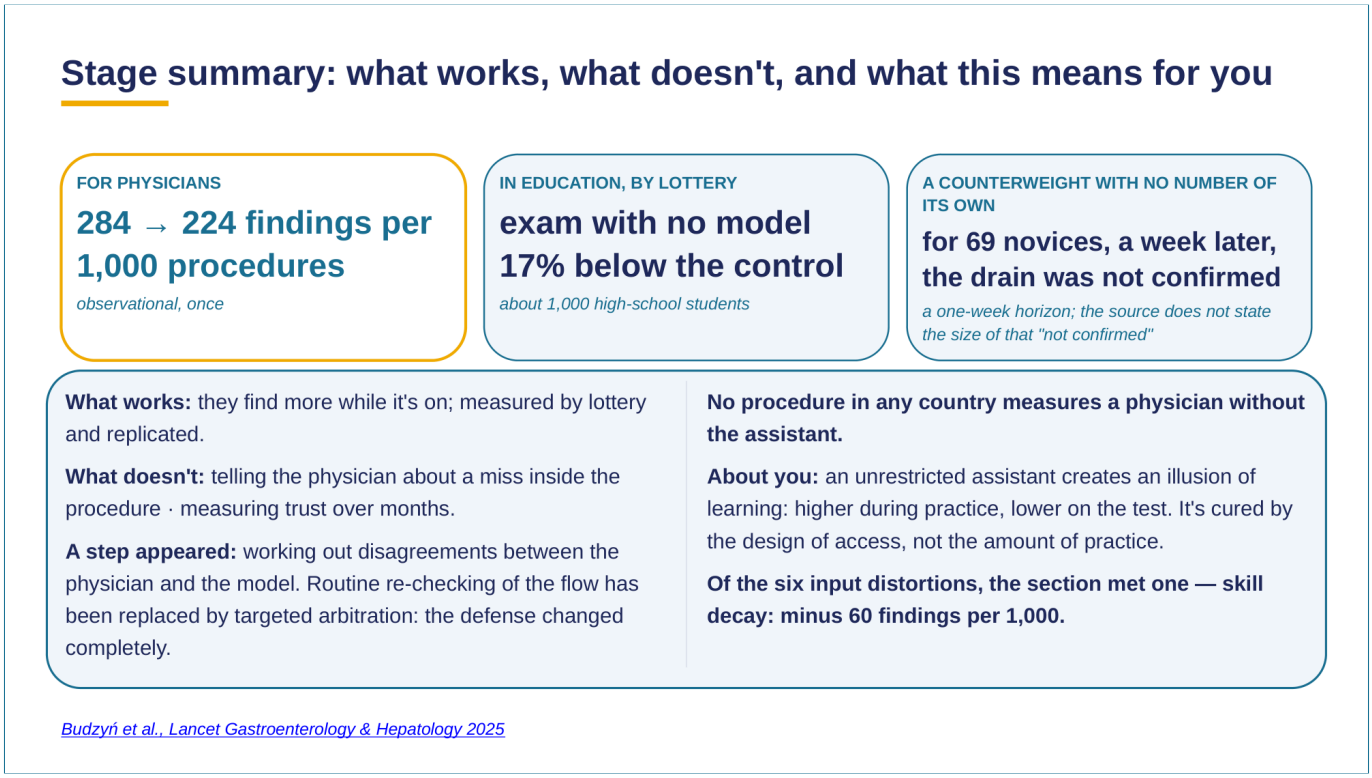
And now the reason this slide stands apart from the summary. The drain on skill at the

second spot is not measured — and cannot be measured. Not because nobody took it on, and not because the measurement is hard: there's nobody there to measure. The person who would have read second, and whose skill could be compared "before" and "after," no longer exists at this spot. This is important to distinguish from every other gap we've named today. Those gaps are a property of our knowledge: there is no procedure for it, but one could be set up. This one is a property of the decision that was made: as long as the spot is occupied by the model, there is nothing to measure, and no future work will fix that.

From this comes a formulation worth stating verbatim: skill is drained where the person formally stayed. As long as he is at the spot and still doing the work, the volume of his own practice quietly declines, and that decline can be caught — the Polish study caught it. Where he was removed, there is nothing left to decline, and worrying about erosion there is pointless: what's worth worrying about instead is who will read if the model is recalled tomorrow, and where the people will come from who had nothing to train on. We already saw this question today, in cytology: the volume dropped, the centers were cut from forty-five to eight, there was nothing left to train on, and there was no model at all in that story.

Separately, about what this slide does not claim. It does not say the second mode is better or safer. It says that the cost of the two modes differs in nature, and it is measured in different quantities: for the first, by the finding rate for the same person; for the second, by how many people are left in the profession and what they're being trained on. These two costs cannot be compared by one number, and attempting such a comparison would be exactly what we've spent the whole evening examining other people's work for.

And last, before the stage summary. Next we will put the whole stage together — what works, what doesn't, what of this transfers to engineering work in general — and the distinction between the two modes is not repeated there: it is made here, and the summary rests on it.



The conclusion of the skill stage: with the assistant on, they find more, and that gain is measured by lottery and replicated; the assistant cannot maintain skill, tell the physician about a miss inside the procedure, or measure his result without itself.

And the part of the stage that touches the audience directly. The same phenomenon is measured by randomization outside the clinic, in education: a field study on about a thousand high-school students, in math, three groups — a control, free access to a model, and a model with teacher-set guardrails that cues instead of answering. During practice, free access gave plus 48 percent over the control, and the guardrailed model plus 127. But on an exam with no access to the model, the free-access group came out 17 percent worse than the control, while the guardrailed group came out about at the control's level. This reads as: practice with an unrestricted assistant creates an illusion of learning. And right alongside it, a remedy is measured: guardrails remove the effect, meaning this is cured by the design of access, not the amount of practice.

Alongside it, in the same genre of transfer, a randomized experiment on 216 members of chess clubs: free access to help gave plus 30 percent of a gain, against plus 64 where the system chose the moment to cue, and the free group played 24 percent fewer training games. The cost is paid by the right to decide for yourself when to ask for help. This is a working paper, not a peer-reviewed publication.

And an honest counterweight, without which the axis turns into a sermon: in a randomized experiment with sixty-nine novices who learned with an assistant and took a test without it a week later, deskilling was not confirmed. Medium strength, teenagers, entry-level tasks, a one-week horizon. This cannot be called a refutation of the clinical finding — just as the school experiment cannot be called a replication of it: a different study, different people, a different task. Agreement of sign in one direction

and divergence in another mean exactly what they mean: the mechanism is similar, the magnitude does not transfer, and the measurement's horizon decides the outcome.

And what is completely absent. There is no replication of the clinical finding on real-world work, and this gap is named with a number: twenty-two separate search attempts — eighteen query phrasings and four targeted visits to pages; a number about the search, which says nothing at all about the subject itself. A longitudinal measurement of skill among those who learned with the assistant from the start and never knew any other way of working does not exist at all. Nobody has data about the people in this room — neither about skill that has dropped, nor about skill that held. The clinical finding does not fill that void: it was taken on specialists with nearly thirty years of experience, which for this audience still lies ahead.

Let's put the stage together by the same four questions, and close with our own gaps.

What works. With the assistant on, they find more: the gain is measured in randomized trials and replicated many times over, and hiding this half would mean doing exactly what vendors were examined for, only in reverse.

What doesn't. Maintain skill — here the direct opposite is measured. Give the physician feedback about a miss — there is none inside the procedure, and that is exactly why the drain is silent. Measure trust over months of real work — nobody has managed that: everything that exists was taken in a single session.

What happened to the work. Two forms from the glossary. The performer is the same, but got worse: the place, the volume, the order of steps, and the hand are unchanged; what changed is the condition of the hand. And a step appeared — twice: a mediator, translating the model's output into language the physician will trust, and targeted arbitration of disagreements.

What holds this in place. The detection rate and withdrawal time — both quantities are taken straight from ordinary work. And an honestly named empty spot: there is no procedure in any jurisdiction that measures a physician's result without the assistant.

There is one practical consequence, and it transfers beyond medicine: when you put an assistant in the loop, you have to measure both the result with it and the result without it, and not just once. The profession has an indicator for such a measurement, but no procedure — and demanding more from one observational study than it carries would be the same error as demanding work out of a claimed accuracy with no denominator.

Sources. Budzyń et al., Lancet Gastroenterology & Hepatology 2025

About the decision flow — what changed in it besides the findings. A mediator role appeared: the model delivers an output, and a person translates it into language the physician will trust, and this role is new for the department. Alongside it is a measured relationship, and its weakness is named right here: trust in the output is higher among those who work with the model by hand — point estimates exist, statistical significance does not. Routine re-checking of every case has dropped away in the meantime, and targeted arbitration of disagreements has taken its place, meaning the structure of

defense against error changed completely, rather than shrinking.

About the change in the structure of defense against error, folded into one line on the slide. Routine re-checking of every case and targeted arbitration of disagreements are not the same thing done cheaper — they are different defense mechanisms: the first catches errors uniformly and cheaply per unit, the second catches errors only where the two sides diverged, and each review there costs more than reading does. The shift from the first to the second changes not the volume of work, but which errors even enter the field of view of the check at all.

And the transition to the evening's wrap-up: next come the nine glossary forms in full, the count of misattributions, the cumulative map, and a named answer to the question the evening began with.

6. Wrap-up

The lecture examined cases one at a time. Here they add up into a tally: each summary answers its own question.

1. **The nine forms in full: what happened to the work**

2. **The count of misattributions: in thirteen of twenty-one cases, the win does not belong to the model**

3. **The map of tasks and data types: four cells out of twelve, and what works in them**

4. **What was paid: attention, skill, the signature, arbitration**

There is one new number here — the count of misattributions. The section adds not a single new case: everything else has already been presented, each with its own base, and it is combined here so that the sum reveals what no single case shows on its own.

There is no answer here to the question the evening began with: the closing section opens with one line about what the evening did not measure, and answers it on the next slide.

Wrap-up. The lecture examined cases one at a time; here they add up into a tally — four summaries: the nine glossary forms in full, the count of misattributions along the through-line, the map of tasks and data types, and a ledger of four payments. The section adds no new case and brings exactly one new number — a count of how many times a win turned out to be misattributed, and not to the trained model. Everything else has already been presented, each with its own base; the section's job is to add up what's been presented so that the sum reveals what no single case shows on its own.

The order of the four summaries carries weight here, and it's worth stating out loud. Four in a row is already a lot, so they come in different forms: a table of glossary forms, a strip of plates, a map of tasks and data types, a payment ledger. The reason isn't aesthetics: an earlier pass of this section ran five summaries of the same type in a row, and the room stopped watching after the third. I'll say plainly what spoils this layout too: three of the four forms are tabular, and only the design of their columns tells them apart, not the genre itself. This is the cost of shortening the evening's tail, and I chose it over ten minutes of ledgers before the finale.

The first answers what happened to the work: the nine glossary forms, which stood as names at the opening of the evening, now each carry a count. Seven of the nine forms carry a number, two stand with a dash, and I call both dashes dashes.

The second answers whose result this is, and it's the only one carrying a new number. Of twenty-one detailed breakdowns, in thirteen the claimed win or the claimed blame belongs not to the trained model: in two, the model isn't in the mechanism at all; in seven, the result belongs to the threshold, the data mix, the target set for training, the implementation, or the volume; in four, the number changed its subject on the way from the primary source to the retelling. There too, as a separate line — seven Russian

breakdowns out of that same twenty-one.

The third lays everything examined onto the course's own grid and immediately cuts it the other way: not by stage, but by task type, because one type runs through several stages, and one section contains several types. The main thing there isn't the occupied cells — it's the eight empty ones out of twelve, three spots outside the window's edge, and one section that occupies nothing in the window at all.

The fourth answers what was paid, and this is the only place all evening where the axis's four addresses are named together, with a measurement under each. It also closes the thread opened at the start with six named distortions: all six were met, and exactly one was named by its introductory mechanism all evening long.

And the seam forward, which has to be said out loud, not swallowed: there is no answer in this section to the question the evening began with — it stands as the first slide of the closing, right after these four, and that is also where the woman-with-the-letter thread is closed.

The nine forms in full: what happened to the work

	form	tally
1	The load moved to another step	note time **−9.5%** vs. control, editing unchanged
2	The step disappeared entirely	**no case encountered**
3	A step appeared	arbitration **3.9** → **9.4** and **11.3** → **13.8** out of 100 reads
4	The load grew at the same spot	**18** alerts per 100 hospitalizations · **1.23** wasted per study
4a	The load fell at the same spot	reading down **44.3%**
5	The performer changed	turnout **22** → **100** out of 100 · findings **413** → **472** per 1,000
6	*The order of work changed*	**no tally**
7	The cost of error shifted	on dark skin, worse **by 9–15** out of 100 · more chronic conditions **by 26** out of 100
8	The performer is the same, but got worse	findings **284** → **224** per 1,000
8a	*The performer is the same, but got better*	*no measurement*

Seven of the nine forms carry a tally; both dashes are named as dashes. Which of the nine, in your view, came up most often this evening — and which never came up at all?

the tally is compiled from the lecture's breakdowns

The nine forms of work relocation — in full, in one table, with a tally under each form. Seven forms carry a number, two don't, and both dashes are named as dashes. The glossary was handed out by name at the start, with no numbers, and has held up every breakdown since: on every breakdown, instead of the image "the load moved rightward," there stood a named form with a number. The image lies wherever the work didn't move but doubled, disappeared, or stayed in place and dropped in quality — and nine lines are needed exactly so these cases don't blur into one.

The glossary splits by question. The first five measure the place and volume of the work: where it is now, and how much of it there is. The second four measure the hand, time, the consequence of an error, and the condition of the hand itself: who does the work, when, and what is paid for a miss. This split was announced on the axis at the start, and is only confirmed here: the form answers what happened to the work; the address answers at which human step it happened.

Let's go through the load-bearing lines. The load moved to another step — recording the visit: note time dropped by nine and a half percent against the control group, while active editing time did not change for either of the two tools. That is, typing got shorter and review stayed exactly the same — the work moved from typing to reading someone else's text, and this is the evening's clearest case of the first form.

A step appeared — arbitration on disagreement, where the model replaces the second reader, and this is the glossary's strongest line. Two National Health Service centers in England, fifty thousand women: without the model, 3.9 percent of reads went to arbitration in one center and 11.3 percent in the other; with the model, 9.4 percent and 13.8 percent. Almost threefold in the first and by a quarter in the second, and this spread is not noise — it is a difference in protocol. Alongside it, a Norwegian

measurement on 441,000 cases: one primary read takes forty-eight seconds, arbitration on disagreement two minutes twelve — that is, one arbitration costs about as much time as roughly five reads.

The performer changed — two pillars here. The first: imaging the fundus at the visit instead of a trip to the ophthalmologist, with turnout rising from 22 out of 100 to 100. The second, domestic: a Russian assistant for colonoscopy takes over part of the search, and procedures with a finding become 472 per 1,000 against 413. This same case also carries the fourth form's second pillar: 1.23 false alarms per study where there used to be zero — the load grew for the same physician at the same spot. The fourth form's first pillar — the sepsis model: an alert comes in for 18 out of 100 hospitalizations, and 12 out of 100 raised alerts are confirmed, and the two are counted against different bases.

The load fell at the same spot — the volume of human reading in the Swedish program is lower by more than forty-four percent, with more cancers found, eighty thousand women. The flip side of this same form — narrowing of visual search under a false cue: the volume of work fell, and that turned out to be a cost, not a win.

The cost of error shifted — two pillars, and both about a subgroup. On dark skin, three models tell sick from healthy worse by nine to fifteen points out of 100, and the fix for that is the composition of the training set, not the algorithm. At an equal screening score, Black patients have more chronic conditions by 26 out of 100 — the score labeled as equally healthy people whose health actually differs. The average across the sample stayed decent in both cases; the subgroup pays.

The performer is the same, but got worse — two pillars, and both have already been voiced. Surgeons with a model's cue on septic shock: 0.712 against 0.820 without the cue. And the one measurement of erosion on real clinical work: for the same nineteen physicians, procedures with a finding became 224 per 1,000 against 284, minus sixty findings per thousand.

Now about the two forms with no tally, and papering over them with a similar case is forbidden. The removal of a step with no new object appearing has not come up once this lecture: three candidates were checked, and in each one a new step appeared in return. The second human reader left the workstation, but in return a step appeared — deciding whether this study needs a second read at all. Routine re-checking of every case went away, and arbitration on disagreement, counted in minutes, took its place. Typing shrank, and reviewing someone else's draft appeared. You cannot claim from this that "the model doesn't remove steps": such a claim would be broader than anything the lecture has measured. The form measuring the order of work has no tally for a different reason: the order in these breakdowns did get rearranged — for example, who reads first, the human or the model — but it was never counted as an order in a single case.

And the distinction the eighth form rests on, because without it you'd confuse it with its neighbors. In the fifth form, the hand changed; here the comparison is built inside the same nineteen physicians, and the measurement directly rules out a change of

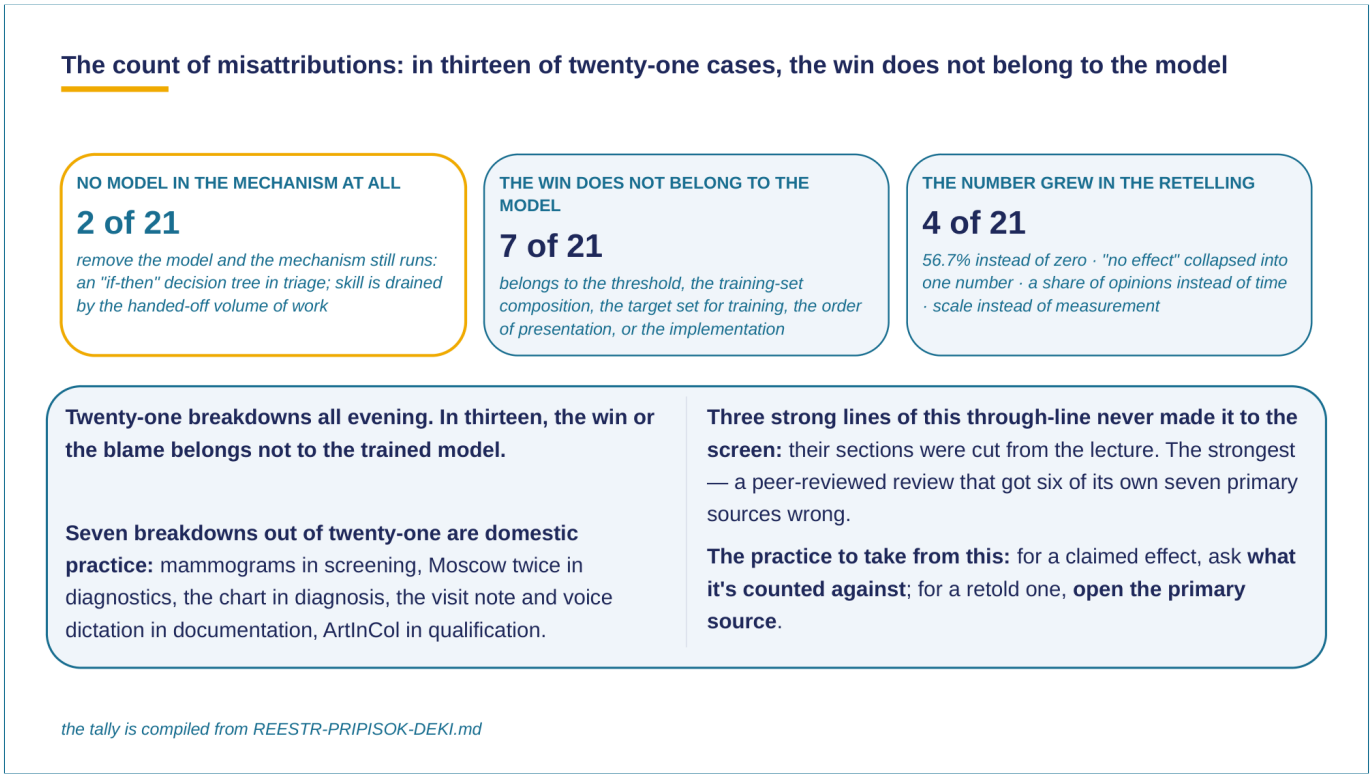
performer. In form four-a, the volume of the operation fell; here the volume is the same, and the finding rate fell — meaning these are different units of measurement. A quick check is the question of the denominator: form four-a has none at all, it's an absolute volume; the eighth has one, and it didn't change. The fifth and eighth stand on the same breakdown, and that is not a contradiction: colonoscopy with the assistant has two states of the loop — while the assistant is on, the model does part of the search, and when it's off, that same hand finds less than it found before the assistant arrived.

The question for the room is worth asking and waiting for an answer to: a form number from your own work gets named in a few seconds, and the named number then carries the rest of the conversation.

Sources — by form, for the ones carrying a tally. Form 1, note time: a three-arm randomized trial, medRxiv preprint 2025. Form 3, the share of reads sent to arbitration and the labor-minutes cost of arbitration: Warren et al., Nature Cancer 2026 and Hovda, Holen, Hofvind, European Radiology 2026. Form 4, alerts fire often and are rarely confirmed: Wong, Otles et al., JAMA Internal Medicine 2021, with an erratum from the same journal. Form 4a, two implementations with opposite signs: Taib et al., Radiology 2026 and Gommers et al., Radiology 2025. Form 5, turnout and reading volume: Wolf et al., Nature Communications 2024 and the MASAI safety analysis, Lancet Oncology 2023. Form 7, both pillars: Daneshjou et al., Science Advances 2022 and Obermeyer et al., Science 2019. Form 8, both pillars: Budzyń et al., Lancet Gastroenterology & Hepatology 2025 and model against surgeon intuition, Journal of Trauma 2023. Forms 4 and 5, the second pillar for both: Achkasov, Shelygin et al., Koloproktologiya 2025; 24(3):12–21. Forms 2 and 6 have no source as a matter of substance: the first has no case to cite, the second has no tally, and that is stated on the slide directly.

What each form measures:

- **The load moved to another step** — place - **The step disappeared entirely** — place - **A step appeared** — place - **The load grew at the same spot** — volume - **The load fell at the same spot** — volume - **The performer changed** — hand - ***The order of work changed*** — *order* - **The cost of error shifted** — cost of error - **The performer is the same, but got worse** — hand - ***The performer is the same, but got better*** — *hand*



Twenty-one detailed breakdowns all evening, and in thirteen of them the claimed win or the claimed blame belongs not to the trained model. This is the one new number of the evening, and it's about us. Every other number in the lecture has already been presented, each with its own base; here what's added up is something no single case shows: how often, while examining a model's deployment, we got down to the mechanism and found that the win or the blame belonged to something else.

I'll state the denominator plainly, because it can be checked. Twenty-one is the detailed breakdowns, the ones each section's divider names by number: four in screening, five in instrumental diagnostics, six in diagnosis, three in documentation, three in qualification. A slide titled "Example X, cost" or "Example X, mechanics" is part of its own breakdown, not a separate case — otherwise the denominator would be double-counted. Thirteen is the breakdowns where the slide itself stated exactly what the result belongs to. You can check this by opening the slide: find the line about the mechanism and compare it against what I'm saying right now.

The first column is the simplest and the most unpleasant. The model isn't in the mechanism at all, yet the conversation is about it. First-line intake triage: an exam score of 81 against 72 for physicians, more than a billion dollars invested, a valuation north of four billion — and inside it, by the account of the person who was responsible there for regulatory matters, an "if-then" decision tree in a spreadsheet. There is one check here, and it always works: remove the trained model. If the mechanism keeps running, then it was running without it all along. The second case in this same column is subtler, and closer to the room: skill decay. We're used to thinking it's the model that drains skill — but in UK cytology, the volume of reading dropped 80 to 85 out of 100 after switching to a different lab test, and the number of centers fell to eight out of forty-five. It is the handed-off volume of work that drains skill. The model is just one

way to hand it off, and far from the most powerful one.

The second column is the most populated, seven breakdowns out of twenty-one, and the model is present in it. But the result does not belong to the model. The savings in the Moscow triage of screening scans did not come from reading getting cheaper: one scan became more expensive for the human reader, and the savings came entirely from that same fifty-five scans out of a hundred that nobody reads at all. The accuracy gap on dark skin belongs to the composition of the training set. The unfairness of selection into the high-attention program belongs to the target variable set for training: the model was trained to predict cost, and the decision made was about illness. The sign of the effect of a cue to a radiologist belongs to the order it was presented in — the same quantity, a different order, the opposite sign. The effect of automatic visit recording belongs to the implementation, not the class of tool: of two tools in one trial, the second showed no change at all. And the sepsis model's class is a penalized logistic regression, not the deep network people assume it to be.

The third column is about those who retell, and for an engineer it is the most practical one. Here the primary source is often careful, and the error is born along the way. 56.7 percent of double-reading workload saved is a recalculation under a scenario the authors themselves called hypothetical, while in the actually deployed practice the model replaced not a single human read — that is, zero. 62.8 percent in the domestic voice-dictation case is the share of physicians who think so, not a measured time. "No effect" for pathologists is the result of a report collapsed into one number, when the group's gain is not significant and the effect runs fourfold in either direction. Scale of deployment, the largest in the world by claimed numbers, is presented as proof of quality, when there is no measurement of the consequences.

I'll say something separately about the seven domestic breakdowns, because this is the very practice part of the room may be working with by hand tomorrow. Moscow, twice: half a million screening scans with autonomous triage, and three and a half thousand head CT scans, where autonomy was specifically withheld. The visit note that a voice assistant fills out, and voice dictation of the radiology report. And the colonoscopy assistant, tested by lottery on more than a thousand patients in Moscow and Ufa. Two were added last: a model in the second-reader spot for mammogram screening, and a model suggesting a final diagnosis from the chart. They sit unevenly across sections, and I'll name that too: two sections have two domestic breakdowns each, three have one, and no section is left without a domestic breakdown.

And about the three lines that never made it to the screen. I name them not for thoroughness's sake, but because this is exactly the measure of how much of the through-line was lost when the lecture was cut down. The strongest — a peer-reviewed review on deskilling that got six of its own seven primary sources wrong: eight residents became thirty-two, 1,443 colonoscopies became "more than twenty-three thousand," an observational study was called randomized. The cost of this line lies in the grade of the source: this is a peer-reviewed review, and it is wrong about its own sources — meaning the error lives higher up than a blog post, at the exact spot a reader usually stops checking. The second — "study-reporting time dropped from twenty minutes to

three seconds": three seconds for a physician to prepare a report is physically impossible; this is, in all likelihood, the model's own processing time, presented as the physician's time. The third — "the early-warning model lowered mortality," where the effect belongs to a new role in the process, a dedicated rapid-response nurse, not to the quality of the model.

The practice the room takes away is worth saying twice — here, and on the last slide. For any claimed effect, ask what it's counted against. For any retold one, open the primary source. These are two different checks, and the second one is cheaper than it looks: in three cases out of four in the third column, it was enough to open the source itself and check the sample.

The map of tasks and data types: four cells out of twelve, and what works in them

Four rows of tasks against three data types — twelve cells. Four are named; nobody went into the remaining eight across five sections.

task, and where it came up	what works	what doesn't
classification — image: screening, diagnostics	a standardized scan, fitness-for-analysis settled before the model	the subgroup underrepresented in the training set: it pays
recognition — image: screening, diagnostics, diagnosis	a flow filter: who gets read twice	grounds to drop a reading: savings from what goes unread
retrieval and generation — text: screening, diagnosis, documentation	a draft ready by the end of the visit	more complete than a human: the same person writes it, but the reader pays
forecasting — structured data: diagnostics, diagnosis	a signal alongside the scale	the decision: the target variable sets the aim, and a document sets the threshold

Three more spots lie outside the window. Speech at the visit: typing by voice works; word endings and other people's speech do not. Video: highlighting during the procedure, with the spot staying with the person. Instrument signal: there is no trained model at all, yet there is bias.

The right column, in not one row, is about the quality of the model — it is about the fitness of the input, the composition of the data, the target set for training, and the assigned threshold. **Qualification occupies nothing in the window:** its assistant works on video.

Four rows of tasks against three data types give twelve cells. Across five sections, four are named; nobody went into the remaining eight. The material is the same as in the five stage conclusions, cut the other way — by task type. One task type runs through several stages: image recognition came up in screening, in instrumental diagnostics, and in diagnosis; forecasting on structured data came up in diagnostics and diagnosis. By stage, each one seems to have its own story; by type, you can see that the boundary between "works" and "doesn't" for a given task type is the same everywhere.

Twelve cells are a window into the course's full forty-two-cell grid, taken exactly to fit what the lecture examines in the clinic: retrieval and generation are combined into one row, there is no planning, just as there is no audio, video, code, or mixed data. The count can be checked right off the screen: each row names exactly one data type, there are four rows — so four cells are occupied, eight are empty.

The eight empty cells carry just as much content as the four occupied ones. The lecture has no classification or recognition on text at all; no retrieval or generation on images or structured data; no forecasting on text or images. There, either there is no tool of that class in the clinic, or there is one, but nobody holds the cost of its error in place with a procedure — and with no procedure, there's nothing to examine. An honest empty cell is better than a made-up occupied one.

The tools of five sections fit into four cells. Classification on images is occupied twice: by screening, which images the fundus, and by instrumental diagnostics, which assesses a skin lesion. Recognition on images, three times: screening with mammograms, diagnostics with triage of the flow in X-ray and CT, diagnosis with gaze tracking over the same mammogram. Retrieval and generation on text, three times: pre-visit intake triage, a language model reading case descriptions, and recording the

visit with the discharge summary. Forecasting on structured data, twice: selection into the high-attention program and the early-warning sepsis model. The class of model does not distinguish the sections; what distinguishes them is the cost of a miss and the procedure built around the person.

Now through the table's rows, with numbers. Recognition on images works as a flow filter: it is capable of deciding what to flag and who gets a second read — the Swedish program measured this by lottery on more than a hundred thousand women, the Moscow triage on half a million scans. What it does not give is grounds to drop a reading, and that is visible in the Moscow case with the arithmetic right on the slide: one scan became more expensive for a human reader, and the thirty-three million in savings came entirely from that same fifty-five scans out of a hundred that nobody reads at all. The savings come from what goes unread, not from accuracy.

Classification on images works where the scan is standardized and the question of its fitness for analysis is settled before the model. Fundus imaging at the visit showed this: turnout rose from 22 out of 100 to 100. But right there stands the boundary too — in the field, a person decides a scan's fitness with no rule, and nobody watches the reject stream. And the second case of the same type, skin lesions, shows a second boundary: the average over the validation set stayed decent, and the subgroup underrepresented in the training set pays.

Retrieval and generation on text works as a draft ready by the end of the visit, and it did remove one real step — the document's structure no longer has to be held in your head while a living person is there. It does not write more completely than a human: in the blinded comparison of a hundred hospitalizations, the machine-written discharge summary came out shorter and more coherent, but less complete. And the cost shifts here: the same person writes it, but the reader pays — the one who opens the document a month later, having never seen the patient.

Forecasting on structured data works as a signal alongside the scale, not in place of it. A comparison against zero gives the wrong picture here: the scale was confirmed by 8 alerts out of 100, the model by 12, and it never left that band. As a decision, forecasting does not work, and the reason is an engineering one: the target is set by the variable it was trained on, and the threshold is set by a document. Selection into the high-attention program is the clearest case: the model was trained to predict cost, and the decision made was about illness. And separately, about why this particular cell is populated: structured data there is already gathered by someone before the model ever gets involved — orders written up, bedside vitals, treatment bills. The model arrives where the numbers are already sitting, and this observation transfers outside medicine almost without edits.

Three spots lie outside the window, and without them it reads as an oversight in selection. Speech recognition — two domestic breakdowns from the documentation section: the visit note that a voice assistant fills out, and voice dictation of the radiology report. It works as typing by voice instead of by keyboard, and forty years before this, the same technology lost to the keyboard: in 1981, voice input took four times as long

and produced twelve errors against zero. It does not work — on word endings and on other people's speech in the room, when more than the physician ends up in the recording; and nobody measured time here — the sixty-two-plus percent is the share of physicians who think it got faster, not a measured time. Recognition on video — real-time highlighting for those who decide: the Polish trial and the domestic ArtInCol. And a measurement with an instrument signal, where there is no trained model at all: bias in a blood-oxygen-saturation sensor arrives before any model, and the system that takes its reading as input inherits the bias silently.

The qualification section is absent from the window for the same reason: the assistant that boxes a finding during the procedure works on video in real time, and video is not among the twelve cells shown — it has a place in the full grid, just outside the edge of this window. What matters here is that the one section all evening that measured the person, not the work, landed on a data type the window doesn't contain.

The third layer of this table is what, in each cell, holds the cost of error in place. The procedures differ across the four cells, and none of them match. For image recognition, what holds it in place is double reading, arbitration on disagreement, and recalculating the true-alarm rate against local prevalence. For image classification, it's the fitness of the input, settled before the model: a system that itself rejects an unfit scan, and a one-year repeat visit for those who came back clean. For forecasting, it's a review of false alarms, counted as working time, and a pre-approved plan for changes to the model. For text, it's physician review and signature, case review, and the stated boundaries of use. From this comes a rule that reads straight off the order of these rows: the later a check sits in the procedure, the longer an error lives. On images, it gets caught within the shift; on text, at the signature; on forecasting, at the next cut of data handed to the regulator. And wear on the hand gets caught nowhere: the review of authorization across three countries showed that there is no procedure anywhere that measures a physician without the assistant — not in the UK, where the work result counts toward authorization as a number, not in the US, where the same indicator exists and is declared off-limits as authorization, and not in Russia, where hours of study are counted. You cannot write "re-certification" in here — no such procedure exists in any jurisdiction.

And the conclusion that transfers outside medicine almost without edits. Look at the whole right column: not one of its rows is about the quality of the model. Fitness of the input, composition of the training data, the target set for training, the assigned threshold, who reads the document afterward — this is all engineering work built around the model, and it is exactly this work that decides whether the deployment succeeds or not. The quality of the model is not mentioned in this list, not once.

What was paid: attention, skill, signature, arbitration

The four payments from the start — by name, with a measurement under each.

payment	what it cost	how it was measured
Attention	scanned area of the image **11.1** → **9.5%**; missed cancers **29** → **61** out of a hundred	eye-tracking study; subgroup of 14 cases
Skill	findings **284** → **224** per thousand procedures, same nineteen doctors	observational, measured once, and disputed
Signature	**9 of 11** machine-written discharge summaries contained an error	one hundred hospitalizations, two blind readers
Arbitration	readings sent to arbitration up **threefold and by a quarter**; one review takes **as long as five readings**	two centers, 50,000 women; 441,276 cases

Paid unevenly, and the order of magnitude matters more than the figure: arbitration has two trials and the evening's largest samples; attention has a subgroup of 14 cases; skill has one unreplicated observation; signature has a share of documents, not an outcome.

Who pays: screening — in arbitration · diagnosis — in attention · documents — in signature · qualification — in skill. **Instrumental diagnostics pays none of the four:** its cost is 290 missed findings, a different plane entirely.

All six distortions from the front door showed up: anchoring and narrowing — in attention, decay — in skill, correction — in signature, disagreement — in arbitration; alert stream got no column of its own. **Exactly one is named by its own name over the evening.**

The four payments of the axis — attention, skill, signature, and arbitration — are collected here by name: under each one sits what it cost, how it was measured, and which section pays it. The vocabulary of forms tells you where the work moved; this ledger tells you what happened to the person doing it. The row order is the same one the addresses were announced in at the start: the promise and the report on it read in the same order.

Attention. Visual search narrows under a wrong suggestion: the scan area covered by eye fixations falls from just over eleven to nine and a half percent, and in human terms that reads as: where the suggestion stayed silent, missed cancers rose to sixty-one out of a hundred that existed, against twenty-nine. The denominator sits right next to the number: a subgroup of fourteen cases, and it's significant. The order of evidence is a lab reading with eye tracking, replicated in direction, though in a different specialty the reading came out reversed. Alongside it sits a second figure for the same address: a correct decision flips to an incorrect one under a colleague's advice, a systematic review. Both are about where the person's eyes go, not about what they think.

Skill. Technique drops in work done without the assistant after people get used to it: from two hundred eighty-four procedures with a finding per thousand down to two hundred twenty-four — minus sixty findings per thousand, among the same nineteen doctors, an observational study, not replicated once. This work doesn't prove causation, and the authors don't claim it does. It's the weakest row in the ledger, and at the same time the only measurement of skill decay in real clinical work — both halves go together.

Signature. Nine out of eleven machine-written discharge summaries carried at least one error, and ten out of nineteen large fabrications; measured by a blind comparison of

one hundred hospitalizations, each document read by two people. What's costly in this row is the price of refusing to sign: a doctor who accepted a suggestion in favor of standard treatment is judged, in a case review, as more reasonable than one who rejected it — and that's the judgment not just of juries but of two hundred forty-eight German doctors, meaning the very side the procedure assigns to do the reviewing. For a non-standard recommendation, there's no difference.

Arbitration. Two centers in England's National Health Service, fifty thousand women: without the model, three point nine percent of readings in one center and eleven point three percent in the other went to arbitration; with the model as second reader, nine point four and thirteen point eight. Growth of nearly threefold in the first center and by a quarter in the second, and that spread is set by protocol: the first center sends only disagreements on recall to arbitration, the second sends every recall at all. A separate Norwegian measurement, on four hundred forty-one thousand two hundred seventy-six cases, counted this in seconds: one primary reading averages forty-eight seconds, resolving one disagreement takes two minutes twelve. One arbitration review, by time, equals about five ordinary readings — a direct answer to the evening's question: after the model, the person performs an operation that costs as much as five of their old ones.

On the unevenness of the bill. The ledger has to be read by its third column, not its second: a figure with no stated order of evidence misleads worse than no figure at all. Arbitration has two independent trials and the evening's biggest samples. Attention has a lab measurement on a subgroup of fourteen cases. Skill has one observational study nobody has repeated. Signature has a share of documents with an error, and what happened to the patients because of those errors, nobody checked. Four rows in one table, and the trust owed to them differs by orders of magnitude.

Who pays. Screening pays, and pays with the evening's best-evidenced number — arbitration. Instrumental diagnostics pays none of the four, and that isn't a gap in the ledger: its cost is measured, and it's large — two hundred ninety missed clinically significant findings across half a million scans — but it sits in the plane of the outcome, not the plane of the person deciding. The four axis addresses measure what happened to the decision-maker; a missed finding measures what happened to the patient. They cannot be put in the same column.

There are two planes missing from this foursome entirely, and both are worth naming so the ledger doesn't look complete where it isn't. The first is trust and consent: explanation raises agreement and doesn't raise decision quality, the visit record gets written faster while the proofreading stays. The second is the working time of a real person who wasn't on the schedule before the tool arrived: the alert stream is paid for in exactly that, not in attention, skill, signature, or arbitration.

And one line, without which the ledger would read as nothing but a list of losses. The difference in error structure between human and model produces a payoff for the pair — a pooled measurement across one hundred six studies, where the sign of the effect flips depending on which side is stronger on the task, and both intervals sit entirely on

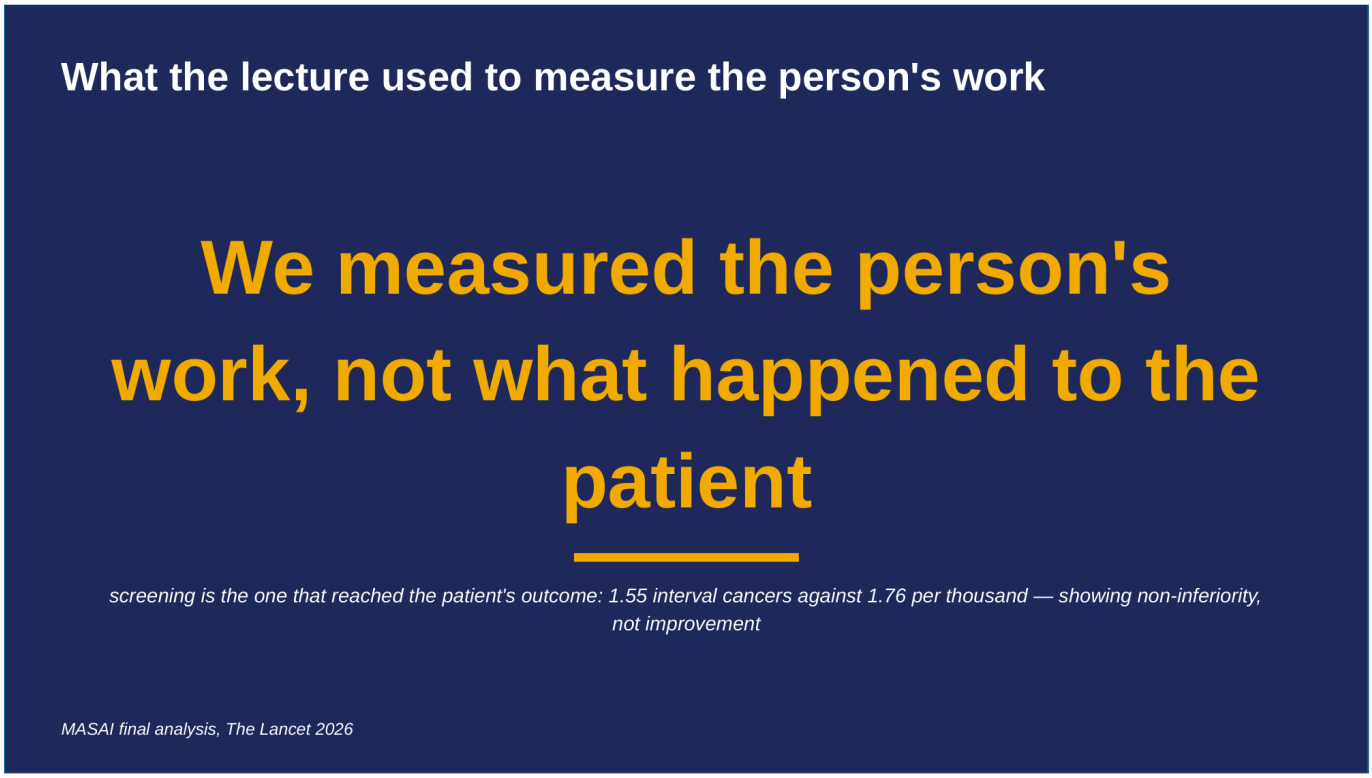
one side of zero. The row has direction, not magnitude: the share of overlapping errors hasn't been measured, and the figures that circulate for it don't hold up against their own primary source.

Significance figures not shown on the slide: the scanned area narrowed at p equals 0.004, and the drop in human terms at p equals 0.002. Both come from the lab eye-tracking measurement; the drop in detection share in the section on skill carries no stated significance in that work, and it should not be presented as if it had been measured.

And one gap that has to be named as a gap. There is no stage of patient consent in this lecture at all: the evening moved through the stages at which decisions are made about a person, and not one of them measured the person the decisions were about.

The thread closes with the same phrase the slide at the front door opened it with, because it hasn't changed over the evening — it has only gathered more grounds. Warning doesn't help: a warning is a message the person notes and then keeps doing what they were doing. What helps is action — considering the opposite before agreeing; arbitration that actually has time on the schedule; a procedure that measures the operator without the tool. Of these three, the lecture has measured the first, counted the second in minutes, and the third doesn't exist in a single jurisdiction.

Sources. The addresses of the measurements are in the breakdowns of the respective sections; volumes, pages, and work addresses stand in the footers of those slides.



Here I stop for a few seconds, and the stop has content, it isn't for breath. The four summaries that just went by counted up the evening: nine forms of relocated work, thirteen attributions out of twenty-one breakdowns, a map of tasks and data types, a ledger of four payments. All four answer the question "what happened to the work." Before answering the question the evening opened with, we need to say out loud what answer we don't have.

Here it is. Almost everything measured tonight measured the person's behavior: where the eye looks and how long it stays in the right area, what share of the scan's area the radiologist covered, how many findings the doctor made without the assistant, what they signed and in how many seconds, how closely their conclusion matched the machine's. These are real measurements, taken by real methods, and half of them are randomized. But their subject is one thing — the work, not the person the work is being done for.

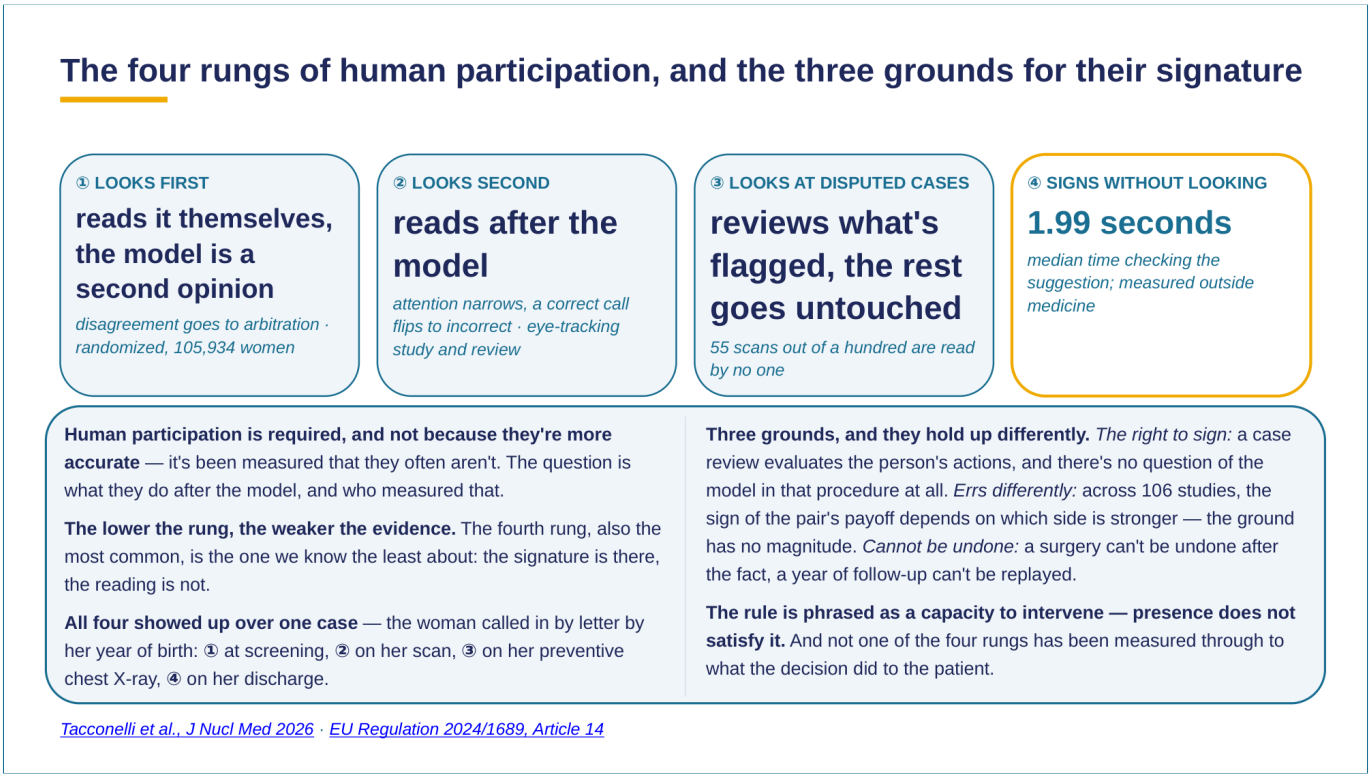
Now the caveat, and it's on the slide, not just spoken, because without it the line above would be exactly the kind of overstatement we've spent the whole evening catching vendors making. Two breakdowns reached the "what did this do to the patient" rung tonight, and the first of them was the Swedish screening trial: one hundred five thousand nine hundred thirty-four women, outcomes pre-registered. The result of that reaching has to be said in full, not halfway. Interval cancers — the ones found between two examinations, meaning the ones the program missed — came to one and fifty-five hundredths per thousand against one and seventy-six hundredths. The rate ratio is zero point eight eight, confidence interval from zero point six five to one point one eight. The interval straddles one. So what's shown is not improvement, but non-inferiority: the model-assisted regime is no worse than the old one, and this trial says nothing more than that.

The second is the surgeons' breakdown, and it, too, declared a primary outcome — one hundred fifty patients, thirty days of prospective follow-up. Its result stood on its own slide: on three outcomes out of four the pair was more accurate than the surgeon; on septic shock it lost, in the one place the human was stronger. So it did reach the outcome rung, and the verdict there was mixed.

Why this pause belongs right here, before the answer. The answer that's about to come is built as four rungs of participation and three grounds for the signature. All seven pieces of that answer talk about the person: what they manage to see, and what holds up the requirement that they stand in the loop at all. Not one of them says what changed for the patient because of it. The room has a right to know the boundary of the answer before the answer, not after.

And one last thing this pause does not mean. It does not mean the evening's measurements are worth nothing: an intermediate quantity is a normal rung of measurement, and the grading scale we've used all evening places it third of four, not at zero. What it means is something else: whenever a vendor says "the model improves outcomes," the question to them is the same one we just asked ourselves on this slide — at which rung was this measured, and does the interval straddle one.

Sources. MASAI final analysis, The Lancet 2026



The four rungs of human participation, and the three grounds this participation rests on, are brought together here into one ledger — first by rung, then by ground. The line about human participation was spoken at the front door as the lecture's own obligation; here it comes back earned.

The woman with the letter, who closes out the ladder of rungs, first appeared at the start of the first section: she was called in on one criterion, her year of birth, none of it her own. From there she has stayed with us the whole evening: the arithmetic of accuracy on two groups of ten thousand was computed about her, the wall between two independent readers held up her program, the report on her scan and the outcome line that will go out to her by letter sit in the fourth section. Her thread closes on the thought that pulls the whole evening together: the four rungs of participation are four positions one decision about one person passes through as it moves along the stages — not four different loops in four hospitals. At screening, the person standing over her case read first. At diagnosis, the person read after a suggestion. On her preventive chest X-ray, the person would have seen only what the model flagged, and fifty-five scans out of a hundred are read by no one. At discharge, a signature under a machine-written draft. And nobody chose her rung on purpose: it came out the way it came out, from how each procedure happens to be built.

The third rung is more dangerous than it looks: whatever the model doesn't flag never reaches the person at all, and they can only learn about what was missed from another source — the next exam, a complaint, a review after the outcome. In Moscow's sorting of preventive chest scans, this has actually been counted: two hundred ninety clinically significant findings turned up only because, within the study, every "normal" result was re-read by people — and in routine work there will be no such layer. You can check your own third rung with one question to the loop: what happens to what the filter didn't

show, and who's counting it.

The strength of evidence for the fourth rung. The median time spent on a suggestion before accepting it is about two seconds, and it was measured on programmers with a code assistant — that is, outside medicine; the transfer is named as a transfer and is not presented as a clinical measurement. What's accepted without a prior check gets corrected afterward in fifty-three percent of cases, against eighteen. The first rung, by comparison, rests on a randomized trial of a screening program with a hundred five thousand women. The order of evidence drops exactly as participation becomes more formal.

Why the fourth rung is the most common is explained by the price of refusing consent: a doctor who accepted a suggestion in favor of standard treatment is judged, in a case review, as more reasonable than one who rejected it — and that's the judgment not just of juries but of two hundred forty-eight German doctors, meaning the very side the procedure assigns to do the reviewing. For a non-standard recommendation, there's no difference.

From here follows a consequence worth naming before moving to the grounds. If the most common rung is the one where the signature is there and the reading is not, and if disagreeing with a suggestion costs more at review than agreeing with it does, then the formal requirement of human participation cancels itself out. It demands the capacity to object while simultaneously making objecting the most expensive move available. The fix is a change to how a case gets reviewed after the outcome: as long as the review rewards agreement, the loop will slide toward the fourth rung under any set of instructions, and training doesn't help here — a separate measurement showed it doesn't raise the rung. The fourth rung isn't anyone's fault: it follows from how the procedure is built, and it's fixed by the procedure. If confirmation has no time on the schedule, the signature will be there, and the reading will not.

Now the three grounds, and each is presented with its own ledger.

The right to sign is granted to the person. In a case review, the committee evaluates the person's actions, and there's no question of the model in that procedure at all. And it has been measured how rarely anyone checks whether the person is even capable of exercising that right: the capacity to intervene — that, not the mere presence of a person nearby — has been checked at 56 of 717 radiology devices, as of June 2024. And it is precisely the person who gets asked for it, which is not the same as saying "the person matters more."

Errs differently. This ground has no magnitude: the share of overlapping errors between human and model has not been measured, and the figures that circulate for it don't hold up against their own primary source. What holds the ground up is mechanism and structure: across 106 studies, the pair performs worse than the model alone where the model is stronger, and better than either side alone where the human is stronger, and both intervals sit entirely on one side of zero. Alongside it is a measured asymmetry in rejections: more accurate findings by the model were rejected more often than less accurate findings of the person's own. Together this confirms the direction — human

and model errors fall on different cases — and does not confirm the magnitude. That's enough for the ground: it claims the errors diverge, not on how many cases.

Cannot be undone. A surgery can't be undone after the fact, a year of follow-up can't be replayed, procedures carried out by a hand that has already slipped have already been carried out. A device, meanwhile, does get recalled, and this has been counted: of 950 cleared devices with a model, 60 were recalled, with 182 recall events among them, and 43.4 percent of recalls fall in the device's first year; as of November 2024. Decisions made on the device during that year are not recalled along with it. The two denominators here, 717 and 950, come from two different studies at two different cutoff dates, which is why each figure carries its own date.

And about the rule itself: it is phrased as a capacity to intervene, not as presence, and it does not require watching every decision. A person's presence near the loop does not satisfy it — which is exactly why "we have a human in the loop" says nothing on its own, until a rung is named.

And one last thing — about the rungs, not about the whole lecture, and the difference here matters. About the evening as a whole, you cannot say "there is no patient outcome in any breakdown": the Swedish screening trial did reach the outcome, and so did the surgeons' breakdown. But not one of the four rungs of participation has been carried through to it. What's been measured is behavior — where the eye looks, what the person signs, how many findings they make without the assistant. What happened to the patient because of it, nobody checked. This isn't a reproach to the researchers — a measurement like that takes years — but the lecture holds itself to the same order-of-evidence standard it demands of vendors. That's why the woman with the letter leaves the evening without an outcome: we know who stood over her case at every stage, and we don't know how it all turned out for her.

Sources. Tacconelli et al., J Nucl Med 2026 · EU Regulation 2024/1689, Article 14 · Lee et al., JAMA Health Forum 2025, device recalls · review of regulatory submission documentation, JAMA Network Open 2025

The device gets recalled; the decisions stay

WHAT GETS RECALLED

the device

the recall procedure exists — the regulator's record sits alongside

WHAT STAYS

the decisions made during its year in service

a procedure can't be redone, a year of follow-up can't be replayed

"In medicine everything is irreversible" is a claim wider than any measurement. **Devices get recalled: the regulator's record sits right here.** Decisions made on a device during its year in service don't get recalled along with it.

What happened to the woman with the letter, nobody measured. The same question for your own loop: what in it gets recalled, and what stays.

Next — an industry where an error can almost always be undone: the dispute there is over who owns the result; the cost of error barely gets a mention.

Lee et al., JAMA Health Forum 2025, cutoff November 2024 · FDA device recall database, record Z-2111-2021, October 2026

The last lecture ended with a line said to the room: a batch can be recalled, a decision about a person cannot. This lecture has worked that claim over with measured material — the claim holds, and there's an amendment that makes it stronger.

The first half holds literally, and even wider than it was stated: not only a batch of parts gets recalled, but a device with a trained model inside it too. The count that confirmed this sat in the oversight section and was cut along with it; what remains here is the checkable half itself — it is specifically the model-equipped device that gets recalled, and a record of such a recall sits right on the slide.

The second half is confirmed by two of the lecture's breakdowns. Decisions made on a device before its recall: they are not recalled along with the device. Technique that degraded in procedures done without the assistant — those procedures have already happened, and they can't be redone.

The amendment worth stating out loud: the recall exists, works, and has been counted. The device gets recalled — what's irreversible are the decisions made on it before the recall. Forty-three percent of recalls fall in a device's first year in service, and that year's decisions stay as they are. The phrasing "in medicine everything is irreversible" is wider than any measurement, and there's nothing to check it against. The phrasing "there's a gap between a device's reversibility and a decision's irreversibility, and it's measured" is checkable, and it can be applied to your own working loop: ask what in it gets recalled, and what stays after a recall.

And an honest caveat about scope: both remaining breakdowns sit inside a medical loop, where the decision was made by the person accountable for it. A third one, where irreversibility concerned a treatment decision about a specific person made by someone other than a doctor, was cut by the course owner's decision along with the

topic it belonged to, and there's nothing to strengthen that pillar with in its place. From here the course moves into an industry where an error can almost always be undone, and the dispute there is over who owns the result rather than over the cost of error: that's the shift, and the change in question is exactly that.

And about the woman the evening began with. What happened to her, nobody measured: not a single measurement of a person's long-term outcome after a decision made with the model's involvement turned up — not in her program, not in any other. Where a sixth step on the path should stand, there's nothing, and that was stated at the start. This is exactly the difference between a device recall and a decision about a person: the first is counted, the second isn't even tracked.

Sources. Lee et al., JAMA Health Forum 2025, device recalls

The number behind this measurement is no longer on the slide: the oversight breakdown that counted it and worked out its denominators was cut along with its own section. If you want to say the share out loud — it was a share of recall EVENTS, not a share of devices, and the two must not be confused; but there's nothing left here to present it with on a slide, and it shouldn't be passed off as shown.

"In medicine everything is irreversible" is a claim wider than any measurement, and it has to be checked the same way we check anyone else's claims. The recall exists as a procedure, and the regulator's record of one sits right on the slide: a device can be taken out of circulation. What's irreversible is something else — the decisions made on that device during its year in service. They aren't recalled along with it, nobody revisits them, and the person something was done to, on their basis, usually isn't notified of the recall at all.

So the question to your own loop is put exactly this way: what in it gets recalled, and what stays. A device, and a batch, can be recalled; a decision made before the recall, and a procedure already carried out, cannot be undone.

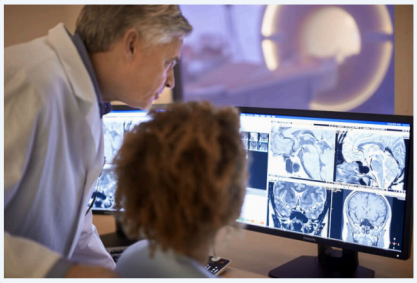
And the transition to the next lecture. From here the course moves into an industry where an error can almost always be undone: roll back a version, rewrite the code, restore data from a backup. The dispute there is over who owns the result — who's the author, whose work, whose rights — and the cost of error is barely heard in it. This comparison is worth stating plainly: not "it's easier there," but "the asymmetry is different there, and it makes different things expensive."

How often this happens, this one hasn't counted.

Three checks for any industry — and a question

Is there an independent source of truth besides the model's own explanation? A different instrument, a different analysis, a different person, or the thing itself. A check against self-explanation is not a check.

Has it been measured that they object — and that they object correctly? Two countable rates. Neither is measured — "the person checks" is no different from "the person signs."



Can they physically keep up? Seconds for the decision against the volume to read. The median time checking a suggestion is the same **1.99 seconds** that came up on the evening's axis; what's accepted without a check gets corrected afterward in **53% of cases against 18**.

If you were putting a model into your own loop tomorrow — which one check would you make mandatory before switching it on?

The three lines on the slide are what the lecture leaves in your hands. Each one is checkable against a specific system, and each has already been applied today in a breakdown — here it's only given its own name.

The first check separates a check from its appearance of being one. If the only thing a person checks the model's output against is the model's own explanation, there's no check at all: an explanation buys agreement and does not buy quality, and that's been measured. A simplified explanation of a wrong recommendation dropped choice accuracy below what the same person gives with no model at all. An independent source of truth is a different instrument, a different analysis, a different person, or the thing itself; self-explanation isn't on that list.

The second check turns the word "control" into a magnitude. Any loop in which a person confirms the model's decision has two countable rates: how often they object, and how often they object correctly. If neither is measured, the claim "the person checks" is no different, in any observable way, from "the person signs." In the lecture's breakdowns, this distinction showed up twice. Where the suggestion arrived before the person's own conclusion, they objected less often and worse. Where the procedure was restructured so the person only looked at disagreements, objecting became a separate piece of work — and its volume turned out to be countable.

The third check is about physics, not diligence. A check has a duration, and it gets compared against the volume that has to be read for the check to actually happen. The measured median for checking a suggestion is about two seconds; a suggestion can't be read in that time, and what's accepted without a prior check gets corrected afterward in 53 percent of cases against 18. The measurement was taken outside medicine, on programmers with a code assistant, and the transfer is named as a

transfer. From here follows a conclusion that carries over to any industry: the requirement "the person confirms" is only achievable when confirmation actually has time on the schedule. If it doesn't, the signature will be there and the reading will not, and that gets fixed by the procedure, not by attentiveness.

As a separate piece of evidence, from a different subject and a different denominator, alongside the third check sits a census of cleared devices: the capacity of a person to intervene has been checked at 56 of 717 radiology devices, as of June 2024. The same point from another angle — the rule requires the possibility of intervening, and does not require that anyone count whether intervention happened.

The question is put about the listener's own working loop, not about medicine: it requires no medical experience, and an automated element deciding something on a person's behalf exists today in almost every working loop. We wait for an answer out loud; if the room stays silent, the first check is named and a direct question is asked — who in the room has an independent source of truth in their loop, and for whom is the screen the only source there is.

In the photo, two people are reading a series on a workstation. The slide proves nothing and confirms not one number from the lecture: it sits here so the question gets asked to a scene of a person looking at a machine's output, not to an empty frame.

These three checks aren't about medicine, and they should be said exactly that way: any one of them applies to a loop at a bank, in manufacturing, in software development — anywhere a person reads a model's output and signs off on the result.

The first check is about an independent source of truth. The model's explanation is not a source of truth: it's produced by the same mechanism as the output itself, and is aligned with it by construction. An independent one is a different instrument, a different analysis, a different person, or the thing itself. A loop where the only way to check the output is the explanation attached to it is closed in on itself.

The second is about two countable rates. How often the person objects, and how often they object correctly. If neither is measured, then "the person checks" is no different, by any observable feature, from "the person signs," and there's nothing to argue about.

The third is about the physics of time. Seconds for the decision against the volume that needs reading. The median time checking a suggestion is about two seconds, and what's accepted without a prior check gets corrected afterward in 53 percent of cases against 18. If confirmation has no time on the schedule, there will be a signature and no reading, and no instruction will change that.

Ask the room the question and wait for an answer: which one check would you make mandatory before putting a model into your own loop. The answer comes within a few seconds, and it's almost always one of these three.

Illustration. Two people reading a series of scans from a workstation — Wikimedia Commons, The Medical Futurist, 2017, CC BY 4.0.